EDO Board Prep Study Guide

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1.1 Command Chains and Governance

1.1.1 Foundations and Goldwater-Nichols

Constitutional basis. The Constitution establishes three branches of government: Article I (Congress) authorizes raising and supporting armies, providing and maintaining a navy, and declaring war; Article II vests civilian control in the President as Commander in Chief; Article III creates the judiciary [1].

Pre-1986 context. Service leaders historically carried both force-provider and force-commander responsibilities. The National Security Act of 1947 formalized the Joint Chiefs and created the National Military Establishment, and the 1949 amendment renamed it the Department of War [1].

Goldwater-Nichols (1986). The Act strengthened the Chairman of the Joint Chiefs of Staff (CJCS) as the principal military advisor, increased Combatant Commander (CCDR) authority, and institutionalized jointness in operations, education, and budgeting [1, 2].

1.1.2 Operational vs. Administrative Chains

Operational chain (OPCON). The operational chain runs President → Secretary of War (SECWAR) → CCDRs; Operational Control (OPCON) authorities flow through that chain, and Service chiefs are not in the operational chain of command [3, 4].

Administrative chain (ADCON). Administrative Control (ADCON) aligns readiness and resourcing: Service Secretaries and chiefs *man, train, and equip* forces; for the Navy this runs through Secretary of the Navy (SECNAV) and Chief of Naval Operations (CNO) down to Systems Commands (SYSCOMs) and operating forces [1, 5].

Figure 1.1 shows the Title 10 chain of command. Figure 1.2 shows the Navy chain from the President to Echelon III commands.

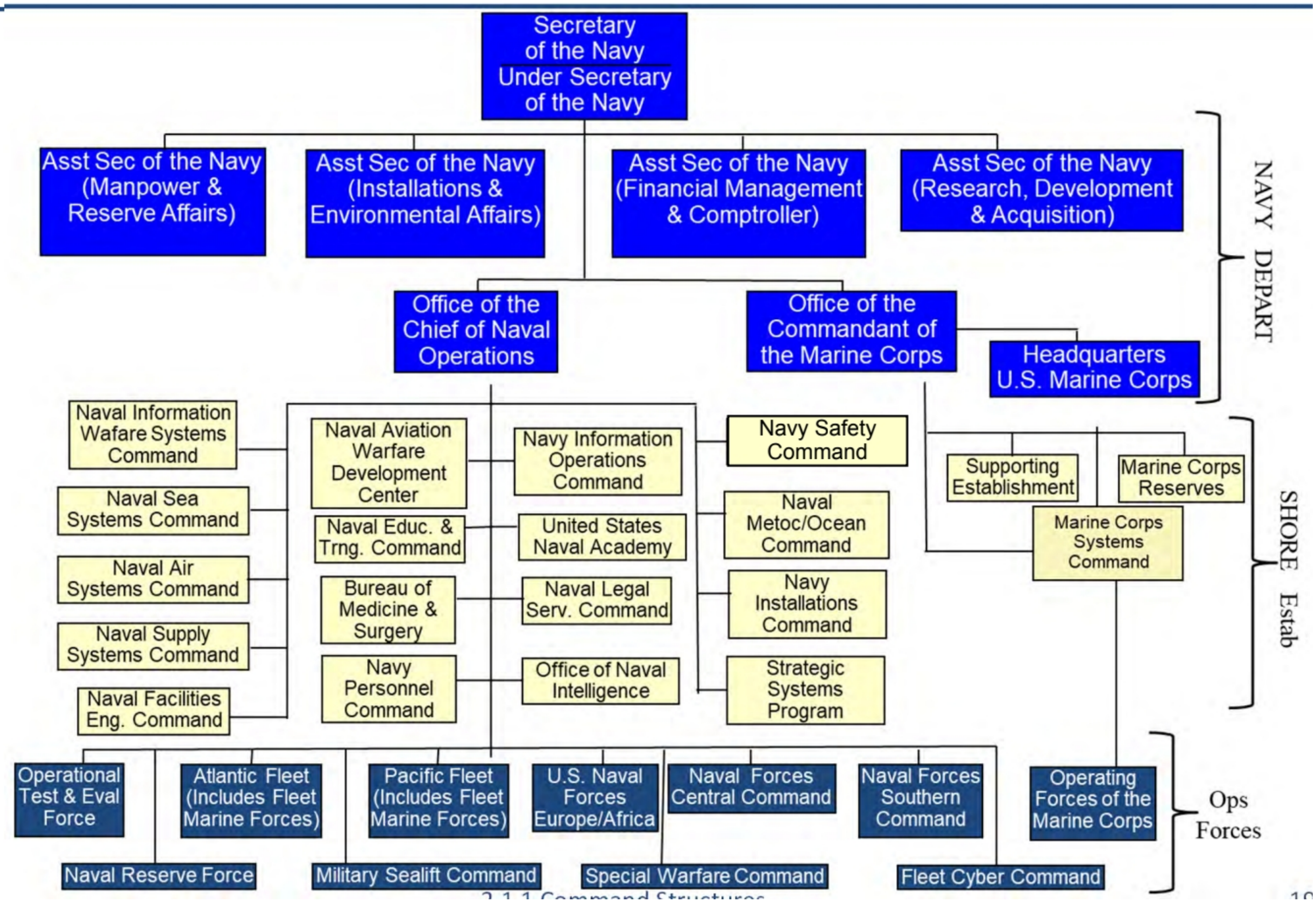

Figure 1.1. *Title 10 Chain of Command. Source:* 2.1.1. Command Structures, *2025* [1]

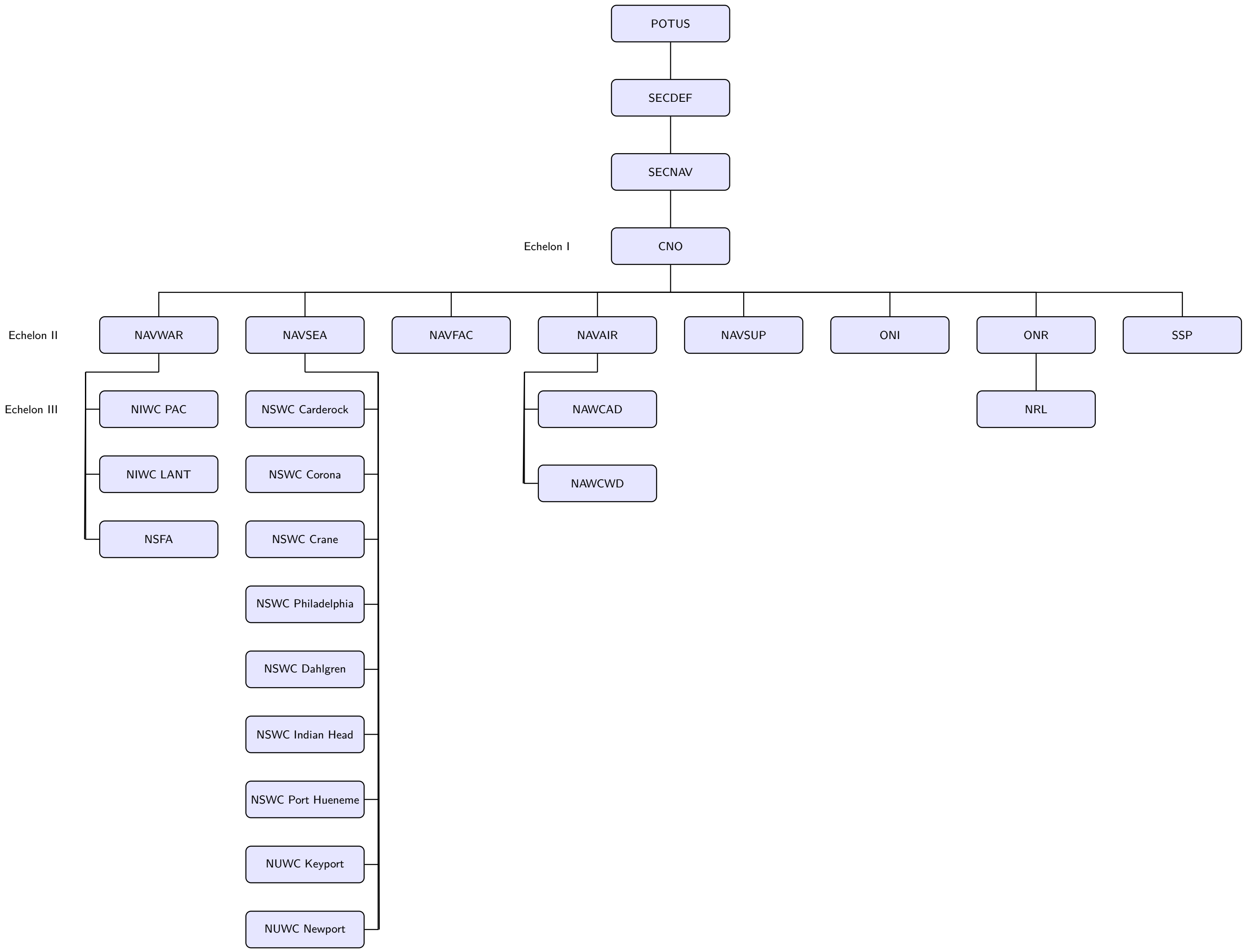

Figure 1.2. *Organization command structure from the President through Echelon III. Source:* 2.1.1. Command Structures, *2025 [1]*

1.1.3 DoW and DoN Components

DoW components. Core elements include the Office of the Secretary of War (OSW), Joint Staff, Military Departments, Unified Combatant Commands (CCMDs), Defense Agencies (e.g., Missile Defense Agency (MDA)), and Department of War (DoW) Field Activities [1].

DoN components. The Department of the Navy (DON) includes the Secretariat (SECNAV and principal advisors), CNO/Office of the Chief of Naval Operations (OPNAV) staff, the Shore Establishment (SYSCOMs and field activities), and the Operating Forces [1].

1.1.4 Strategic Direction (NSS and NDS)

The National Security Strategy (NSS) sets national-level priorities; the National Defense Strategy (NDS) translates them into defense priorities, posture, and force-development guidance. The 2022 NDS emphasizes defending the homeland, deterring strategic attacks, deterring aggression while prepared to prevail, and building a resilient joint force and defense ecosystem; it advances these priorities through integrated deterrence, campaigning, and building enduring advantage. Current as of 2026-01-11 [6, 7].

1.1.5 SECNAV Principal Advisors

Table I: SECNAV principal advisors

Office	Focus and responsibilities
Under Secretary of the Navy	Leads the Secretariat and serves as the principal advisor to the Secretary of the Navy for all Department of the Navy matters.
Assistant Secretary of the Navy (Manpower and Reserve Affairs)	Manpower policy, personnel readiness, quality of life, and workforce programs across the Navy and Marine Corps.
Assistant Secretary of the Navy (Installations and Environment)	Real property, housing, facilities, environmental protection, and BRAC processes.
Assistant Secretary of the Navy (Financial Management and Comptroller)	Budget formulation and execution; principal financial advisor and comptroller for the Department of the Navy.
Assistant Secretary of the Navy (Research, Development and Acquisition)	Acquisition policy, program oversight, and life-cycle management for Navy and Marine Corps systems.

Source: *2.1.1. Command Structures*, 2025 [1]

The Assistant Secretary of the Navy for Research, Development and Acquisition (ASN(RD&A)) is the DON Component Acquisition Executive (CAE) and the Navy Acquisition Executive (NAE), with authority and accountability for acquisition policy and program oversight [8].

1.1.6 Acquisition Governance (Department of the Navy)

Policy authority. ASN(RD&A) is the Service Acquisition Executive (SAE) for the DON and sets policy for Program Executive Offices (PEOs) and Direct Reporting Program Managers (DRPMs) [8, 9].

Current portfolio authority. Since Dec 2025 and the 2026 DON acquisition restructuring, Portfolio Acquisition Executives (PAEs) are the board-relevant portfolio layer under ASN(RD&A). Public DON releases describe PAEs as accountable officials for whole portfolios, with authority over program offices and associated technical, contracting, sustainment, and industrial-base functions; about 70 percent of those associated functions and personnel are moving from SYSCOMs into PAEs as the construct matures [10, 11].

Matrix execution. The older PEO/SYSCOM matrix remains important for board recall and for live execution during transition: PEOs and Program Managers (PMs) still execute named programs where public pages have not fully caught up; SYSCOMs (Naval Sea Systems Command (NAVSEA), Naval Information Warfare Systems Command (NAVWAR), Naval Air Systems Command (NAVAIR)) still provide technical authority, engineering standards, warfare-center depth, contracting infrastructure, and sustainment support. The PAE construct changes accountability and alignment; it does not give a PM permission to ignore Technical Authority (TA) or contracting authority [8, 10, 11].

Flag/PEO billets (current).

EDO-led today: ASN(RD&A)'s Principal Military Deputy (PMILDEP) (Vice Admiral (VADM) Seiko Okano), Commander, Naval Information Warfare Systems Command (COMNAVWAR) (Rear Admiral (Upper Half) (RADM) J. Kurt Rothenhaus), Program Executive Office, Ships (PEO Ships) (Rear Admiral (Lower Half) (RDML) Brian Metcalf), Program Executive Office, Aircraft Carriers (PEO CVN) (RADM Casey Moton), and Program Executive Office, Attack Submarines (PEO SSN) (RADM Jonathan Rucker) anchor TA and acquisition governance for the community [12, 13, 14, 15, 16, 17].

Non-EDO leaders (awareness): NAVSEA 08 is currently Admiral (ADM) William J. Houston (Submarine officer), Program Executive Office, Integrated Warfare Systems (PEO IWS) is RDML Thomas J. Dickinson (Surface Warfare Acquisition Professional), Program Executive Office, Manpower, Logistics and Business IT (PEO MLB) is RDML Kevin P. Biehn (Surface Warfare Acquisition Professional), NAVSEA 05 is RDML Jay A. Seif (Submarine Acquisition Professional), and Program Executive Office, Unmanned and Small Combatants (PEO USC) is filled by an Senior Executive Service (SES) (formerly RDML Kevin R. Smith) [18].

(Appendix C.2: See Appendix C.2 for the current EDO flag billet roster.)

Figure 1.3 shows the legacy Navy acquisition governance chain of command. Use it to understand the PEO/SYSCOM matrix, then apply the PAE update in Figure 1.4 and Table III.

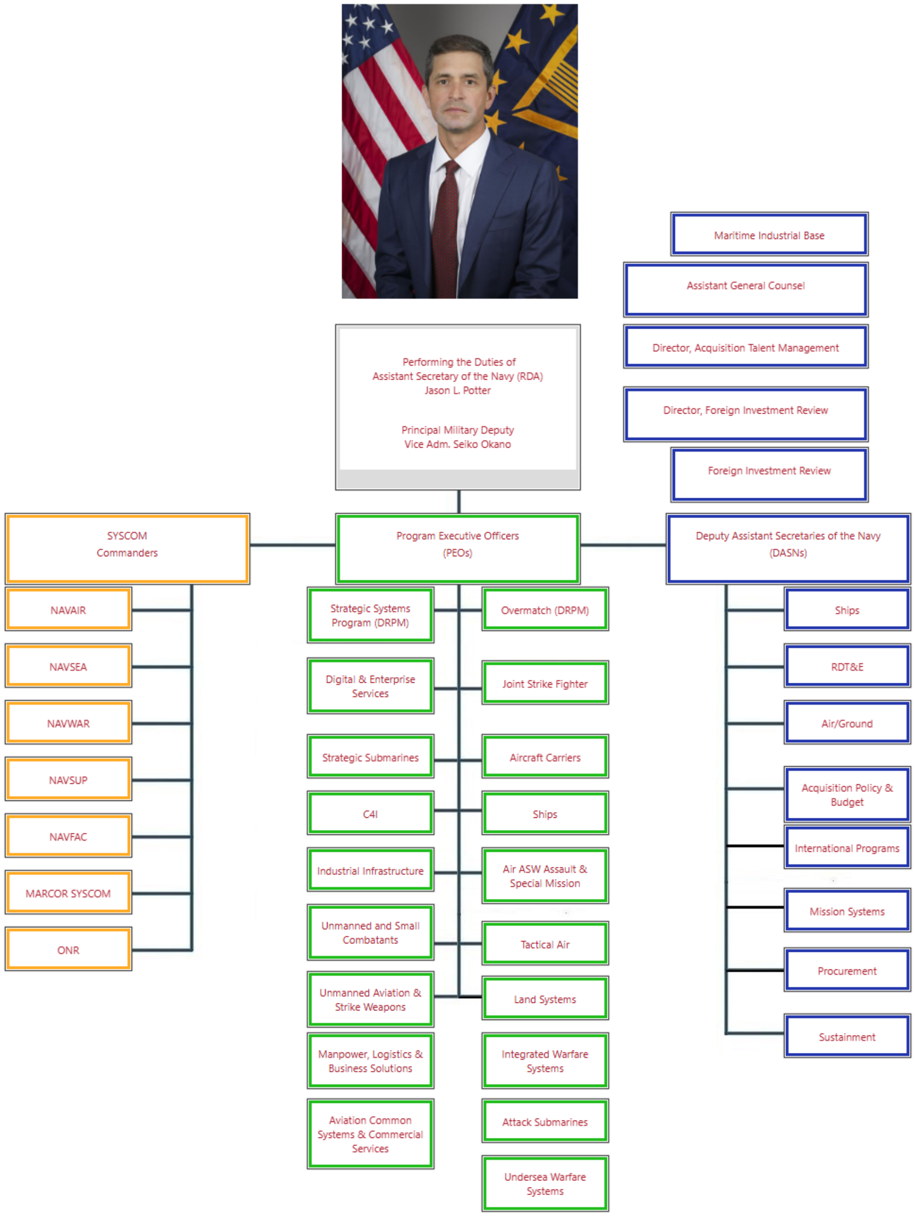

Figure 1.3. *Acquisition Governance Chain of Command. Source:* 2.1.1. Command Structures, *2025 [1]*

1.1.7 ASN (RD&A) Organization and Portfolio Acquisition Executives

Current as of 2026-06-04, Mr. William F. Mahan is performing the duties of ASN(RD&A) and serves as the senior acquisition executive for the Navy and Marine Corps. Mr. Jason L. Potter returned to Principal Civilian Deputy ASN(RD&A), and PMILDEP VADM Seiko Okano remains the senior uniformed acquisition deputy supporting DON governance [19]. Figure 1.4 captures the board-level PAE organization, Table II captures top roles, and Table III lists the publicly announced PAEs.

PAE purpose. PAEs align authority with responsibility for outcomes across like portfolios. The reform target is faster delivery, fewer non-value-added reviews, disciplined cost/schedule/performance trades, direct industrial-base management, and cradle-to-grave control across program, technical, contracting, and sustainment functions [10, 11].

Board answer. Draw SECNAV at the top, ASN(RD&A) beneath SECNAV, Principal Civilian Deputy and PMILDEP directly under ASN(RD&A), and the PAEs as the current acquisition portfolio owners. Keep CNO and Commandant of the Marine Corps (CMC) in the service chain for requirements/readiness and keep SYSCOMs as the technical, contracting, sustainment, and workforce infrastructure that is being realigned into or in support of the PAEs.

Transition caution. The user's PAE organization chart is a useful transition snapshot, but it predates the May 2026 public announcement of PAE Aviation, PAE Mission Systems, and PAE Munitions. Use the current count of nine PAEs unless a newer official chart supersedes it [11].

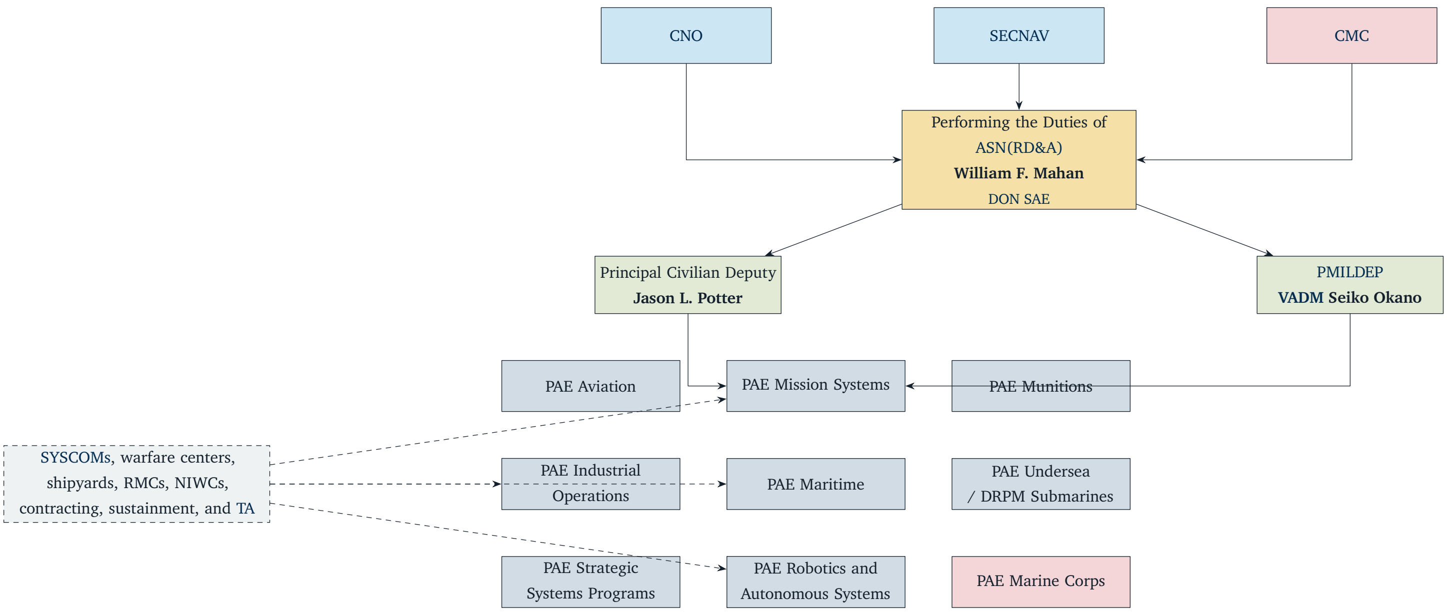

Figure 1.4. *Board-level ASN (RD&A) and PAE organization transition, updated from the user-provided PAE chart with public announcements through 2026-06-04 [10, 11, 19].*

Table II lists the current top ASN(RD&A) roles.

TABLE II: ASN (RD&A) LEADERSHIP ROLES

Role	Person	Roles and responsibilities
PTDO ASN (RD&A)	William F. Mahan	Serves as the DON senior acquisition executive for the Navy and Marine Corps while performing the duties of ASN (RD&A).
Principal Civilian Deputy	Jason L. Potter	Provides senior civilian acquisition continuity and returned to this role after serving as PTDO ASN (RD&A) from July 2025 to May 2026.

Continued on next page

Table II: ASN (RD&A) leadership roles **(Continued)**

Role	Person	Roles and responsibilities
PMILDEP	VADM Seiko Okano	Provides senior military advice to ASN (RD&A), helping manage DON research, development, acquisition governance, and the operational perspective across the acquisition enterprise.

Source: *Department of the Navy Names New Service Acquisition Executive*, 2026 [19]

Table III summarizes the public PAE portfolio map. Use Appendix B.2 for detailed PAE-to-legacy-organization mapping.

Table III: Current public PAE roster and board cues

PAE	Public leader / cue	Board-level scope
Aviation	VADM John Dougherty	Naval aviation portfolio; public deputy lanes are Carrier Strike, Marine Corps Aviation, and Maritime ISR/NC3.
Industrial Operations	VADM James P. Downey	Industrial capacity, naval shipyards, regional maintenance, maintenance contracting, and industrial-base throughput.
Marine Corps	LtGen Eric Austin	Marine Corps ground systems; combines former PEO Land Systems and Marine Corps Systems Command acquisition support functions.
Maritime	Christopher Miller	Conventional surface-ship delivery, including aircraft carriers, combatants, expeditionary and mine warfare, auxiliaries, and modernization/sustainment.
Mission Systems	Jim Day	Central mission integrator for kill-chain closure across PEO C4I, PEO Digital, PEO IWS, PEO MLB, DRPM Overmatch, DRPM LRNFO, DRPM MILE, Minotaur, and mission-system elements across SYSCOMs.
Munitions	Paul Mann	Munitions production and weapons portfolio; public pillars are Weapons Industrial Base, Air Weapons, Surface Weapons, and Advanced Capabilities and Innovation.
Robotics and Autonomous Systems	Public leader not confirmed in this refresh	Hedge and unmanned/autonomous capabilities; current public example is MUSV marketplace at-sea demonstrations.
Strategic Systems Programs	VADM Johnny Wolfe	Sea-based strategic deterrence and strategic weapons; SSP aligned inside the PAE construct.
Undersea / DRPM Submarines	VADM Robert Gaucher	Submarine and undersea delivery; dual-hatted DRPM construct focused on SSN, SSBN, and SSGN production and undersea industrial-base execution.

Source: *Navy Reshapes Warfighting Acquisition System*, *Navy Advances Acquisition Reform Strategy: Appoints Three New Portfolio Acquisition Executives*, *Portfolio Acquisition Executive Maritime*, *Department of the Navy Launches PAE Mission Systems to Accelerate Warfighting Capability at Speed*, *Navy Establishes Portfolio Acquisition Executive for Munitions*, *Navy Establishes Portfolio Acquisition Executive for Aviation*, *About Us*, *U.S. Navy Announces Seven Companies Selected for MUSV Marketplace At-Sea Demonstrations*, 2026, 2026, 2026, 2026, 2026, 2026, 2026, 2026 [10, 11, 20, 21, 22, 23, 24, 25]

1.1.8 OPNAV Organization and CNO Principal Assistants

OPNAV role. OPNAV is the CNO's staff within the administrative chain, responsible for translating naval policy, requirements, and resourcing guidance into executable direction across the Fleet and Shore Establishment [1, 5].

Table IV: CNO principal assistants (N-codes)

N-code	Primary responsibilities
N1	Manpower, personnel, training, and education policy and readiness.
N2/N6	Information warfare, intelligence, cyber, command and control, and related information capabilities.
N3/N5	Operations, plans, and strategy; joint operations and campaign planning.
N4	Readiness and logistics, including supply, ordnance, energy, and mobility.
N7	Warfighting development, concepts, and integration.
N8	Integration of capabilities and resources; PPBE resourcing and assessment.
N9	Warfare systems requirements and resourcing across major warfighting domains.

Source: *2.1.1. Command Structures*, 2025 [1]

1.1.9 Fleet Forces, TYCOMs, and Numbered Fleets

USFF. U.S. Fleet Forces (USFF) executes the Prepare the Force and Provide the Force functions, generating ready forces for CNO and CCDRs, and delivering trained, equipped forces to operational commanders [1, 26].

TYCOMs. Type Commanders (TYCOMs) supervise specific categories of forces and balance readiness, manning, training, and equipment; they are in the administrative chain of command [1].

Numbered Fleets. Numbered Fleet Commanders report to the appropriate CCDRs; 2nd, 3rd, and 4th Fleets are home-ported in the U.S., 5th, 6th, and 7th Fleets are forward-based, and 10th Fleet operates in cyberspace [1].

1.1.10 Echelons and Enterprise Construct

Echelons. The CNO is Echelon I; commands reporting to the CNO are Echelon II; commands reporting to Echelon II are Echelon III, and so on [1].

Enterprise construct. Requirements are set by warfighters; providers (PEOs, SYSCOMs, and warfare centers) design, acquire, build, and maintain; resources are programmed and budgeted to deliver the capability [1].

1.1.11 Approving Authorities

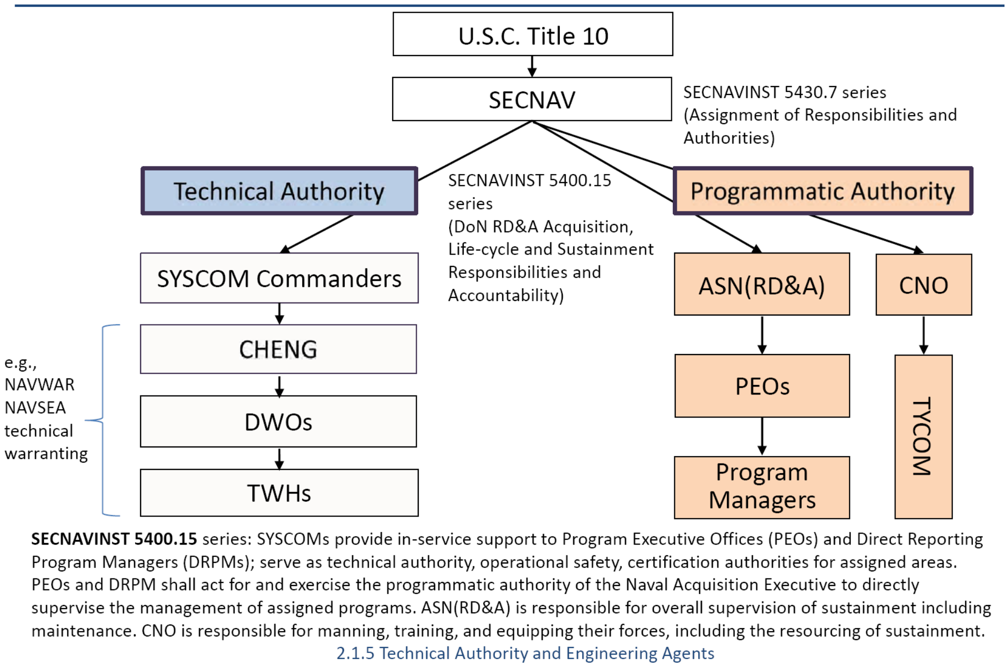

Figure 1.5. *Technical Authority vs. Program Authority alignment. Source:* 2.1.5. Technical Authority and Engineering Agents, *2025 [27]*

Figure 1.5 contrasts the roles of TA and Program Authority (PA) in Navy acquisition.

1.1.12 Coursebook cue: EDO flag billets

Coursebook 2.1.1 asks students to identify the EDO flag billets by name and responsibility. Use Appendix C.2 as the live roster for that board cue because the coursebook roster is a date-stamped training snapshot, while the appendix is maintained against public Navy flag-officer sources [1, 28].

1.2 Technical Authority and Engineering Agents

1.2.1 Definition of technical authority

Definition. TA is the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance with higher authority policy, requirements, architectures, and standards [27].

Process focus. The exercise of TA establishes and assures adherence to technical standards and policies, providing a range of technically acceptable alternatives with risk and value assessments [27].

1.2.2 Ultimate technical authority

Inherently governmental. Technical authority is an inherently governmental function assigned to SYSCOM commanders by SECNAV [27].

Ships and most weapons systems. Commander, NAVSEA (COMNAVSEA) is the ultimate technical authority for ships and most weapons systems [27].

C4ISR, cybersecurity, and IT. Commander, NAVWAR (COMNAVWAR) is the ultimate technical authority for Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance (C4ISR), cybersecurity, and Information Technology (IT) systems [27].

Policy basis. Title 10 and SECNAVINST 5400.15 series underpin the technical authority structure [27].

1.2.3 Responsibilities of technical warrant holders

Responsibilities of Technical Warrant Holders (TWHs) include the following technical competencies [27].

Set technical standards. Create standards and tailor them to specific program applications while balancing requirements with intended missions.

Technical area expertise. Provide expert testimony (risk assessments, technology readiness evaluations, critiques, mishap investigations, trouble reports, test plans, and test evaluation reviews).

Safe and reliable operations. Ensure safety and reliability are addressed in technical documentation and act as certification authority when required.

Effective systems engineering. Ensure engineering and technical products meet Navy needs, considering full life-cycle and system-of-systems relationships.

Unbiased technical judgment. Provide leadership and accountability for engineering decisions; promote communication so technical requirements, issues, and risks are identified early.

Stewardship. Forecast, shape, and develop the capability and capacity of engineering support networks.

Accountability and integrity. Balance speed to decision, expertise of empowered organizations, and the need to elevate significant issues.

1.2.4 Technical authority interfaces

Risk discussion. Technical decisions are part of the cost, schedule, and performance risk discussion [27].

Stakeholder balance. Stakeholders (Program Management Office (PMO), Naval Shipyard (NSY), Fleet) balance risk with technical requirements [27].

Recommendations. Technical authorities assess risk and provide recommendations and acceptable alternatives to programmatic authorities when operational or programmatic challenges arise [27].

1.2.5 Need for technical authority

USS Dolphin (AGSS-555) case

Incident. On 21 May 2002, USS Dolphin was operating 100 miles off San Diego when the ship began flooding while surfaced to recharge the battery; fires broke out and most of the crew prepared to abandon ship, but two crew members prevented total loss [27].

Technical risk. A black D-shaped sponge gasket was substituted for the specified red silicone without technical authority authorization; the Mk 54 shield side door opened in heavy seas, water entered the submarine, flooded spaces, and power was lost for de-watering [27].

Why technical authority is required

Disciplined processes. Develop and employ consistent, disciplined, collaborative engineering processes that provide safe, reliable, effective, integrated, timely, and affordable products for the Navy [27].

Standards and tools. Establish, monitor, and approve standards, tools, and processes to provide risk-based solutions to the Fleet [27].

Core process oversight. Oversee technology development, systems engineering, cost estimating, and test and evaluation [27].

Technical infrastructure. Operate and sustain the most efficient technical infrastructure to support acquisition, fielding, and in-service support [27].

Senior support. Support ASN(RD&A) and CNO analysis of mission areas, systems, and requirements [27].

1.2.6 Technical authorities and warrant holders

Technical authorities

Chief Systems Engineer (CSE). Facilitates, collaborates, and coordinates with Technical Domain Managers within their warranted area to ensure proper oversight of technical authority (for example, ballistic missile defense, C4ISR, space) [27].

Technical Domain Manager. Provides deep technical expertise to programs and activities (marine engineering, structures, fluid mechanics) [27].

Senior Technical Authority (STA). Certifies sufficient risk mitigation and maturity for critical systems to proceed past Milestone B; Title 10 USC §8669b requires SECNAV to establish an Senior Technical Authority (STA) for each naval vessel class; STAs are dual-hatted as Chief Systems Engineer (CSE) Deputy Warranting Officers [27].

Technical warrant holders

Ship Design Manager (SDM). Manages systems engineering and design efforts for assigned platforms (in-service, new construction, future design) [27].

Systems Integration Manager (SIM). Assists Ship Design Managers (SDMs) in systems engineering integration of complex warfare systems into platforms (for example, ASW or DDG 51/CG 47 combat systems) [27].

Cost Engineering Manager (CEM). Ensures independent cost engineering and estimating for Navy programs (CVN, submarines, surface platforms) [27].

Technical Area Expert (TAE). Navy expert in the assigned product technical area (propulsion, missiles, shock, combat systems) [27].

Technical Process Owner (TPO). Provides definition and documentation for assigned technical processes [27].

Waterfront Chief Engineer (WFCHENG). Leads and focuses technical efforts of SYSCOMs from the waterfront (NSY, regional maintenance centers, Supervisor of Shipbuildings (SUPSHIPs), NAVWAR) [27].

1.2.7 NAVSEA warranting structure (overview)

Leadership. COMNAVSEA; NAVSEA Chief Engineer (SEA 05); Deputy Warranting Officers [27].

Deputy Warranting Officer domains. Cost Engineering and Industrial Analysis (SEA 05C); Explosive Ordnance Engineering (SEA 05E); Warfare Systems Engineering, Littoral and Mine Warfare (SEA 05M); Surface Ship Design and Systems Engineering (SEA 05D); Integrated Warfare Systems Engineering (SEA 05H); Undersea Warfare Systems Engineering (SEA 05N); Ship Integrity and Performance Engineering (SEA 05P); Readiness and Logistics (SEA 05R); Submarine and Submersible Design and Systems Engineering (SEA 05U); Aircraft Carrier Design and Systems Engineering (SEA 05V); Surface Warfare Systems Engineering (SEA 05W); Marine Engineering (SEA 05Z) [27].

Warrant holders. See Section 1.2.6 [27].

Engineering agents. Engineering agents execute delegated technical tasks in support of warrant holders [27].

1.2.8 Resolving waterfront technical issues

TA Principles. Technical authority is exercised independently of programmatic cost, schedule, and performance constraints. It provides safe, reliable, and standardized solutions, using risk-informed engineering judgments to support the fleet.

DFS. When a repair or modification departs from specifications, drawings, or standards, a Departure from Specification (DFS) must be submitted. The DFS is routed through the Lead Technical Authority (LTA) for technical review and disposition, and is tracked in the ship's Current Ship's Maintenance Project (CSMP) [27].

Escalation Chain for Repairs. If a waterfront technical issue or DFS cannot be resolved locally, or exceeds local authority limits, it is elevated through the following chain:

1. **Waterfront Engineer / Project Team**: Identifies the deviation or issue.

2. **Local Technical Authority (LTA) / Waterfront Chief Engineer (WFCHENG)**: Adjudicates within delegated local limits (e.g., at the Shipyard, RMC, or SUPSHIP).

3. **Technical Warrant Holder (TWH)** (e.g., Ship Design Manager (SDM) or Technical Area Expert (TAE) at NAVSEA 05): Resolves issues crossing domain, design, or platform boundaries.

4. **Deputy Warranting Officer (DWO)**: Reviews issues that cut across domains or have significant program impact.

5. **NAVSEA Chief Engineer (CHENG, SEA 05)**: Resolves high-level disputes and sets overall engineering policy.

6. **Commander, NAVSEA (COMNAVSEA)**: Serves as the ultimate technical authority for ships and weapons systems.

Programmatic vs. Technical Disputes. Program Managers (PMs/PEOs) possess programmatic authority (cost, schedule, performance), while SYSCOMs/TWHs possess technical authority (safety, standards). A PM cannot override a technical authority's disapproval of a deviation. Disagreements must be elevated through their respective chains to the PEO and COMNAVSEA level, and ultimately to the ASN(RD&A) for final adjudication.

Get to yes safely. Avoid saying no when possible, avoid analysis paralysis, and use engineering judgment when data is not available; clearly explain why an operational restriction is needed [27].

Underway authority. The ship's Commanding Officer (CO) retains technical authority while underway to support mission accomplishment and survivability [27].

1.2.9 Approving authorities

Policy chain. U.S.C. Title 10, SECNAVINST 5430.7 series, and SECNAVINST 5400.15 series define assignment of responsibilities and authorities [27].

SYSCOM role. SYSCOMs provide in-service support to PEOs and direct reporting program managers; they serve as technical authority and operational safety and certification authorities for assigned areas [27].

Programmatic authority. PEOs and direct reporting program managers exercise programmatic authority for assigned programs on behalf of the NAE [27].

Sustainment oversight. ASN(RD&A) is responsible for overall supervision of sustainment; CNO is responsible for manning, training, and equipping forces, including sustainment resourcing [27].

1.2.10 Engineering Agents (EAs)

Definition. Engineering Agents (EAs) are organizations (government or private contractor) with responsibilities delegated by assigning documents and tasking for functions within their assigned technical area, system, or mission [27].

Lifecycle support. EAs provide technical services to TWHs, program managers, and Fleet customers throughout a system's life-cycle [27].

1.2.11 Types of Engineering Agents

Five types. Technical Direction Agent (TDA), Design Agent (DA), Systems Integration Agent (SIA), Alteration Engineering Agent (AEA), and In-Service Engineering Agent (ISEA) [27].

Lifecycle coverage. Engineering agents exercise acquisition responsibilities across the acquisition life-cycle [27].

1.2.12 Technical Direction Agent (TDA)

Responsibilities. Develop and analyze engineering concepts; develop system specifications; evaluate design agent proposals; assist in developing the Test and Evaluation Master Plan (TEMP) [27].

Purpose. Assist in establishing initial program concepts, performance specifications, Research, Development, Test, and Evaluation (RDT&E), and alternative technical approaches [27].

1.2.13 Design Agent (DA)

Responsibilities. Develop initial and complete engineering design and prototypes; prepare technical data; prepare maintenance support documentation; review engineering change proposals and waivers or deviations for impact on design, performance, safety, producibility, maintainability, life-cycle cost, training, and interfaces [27].

Purpose. Translate performance specifications into engineering specifications for the ship and its systems [27].

1.2.14 System Integration Agent (SIA)

Responsibilities. Ensure equipment and software compatibility; advise the system manager of interface problems; coordinate with other system managers; support TDA in Test and Evaluation (T&E) as required; review engineering change proposals for impact on interfaces [27].

Purpose. Ensure compatibility of all elements that make up a system and its interface requirements [27].

1.2.15 Acquisition Engineering Agent (AEA)

Responsibilities. Review technical manuals and Maintenance Requirement Cards (MRCs) for accuracy; analyze and report production costs and issues; assist in developing maintenance concepts and criteria for all levels of maintenance [27].

Purpose. Provide logistics support prior to production or major modification improvement programs [27].

1.2.16 In-Service Engineering Agent (ISEA)

Responsibilities. Support and provide life-cycle maintenance; recommend corrections to deficiencies in all areas of life-cycle support; manage configuration of equipment; provide engineering assistance to the Fleet [27].

Purpose. Delegated functions for overall engineering, test, maintenance, and logistics of in-service equipment [27].

1.3 NAVSEA Organization and Warfare Centers

1.3.1 Mission, priorities, and vision

Mission. Design, build, deliver, and maintain ships, submarines, and systems reliably, on time, and on cost for the United States Navy. NAVSEA is the force behind the fleet, with the primary mission of life-cycle engineering of ships and ship systems [29].

Mission priorities. Deliver combat power, transform digital capabilities, and build a team to compete and win [29].

Vision. "Expand the advantage" by ensuring the United States Navy can protect and defend America, operating with a modern industrial base, serving as a world-class employer of choice, and setting the value-added standard for acquisition, engineering, business, and maintenance [29].

1.3.2 Major functions and scope

Major functions. Oversee core processes for acquisition, in-service support, weapon systems, IT, engineering systems, cybersecurity systems, demilitarization, and disposal. Core functions include cost estimating, technical development and readiness assessment, systems engineering, manufacturing, integrated logistics, maintenance and modernization, and comptroller/legal/contracting support services [29].

Assigned ships and systems. Surface ships, aircraft carriers, submarines, submersibles, Hull, Mechanical, and Electrical (HM&E) systems, combat systems, ordnance, chemical/biological/radiological defense material, and towing/salvage equipage [29].

Technical authority. Exercises technical and certification authority across ships and ship systems [29].

1.3.3 Corporate overview

Organization. Headquarters command staff and directorates at Washington Navy Yard provide policy, guidance, oversight, and support across financial management, contracting, logistics/maintenance/industrial operations, engineering, undersea warfare, corporate operations, IT, legal support, and security. The coursebook snapshot lists 42 field activities and 30+ detachments (naval shipyards, regional maintenance centers, SUPSHIPs, warfare centers, and other field activities); verify exact counts against current NAVSEA public pages before using them as live figures [29].

Budget. Coursebook snapshot: approximately $64B (Fiscal Year (FY) 2023), about one-fourth of the Navy budget; verify current budget figures before quoting them as live data [29].

People. Coursebook snapshot: 86,886 civilian and military personnel; verify current manning before quoting it as live data [29].

Acquisition scope. Coursebook snapshot: 150+ acquisition programs across NAVSEA and seven affiliated PEOs; verify current program counts before quoting them as live data [29].

1.3.4 Organizational management approaches

Functional organization. Expertise grouped by discipline (e.g., engineering, contracting, logistics), emphasizing depth and standardization [29].

Product/project organization. Resources organized around a specific platform or product line to drive ownership, schedule focus, and rapid decisions [29].

Matrix organization. Functional resources support projects as needed; NAVSEA and affiliated PEOs operate as a matrix to balance technical authority with program execution [29].

1.3.5 Program management relationships

PEO reporting. PEOs report to the ASN(RD&A) for acquisition matters and to Commander, Naval Sea Systems Command (COMNAVSEA) for in-service support and modernization [29].

SPM and PARM. The Ship Program Manager (SPM) owns platform acquisition and life-cycle management; the Participating Acquisition Resource Manager (PARM) manages acquisition and life-cycle support for major subsystems (e.g., combat systems) [29].

SHAPM and SLM. Ship Acquisition Program Manager (SHAPM) focuses on acquisition for the ship/submarine platform; Ship Lifecycle Manager (SLM) manages in-service life-cycle responsibilities for the platform [29].

SPD. The Ship Project Directive (SPD) is the contract-like agreement between the platform manager (SPM or SHAPM/SLM) and the PARM; it may be fulfilled by a 7300 document [29].

1.3.6 Directorates and major commands

Table V summarizes the major NAVSEA directorates and commands shown on the December 2025 organization chart.

Table V: NAVSEA directorates and major commands.

Directorate or command	Core functions
SEA 01, Comptroller	Provides financial management accountability for NAVSEA, executes 1301/1517 authorities, manages the chart of accounts and allocations for headquarters and PEO appropriated funds, serves as the BSO, and owns enterprise financial management processes and closeout actions.
SEA 02, Contracts	Serves as contracting officer support to NAVSEA, PEOs, and PMs, including new construction (SEA 022), HME/ocean engineering and regional maintenance (SEA 024), surface ship systems (SEA 025), and undersea, electronic surveillance, and special operations (SEA 026).
SEA 03, Cyber Engineering and Digital Transformation	NAVSEA cybersecurity technical authority, with NAVWAR as lead SYSCOM; provides infrastructure and support services for programs and developers, including cybersecurity guidance, IT services, and digital transformation across the enterprise.

Continued on next page

Table V: NAVSEA directorates and major commands. **(Continued)**

Directorate or command	Core functions
SEA 04, Industrial Operations	Exercises management control of naval shipyards, sets NAVSEA logistics policy, provides programmatic logistics support to PMs, PEOs, and DRPMs, manages logistics RDT&E, and audits quality assurance performance. SUPSHIPs execute shipbuilding contracts on-site, overseeing cost, schedule, and quality while providing contracting, engineering, quality assurance, and resource management services.
SEA 05, Naval Systems Engineering	Provides technical authority for ship design, integration, certification, and systems engineering; resolves cross-platform performance issues; leads concept development and technology insertion strategy; and manages the engineering and science workforce across NAVSEA.
SEA 06, Sustainment	Reestablished in 2024 to consolidate logistics and sustainment support functions and improve fleet readiness through integrated sustainment policy and execution.
SEA 07, Undersea Warfare	Dual-hatted as PEO SSBN; provides life-cycle support to in-service submarine and undersea forces, including SUBSAFE and deep submergence systems (PMS 07Q) and logistics and product support management (PMS 07L).
Team Submarines	A submarine-centric enterprise that unites research, acquisition, and life-cycle support across PEO SSN, PEO SSBN, PEO UWS, and related integration offices, including the AUKUS direct reporting program office, to remove stovepipes and improve submarine force readiness.
SEA 09, Safety and Regulatory Compliance	Consolidates explosives safety, weapon systems safety, industrial and occupational safety, environmental compliance, and radiological controls, excluding Naval Reactors systems, to align safety oversight and reporting.
SEA 10, Total Force and Corporate Operations	Provides enterprise operations support for NAVSEA directorates, field activities, and PEOs, including manpower and budget management, corporate planning, acquisition support, foreign military sales and security assistance, human resources and equal employment opportunity, facilities, directives, security, and customer services.
SEA 21, Surface Ship Maintenance, Modernization and Sustainment	Manages in-service surface ship modernization and sustainment for non-nuclear ships, including PMS 321, PMS 326, PMS 339, PMS 421, PMS 443, PMS 451, SEA 21I, and the Surface Maintenance Engineering Planning Program activity.

Source: *2.1.2. NAVSEA Organization*, 2025 [29]

Note

Directorate names and alignments are subject to change; use the official NAVSEA organization chart for the current structure.

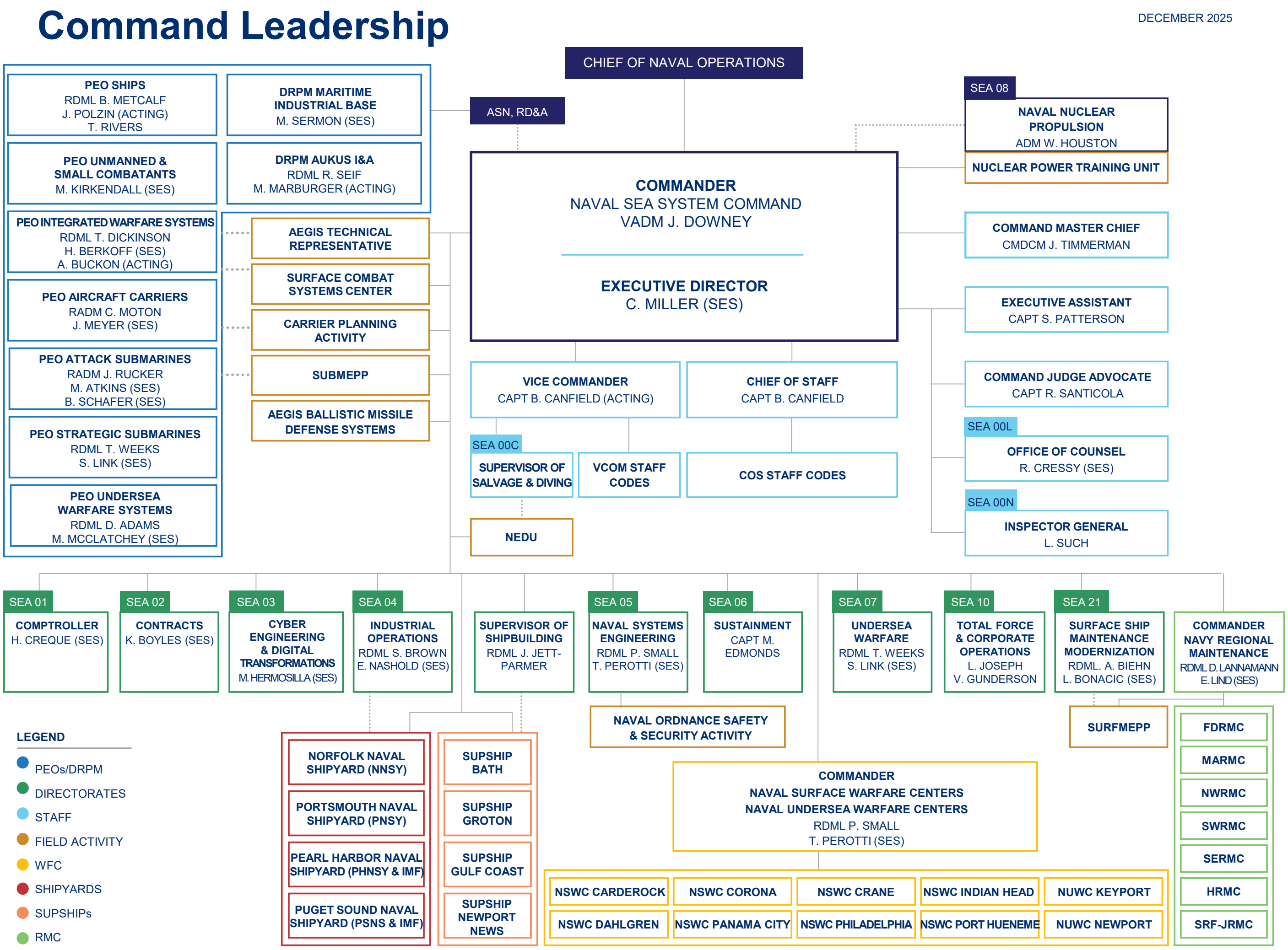

Figure 1.6. *NAVSEA Organization Chart, December 2025. Source:* NAVSEA Organization Chart (December 2025), *2025 [18]*

1.3.7 Program Executive Offices (PEOs) affiliated with NAVSEA

PEO (role). Directs multiple acquisition programs and major system portfolios; reports to ASN(RD&A) for acquisition matters and to COMNAVSEA for in-service support. PMs report to the PEO and Milestone Decision Authority (MDA) for program execution [29].

PEO Ships. Manages acquisition for surface combatants (destroyers and next-generation destroyer programs), amphibious ships (Landing Helicopter Dock (LHD)/Landing Helicopter Assault (LHA)/Landing Platform Dock (LPD)), and auxiliary/support ships, plus related activities including Foreign Military Sales (FMS) and the Electric Ships Office [29].

PEO USC. Delivers unmanned maritime systems, mine warfare systems, and small surface combatants such as Littoral Combat Ship (LCS) and the Constellation-class frigate, emphasizing rapid delivery to the fleet [29].

PEO SSN. Focal point for attack submarine design, construction, and in-service support (e.g., Virginia-class new construction and in-service programs, advanced undersea systems, and Submarine Maintenance Engineering Planning Program (SUB-MEPP)) [29].

PEO SSBN. Manages Columbia-class ballistic missile submarine design, construction, and in-service support, plus Ohio-class service life extension [29].

PEO UWS. Oversees submarine combat, weapons, acoustic, electromagnetic, surveillance, communications, training, and safety systems to improve undersea warfighting and cybersecurity across submarine platforms [29].

PEO IWS. Responsible for surface ship combat systems (combat management, sensors, missiles, launchers, electronic warfare, gun systems), open-architecture integration, and field activities such as the Aegis Technical Representative and Surface Combat Systems Center [29].

PEO CVN. Manages carrier RDT&E, design, construction, midlife refueling/overhauls, and life-cycle support, including PMS 312 (in-service), PMS 378 (CVN 78), and PMS 379 (CVN 79/80) [29].

Note

Current as of 2026-06-04: the DON publicly established PAE Mission Systems and PAE Munitions on 11 May 2026. Mission Systems pulls together PEO IWS, PEO C4I, PEO Digital, PEO MLB, DRPM Overmatch, DRPM LRNFO, DRPM MILE, Minotaur from PMA-290, and mission-system elements of NAVWAR, NAVSEA, NAVAIR, and Marine Corps Systems Command; Munitions pulls in IWS weapons offices IWS 3.0, IWS 11.0, and IWS 12.0. For board recall, this is acquisition-governance realignment; it does not erase the need to know current NAVSEA public PEO/IWS labels or technical authority lanes [11, 21, 22].

1.3.8 PEO IWS Program Offices

PEO IWS is tasked with developing, delivering, and sustaining operationally dominant integrated combat systems, encompassing surface radars, missiles, launchers, electronic warfare suites, undersea sensors, and integrated shipboard computing environments. PEO IWS leadership and billet incumbency should be confirmed from the current official NAVSEA PEO IWS page at board-prep time; if not verified from the live source, answer with office role and portfolio rather than a name [30]. The Engineering Duty Officer (EDO) coursebook and current NAVSEA public pages divide the numbered offices into Integrated Combat Systems (ICS) Major Program Managers and Product Line Major Program Managers. ICS offices plan, upgrade, integrate, and sustain surface combat-system baselines for ship or fleet groupings; Product Line offices develop and sustain combat-system components such as sensors, weapons, undersea systems, command and control, and terminal-defense systems [29, 31, 32].

Table VI: PEO IWS Program Offices (official-page snapshot; verify at board-prep time).

Category	Program Office	Office Name / Role	Primary Products / Board Cue
ICS	IWS 1.0	AEGIS Combat Systems	AEGIS combat-system cruiser and destroyer modifications, new construction, AEGIS Ashore, CPS/CDS, JTM, and STM
ICS	IWS 1.0F	AEGIS Fleet Readiness	Fleet liaison for AEGIS in-service readiness, sustainment, maintenance, and baseline upgrades
Product Line	IWS 2.0	Above Water Sensors	Radars, electronic warfare, EO/IR sensors, offboard decoys, and directed energy; examples include AMDR/SPY-6, SEWIP, Nulka, HELIOS, and ODIN
Product Line	IWS 3.0	Surface Ship Weapons	Naval surface weapons and launchers; examples include OTH-WS, SM-2, SM-6, VLS, Mk 110, and gun ammunition
ICS	IWS 4.0	International and Foreign Military Sales	Integrated combat systems and combat-system elements for international partners; international initiatives and technology-transfer policy coordination
Product Line	IWS 5.0	Undersea Systems	Acquisition and life-cycle support for submarine and surface-ship undersea sensors and command-and-weapons-control systems; examples include AN/SQQ-89 and USW-DSS
Product Line	IWS 6.0	Command and Control	Combat-system command and control, sensor netting, tracking management, and precision navigation; examples include CEC, MIPS, AN/BSN-2, AN/WSN series, and Scalable ECDIS-N
ICS	IWS 9.0	Zumwalt Integrated Combat Systems	TSCE and integrated combat system integration for DDG 1000-class ships; legacy board cues may also mention LCS, USCG Navy-type/Navy-owned combat-system elements, PCs, and FFG in-service support
Product Line	IWS 11.0	Terminal Defense Systems	Inner-layer defense against anti-ship cruise missile threats; examples include RAM, SeaRAM, CIWS, and RAM Block 1/2
Product Line	IWS 12.0	NATO Seasparrow	NATO Seasparrow and ESSM portfolio; examples include ESSM Block 1/2 and RIM-7/Mk 57 NSSMS
ICS	IWS 80.0	Atalanta Combat Systems	Integrated combat systems for carrier, amphibious, littoral, and unmanned fleets; includes LCS, SSDS platforms, USV integrated combat systems, and Navy/Marine Corps C2 afloat

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Table VI: PEO IWS Program Offices (official-page snapshot; verify at board-prep time). **(Continued)**

Category	Program Office	Office Name / Role	Primary Products / Board Cue
ICS	IWS X	Integrated Combat System	Technical governance and driving force for a common, tailorable Integrated Combat System strategy and modern computing processes that enable rapid, continuous capability delivery across the surface fleet

Note: Use the coursebook/NAVSEA office names for board recall. NAVSEA's Industry Engagement page uses "Atalanta Integrated Combat Systems" for IWS 80.0, while the coursebook and About page use "Atalanta Combat Systems."

Source: *2.1.2. NAVSEA Organization*, *About PEO IWS*, *PEO IWS Industry Engagement*, *PEO IWS X Program Overview and ICS Development*, 2025, 2026, 2026, 2022 [29, 31, 32, 33]

Note

Program-office labels and responsibilities change periodically; confirm against current NAVSEA public pages before a board. Do not map IWS 4.0 to launchers or IWS 5.0 to gun systems; current coursebook and NAVSEA sources identify them as International/FMS and Undersea Systems respectively. Current public NAVSEA pages list IWS 80.0 and IWS X rather than standalone IWS 8.0 or IWS 10.0; a 2022 IWS X brief describes IWS X as the Integrated Combat System technical-governance office and shows IWS 8.0 and IWS 10.0 as legacy platform-oriented combat-system offices in the transition toward a common, tailorable Integrated Combat System [33]. Field activities include AEGIS TECHREP and SCSC Wallops Island, but they are not numbered PEO IWS program offices [31]. Distinguish IWS 6.0 from NAVWAR PMW 150: IWS 6.0 owns combat-system C2 and sensor-netting products such as CEC, while PMW 150 owns Navy operational/tactical C2 applications such as GCCS-M.

Key ACAT I Programs within PEO IWS

IWS 1.0 (AEGIS Combat Systems): The AEGIS Modernization (AMOD) program ensures the combat system outpaces evolving threats via the Advanced Capability Build (ACB) process. Through ACB methodology, IWS 1.0 continuously integrates commercial hardware refreshes—specifically the Common Processor System (CPS) and Common Display System (CDS)—with software capability expansions. The DDG 51 Flight III upgrade program elevated the DDG 51 program to ACAT 1D status due to extreme cost and strategic impact.

IWS 2.0 (Above Water Sensors): The Air and Missile Defense Radar (AMDR), designated AN/SPY-6, represents a generational leap over legacy AN/SPY-1. Utilizing AESA technology built upon scalable Radar Modular Assemblies (RMAs), SPY-6 provides unprecedented sensitivity, range, and target discrimination. It is the mandatory sensor enabling Flight III DDG 51 class to execute Integrated Air and Missile Defense (IAMD) against ballistic missiles, cruise missiles, and advanced aircraft.

IWS 3.0 (Surface Ship Weapons): The SM-6 (Standard Missile 6) is an ACAT I major defense acquisition program. By integrating the SM-2 block airframe with the active radar seeker of the AIM-120 AMRAAM, SM-6 provides over-the-horizon air defense, anti-surface warfare (ASuW), and terminal-phase ballistic missile defense (BMD) capabilities. Its active seeker enables "engage-on-remote" (EoR) tactics that vastly expand strike group engagement envelopes.

IWS 6.0 (Command and Control): The Cooperative Engagement Capability (CEC) is an ACAT IC program that shares raw, real-time radar measurement data among surface ships and aircraft. This sensor netting creates a single, integrated, fire-control-quality composite track picture, transcending traditional tactical data links such as Link 16. CEC is the ultimate equalizer against radar horizon limitations.

1.3.9 Warfare Centers (alphabetical)

Table VII lists NAVSEA Warfare Centers with locations and descriptions. Figure 1.7 shows the geographic locations of Naval Warfare Centers (NWCs). For the EDO board, most recommend drawing the US to show where the NWCs are located. The other option is to list them out, but a map is more visual.

Table VII: NAVSEA Warfare Centers with locations and descriptions.

Center	Location	Description
NSWC Carderock	West Bethesda, MD	Carderock Division is the U.S. Navy's state-of-the-art *research, engineering, modeling, and test center for ships and ship systems*; the Navy and maritime communities rely on it for advanced platforms, analytics, and cost-conscious performance improvements.
NSWC Corona	Corona, CA	NSWC Corona Division is the Navy's independent *assessment agent*, leading NAVSEA *data analytics* and providing transparency for warfighting readiness, fleet LVC training integration, and metrology/calibration accuracy.
NSWC Crane	Crane, IN	NSWC Crane provides technical engineering solutions and total lifecycle leadership for many of the systems that protect and enable the Warfighter. Every decision we make, technology we create or business relationship we foster is performed with the fighting man and woman in mind. NSWC Crane has concentrated its resources and core competencies in the 3 mission areas which best support the Warfighter. The mission areas are *Strategic Mission, Electromagnetic Warfare, and Expeditionary Warfare*.
NSWC Dahlgren	Dahlgren, VA	The Naval Surface Warfare Center Dahlgren Division originally established itself as the major testing area for naval guns and ammunition. Today, it continues to provide the military with the *development and integration of warfare systems* for the warfighter, warfighting and the future fleet.
NSWC Indian Head	Indian Head, MD	As a field activity of the Naval Sea Systems Command, and part of the Navy's Science and Engineering Establishment, NSWC IHD is the Navy's premier facility for *ordnance, energetics and explosive ordnance disposal solutions*. We support the warfighter of today and tomorrow through discoveries that anticipate the next generation's future needs.
NSWC Panama City	Panama City, FL	The mission of NSWC, Panama City Division is to conduct research, development, test and evaluation, and in-service support of *mine warfare systems, mines, naval special warfare systems, diving and life support systems, amphibious and expeditionary maneuver warfare systems*, and other missions that occur primarily in coastal (littoral) regions.

Continued on next page

Table VII: NAVSEA Warfare Centers with locations and descriptions. **(Continued)**

Center	Location	Description
NSWC Philadelphia	Philadelphia, PA	Provide research, development, test and evaluation, acquisition support, engineering, systems integration, in-service engineering and fleet support with cyber-security, comprehensive logistics, and life-cycle savings through commonality for *surface and undersea vehicle machinery, ship systems, equipment and material*.
NSWC Port Hueneme	Port Hueneme, CA	*Integrate, Test, Evaluate, and Provide Life-Cycle Engineering and Product Support for Warfare Systems*.
NUWC Keyport	Keyport, WA	Keyport's mission is focused on developing and applying advanced technical capabilities to test, evaluate, field, and maintain undersea warfare systems and related defense assets. These advanced technical capabilities directly support the full spectrum of Navy undersea programs.
NUWC Newport	Newport, RI	The Naval Undersea Warfare Center Division Newport provides research, development, test and evaluation, engineering, analysis, and assessment, and fleet support capabilities for submarines, autonomous underwater systems, and offensive and defensive undersea weapon systems, and stewards existing and emerging technologies in support of undersea warfare.

1.4 NAVWAR Enterprise

1.4.1 Team NAVWAR overview

Organization. The NAVWAR enterprise consists of the NAVWAR systems command (Echelon II), three affiliated PEOs (Program Executive Office, Command, Control, Communications, Computers, and Intelligence (PEO C4I), Program Executive Office, Digital and Enterprise Services (PEO Digital), PEO MLB), and three Echelon III activities: Naval Information Warfare Center Atlantic (NIWC LANT), Naval Information Warfare Center Pacific (NIWC PAC), and NAVWAR Space Field Activity (NSFA) [35].

Current PAE alignment. Current public structure also shows DRPM Project Overmatch and the DON's PAE Mission Systems, publicly established on 11 May 2026. PAE Mission Systems pulls together PEO C4I, PEO Digital, PEO IWS, PEO MLB, DRPM Overmatch, DRPM LRNFO, DRPM MILE, Minotaur from PMA-290, and mission-system elements of NAVWAR, NAVSEA, NAVAIR, and Marine Corps Systems Command [11, 21]. Treat this as current acquisition-governance realignment while official NAVWAR pages continue to present the core NAVWAR enterprise and PEO pages unevenly.

Team NAVWAR. More than 11,000 employees support fleet concentration areas and deployed operations worldwide [35].

1.4.2 Mission

Mission statement. Identify, develop, deliver, and sustain information warfighting capabilities and services that enable naval, joint, coalition, and other national missions from seabed to space, while providing research and development, systems engineering, test and evaluation, technical, in-service, and support services to PEOs across program life cycles [35].

Figure 1.7. *Geographic locations of Navy Warfare Centers. Source:* NAVSEA Organization Chart (December 2025), 2.1.4. Naval Warfare Centers, *2025, 2025 [18, 34].*

1.4.3 NAVWAR directorates (competencies)

Table VIII summarizes the NAVWAR directorates and their core functions. Figure 1.8 shows the organizational structure of the NAVWAR enterprise.

Table VIII: NAVWAR directorates and core functions.

Code	Core functions
1.0, Comptroller	Financial management and cost estimating; chief financial advisor to NAVWAR leadership; directs PPBE inputs for NAVWAR and PEO programs.
2.0, Contracts	Acquisition contracting for ACAT and non-ACAT programs; contracting policy and strategic initiatives; cradle-to-grave contract support from RDT&E through sustainment and phase-out.
3.0, Counsel	Command legal support, intellectual property, environmental law, civilian personnel law, and acquisition legal support.
4.0, Logistics & Fleet Support	Programmatic logistics support to PMs and PEOs; develops and maintains NAVWAR logistics policy and procedures.
5.0, Chief Engineer	Technical authority for IT systems on ships, submarines, and shore installations; cybersecurity technical authority; resolves cross-platform system performance issues; manages the engineering/science workforce.
6.0, Program Management	Acquisition policy, process, and tools; acquisition workforce management; program management support and staffing.
7.0, Science & Technology	Research and applied science; forecasting, assessment, and technology transition.
8.0, Corporate Operations	Resource management and tracking; human resources (HRO), manpower, facilities, knowledge management, corporate strategy, enterprise IT policy, and Inspector General functions.

Source: *2.1.3. NAVWAR Enterprise*, 2025 [35]

1.4.4 Echelon III activities and core competencies

NIWC Pacific

Mission. Conduct research, development, prototyping, engineering, test and evaluation, installation, and sustainment of integrated C4ISR, cyber, and space systems across all warfighting domains [35].

Technology areas. Cyber operations; command and control communications and networks; production, installation, and in-service engineering; intelligence, surveillance, reconnaissance and information operations; science and technology; marine mammal programs; autonomous unmanned systems [35].

Key capability areas. Full life-cycle engineering for information warfare; cloud-based digital development with Development, Security, and Operations (DevSecOps) and digital twins; rapid prototyping and delivery; Artificial Intelligence (AI)/data science; applied electromagnetics and photonics; offensive/defensive cyber; positioning, navigation, and timing labs; and unique test facilities including San Clemente Island and integrated data-link test beds [35].

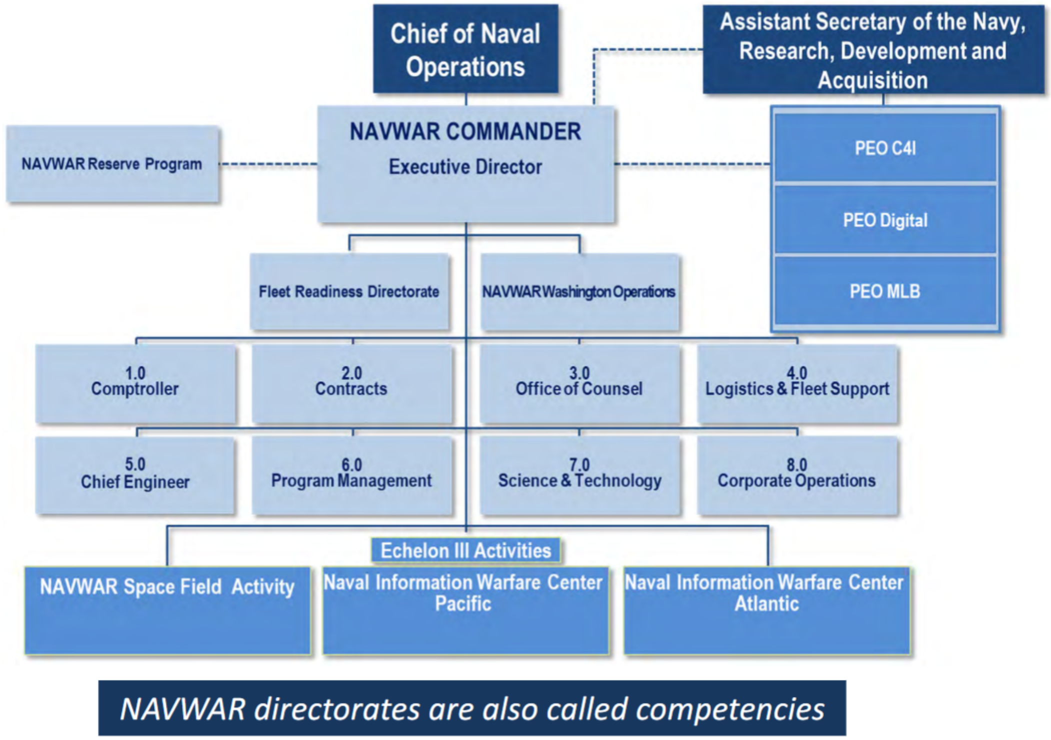

Figure 1.8. *NAVWAR Organizational Chart. Source:* 2.1.3. NAVWAR Enterprise, *2025 [35]*

Figure 1.9. *NIWC Pacific locations. Main locations are: San Diego, Everett, Hawaii, Guam, and Japan. Source:* 2.1.3. NAVWAR Enterprise, *2025 [35]*

NIWC Atlantic

Mission. Deliver information warfare solutions that protect national security, including communications systems, networking, cyber operations, intelligence and decision support applications, business systems, and information security [35].

Five departments. Fleet C4I and readiness; expeditionary warfare; enterprise systems; shore C2ISR and integration; science and technology [35].

Key capability areas. Surface ship and submarine C4I test and integration, cyber forensics and vulnerability assessment, enterprise data centers, satellite communications production and installation, electromagnetic interference testing, global distance support, air traffic control systems, and electronic security systems [35].

Figure 1.10. NIWC Atlantic locations. Main Locations are: Charleston, National Capitol Region, Norfolk, New Orleans, Naples, Rota, and Bahrain. Source: 2.1.3. NAVWAR Enterprise*, 2025 [35]*

NAVWAR Space Field Activity (NSFA)

Mission. Lead the U.S. Navy's participation in the National Reconnaissance Office (NRO) by providing naval warfare and acquisition expertise, coordinating naval space RDT&E with national reconnaissance programs, and enabling fleet use of national space capabilities [35].

Responsibilities. Facilitate interaction between the DON and NRO; coordinate research, development, and acquisition activities that support space-based intelligence collection [35].

1.4.5 Affiliated program executive offices

PEO MLB. DON manpower, logistics, and business IT acquisition agent; delivers MyNavy HR IT, Ready Relevant Learning, Navy ERP financials, logistics IT, Marine Corps logistics IT, naval applications/business services, and Marine Corps manpower IT modernization portfolios [35, 36].

PEO Digital. DON enterprise-wide IT acquisition agent; portfolios include naval enterprise networks, special networks and intelligence mission applications, naval commercial cloud services, special access program networks, enterprise IT strategic sourcing, and Marine Corps supporting establishment systems [35].

PEO C4I. Delivers command and control, intelligence sharing, terminals, and integration programs through capability development offices (PMW 120/130/150/160/170) and platform integration offices (PMW 740/750/760/770/790) to provide affordable, integrated, and interoperable Command, Control, Communications, Computers, and Intelligence (C4I) capability to the fleet [35].

1.4.6 NAVWAR Legacy ACAT I and Current Portfolio Examples

NAVWAR manages a vast portfolio through its affiliated PEOs. Official NAVWAR and PEO pages should be treated as authoritative for current portfolio counts. This section preserves legacy coursebook examples for board discussion, but current PMW alignments and acquisition-category designations require current-source confirmation before quoting exact numbers. Current public sources verified on 2026-05-27 show some legacy Acquisition Category (ACAT) examples have shifted into software, service, or business-system framing, so the following table is a board-prep orientation aid, not a live portfolio ledger [36, 37, 38, 39, 40, 41, 42, 43, 44, 45].

Table IX: NAVWAR legacy ACAT I and current portfolio examples.

Program Office	Office / Portfolio	Program or Capability	Board-status cue
PMA/PMW 101	Multifunctional Information Distribution Systems	MIDS / Link 16	Dual-designated MIDS program office; use the published PMA/PMW 101 or PMA/PMW-101 styling.
PMW 120	Battlespace Awareness and Information Operations	DCGS-N FoS	Current public status: Software Acquisition Pathway; legacy coursebook ACAT IAC/IAM.
PMW 150	Naval Command and Control Systems	GCCS-M	Current public status: ACAT IAC.
PMW 150	Naval Command and Control Systems	NTCSS	Legacy coursebook ACAT IAC; not confirmed on current PMW 150 top-program sheet.
PMW 160	Tactical Networks	CANES	Current public status: ACAT IAC.
PMW 160	Tactical Networks	ADNS	Current public status: ACAT II; tactical WAN and ship-to-shore routing layer.
PMW/A 170	Communications and GPS Navigation	NMT	Current public status: ACAT IC; use the published PMW/A 170 or PMW/A-170 styling.
PEO Digital / NEN	Naval Enterprise Networks	NMCI, ONE-Net, SMIT, EUHW	Current public framing: enterprise network service/portfolio; do not teach NGEN/NGEN-R ACAT I/MAIS as current unless separately sourced.

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Table IX: NAVWAR legacy ACAT I and current portfolio examples. **(Continued)**

Program Office	Office / Portfolio	Program or Capability	Board-status cue
PMW 790	Shore and Expeditionary Integration	DJC2	Current public status: ACAT IAC.
PEO MLB / PMW 220	Manpower, Logistics and Business Solutions	Navy ERP Financial IT Services / Navy ERP+	Current public and DAU status: BCAT I.

PMA/PMW 101: Multifunctional Information Distribution Systems

Program: Multifunctional Information Distribution Systems (MIDS)

PMA/PMW 101 develops, produces, and sustains tactical data-link products and waveforms, including Link 16 and Tactical Targeting Network Technology, to provide interoperable tactical data exchange across air, surface, shore, coalition, and international users. The current public tear sheet lists the office at Naval Base Point Loma, Harbor Drive Annex, 33050 Nixie Way, Building 17A, Suite 416, San Diego [46].

Board cue: PMA/PMW 101 is the MIDS/Link 16 office. It is adjacent to C4I recall, but do not confuse it with PMW 160: MIDS/Link 16 is the tactical data-link/waveform lane, while CANES/ADNS are the afloat network and routing lanes. Do not normalize every cross-NAVWAR/NAVAIR office to the same slash convention. Current PEO C4I and NAVAIR sources style the MIDS office as PMA/PMW 101 or PMA/PMW-101. Use that exact form for MIDS/Link 16[46, 47].

PMW 120: Battlespace Awareness and Information Operations

Program: Distributed Common Ground System-Navy (DCGS-N)

Status cue: Current public status is Software Acquisition Pathway; legacy coursebook status was Increment 1 (ACAT IAC) and Increment 2 (ACAT IAM) [39].

PMW 120 focuses on creating and breaking “kill webs” for the warfighter by delivering integrated intelligence, surveillance, reconnaissance (ISR), targeting, and electronic warfare capabilities. DCGS-N is the Navy’s flagship intelligence, surveillance, reconnaissance, and targeting support capability and is part of the broader DoD DCGS ecosystem [39].

DCGS-N is designed to provide multiservice integration of ISR and targeting capabilities. The legacy board cue is that DCGS-N consolidated intelligence analytical tools into an integrated computing environment and then evolved toward more agile software delivery. For current-status answers, call it DCGS-N Family of Systems under PMW 120 and state the Software Acquisition Pathway unless a current source supports a different designation. Do not expand the acronym as “Digital Consolidated” or “Maritime”; the board-safe expansion is *Distributed Common Ground System-Navy*.

Strategic Insights: The evolution of DCGS-N highlights a broader trend within NAVWAR ACAT I programs: the migration from hardware-heavy, monolithic deployments to agile, cloud-based software architectures. PMW 120’s strategy involves transitioning from sustaining isolated systems to deploying afloat and ashore cloud environments. However, this transition introduces significant challenges in software growth, with instances of up to 30-percent increases in Source Lines of Code (SLOC) growth, requiring careful cost estimation and rigorous continuous testing.

PMW 150: Naval Command and Control Systems

PMW 150 is the Navy's preeminent provider of operational and tactical command and control solutions. Current public materials keep GCCS-M as an ACAT IAC program; NTCSS remains useful legacy coursebook context, but it is not confirmed on the current PMW 150 top-program sheet [40].

Program 1: Global Command and Control System-Maritime (GCCS-M)

ACAT Status: ACAT IAC [40]

GCCS-M is the Navy's primary Command and Control (C2) program of record and the maritime iteration of the DoD's Joint C2 Family of Systems. It provides near-real-time tactical and operational situational awareness by generating the Common Operational Picture (COP) and Common Tactical Picture (CTP). GCCS-M processes and fuses tracks from ground, air, and maritime sensors, allowing commanders at Maritime Operations Centers (MOCs) ashore and aboard carrier strike groups to synchronize decentralized execution across the battlespace.

Over time, PMW 150's mandate has been to evolve GCCS-M into a more agile, Service-Oriented Architecture (SOA). This evolution involves integrating Maritime Tactical Command and Control (MTC2) software applications, which dynamically plan, direct, monitor, and assess Distributed Maritime Operations (DMO) in real-time.

Program 2: Naval Tactical Command Support System (NTCSS)

Status cue: Legacy coursebook ACAT IAC; not confirmed on the current PMW 150 top-program sheet [40].

While GCCS-M commands the tactical fight, NTCSS commands the logistics required to sustain it. In legacy coursebook framing, NTCSS was treated as a Mission Essential Major Automated Information System (MAIS) (ACAT IAC), a comprehensive suite of software applications providing shipboard and aviation maintenance management, inventory control, and financial business management. Established in 1995 through the merger of legacy systems (SNAP, NALCOMIS, and MRMS), NTCSS is installed on vessels and ashore to support the unit commanding officer in tracking repair work packages, aviation repairables, and critical parts inventory.

The sustainment of the legacy NTCSS ACAT requires continuous configuration management and integration support. As PMW 150 shifts its focus toward modernized logistics frameworks, such as the Naval Maintenance, Repair, and Overhaul (N-MRO) and Naval Operational Supply System (NOSS) initiatives, maintaining the stability of the NTCSS backbone is paramount to prevent operational degradation and supply chain failures during the transition period.

PMW 160: Tactical Networks

Program: Consolidated Afloat Networks and Enterprise Services (CANES)

ACAT Status: ACAT IAC [41]

PMW 160 is tasked with providing mission-effective and cyber-resilient tactical networks to forces afloat. Its premier ACAT I program is CANES, a Chief of Naval Operations-directed initiative that serves as the indispensable core of the Navy's Information Warfare Platform.

CANES was initiated to collapse and replace five disparate legacy network systems into a single, consolidated afloat computing environment. Current PMW 160 public material groups the predecessor set as Legacy Network Systems (LNS): Integrated

Shipboard Network System (ISNS), Sensitive Compartmented Information networks, CENTRIXS-M, and SubLAN [41]. It provides the necessary data transport, edge computing, voice, video, systems management, enterprise services, and cybersecurity functionality across multiple security domains for the fleet. Achieving a Full Deployment Decision (FDD) in October 2015, CANES operates under ASN (RD&A) oversight as an ACAT IAC program.

Related program: Automated Digital Network System (ADNS) is PMW 160's ACAT II tactical wide area network capability. It supplies surface ship, submarine, airborne, tactical-shore, and shore-based WAN gateway services and is the board cue for ship-to-shore IP routing [41]. In an afloat email example, the clean flow is: sailor application or workstation, CANES shipboard LAN/compute services, ADNS routing and traffic engineering, SATCOM/RF bearer, shore gateway/Fleet Network Operations Center/DISN path, then the destination shore enterprise service.

Strategic Insights: CANES represents arguably the most foundational enabler of maritime combat power within NAVWAR. Without the hardware and software infrastructure provided by CANES, advanced analytical applications developed by PMW 120 (DCGS-N) and PMW 150 (GCCS-M) simply cannot function afloat. Recognizing the pace of cyber threats, PMW 160 is aggressively driving CANES toward a DevSecOps framework, decoupling software development from hardware procurement and ensuring that commercial-off-the-shelf (COTS) baselines are continuously evolved and patched through cyclical Service Pack updates.

PMW/A 170: Communications and GPS Navigation

Program: Navy Multiband Terminal (NMT)

ACAT Status: ACAT IC [42]

PMW/A 170 serves as the premier provider of resilient communications and assured Position, Navigation, and Timing (PNT). The office's flagship ACAT I program is the Navy Multiband Terminal (NMT).

NMT is the Navy's sole protected satellite communications terminal, providing critical, secure, survivable, and interoperable military satellite communications (MILSATCOM). NMT enables high-capacity communication by providing simultaneous access to a variety of existing and future satellite constellations, including Advanced Extremely High Frequency (AEHF), Wideband Global SATCOM (WGS), MILSTAR, Defense Satellite Communications System (DSCS), and the Enhanced Polar System (EPS).

PMW/A 170 is also cross-domain, but current PEO C4I and NAVAIR sources style it as PMW/A 170 or PMW/A-170, not PMA/PMW 170. Use the published office designator rather than inferring a convention from PMA/PMW 101[42, 48].

Strategic Insights: In an era of great power competition, the electromagnetic spectrum is a highly contested domain. NMT is designed specifically to operate in anti-access/area-denial (A2/AD) environments, featuring extreme anti-jam, anti-scintillation, and low probability of intercept (LPI)/LPD capabilities using Extended Data Rate (XDR) waveforms. The strategic importance of this ACAT IC program cannot be overstated; it supports the President's Ballistic Missile Defense priorities and the National Command Authority by ensuring the uninterrupted flow of critical strategic data, even under sustained electronic and physical attack.

PEO Digital: Naval Enterprise Networks

Portfolio: Naval Enterprise Network (NEN), including Navy Marine Corps Intranet (NMCI), ONE-Net, SMIT, and enterprise user services.

Status cue: Current public framing is enterprise network service/portfolio; do not teach Next Generation Enterprise Network (NGEN)/NGEN-Recompete (NGEN-R) ACAT I/MAIS as current unless separately sourced [44].

Operating under the PEO Digital, the Naval Enterprise Networks portfolio is responsible for enterprise-wide networks that provide secure, seamless, and global computer connectivity for the Department of the Navy ashore.

Legacy courseware often used NGEN and NGEN-R as the acquisition shorthand for the follow-on to NMCI. Current PEO Digital public material emphasizes the broader NEN portfolio, including NMCI, ONE-Net, SMIT, and end-user hardware services, rather than presenting NGEN-R as a current ACAT I/MAIS program.

Strategic Insights: The board-level point is not the legacy label; it is that ashore enterprise networks are acquisition-critical infrastructure. The seamless operation of NEN services is a prerequisite for shore-based administrative, logistical, and acquisition planning operations.

PMW 790: Shore and Expeditionary Integration

Program: Deployable Joint Command and Control (DJC2)

ACAT Status: ACAT IAC [43]

PMW 790 delivers resilient, interoperable shore and expeditionary C4I capabilities, ensuring that the infrastructure ashore matches the capacity of deployed fleets. The office's primary ACAT I capability is the DJC2 system.

Originating as a Secretary of Defense priority transformation initiative, DJC2 provides a standardized, self-contained, rapidly deployable, and scalable joint command and control infrastructure to Geographic Combatant Commands (GCCs) and Combined/Joint Task Force (C/JTF) Commanders. By integrating computer-network-enabled C2 applications with self-powered tactical shelters, DJC2 allows commanders to establish a fully functional operations center in austere expeditionary environments rapidly. PMW 790 integrates this capability with over 60 unique shore and expeditionary networks, including the Shore Tactical Assured C2 (STACC) enterprise, to maintain global reach.

Board cue: The PMW 790 ACAT I answer is DJC2, not D2GCS or DG2S.

PEO MLB: Manpower, Logistics and Business Solutions

Program: Navy Enterprise Resource Planning (Navy ERP)

ACAT Status: BCAT I [36, 45]

While strictly categorized as a Business Capability Acquisition Category (BCAT I) program, Navy ERP represents an enterprise IT investment equivalent in scale, complexity, and financial footprint to a traditional kinetic ACAT I weapon system. DAU's 2024 MDAP change notice associates Navy ERP Financial IT Services with PMW 220, and current PEO MLB public materials describe Navy ERP+ modernization following a June 2024 ASN(RD&A) acquisition decision memorandum. The board cue is that Navy ERP standardizes Department of the Navy business processes for acquisition, financial management, and logistics operations.

Current PEO MLB public material organizes the office around capability portfolios rather than the PMW 120/150/160-style construct used by PEO C4I. The board-safe portfolio list is MyNavy HR IT Solutions, Ready Relevant Learning, Navy ERP Financial IT Services, Logistics IT Services, Marine Corps Logistics Integrated Information Solutions, Naval Applications and Business Services, and Marine Corps Manpower IT Systems Modernization [36].

Combining commercial off-the-shelf software (primarily SAP architecture) with extensive business process reengineering, Navy ERP utilizes a single database to manage shared common data across functional communities. As of recent reporting, Navy ERP controls over 50 percent of the Navy's financial spending, manages global supply operations, and provides workforce management for over 65,000 personnel. By modernizing its code and upgrading software capabilities, Navy ERP serves as the fundamental enabler for the DoD's strategic goal of reforming business practices for greater performance and affordability, providing the auditability and financial tracking required for every other ACAT I program in the Navy's portfolio.

1.4.7 Coursebook cue: Fleet Readiness Directorate

For NAVWAR organization questions, distinguish program offices from the Fleet Readiness Directorate (FRD). FRD is NAVWAR's fleet-facing readiness integrator: it is the fleet single point of entry, NAVWAR single point of accountability, and the organization that drives execution rigor across the enterprise for fleet support and installations. Coursebook 2.1.3 identifies the two core branches as FRD 100 Fleet Support and FRD 200 Fleet Installations [35].

Board cue. FRD is the fleet-facing accountability and execution-rigor arm, not a PEO. Name FRD 100 Fleet Support and FRD 200 Fleet Installations if asked for the branches.

1.5 Naval Warfare Centers

Warfare centers provide critical technical expertise to ensure current and future program success and fleet readiness [34].

1.5.1 Roles of the Warfare Centers

Industry will not do. Not profitable or accepts too much liability.

Industry should not do. Technical authority and certification work (for example, shock tests).

Industry cannot do. Specialized facilities such as ranges and unique equipment.

Program success. Provide technical advice and engineering leadership to make naval technical programs successful.

Capability development. Help determine and develop the capabilities the Navy and Marine Corps need as trusted technical advisors.

Quality and safety. Verify quality, safety, and effectiveness of platforms and systems through rigorous systems engineering.

Urgent needs. Design, develop, and field solutions for urgent operational fleet needs.

Warfighter bridge. Translate warfighter needs into technical requirements.

1.5.2 Reporting relationships

Warfare centers and divisions align under the Secretary of War and Under Secretary for Acquisition through the CNO (acquisition and fleet support) and the Navy Acquisition Executive ASN(RD&A) (tech base) [34].

Table X: Warfare center reporting relationships by organization.

Organization	Reporting units
NAVWAR	NIWC Atlantic, NIWC Pacific, NSFA.
NAVAIR	NAWCAD; NAWCWD.
NAVSEA	NSWC divisions: Dahlgren, Corona, Crane, Carderock, Port Hueneme, Philadelphia, Panama City, Indian Head; NUWC divisions: Keyport, Newport; Naval Sea Logistics Center (NSLC).
NAVFAC	Engineering and Expeditionary Warfare Center (EXWC).
ONR	NRL.

Source: *2.1.4. Naval Warfare Centers*, 2025 [34]

1.5.3 Warfare center missions

Full-spectrum role. The Navy's full-spectrum RDT&E, engineering, and fleet support centers for platforms, vehicles, systems, and facilities associated with a particular warfare area.

NAVWAR. C4I systems and information assurance.

NAVAIR. Aircraft and aircraft systems.

NAVSEA. Ships and ship systems.

NAVFAC. Facilities.

1.5.4 NAVWAR activities

NIWC Pacific

Mission. Navy's premier RDT&E laboratory supporting C4ISR; supports the development, acquisition, and fielding of leading-edge C4ISR systems for Navy, joint, and allied customers [34].

Locations. See Figure 1.9.

NIWC Atlantic

Mission. Primary NAVWAR provider for joint and homeland security C4ISR solutions; developer and employer of life-cycle logistics support solutions in a web-enabled portal environment; Navy provider of critical engineering and acquisition expertise for Navy and joint commands; rapid integrator of C4ISR technology providing interoperability to the Navy, federal agencies, and the joint warfighter [34].

Locations. See Figure 1.10.

NAVWAR Space Field Activity (NSFA)

Background. Established to coordinate naval space research activities within the NRO [34].

Mission. Provide line management staffing of naval space research at NRO [34].

Responsibilities. Facilitate interaction between the DON and NRO; coordinate all naval space research, development, and acquisition functions and activities [34].

1.5.5 NAVAIR warfare centers and divisions

NAWC Aircraft Division (NAWCAD)

Location. Patuxent River, MD; detachments at Lakehurst, NJ and Orlando, FL [34].

Mission. The Navy's principal RDT&E, engineering, and fleet support activity for naval aircraft, engines, avionics, aircraft support systems, aircraft launch and recovery, and ship, shore, and air operation [34].

Technical leadership areas. Airworthiness and engineering analysis; trainers and training development; Test Pilot School; test ranges; carrier support and aircraft launch and recovery equipment (ALRE); propulsion systems [34].

NAWC Weapons Division (NAWCWD)

Location. China Lake and Point Mugu, CA [34].

Mission. Provide Navy and Marine Corps warriors with effective, affordable, integrated warfare systems and life-cycle support to ensure battlespace dominance [34].

Technical leadership areas. Perform RDT&E, logistics, and in-service support for guided missiles, free-fall weapons, targets, support equipment, crew systems, and electronic warfare; operate the Navy's western land and sea range test and evaluation complex; integrate weapons and avionics on tactical aircraft [34].

1.5.6 NAVSEA warfare centers and divisions

Naval Surface Warfare Center (NSWC)

Location. Headquarters at Washington, DC [34].

Mission. Provide the right technology, the right capabilities, and the specialized research and development facilities to support all aspects of surface warfare [34].

Technical leadership areas. Ships and ship systems; surface ship combat systems; littoral warfare systems; Navy strategic weapon systems; ordnance; homeland and force protection; surface warfare logistics and maintenance; force level warfare systems [34].

NSWC Carderock Division

Location. West Bethesda, MD [34].

Mission. Provides research, engineering, modeling, and testing for ships and ship systems for advanced platforms, analytics, and cost-conscious performance improvements [34].

Technical leadership areas. Hydromechanics (hull form, resistance measurement, prop design, maneuvering, seakeeping analysis); surface and submarine survivability; ship and submarine electromagnetic and optical signatures [34].

Life-cycle focus. Early life-cycle [34].

NSWC Corona Division

Location. Corona, CA [34].

Mission. Gauges the warfighting capacity of ships and aircraft, from unit to battlegroup level, by assessing the suitability of design, the performance of equipment and weapons, and the adequacy of training [34].

Technical leadership areas. Fleet exercise assessments; weapons and combat systems performance analysis; missile flight analysis; data collection, processing, reduction and display; trend and failure pattern analysis; test systems availability assessment; metrology systems engineering; material readiness assessment [34].

Life-cycle focus. Entire life-cycle [34].

NSWC Crane Division

Location. Crane, IN [34].

Mission. Advance multi-domain systems within electronic warfare, strategic systems hardware, and special and expeditionary warfare; conduct RDT&E, acquisition, and in-service engineering across the entire life-cycle [34].

Technical leadership areas. Electronic warfare systems RDT&E, acquisition, and sustainment; special warfare and expeditionary system hardware; strategic systems hardware; infrared countermeasures, pyrotechnic RDT&E, and life-cycle support; advanced electronics and energy systems; sensors and surveillance systems [34].

Life-cycle focus. Entire life-cycle [34].

NSWC Dahlgren Division

Location. Dahlgren, VA [34].

Mission. Research, develop, test and evaluate, analyze, engineer, integrate, and certify complex naval warfare systems related to surface warfare, strategic systems, combat and weapons systems [34].

Technical leadership areas. Surface ship combat systems software engineering (theater air defense, Aegis combat system, DDG 1000); certifies each strike group for interoperability; develop and certify targeting and fire control systems for Trident submarines [34].

Life-cycle focus. Early life-cycle [34].

NSWC Indian Head Explosive Ordnance Disposal Technology Division (IHEODTECHD)

Location. Indian Head, MD [34].

Mission. Research, develop, test, evaluate, manufacture, and provide in-service support of energetics and energetic systems; provide warfighters with technology to detect, locate, access, identify, render safe, recover, exploit, and dispose of explosive threats [34].

Technical leadership areas. Energetics research, detonation science, chemical and physical characterization; weapons product development; ordnance test and evaluation; NAVSEA lead for weapon and combat systems packaging, handling, storage, and transportation; support the Joint Explosive Ordnance Disposal (EOD) technician; counter radio-controlled improvised explosive device electronic warfare (CREW) science and technology; only US source and DoW facility for many types of propellant and high-energy chemicals; one of two solvent-less extrusion plants in the US; model and scale-up warheads [34].

Life-cycle focus. Entire life-cycle [34].

NSWC Panama City Division

Location. Panama City, FL [34].

Mission. Conduct RDT&E and in-service support of mine countermeasure systems, naval sea mine systems, naval special warfare systems, diving and life support systems, amphibious and expeditionary maneuver warfare systems, and other systems that occur primarily in coastal (littoral) regions [34].

Technical leadership areas. Mine warfare; LCS mission packages; expeditionary maneuver warfare; assault craft vehicles; diving and life support systems; unmanned systems (air and ground); chemical and biological individual protection; special warfare [34].

Life-cycle focus. Entire life-cycle [34].

NSWC Philadelphia Division

Location. Philadelphia, PA [34].

Mission. Provides RDT&E, fleet support, and in-service engineering for surface and undersea vehicle HM&E and propulsors; provides logistics Research and Development (R&D) [34].

Technical leadership areas. Surface and submarine machinery R&D (power systems, ship automation control, dynamics and silencing, shipboard energy availability and conservation, and electrical machinery integration); machinery ISEA for propulsion, electric and mechanical systems; Integrated Propulsion System development; control systems and HM&E cybersecurity [34].

Life-cycle focus. Post-production [34].

NSWC Port Hueneme Division

Location. Port Hueneme, CA [34].

Mission. Provide test and evaluation, in-service engineering, and integrated logistics support for surface warfare combat systems and ordnance of the Navy surface fleet [34].

Technical leadership areas. Surface missile launcher systems; UNREP test site and fleet support; Self Defense Test Ship; surface ship combat systems; land-based and at-sea T&E [34].

Life-cycle focus. Post-production [34].

Naval Undersea Warfare Center (NUWC)

Location. Headquarters in Newport, RI [34].

Mission. The Navy's full-spectrum RDT&E, engineering, and fleet support center for submarines, autonomous underwater systems, and offensive and defensive weapons systems associated with undersea warfare [34].

Technical leadership areas. Undersea warfare modeling and analysis; submarine combat and combat control systems; surface ship and submarine sonar systems; submarine electronic warfare; submarine unique on-board communication systems and communication nodes; undersea ranges; submarine electromagnetic, electro-optic and non-acoustic effects reconnaissance, and search and tracking systems; undersea vehicle active and passive signatures (except HM&E); submarine vulnerability and survivability (except HM&E); torpedoes and torpedo countermeasures [34].

NUWC Keyport Division

Location. Keyport, WA [34].

Mission. Provide advanced technical capabilities for test and evaluation, in-service engineering, maintenance and industrial base support, fleet material readiness, and obsolescence management for undersea warfare [34].

Technical leadership areas. Test and evaluation and training (National UUV Test and Evaluation Center (NUTEC), 3D undersea warfare tracking range, range alternatives); life-cycle systems supportability (fielded equipment, systems, subsystems); fleet material readiness (material support, modernization, prototype development, testing); depot level torpedo repair and product proofing [34].

Life-cycle focus. Post-production [34].

NUWC Newport Division

Location. Newport, RI [34].

Mission. Provide the technical foundation which enables the conceptualization, RDT&E, fielding, modernization, and maintenance of systems that ensure the Navy's undersea superiority [34].

Technical leadership areas. Product-oriented research and advanced development; undersea warfare modeling and analysis; unmanned underwater vehicles (UUVs); torpedoes, torpedo launchers, and torpedo countermeasures (submarine and surface) as in-service engineering agent and design agent; operational testing support (Operational Test and Evaluation Force (OPTEVFOR), Combat Systems Ship's Qualification Trials (CSSQT)) – Atlantic ranges; operate Atlantic Undersea Test and Evaluation Center (AUTEC); submarine cybersecurity [34].

Life-cycle focus. Early life-cycle [34].

1.5.7 Technical capability focus by division

Table XI: Technical capabilities by warfare center division.

Division	Technical capability focus
NSWC Carderock	17 TCs in naval architecture and marine engineering for surface and undersea vehicles and associated ship systems.
NSWC Corona	8 TCs in performance assessment of weapons and combat systems from unit to force level.
NSWC Crane	10 TCs in electronic warfare, special warfare weapons and devices, and strategic systems components and hardware.
NSWC Dahlgren	29 TCs in surface ship weapons system development and integration, missile defense, strategic systems, and joint and homeland defense.
NSWC Indian Head	10 TCs in energetic systems and energetic materials focused on ordnance disposal technology and counter-IED tools.
NSWC Panama City	12 TCs for mine warfare systems and other littoral warfare systems, including mines, special warfare systems, and diving and life support systems.
NSWC Philadelphia	13 TCs for surface and undersea vehicle machinery, ship systems, equipment and material, including cybersecurity and life-cycle commonality.
NSWC Port Hueneme	10 TCs for surface ship T&E, in-service engineering and logistics, and integration of weapons and combat systems as primary interface with the surface fleet.
NUWC Keyport	16 TCs for undersea warfare test and evaluation, in-service systems integration and supportability, and industrial base maintenance and material support.
NUWC Newport	20 TCs for undersea warfare systems development and integration for sensors, weapons, vehicles, payloads, communications, training, and combat systems.

Source: *2.1.4. Naval Warfare Centers*, 2025 [34]

1.5.8 Warfare center organization

Leadership. Warfare Center Board of Directors; Naval Surface Warfare Center (NSWC) and Naval Undersea Warfare Center (NUWC) Commander and Executive Director; NSWC and NUWC staff [34].

Division leadership. Each NSWC/NUWC division has a Commanding Officer, Technical Director, and Senior Civilian [34].

Dam Neck Activity. NSWC Dahlgren Dam Neck Activity includes a Commanding Officer and Senior Civilian; O-5 command [34].

NSLC. Naval Sea Logistics Center (Mechanicsburg) reports within the warfare center enterprise [34].

1.5.9 Office of Naval Research and Naval Research Laboratory

Office of Naval Research (ONR)

Leadership. Typically headed by a Navy two-star; mostly civilian staff; reports to ASN(RD&A) [34].

Research channels. Science and technology; Naval Research Laboratory; Office of Naval Research (ONR) Global (fleet, international, and Science and Technology (S&T) community information sharing); Naval Reserve S&T Program; Commercial Technology Transition Office for rapid transitions [34].

Naval Research Laboratory (NRL)

Location and reporting. Washington, DC; headed by an O-6; reports to ONR [34].

Mission. The Navy's full-spectrum corporate laboratory to conduct broad-based multidisciplined scientific research and advanced technology development for maritime applications of new and improved materials, techniques, equipment, systems, and ocean, atmospheric, and space sciences [34].

Technical leadership areas. Primary in-house research for physical, engineering, space, and environmental sciences; broad-based exploratory and advanced development for anticipated Navy needs; broad multidisciplinary support to the Naval Warfare Centers [34].

Life-cycle focus. Early life-cycle [34].

1.5.10 Coursebook cue: warfare-center customers and products

Coursebook 2.1.4 expects students to recognize that naval warfare centers are not single-customer laboratories. Their customer base includes NAVSEA and NAVAIR program offices, fleet type commanders, resource sponsors, other SYSCOMs, joint and allied partners, and field activities that need technical authority, test, evaluation, in-service engineering, fleet support, certification, and life-cycle problem solving. The product is often not just hardware; it can be a technical decision, test result, certification basis, engineering agent action, fleet advisory, modernization package, or sustainment recommendation [34].

Board cue. Answer with customers plus products: PEOs and fleet customers buy decisions, data, certification, test, engineering support, modernization, and sustainment outcomes.

2 PPBE

2.1 PPBE (Planning, Programming, Budgeting, and Execution)

2.1.1 Role of the PPBE process

Why PPBE exists. The DoW uses Planning, Programming, Budgeting and Execution (PPBE) to identify fiscal needs and allocate resources across the Services and DoW agencies.

Scale. Annual budget exceeds $885B and supports about 3 million active duty and civilian employees.

Balancing act. Weighs competing requirements for limited funds while reconciling internal priorities and external direction from Office of Management and Budget (OMB) and Congress.

Output. The end product is the DoW portion of the President's Budget.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.2 PPBE overview

Planning. Predict the strategic environment and define the capabilities it requires.

Programming. Allocate resources to missions and integrate programs to minimize shortfalls and duplication.

Budgeting. Produce an annual budget for Congress that links missions to requested funding.

Execution. Operate in the real world and make adjustments.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.3 PPBE timeline and the DAS

PPBE is calendar-driven, while the Defense Acquisition System (DAS) is event-driven. Once a program is initiated, PPBE resource allocations can reshape DAS execution plans. Figure 2.1 shows the PPBE timeline across multiple fiscal years [49].

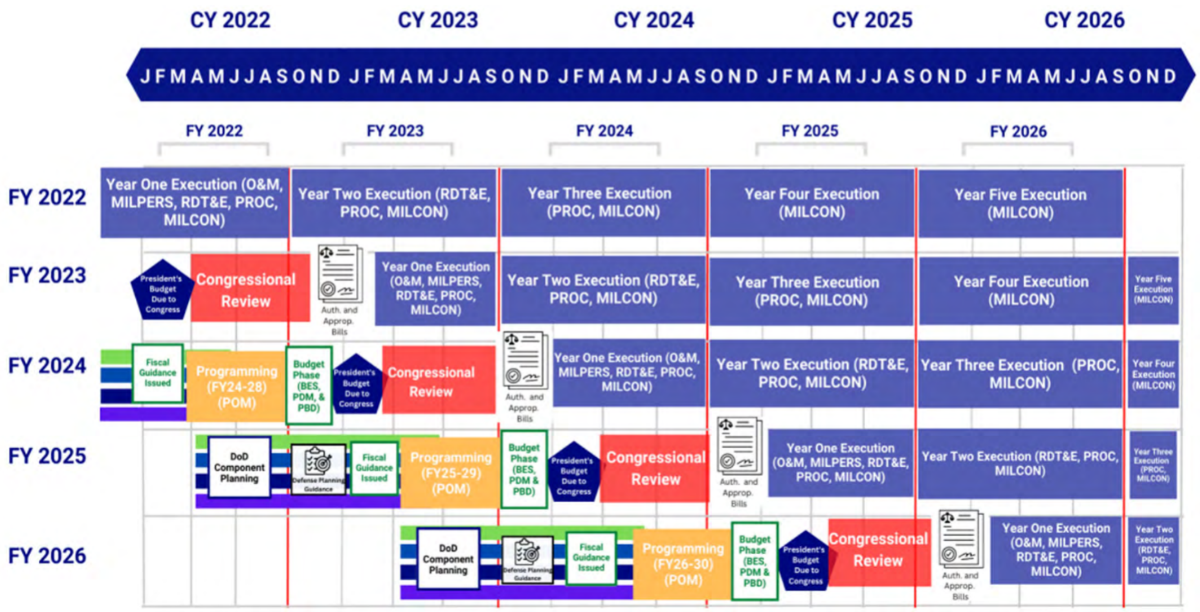

Figure 2.1. *PPBE timeline across fiscal years. Source:* 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), *2026 [49]*

Figure 2.2 shows the Navy PPBE timeframes from the Defense Acquisition University (DAU) Platinum Card.

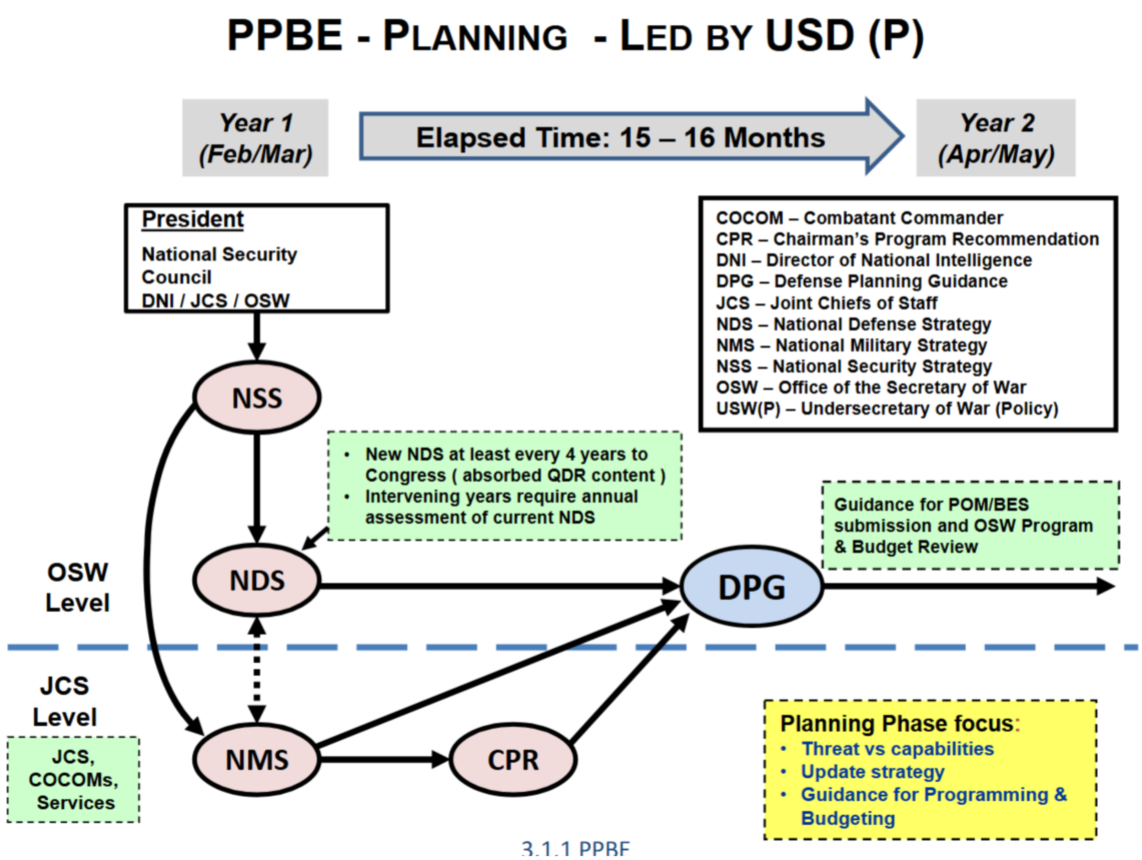

Figure 2.2. *Timeframes of the Navy PPBE process. Source:* 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), *2026 [49]*

2.1.4 Future Years Defense Program (FYDP)

Definition. The DoW's internal database for tracking resource allocations.

Navy access. The Navy uses the Program Budget Information System (PBIS) for controlled access.

Time horizon. Prior Year (PY), Current Year (CY), Budget Year (BY), plus four out-years (POM) and additional force-structure-only years.

Purpose. Track DoW funds, force structure, and personnel strength throughout PPBE.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.5 FYDP structure

The Future Years Defense Program (FYDP) lets users identify, sort, and display DoW plans by three dimensions: Major Force Program, appropriation, and component/agency [49].

Major Force Programs

Twelve MFPs. The FYDP organizes data into 12 Major Force Programs (MFPs).

Combat vs support. Six MFPs are combat force programs and six are support programs.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

Appropriations

Appropriation categories are the major funding classes by which Congress provides funds for all federal agencies, including the DoW [49].

Component or agency

The FYDP tracks funding for each military service and other DoW agencies; each data set may include multiple appropriations and MFPs [49].

Program elements

PE codes. Resources are assigned to a unique MFP using Program Element (PE) codes.

Granularity. PEs can be as narrow as a single platform or as broad as strategic planning.

Multiple appropriations. A single PE can include multiple appropriation types (for example, procurement plus Operation and Maintenance, Navy (OMN) for sustainment).

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.6 Planning phase

Lead. Led by Under Secretary of War for Policy (USW(P)) with inputs from CJCS, CCMDs, and OSW staff.

Key products. Chairman's Program Recommendation (CPR) and Defense Planning Guidance (DPG).

Focus. Threat versus capability gaps, strategy updates, and guidance for programming and budgeting.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

Planning phase key documents

CPR. CJCS coordinates with the Joint Requirements Oversight Council (JROC) and CCMDs to identify priorities for DPG input; signed by CJCS.

DPG. Drafted by USW(P) and signed by SECWAR; provides force-structure and resource priorities for program development and is the end product of the planning phase.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.7 Programming phase

POM development. Each department and DoW agency creates a Program Objective Memorandum (POM) that allocates resources across programs over five years starting with the budget year.

OSW program review. OSW conducts a program review; Cost Assessment and Program Evaluation (CAPE) adjudicates issues and CJCS signs the Chairman's Program Assessment (CPA).

Decision. Deputy SECWAR issues a Program Decision Memorandum (PDM) approving POMs as modified by the review.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.8 Navy programming process

Top-down fiscal guidance. Fiscal controls flow from OMB to OSW, DON, OPNAV, resource sponsors, and then programs.

Bottom-up POM. Units propose resources; OPNAV Integration of Capabilities and Resources (N8) integrates sponsor submissions into the Navy tentative POM; DON integrates Navy and USMC inputs and SECNAV signs the DON POM.

Issues. Proposed changes from previous funding levels are tracked and adjudicated as issues.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

Resource sponsors

Table XII: Navy PPBE resource and integration responsibilities.

Sponsor	Primary focus
N1	Manpower, personnel, training, and education.
N2/N6	Information warfare.
N3/N5	Navy warfare development.
N4	Afloat and ashore readiness.
N8	Integration of capabilities and resources; Navy PPBE integrator, not a platform sponsor.
N9	Warfare systems resource-sponsor family.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *Activity Manpower Management Guide*, *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, 2026, 2025, 2024 [49, 50, 51]

Resource integrator N8

Role. N8 integrates Navy capabilities and resources through PPBE; for EDO board purposes, treat N8 as the "checkbook" integrator that balances sponsor submissions rather than as a single warfare platform sponsor.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *Activity Manpower Management Guide*, *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, *OPNAVINST 3050.27A*, 2026, 2025, 2024, 2024 [49, 50, 51, 52]

Table XIII: N8 offices most relevant to EDO PPBE and readiness work.

Code	Office title / common label	Core function	Responsibilities / what to know
N8	Deputy Chief of Naval Operations, Integration of Capabilities and Resources	Navy-wide integration of capabilities, resources, assessments, and PPBE	N8 is the integrator, not a platform sponsor. In official laydown guidance, N8 provides the Navy Force Structure Assessment, POM planning memoranda, warfighting capability/capacity assessments, campaign analysis support, and independent assessment of approved COAs.
N80	Programming Division	Builds and integrates the Navy program / POM	Treat N80 as the **programming engine**: sponsor inputs, fiscal controls, and priorities are reconciled into the Navy program. N8 is the integrating sponsor for submissions into the Navy tentative POM.

Continued on next page

Table XIII: N8 offices most relevant to EDO PPBE and readiness work. **(Continued)**

Code	Office title / common label	Core function	Responsibilities / what to know
N81	Assessment Division	Force-structure and warfighting-capability assessments	N81 is the analytic/assessment side: force mix, capability gaps, warfighting capacity/capability, and assessment products that inform POM choices. OPNAVINST 3111.17C assigns N8 warfighting capability and capacity assessment responsibilities.
FMB / N82	DASN Budget / OPNAV Fiscal Management Division	Budget formulation, execution, and fiscal controls	Treat the FMB/N82 lane as **budgeting/execution controls**, not POM integration. Technically, FMB is the Deputy Assistant Secretary of the Navy for Budget office/title, while N82 is OPNAV's Fiscal Management Division; the same leader may be dual-hatted. See Section 2.1.8.
N83	Fleet Readiness	Fleet readiness and readiness-related integration	The study guide calls N83 the OPNAV fleet-readiness point of contact; an official OPNAV listing identifies "Director, Fleet Readiness (OPNAV N83)." Be careful: older 2012 realignment language used N83 differently.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, *OPNAVINST 3050.27A*, 2026, 2024, 2024 [49, 51, 52]

FMB and Fiscal Management Division (N82)

Organization distinction. Deputy Assistant Secretary of the Navy for Budget (FMB) refers to the Deputy Assistant Secretary of the Navy for Budget office/title; OPNAV Fiscal Management Division (N82) is OPNAV's Fiscal Management Division. A War Department flag-officer assignment notice identifies the same officer as both DASN Budget (FMB) and Director, Fiscal Management Division, N82, which is why board discussions often say "FMB/N82" as a combined budget lane [53].

Role. The FMB/N82 lane is the Navy budgeting and fiscal-control arm, responsible for turning the Navy program (POM) into an executable budget and managing its execution.

PPBE space. FMB/N82 operates in the **Budgeting + Execution** phases, distinct from N80's programming role.

Budget formulation. Translates the first budget year of the POM into the Budget Estimate Submission (BES), aligns Navy funding with OSW/OMB guidance, and prepares and defends the Navy budget through OSW review, OMB review, and Congressional submission.

Fiscal controls and execution. Manages appropriation execution, enforces obligation/expenditure controls and reprogramming rules, and tracks whether programs are spending as planned or deviating from budget intent.

Budget defense. Leads Navy response to Program Budget Decisions (PBDs) from OSW and Congressional marks by drafting reclamas, building business case and operational impact arguments, and coordinating with resource sponsors (N9, etc.).

Integration with resource sponsors. Works with N9 sponsors (N95–N99, N9I, N94) to refine budget submissions, adjust funding based on marks, and ensure affordability within constraints.

Appropriation structure. Operates heavily in appropriation categories (RDT&E, Procurement, O&M, MILPERS, etc.), PEs, and budget exhibits.

Key distinction. Programming (N80) = capability + trade space; Budgeting (FMB/N82) = appropriation control + executable funding.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, *Flag Officer Assignments*, 2026, 2024, 2020 [49, 51, 53]

FMB responsibilities (high-yield list)

For board answers, the FMB/N82 budget lane is responsible for:

- Budget formulation (BES)
- Budget justification and defense (OSW, OMB, Congress)
- Managing fiscal controls and execution
- Responding to PBDs and Congressional marks
- Ensuring compliance with appropriation law and policy
- Translating Navy priorities into executable funding
- Monitoring execution performance

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, 2026, 2024 [49, 51]

FMB common traps

Table XIV: Common misconceptions about FMB (N82).

Incorrect assumption	Correction
FMB builds the POM	N80 builds the POM
FMB validates requirements	JCIDS / N9 sponsors validate requirements
FMB is a warfare sponsor	N9 sponsors are warfare sponsors
FMB owns force structure	N81 / N8 assessments own force structure
FMB is just accounting	FMB is strategic budget control + defense
FMB and N82 are the same acronym	FMB is DASN Budget; N82 is OPNAV Fiscal Management Division

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, 2026, 2024 [49, 51]

PPBE core organizations

Table XV: Core PPBE organizations and board hooks.

Entity	Purpose	Duties / responsibilities	Board hook
CAPE	Independent cost and program-analysis office.	Conducts independent cost analysis and reviews Component POMs for strategic alignment, affordability, force balance, analytic support, and tradeoffs.	CAPE checks the program.
N8 / N80	Navy resource integration and programming engine.	Integrates Navy resource-sponsor inputs, fiscal controls, offsets, and issue papers into the Navy program and tentative POM.	N8/N80 builds the Navy POM.
OMB	President's budget-control office.	Reviews agency budget submissions with OUSD(C); aligns the defense request to presidential fiscal policy; helps shape the President's Budget submitted to Congress.	OMB shapes the PB.
FMB / N82	Navy budget and execution-control lane.	Converts the Navy program into the BES and budget exhibits; defends estimates; supports reclamas, marks, reprogramming, execution tracking, and budget justification.	FMB/N82 builds and defends the Navy budget.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *Flag Officer Assignments*, *3.1.6. Cost Estimating*, *10 U.S.C. § 139c*, *OMB Circular A-11: Preparation, Submission, and Execution of the Budget*, 2026, 2020, 2026, 2023, 2025 [49, 53, 54, 55, 56]

Resource sponsor N9

Role. OPNAV Warfare Systems (N9) advocates the largest portion of Navy Total Obligational Authority (TOA) each year, integrating expeditionary, surface, undersea, air, unmanned, and warfare-integration requirements with associated systems, manpower, training, and readiness.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *Activity Manpower Management Guide*, *OPNAVINST 3111.17C*, *OPNAVINST 5420.117A*, 2026, 2025, 2024, 2021 [49, 50, 57, 58]

Table XVI: N9 warfare systems directorates.

Code	Office title / resource sponsor description	Core function	Responsibilities / what to know
N9	DCNO for Warfighting Requirements and Capabilities / Warfare Systems	Warfare-system requirements and resource sponsorship	Official excerpts say N9 establishes requirements, sets priorities, and directs planning/programming for expeditionary, surface, undersea, air, unmanned, and warfare systems, including associated manpower, support, training, and readiness.

Continued on next page

Table XVI: N9 warfare systems directorates. **(Continued)**

Code	Office title / resource sponsor description	Core function	Responsibilities / what to know
N9B	Assistant DCNO for Warfighting Requirements and Capabilities	Senior civilian/assistant advisory role to N9	A DON senior executive bio says N9B advises N9 on warfare requirements, naval warfare system integration, budget matters, and centralized coordination of Navy programs/plans.
N9I / 9I	Integrated Warfare / Warfare Integration	Cross-domain integration across warfare portfolios	NAVMAC AMMG lists RS code **9I = Warfare Integration**. Use N9I/9I as the current board answer; N91 appears in older or alternate public references and should not be treated as the current resource-sponsor code. N9I is the cross-domain integrator when the issue cuts across surface, air, undersea, expeditionary, and unmanned portfolios.
N9DW	Digital Warfare Office	Naval operational architecture, digital integration, kill-chain integration	OPNAVINST 5400.43C states that CNO N9 and CNO N2N6 maintain a cooperative **Director, Digital Warfare Office (OPNAV N9DW)** to lead naval operational architecture development/prototyping, drive digital integration, and identify requirements to close long-range kill chains.
N94	Test & Evaluation	T&E resource sponsor	NAVMAC lists RS code **94 = Test & Evaluation, org code N94**. OPNAVINST 5450.352B excerpt says CNO N9 is resource sponsor for S&T, major range and test facility base infrastructure, Commander Operational Test and Evaluation Force personnel, and oversight of Navy T&E programs.
N95	Expeditionary Warfare	Expeditionary warfare resource sponsor	NAVMAC lists **95 = Expeditionary Warfare, org code N95**. NAVEDTRA 135D also maps N95 to expeditionary-related learning centers such as EOD/diving, security forces, Seabees/facilities engineering, and SEAL/SWCC training centers.
N96	Surface Warfare	Surface warfare resource sponsor	NAVMAC lists **96 = Surface Warfare, org code N96**. NAVEDTRA 135D maps N96 to Center for Surface Combat Systems and Surface Warfare Officers School Command.
N97	Undersea Warfare	Undersea warfare resource sponsor	NAVMAC lists **97 = Undersea Warfare, org code N97**. NAVEDTRA 135D maps N97 to the Submarine Learning Center.
N98	Air Warfare	Air warfare resource sponsor	NAVMAC lists **98 = Air Warfare, org code N98**. NAVEDTRA 135D maps N98 to the Center for Naval Aviation Technical Training.

Continued on next page

Table XVI: N9 warfare systems directorates. **(Continued)**

Code	Office title / resource sponsor description	Core function	Responsibilities / what to know
N99	Unmanned Warfare Systems	Unmanned systems resource sponsor / early integration of unmanned capabilities	NAVMAC lists **99 = Unmanned Warfare Systems, org code N99**. NAVADMIN 255/15 established N99 to identify, develop, field, and integrate unmanned technologies, accelerate prototypes and experimentation, align unmanned investment strategy, and assess future unmanned warfighting capabilities.
N9SP	Special Programs	Special programs	Older realignment moved Special Programs from N8/N89 to N9 as N9SP; treat N89 as legacy and N9SP as the current public OPNAV special-programs label when this lane matters.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, *Activity Manpower Management Guide*, *OPNAVINST 3111.17C*, *OPNAVINST 5420.117A*, *SECNAVINST 5460.4*, 2026, 2025, 2024, 2021, 2018 [49, 50, 57, 58, 59]

2.1.9 POM resources

Allocation. Future resources are allocated to buy programs that deliver planning phase capabilities across the FYDP.

Resource types. Quantities, end strength, and money.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

Table XVII: Example of POM resource types for one fiscal year (F/A-18).

Resource type	Quantity	Unit cost	Subtotal
Aircraft quantity	3	$75,000K each	$225,000K
End strength	25	$120K each	$3,000K
Flight hours	3,000	$5K per hour	$15,000K
Total			$243,000K

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.10 Navy POM exercise

Baseline and new requirements. Start with the current funded program of record, add new strategic requirements, and create offsets to balance the tentative POM.

Integration. N8 integrates resource sponsor inputs into the Navy tentative POM and the DON POM.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.11 POM example (DDG 1000 combat systems)

Stakeholders. Resource sponsor OPNAV N96E, ship PM (PMS 500), and combat systems PM (PEO IWS 9.0).

Content. Detailed POM slides across four program elements over five fiscal years, covering cybersecurity through spares.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.12 Programming versus budgeting

TABLE XVIII: PROGRAMMING VERSUS BUDGETING.

	Programming	Budgeting
Purpose	Balanced allocation of limited resources within DoW	Communicate DON and DOD intent outside DOD and inform the President's Budget
Process	OSD-developed; differs by Service	Constitutionally derived; mostly common across Services
Timeframe	Budget year plus four out-years (FYDP)	Budget year only (first year of FYDP)

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

Figure 2.3 illustrates the concurrent program and budget review cycle.

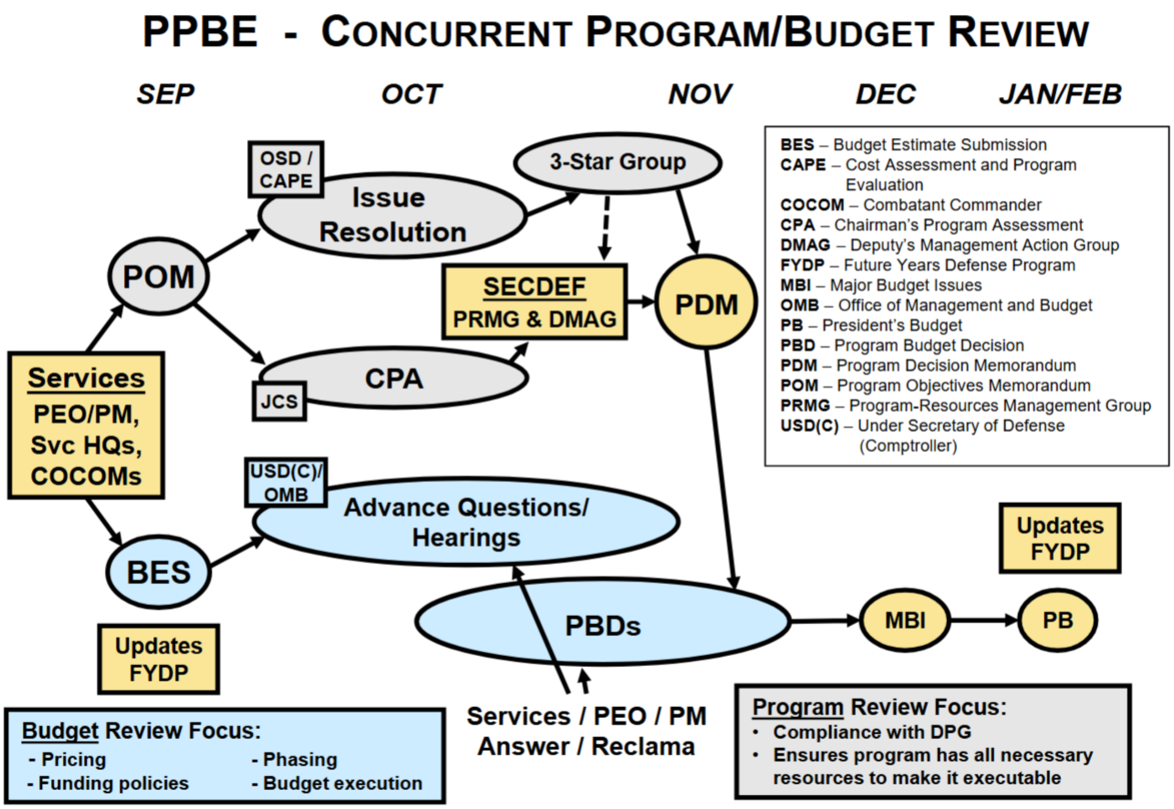

Figure 2.3. *Concurrent program and budget review. Source:* 3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), *2026 [49]*

2.1.13 Program and budget review artifacts

Table XIX: Major PPBE program and budget review artifacts.

Item	Purpose	Inputs	Outputs	Main players
POM	Five-year Component program proposal across the FYDP.	DPG, fiscal guidance, sponsor proposals, force structure, manpower, quantities, costs, offsets.	Balanced Component program request, updated FYDP data, and issues for OSD/CAPE review.	Navy sponsors, N8/N80, DON, CAPE, Secretariat.
CAPE review	Tests whether the POM is affordable, balanced, analytically supportable, and aligned to strategy.	Component POMs, DPG, CPA, issue papers, cost and force analysis, FYDP data.	Program-review issues and recommended decisions that feed the PDM.	CAPE, OSD staff, Joint Staff, Components, sponsors.
CJCS review / CPA	Military assessment of whether submitted POMs conform to strategy and combatant-command priorities.	Component POMs, NMS, combatant-command priorities, capability gaps, readiness and risk assessments.	CPA identifying risk, gaps, and alignment problems for senior decision makers.	CJCS, Joint Staff J-8, JROC/FCBs, CCMDs.
3-star group review	Internal Navy senior review that adjudicates unresolved program issues before the top-level decision forum.	Sponsor submissions, issue papers, offsets, N8 integration products, risk trades.	Recommended issue decisions or escalations to the 4-star/Secretariat level.	OPNAV 3-stars, N8/N80, resource sponsors, FMB/N82 as needed.
Secretariat / 4-star review	Senior decision forum that settles major program trades before final submission or defense.	3-star recommendations, unresolved issues, fiscal controls, strategic priorities.	Final Navy/DON position or elevated issue set.	SECNAV Secretariat, CNO/VCNO or senior Navy leadership, N8, sponsors, FMB/N82.
BES	Budget-year submission in appropriation format.	First year of the POM, pricing, phasing, execution data, fiscal controls, budget exhibits.	Component budget request for OUSD(C)/OMB review.	FMB/N82, comptrollers, sponsors, PMs, OUSD(C), OMB.
PDM	Programming decision document.	POM, CAPE/program review, CPA, issue papers, senior decisions.	Approved programmatic changes to the POM/FYDP.	CAPE and OSD staff support; Deputy SecDef issues.
PBD	Budgeting decision document.	BES, OUSD(C)/OMB hearings, budget-analyst review, execution data, reclamas.	Approved budget adjustments for the President's Budget.	OUSD(C), OMB, Components, FMB/N82; Deputy SecDef final decision.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution), 10 U.S.C. § 139c, OMB Circular A-11: Preparation, Submission, and Execution of the Budget*, 2026, 2023, 2025 [49, 55, 56]

The flow is: Planning produces the DPG; the DPG tells Components what strategic priorities and fiscal constraints to program against; Navy sponsors propose requirements and resources; N8/N80 integrates them into the Navy POM; CAPE reviews the POM; and CJCS separately provides the CPA on strategy, risk, and combatant-command alignment. In parallel or near-parallel, FMB/N82 turns the budget year of the program into the BES; Office of the Under Secretary of War (Comptroller) (OUSW(C)) and OMB review the BES for pricing, phasing, funding policy, execution, and defendability. Programming decisions become PDMs; budget decisions become PBDs. If a Service strongly disagrees with a draft budget decision, it submits a reclama; if still unresolved and important enough, the issue may become an Major Budget Issue (MBI) before final President's Budget (PB) lock.

2.1.14 Programming phase key documents

POM. Time-phased resource allocation to support the DPG; reviewed for compliance with DPG and the National Military Strategy (NMS); signed by Service Secretaries (SECNAV for the Navy).

PDM. SECWAR direction that summarizes decisions and approves the component POM as modified by the program and budget review; issued by the Deputy SECWAR.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.15 Budgeting phase key events and documents

Budget Estimate Submission. Services submit BES or Budget Change Proposals in appropriation format; signed by Service Secretaries.

Budget hearings. Joint OSW and OMB hearings with advance questions on programs.

Program Budget Decisions. PBDs direct budget adjustments; Services prepare reclamas; signed by the Deputy SECWAR.

Major Budget Issues. MBI provide the last chance to raise issues to OSW before the PB.

President's Budget. PB is submitted to Congress (nominally the first Monday in February).

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

PBDs and reclamas

PBD origin. Issued by USW(FM&C) analysts after review of the POM/BES and budget hearing responses.

Reclama. Draft PBDs are vetted for comment; affected parties submit reclamas with operational impacts and business case justification.

Final decision. USW(FM&C) recommends to the Deputy SECWAR, who signs the final PBDs.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.1.16 Execution phase

Purpose. Assess programs of record against actual performance and adjust plans and budgets to minimize near-term financial risk.

Activities. Mid-year review, reprogramming, supplemental appropriations, and financial closeout.

Source: *3.1.1. PPBE (Planning, Programming, Budgeting, and Execution)*, 2026 [49]

2.2 Field Activity Financial Management

2.2.1 Cost categories

Direct costs. Costs directly attributable to a project (for example, materials, labor, and travel tied to a specific job) [60].

Indirect costs. Costs that cannot be attributed to a single project (for example, security, executive salaries, and shared services) [60].

Note

Direct costs typically include labor, material, and Other Direct Cost (ODC); indirect costs include overhead and General and Administrative (G&A) [60].

2.2.2 Financial management systems

Mission funded (General Fund). Appropriated funds flow down the chain of command to execute mission needs; the goal is to budget adequate resources without excess [60].

Working capital funded (NWCF). A reimbursable system where activities are not directly appropriated; they recover costs from customers and aim to break even over time [60].

Note

General Fund activity management is a *"checking account"* model; Navy Working Capital Fund (NWCF) is a *"credit card"* model. Some NWCF commands have mission-funded units, but funds cannot be mixed [60].

2.2.3 Activity examples

Table XX: Mission-funded vs NWCF activity examples

Mission funded (General Fund)	NWCF activities
• Naval Shipyards, Regional Maintenance Centers, Ship Repair Facilities • Supervisors of Shipbuilding, Program Executive Offices • Systems commands (HQ only) • Operating forces (fleet commanders, TYCOMs, ships) • Most other activities (e.g., EDO School)	• SYSCOM Echelon III activities (NAVSUP FLCs; NAVAIR/NAVSEA/NAVWAR warfare centers; NAVFAC facility engineering commands and expeditionary warfare centers; Naval Research Laboratory) • Military Sealift Command • Depot maintenance (non-ship): Fleet Readiness Centers (aircraft), Marine Corps depots

Source: *3.1.5. Field Activity Financial Management*, 2026 [60]

2.2.4 General Fund operations

Appropriation flow. Assistant Secretary of the Navy (Financial Management and Comptroller) (ASN(FM&C)) (Echelon I) allocates funds to Echelon II Budget Submitting Offices (BSOs); BSOs distribute allotments and Expense Operating Budgets (EOBs). Echelon III field activities issue allowances and Operating Targets (OPTARs) to subordinate units at Echelon IV [60].

2.2.5 Appropriated versus reimbursable funds for mission-funded activities

Appropriated funds. Pay for mission labor (civilian salaries), mission non-labor (materials, travel, contracts), and all indirect costs (overhead) [60].

Reimbursable funds. Used when a mission-funded activity performs work for a customer; reimbursements cover direct labor and other direct costs, but not indirect costs [60].

2.2.6 General Fund budgeting cycle (EOB)

1. Mid-year review of the current year budget [60].

2. Answer budget data calls [60].

3. Submit next year's (future) budget [60].

4. Marks and reclamas (summer review) [60].

5. Sept/Oct: end of current FY, start of next FY; receive EOB and execute [60].

2.2.7 What is the NWCF?

Definition. NWCF is a reimbursable operation that provides goods or services to customers and recovers the full cost of output [60].

Criteria to join. Output costs must be identifiable, customers must be known, and the advantages and disadvantages of the buyer-seller relationship must be evaluated [60].

2.2.8 NWCF cycle of operations

1. Congress provides corpus to begin operations [60].
2. The NWCF activity receives a reimbursable work request via a funding document [60].
3. The activity performs work; direct and indirect costs are financed by the corpus [60].
4. Costs are priced using stabilized rates held constant during execution [60].
5. The customer is billed for costs incurred; payment reimburses the corpus [60].

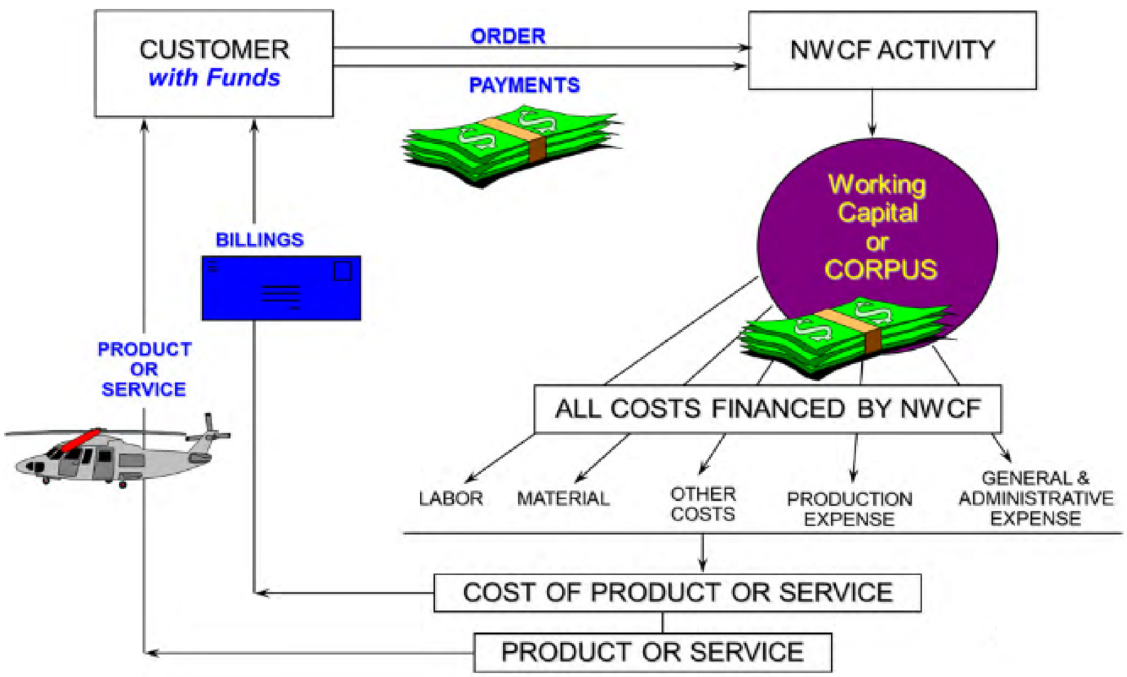

Figure 2.4. NWCF Cycle of Operations. Source: 3.1.5. Field Activity Financial Management, *2026 [60].*

2.2.9 Stabilized labor rates (SLR)

Purpose. Stabilized Labor Rate (SLR) reimburses NWCF activities for work performed and is established during the budget process to recover full costs, prior period gains/losses, and approved capital surcharges [60].

Formula.

$$\text{SLR} = \frac{\text{Direct labor} + \text{Indirect (overhead/G\&A)} + \text{AOR recoupment}}{\text{Direct hours}}$$

Components. Direct labor, indirect expenses (including G&A), and Accumulated Operating Result (AOR) recoupment to achieve break-even [60].

2.2.10 Operating results: NOR and AOR

NOR. Current-year profit or loss based on revenue minus expenses [60].

AOR. Cumulative gains and losses since inception; AOR is managed at Navy level and used to set rates [60].

Rate adjustment. Gains reduce rates two years out; losses increase rates two years out to drive AOR toward zero in the budget year [60].

2.2.11 NWCF budget timeline

Feb–Apr. Internal budget formation at the activity level [60].

Apr–May. Activity budgets consolidate to activity group and submit to the BSO [60].

May–Jun. BSO submits the activity group budget to FMB [60].

Jul–Aug. FMB submits the activity group budget to DoW for the BES [60].

Sep. DoW submits to Congress [60].

Feb. President signs the budget (first Monday in February); rates are set 18–24 months in advance [60].

2.2.12 Funding documents and cost stages

Reimbursable vs direct cite. Reimbursable work charges costs to the NWCF corpus and then bills the customer; direct cite (commercial contracts only) charges the customer's line of accounting directly [60].

Cost stages. Committed (funding document received), obligated (contract awarded), costed (work performed/material received) [60].

2.2.13 Carryover

Definition. Dollar value of reimbursable work ordered and funded but not yet completed at FY end [60].

Formula. Carryover = Available balance (cash) + commitments + obligations (costed transactions excluded) [60].

Rules. Some carryover is required to start the new FY; excess carryover can trigger congressional marks and future budget reductions [60].

2.2.14 Work Requests (WR) (Economy Act Orders)

Specificity. Used to request routine or recurring day-to-day services within the Navy [60].

Timing. Work must be completed before expiration of funds cited [60].

Restrictions. Cannot be used to fund the basic mission of the performing activity [60].

2.2.15 Project orders (PO)

Specificity. Description of work and period of performance must be specific and produce a tangible end product [60].

Timing. Work may extend past fund expiration but not beyond cancellation; Project Orders (POs) cannot be issued solely to extend funds and scope cannot expand after expiration [60].

51/49 rule. At least 51% of reimbursable work must be performed in-house; no more than 49% can be out-house [60].

Restrictions. POs are reimbursable only and limited to DoW activities; they extend the cited funds beyond expiration and carry additional restrictions [60].

2.2.16 Other funding documents

MIPR. Transfers funds between military organizations for supplies or services; can be accepted on a direct cite or reimbursable basis [60].

IPR. Used when requiring and performing activities are from different departments (outside DoW); cannot cite the PO statute but may cite Work Request (WR) or other statutory authorities [60].

FS Form 7600B. Treasury G-Invoicing form for reimbursable Intragovernmental Transactions (IGTs); applicable to manual or electronic submissions and not limited to WR or PO authorities [60].

2.3 Congressional Enactment

2.3.1 Constitutional basis for appropriations

Article I authority. Congress provides for the common defense and general welfare and controls the Treasury.

Section 8.

"The Congress shall have power to... provide for the Common Defense... and general welfare..."

Section 9.

"No money shall be drawn from Treasury, but in consequence of Appropriations made by law..."

Appropriations. All funding to operate the federal government is appropriated by Appropriations Acts.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.2 Key terms

What is a budget?

Definition. A financial plan for accomplishing an organization's objectives.

Management tool. An instrument of planning, decision making, and management control.

Priorities. A statement of priorities and fiscal policy.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

Budget definitions

Budget authority. Authority granted by appropriation law to enter obligations that will result in immediate or future outlays.

Commitment. Administrative reservation of funds, usually by the local comptroller, in anticipation of a future obligation.

Obligation. Legal reservation of funds to make a future payment of money.

Expenditure. Charge against available funds resulting from a voucher, claim, or other document approved by competent authority.

Outlay. Occurs when the vendor cashes the expenditure check and money flows from the Treasury to the vendor or supplier.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

Authorization versus appropriation

Authorization. A statute providing DoW legal authority to do something (for example, procure a ship or conduct operations). It authorizes or modifies programs and sets spending ceilings, but does not provide funding; drafted by House Armed Services Committee (HASC)/Senate Armed Services Committee (SASC).

Appropriation. A statute providing Budget Authority (BA) for federal agencies; drafted by House Appropriations Committee (HAC)/Senate Appropriations Committee (SAC).

Rule. Defense programs must have authorization to conduct activities and appropriated funding to pay for them.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.3 Congressional roles and responsibilities

Personal versus professional staff

Personal staff. Employees of individual members of Congress; provide constituent support and policy expertise with limited DoW interaction.

Professional staff members. Committee staff who review the President's Budget down to the program element; the primary congressional interface with DoW; PMs engage directly with Professional staffers.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

Oversight and analysis

Government Accountability Office. Congress' chief investigator, auditor, and program evaluator.

Congressional Budget Office Supports Budget and Appropriations Committees; analyzes the President's Budget, revenues, and outlays.

Congressional Research Service. Provides confidential, impartial research and analysis for Congress.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

Key committees

Budget Committees. House Budget Committee (HBC)/Senate Budget Committee (SBC) craft the annual Budget Resolution.

Armed Services Committees. HASC/SASC craft the annual National Defense Authorization Act (NDAA).

Appropriations Committees. HAC/SAC craft annual Appropriations Bills.

Conferences. Joint House/Senate meetings reconcile differences; members appointed by Speaker and Senate Majority Leader.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.4 Congressional enactment process

Budget Resolution. Blueprint that sets top-line spending ceilings for 20 major government functions.

Authorization Act. Approves programs, quantities, end strength, and policy (NDAA).

Appropriation Bills. Provides BA (legal authority to spend).

Source: *3.1.2. Congressional Enactment*, 2026 [61]

Congressional enactment timeline

Figure 2.5 shows the budget resolution, authorization, and appropriations flow across the fiscal year.

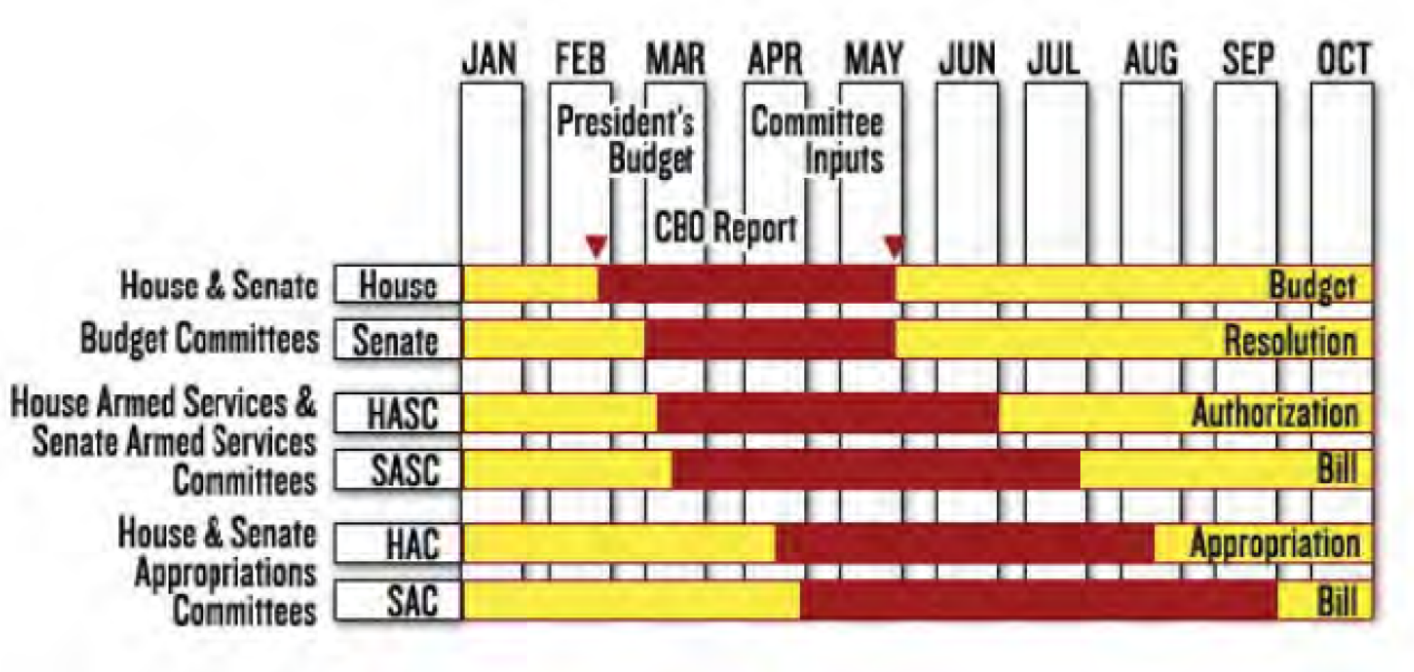

Figure 2.5. *Congressional Enactment Timeline. Source:* 3.1.2. Congressional Enactment*, 2026 [61].*

2.3.5 Budget Resolution

Drafted by. HBC/SBC; process established by the Budget and Impoundment Control Act of 1974.

Purpose. Sets spending ceilings for 20 major government functions; not a law and independent of the President's Budget.

Process. Congressional Budget Office (CBO) guidance; House and Senate develop and debate resolutions; conference resolves differences; concurrent budget resolution returned to floors.

Goal. Budget Resolution by 15 April; appropriation sub-allocations follow the ceilings.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.6 Authorization

NDAA. Drafted by HASC/SASC; allows programs to be established or continued; provides authorization for each DoW appropriation account.

Annual authorization law. Program approval, procurement quantities, personnel end strength, funding ceilings, and policy updates.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

Authorization subcommittees

Table XXI: HASC and SASC subcommittees.

HASC	SASC
Tactical Air and Land Forces	Cybersecurity
Military Personnel	Emerging Threats and Capabilities
Oversight and Investigations	Personnel
Readiness	Readiness and Management Support
Seapower and Projection Forces	Seapower
Strategic Forces	Strategic Forces
Emerging Threats and Capabilities	

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.7 Appropriations

Committees. HAC/SAC; 12 annual appropriation bills to operate the federal government.

Subcommittees. Each bill has its own subcommittee; bills may be combined into consolidated or omnibus bills.

Timing. Hearings kick off around March; committees should follow authorizer guidance.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

HAC/SAC subcommittees

List.
- Agriculture, Rural Development, Food and Drug Administration;
- Commerce, Justice, Science;
- Defense;
- Energy and Water Development;
- Financial Services and General Government;
- Homeland Security;
- Interior and Environment;
- Labor, Health and Human Services, Education;
- Legislative Branch;
- Military Construction and Veterans Affairs;

- State and Foreign Operations;
- Transportation, Housing and Urban Development.

Navy focus. Most Navy interaction is with the Defense subcommittees (HAC-D/SAC-D).

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.8 Appeals to congressional marks

Definition. An appeal requests reconsideration of congressional adjustments (authorization or appropriation) to dollars, quantities, manpower, or language.

Committee routing. Appeal to the next committee in the phase (for example, SASC, SAC, or the conference committee).

Dollar rule. Appeal to the dollar amount closest to the President's Budget; generally to the higher of the two prior marks unless it exceeds the President's position.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.9 Continuing Resolution

Definition. Special stopgap appropriations that keep government operating when annual appropriations are not enacted on time.

Typical rules. Operate at current rates; new programs cannot start unless specified; funding limited to last year's levels or current committee marks.

Impacts. Restrictive apportionment, limited obligation rates, and cost/schedule risk; shipbuilding may receive specific anomalies.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.3.10 Sequestration

Definition. Automatic spending cuts through withdrawal of funding for certain government programs.

Roles. CBO estimates statutory caps and sequestration risk; OMB determines if sequestration is required and its size.

Source: *3.1.2. Congressional Enactment*, 2026 [61]

2.4 Program Funding and Execution

2.4.1 Program Funding

Financial terms

Budget authority. BA is authority granted by appropriation law to enter obligations that will result in immediate or future outlays [62].

Commitment. Administrative reservation of funds (usually by the local comptroller) in anticipation of a future obligation [62].

Obligation. Legal reservation of funds to make a future payment of money [62].

Expenditure. Charge against available funds resulting from a voucher, claim, or other document approved by competent authority [62].

Outlay. Occurs when the vendor cashes the expenditure check and money flows from the Treasury to the vendor or supplier [62].

Credit-card analogy. BA is the line of credit, commitment is the intent to purchase, obligation is the charge slip, expenditure is the bill/check, and outlay is the cash transfer from Treasury [62].

Appropriation categories and obligation windows

Table XXII: Appropriation categories, typical uses, and obligation windows

Category	Typical use	Obligation window
RDT&E	Research, development, test, and evaluation	2 years
Procurement	End items, modernization, and spares (e.g., APN/WPN/OPN)[a]	3 years
O&M	Operations, maintenance, training, minor construction, travel	1 year
MILPERS	Military personnel pay and allowances	1 year
MILCON	Facilities design and construction	5 years

[a] SCN is a procurement appropriation with a 5-year obligation window.

Source: *3.1.3. Program Funding*, 2026 [62]

DoW appropriation accounts (DoN examples)

Table XXIII: DoN appropriation accounts (examples)

Appropriation	Accounts
Military Personnel	MPN; RPN; MPMC

Continued on next page

Table XXIII: DoN appropriation accounts (examples) **(Continued)**

Appropriation	Accounts
Operation and Maintenance	O&M,N; O&MNR; O&MMC
Research, Development, Test & Evaluation	RDT&E,N
Procurement	APN; WPN; SCN; OPN; PMC
Military Construction	MCN; Family Housing, Navy and Marine Corps (FH,N&MC)

Source: *3.1.3. Program Funding*, 2026 [62]

Appropriation life-cycle and expired funds

Current. Funds are available for new obligations during the appropriation's obligation window (1–5 years depending on category) [62].

Expired. A 5-year period where funds are no longer available for new obligations but may be used for obligation adjustments, expenditures, and outlays [62].

Canceled. Funds are closed; remaining balances return to Treasury [62].

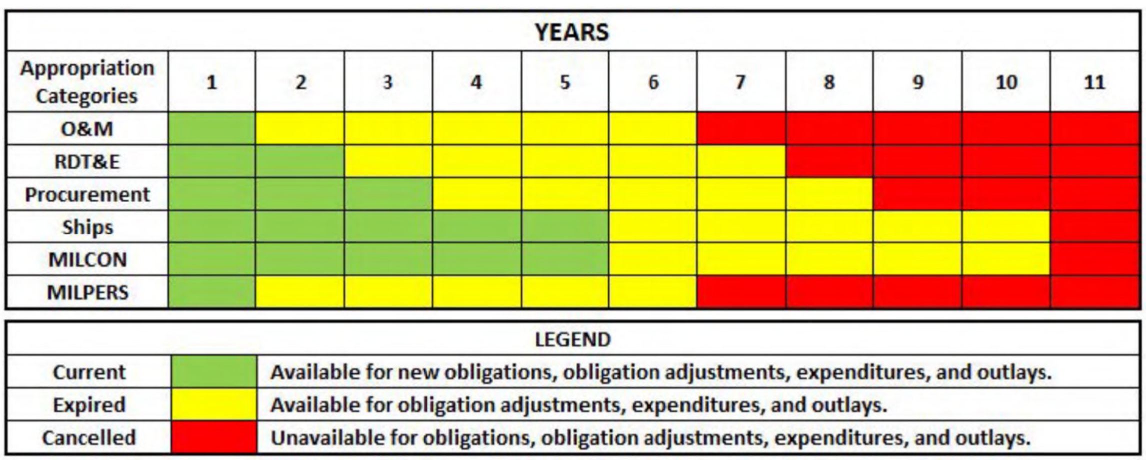

Figure 2.6. *Appropriations life-cycle (Current, Expired, Canceled). Source:* 3.1.3. Program Funding, *2026 [62]*

Expired funds example 1. In Aug 2025, FY22 Other Procurement, Navy (OPN) funds are expired (FY22–FY24 obligation window), so a Procuring Contracting Officer (PCO) cannot sign a new contract; the funds are limited to obligation adjustments, expenditures, and outlays [62].

Expired funds example 2. In Jan 2025, FY24 RDT&E funds are still current (FY24–FY25), so a PCO may sign a new contract for work starting in FY25 [62].

Note

Obligation windows: Military Personnel (MILPERS) and OMN are 1 year; RDT&E is 2 years; procurement is 3 years (Shipbuilding and Conversion, Navy (SCN) is 5); Military Construction (MILCON) is 5 years [62].

Funding policies

Annual funding. Applies to expense appropriations (OMN, MILPERS); request BA only for the goods or services needed in a given fiscal year, with a major exception for non-severable service contracts that cross fiscal years [62].

Incremental funding. Applies to RDT&E; request funding only for work expected to be performed in the given fiscal year. The Navy is authorized to use incremental funding for SCN on certain ship programs (Nuclear-Powered Aircraft Carrier (CVN), Ballistic Missile Submarine, Nuclear (SSBN), LHA/LHD) [62].

Full funding. Applies to procurement and MILCON; annual budget requests must cover the total cost of complete, usable end items delivered within a 12-month period. If multiple end items are bought, do not budget for more items than can be delivered within 12 months; options can extend procurement across years [62].

Note

Incremental funding example: A 28-month, $260M RDT&E effort with FY24–FY26 costs of 35%/45%/20% would request $91M (FY24), $117M (FY25), and $52M (FY26) [62].

Note

Full funding practical: A Weapons Procurement, Navy (WPN) contract with deliveries accepted over two years violates full funding; fix by structuring lots or options so each lot delivers within 12 months and is funded with the year-of-delivery appropriation [62].

Table XXIV: Funding policy summary

Appropriation	Policy
RDT&E	Incremental
Procurement[i]	Full
O&M	Annual
MILPERS	Annual
MILCON	Full

[i] Select SCN programs with congressional approval will have incremental funding.

Source: *3.1.3. Program Funding*, 2026 [62]

Program funding profiles by acquisition phase

Table XXV: Primary appropriations by acquisition phase

Phase	Primary appropriations
MSA	RDT&E (analysis and early prototypes)
TMRR	RDT&E (risk reduction, development, test)
EMD	RDT&E (build/test/qualify)
P&D	Procurement (WPN/OPN/APN/SCN)
O&S	O&M

Source: *3.1.3. Program Funding*, 2026 [62]

Exceptions to full funding

Advance procurement (long lead items). An exception to full funding for long-lead components; documented in the acquisition strategy, approved at milestone, and funded as a separate line item to protect schedules or critical skills [62].

Multiyear procurement (MYP). A single contract for a specific quantity of usable end items delivered over more than one year (no more than five). Congress authorizes the full contract scope at award (NDAA), but funding and appropriations remain annual [62].

Benefits. Stability in acquisition and budgeting, productivity gains, cost growth reduction, Economic Order Quantity (EOQ) opportunities, and reluctance to cancel once in place [62].

Challenges. Termination liability if funding is not provided, inability to budget for termination liability, and political controversy (Congress and OSW) [62].

Required criteria. Savings relative to annual contracting (guidance 10%), stable requirements/design, confidence in cost estimates and production capability, and commitment to funding by Service, OSW, and Congress [62].

Initiation. Proposed in the President's Budget; if approved, authorized by the NDAA and funded through annual appropriations on a year-by-year basis [62].

Note

DDG-51 Multiyear Procurement (MYP) example: The FY98–FY01 and FY02–FY05 MYP contracts saved over $1B; the FY13–FY17 MYP saved about $1.5B and enabled a 10th ship option [62].

Other exceptions. Block Buy (BB) (fewer restrictions than MYP), cost-to-complete (prior-year completion), and incremental funding for Refueling and Complex Overhaul (RCOH) [62].

Funding product improvements

Phase 1: Development and testing. RDT&E when the modification enhances performance or requires independent Developmental Test (DT)/Initial Operational Test and Evaluation (IOT&E); otherwise Procurement for in-production systems, or OMN if the system is no longer in production [62].

Phase 2: Fabrication. Procurement [62].

Phase 3: Installation. Procurement [62].

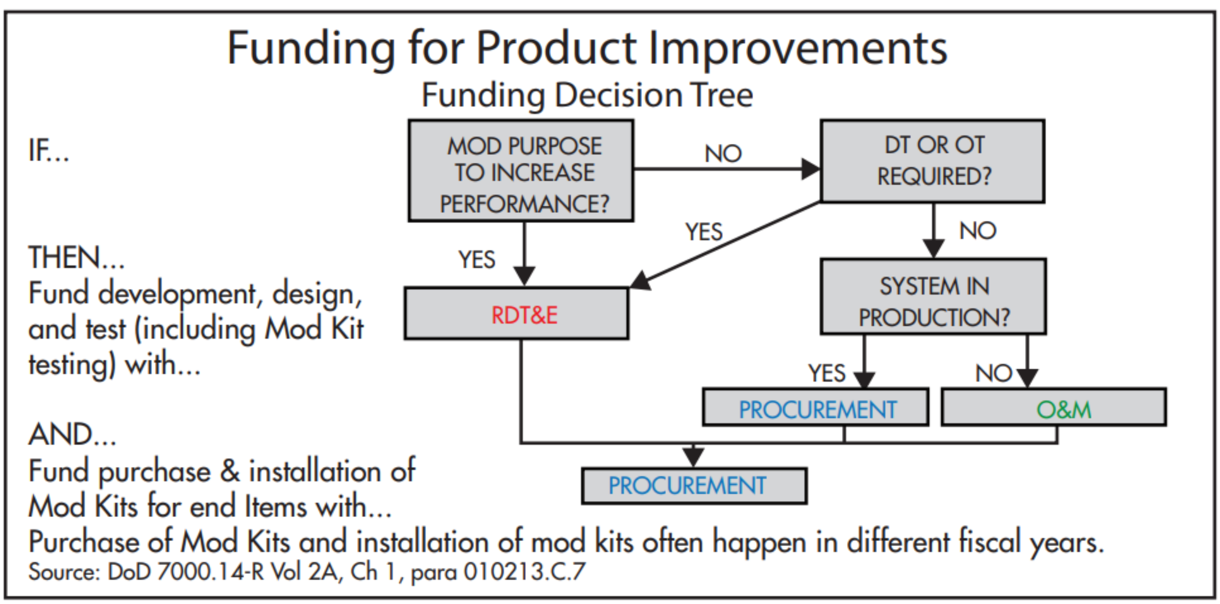

Figure 2.7. *Product improvement funding decision tree. Source:* 3.1.3. Program Funding, *2026 [62]*

MILCON funding

Program. The acquisition budget culminates in the annual Military Construction Program submitted to Congress for authorization and appropriation [62].

Threshold. MILCON is the acquisition (design and/or construction) of facilities costing more than $4,000,000 in support of active and reserve requirements [62].

Sustainment. Maintenance and repair activities funded with OMN [62].

Modernization. Alteration or replacement of facilities to implement new standards, new functions, or components with 50+ year service life; funded with OPN [62].

Rule. It is expressly forbidden to incrementally fund a project to avoid MILCON triggers [62].

Appropriation summary

Table XXVI: Appropriation abbreviations and obligation windows

Appropriation	Abbreviation	Obligation window (years)
Military Construction, Navy	MCN	5
Military Construction, Naval Reserve	MCNR	5
Shipbuilding & Conversion, Navy	SCN	5
Aircraft Procurement, Navy	APN	3
Weapons Procurement, Navy	WPN	3
Other Procurement, Navy	OPN	3
Research, Development, Test & Evaluation, Navy	RDT&EN	2
Operations & Maintenance, Navy	O&MN	1
Operations & Maintenance, Naval Reserve	O&MNR	1
Military Pay, Navy	MPN	1
Reserve Pay, Navy	RPN	1

Source: *3.1.3. Program Funding*, 2026 [62]

Summary check

Policies. Annual (OMN, MILPERS); Incremental (RDT&E, select SCN); Full Funding (Procurement, MILCON) [62].

Full funding exceptions. Advance Procurement and MYP are the primary exceptions; BB, cost-to-complete, and RCOH are additional cases [62].

Product improvements. Phase 1 uses RDT&E/Procurement/OMN depending on testing and production status; Phases 2–3 use Procurement [62].

Expired funds rule. Expired funds cannot create new obligations; they are limited to obligation adjustments, expenditures, and outlays [62].

2.4.2 Budget Execution

Execution terminology and flow

Budget authority. Authority granted by appropriation law to enter obligations that will result in immediate or future outlays [63].

Commitment. Administrative reservation of funds (usually by the local comptroller) in anticipation of a future obligation [63].

Obligation. Legal reservation of funds to make a future payment of money [63].

Expenditure. Charge against available funds resulting from a voucher, claim, or other document approved by competent authority [63].

Outlay. Occurs when the vendor cashes the expenditure check and money flows from Treasury to the vendor or supplier [63].

Reprogramming. Redirecting funds for purposes other than those contemplated at the time of appropriation; rules are based on agreements between DoW and Congress, and DoW reports reprogramming semi-annually to Congress [63].

Obligation plan. Month-by-month projection of available funds for new obligations for each fiscal year; prepared by the PMO [63].

Expenditure plan. Month-by-month projection of expenditures for each fiscal year of funding; required for all appropriation years not fully expended, even after the obligation window closes [63].

Execution flow. PMO submits a purchase request, the comptroller certifies funds (commitment), the contracting officer signs the contract (obligation), the finance office issues payment (expenditure), and the Treasury disburses cash (outlay) [63].

Execution laws (purpose, amount, time)

Misappropriation Act / Purpose Statute (Purpose). Funds must be used only for the purposes and programs for which the appropriation was made unless otherwise provided by law. The board-safe DoD acquisition term is “Misappropriation Act”; federal appropriations-law references also commonly call 31 U.S.C. § 1301(a) the “Purpose Statute”[63, 64, 65].

Anti-Deficiency Act (Amount). Prohibits obligations or expenditures in excess of available appropriations, before appropriations are available, or beyond apportionment/allotment authority. Amount violations create formal accountability and reporting requirements under 31 U.S.C. § 1341 and § 1517[63, 66].

Bona Fide Need Rule (Time). Funds must meet a bona fide need arising during the appropriation’s period of availability. It is not enough that funds are convenient or still on hand; the requirement must belong to the proper fiscal-year obligation window under 31 U.S.C. § 1502(a)[63, 67, 68].

Which law? Obligating funds before an appropriation act violates the Anti-Deficiency Act; using Procurement funds for an RDT&E purpose violates the Misappropriation Act/Purpose Statute; obligating FY04 RDT&E for FY08 services violates the Bona Fide Need Rule[63].

Board cue. Purpose asks whether the color of money is legally available for the object of the expense; time asks whether the need belongs to the appropriation's period of availability; amount asks whether sufficient budget authority is available and not exceeded. A purpose violation is not automatically an Anti-Deficiency Act violation, but correcting the wrong color of money can create an Anti-Deficiency Act problem if the proper appropriation has no available funds.

Spend plans (obligation and expenditure)

Required for. Procurement line items, RDT&E program elements, and OMN sub-activity groups each require spend plans [63].

Obligation plan. Month-by-month projection of new obligations; prepared by the PMO [63].

Expenditure plan. Month-by-month projection of expenditures; maintained for all appropriation years not fully expended [63].

Basis. Spend plans are essential tools for managing execution and are based on the BES [63].

Failure to execute. Programs that miss obligation or expenditure goals become targets for reprogramming, risk losing funding, and face future budget adjustments [63].

Reprogramming types

Congressional prior approval (PA). Required for congressional special interest items, major end item quantity increases, use of transfer authority, actions above Below-Threshold Reprogramming (BTR) thresholds, and program starts/cancellations. Threshold guidance is $15M for MILPERS, OMN, Procurement, and RDT&E. SECWAR requests approval from HAC/SAC/HASC/SASC and, when applicable, intelligence committees [63].

Congressional notification. Certain actions require notification to Congress even if they do not need prior approval [63].

Internal reprogramming (IR). Creates an audit trail for actions not requiring congressional approval; used to reclassify funds between line items or program elements without changing program substance and to execute transfer authority accounts [63].

Below-threshold reprogramming (BTR). Component-level reprogramming below congressional thresholds; controlled by comptrollers and generally does not require congressional involvement. BTR cannot change appropriation categories or execute actions requiring congressional approval [63].

Letter transfer. Used to process congressionally directed transfers (e.g., Defense Health Program to Department of Veterans Affairs) [63].

Level of control. Transfers below the appropriation's level of control are not considered reprogramming and can be approved by the program manager [63].

Note

Congress recognizes the need for flexibility in execution, but reprogramming rules are based on DoW/Congress agreements and reported to Congress semi-annually [63].

Note

Example (PA): Virginia-class submarine reprogramming realigned $29.8M of SCN 04/08 and transferred $10.2M (OPN 04/06) and $9.0M (RDT&E 04/05) to SCN [63].

Funding cuts and impact statements

Impact statement. A written assessment of the effects a budget cut will have on a program [63].

Rules. Provide specific, credible impacts; use simple language and spell out acronyms; be prepared to follow through with program changes cited [63].

Contents. Operational impact (capability and delay), business case (cost growth), policy non-compliance risks, and program impact tied to operational impact and Acquisition Program Baseline (APB) deviations [63].

Note

Example rewrite: A vague staffing loss becomes a quantified operational impact (e.g., satellite launch delayed two months, reducing coverage to 65%) [63].

Funding cuts: impacts on industry

Proposals. Bidders may need to revise price, performance, or schedule [63].

Workforce impacts. Contractors may shift resources and personnel; overhead rates can rise as workload declines [63].

Cost case. Will-cost/should-cost analysis helps justify the current budget by showing savings already identified and implemented [63].

Appropriation subdivisions

Table XXVII: Appropriation subdivisions and level of control

Appropriation	Subdivisions and level of control
RDT&E	Budget activities, then program elements, then projects; Congress appropriates at the program element level.
Procurement	Budget activities, then line items; Congress appropriates at the line item level.
O&M	Budget activities, then activity groups, then sub-activity groups; Congress appropriates at the budget activity level.

Continued on next page

Table XXVII: Appropriation subdivisions and level of control **(Continued)**

Appropriation	Subdivisions and level of control
MILPERS	Budget activities, then budget sub-activities; Congress appropriates at the budget activity level.
MILCON	Budget activities, then subordinate accounts, then projects; Congress appropriates at the project level.

Source: *3.1.4. Budget Execution*, 2026 [63]

Summary check

Execution laws. Misappropriation Act (purpose), Anti-Deficiency Act (amount), and Bona Fide Need Rule (time) [63].

Funding cuts. Programs respond with impact statements; cuts do affect industry [63].

2.4.3 Coursebook cue: apportionment, allocation, and allotment

Budget execution authority moves in controlled steps. An appropriation gives Congressional authority; an apportionment limits how the executive branch may use that authority over time, program, or activity; an allocation distributes authority to subordinate organizations; and an allotment is the formal authorization that permits an accountable activity to incur obligations [63].

Table XXVIII: Budget execution authority chain.

Step	Board cue
Appropriation	Congress provides budget authority for a purpose, time, and amount.
Apportionment	OMB controls the rate or category of obligation to prevent overexecution.
Allocation	A higher headquarters distributes authority to a subordinate command or account.
Allotment	The accountable activity receives authority to incur obligations within the stated limits.

Board cue. Anti-deficiency risk is measured against the legally controlling limit, not just the headline appropriation total.

2.5 Cost Estimating

2.5.1 Life-cycle cost composition

Flyaway/Sailaway/Rollaway cost. Cost of procuring only the prime mission equipment (aircraft, ship, tank) [54].

Weapon system cost. Prime mission equipment plus support items required to deliver a functioning system (e.g., manuals, test equipment) [54].

Procurement cost. Weapon system cost plus support items and initial spares required to stock the supply system [54].

Program acquisition cost. Costs to develop, procure, and house the system, including facilities [54].

Operating and support cost. All costs to operate, modify, maintain, and support the system in the inventory [54].

Disposal cost. Demilitarization and disposal after useful life [54].

> **Note**
>
> Life-Cycle Cost (LCC) equals program acquisition cost plus operating and support plus disposal; Operations and Support Phase (O&S) is typically the largest share (about two-thirds) for most programs [54].

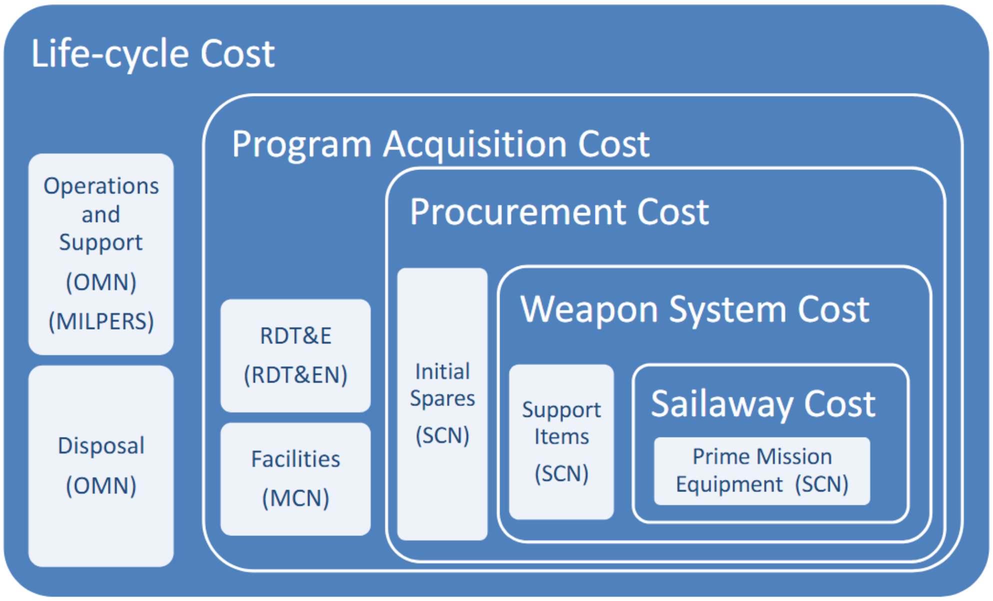

Figure 2.8. Life-cycle cost composition. Source: 3.1.6. Cost Estimating, *2026 [54]*

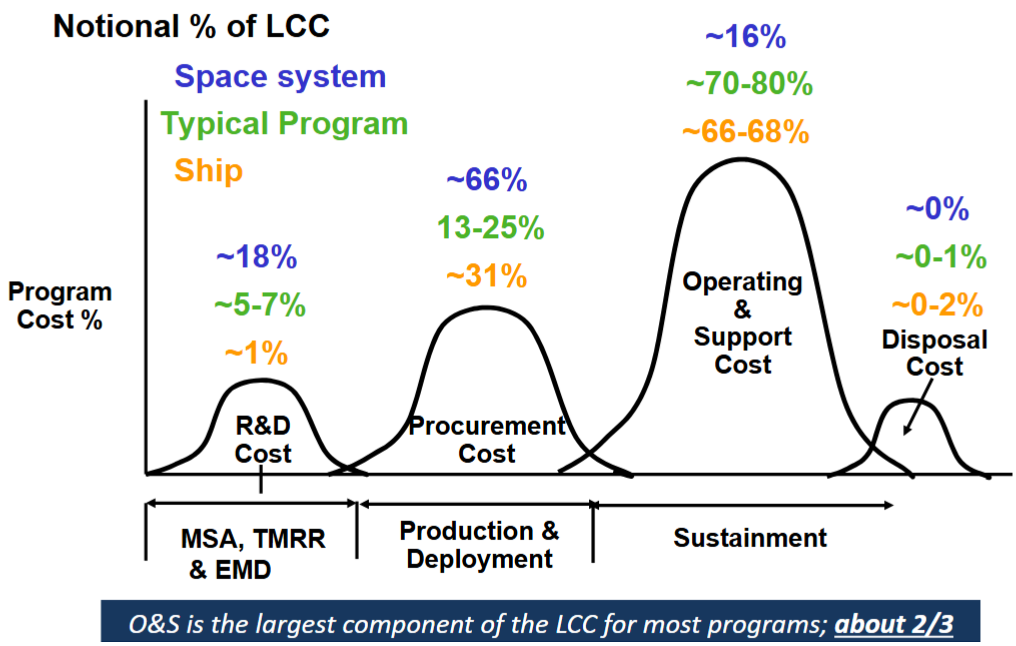

Figure 2.9. *Notional life-cycle cost distribution over time. Source:* 3.1.6. Cost Estimating, *2026 [54]*

2.5.2 Total ownership cost (TOC)

TOC. Total Ownership Cost (TOC) includes LCC plus common spares/support systems, infrastructure, and distributed indirect costs across the DoW enterprise [54].

TOC vs LCC. LCC is a subset of TOC and focuses on direct program costs; TOC seeks to allocate additional indirect and infrastructure costs to the program [54].

2.5.3 Cost estimating methods

Analogy. Compares to a similar system and adjusts for differences; used very early when data are limited [54].

Parametric (CER). Uses statistical cost-estimating relationships (e.g., weight, speed, schedule) when key parameters are known [54].

Engineering (bottom-up). Builds from a detailed Work Breakdown Structure (WBS); most accurate but time-consuming and data-intensive [54].

Actuals (extrapolation). Uses actual costs from the same system or close predecessor; strongest in production [54].

Table XXIX: Cost estimating methods by acquisition phase

Phase	Most applicable methods
MSA	Analogy, Parametric
TMRR	Parametric, Engineering
EMD	Engineering, Actuals
P&D	Engineering, Actuals
O&S	Actuals

Source: *3.1.6. Cost Estimating*, 2026 [54]

Note

Pool example: analogy (neighbor costs), parametric (cost per square foot), engineering (detailed materials and labor), actuals (later builds in a housing development) [54].

2.5.4 Cost estimating products and players

Table XXX: Cost estimating players and products

Player	Product
Program Management Office / SYSCOM	CARD, economic analysis, POE
Component cost agency (e.g., NCCA)	ICE, CCE, CCP
CAPE	ICE

Source: *3.1.6. Cost Estimating*, 2026 [54]

Program management office. Defines the program and develops the CARD and POE [54].

Component cost agency. Leads cost analysis independent of the acquisition decision chain; prepares Component Cost Estimate (CCE) for ACAT IC/ID programs and may serve as the Independent Cost Estimate (ICE) [54].

CAPE. Conducts or approves ICEs for ACAT ID/IC programs, participates in PPBE reviews, and advises SECWAR [54].

2.5.5 Cost estimating products

CARD. Developed by the program office; provides quantitative technical and programmatic descriptions for costing; updated for all milestone reviews (DoW Manual 5000.04 series) [54].

CCE. Developed by the component cost agency using input from the POE; reviewed by CAPE for Major Defense Acquisition Programs (MDAPs) and MAISs [54].

POE. Developed by the SYSCOM cost organization on behalf of the PMO; estimates full life-cycle resources and costs based on the CARD [54].

CCP. Reconciliation between the CCE and POE; official component cost position (DoN typically uses the POE in lieu of a CCP) [54].

ICE. Independent LCC required for ACAT I programs; conducted by an organization independent of the PMO [54].

Economic analysis. Evaluates relative costs of technical alternatives and acquisition strategies; developed by the PMO for MAISs [54].

2.5.6 Typical CARD elements

Table XXXI: Typical CARD elements

System and program description	Support and planning
System description and WBS	System logistics and support concepts
Subsystem descriptions and tech maturity	Hardware/software maintenance and support
Quality factors and R/M/A	Training concept and manpower requirements
Risk assessment and mitigation	Time-phased quantities and activity rates (OPTEMPO)
Operational concept and basing	Milestone schedule and acquisition strategy

Source: *3.1.6. Cost Estimating*, 2026 [54]

2.5.7 Learning curve theory

Definition. When production quantity doubles, per-unit labor hours decrease by a constant percentage [54].

Drivers. Experience, uninterrupted production, consistent design, tooling improvements, and productivity focus [54].

Rule. Learning curves apply to production labor only and to repeatable processes (e.g., a 90% curve makes unit 4 labor 90% of unit 2) [54].

2.5.8 Escalation

Purpose. Escalation accounts for inflation and outlay rates to convert constant-year estimates into then-year budget dollars [54].

Indices. Raw (compound) index uses inflation only; weighted (composite) index combines inflation and outlay rates and is unique to each appropriation [54].

Source. OSW provides annual inflation and outlay rates; services compute indices for budget submissions [54].

2.5.9 Will cost / should cost

Will cost. Funded-to level without additional cost-saving initiatives [54].

Should cost. Desired cost after management initiatives; each savings opportunity must be tied to a quantifiable engineering or business change [54].

Opportunities. Learning curve gains, process improvements, contracting changes, competition, incentives, and Government-Furnished Equipment (GFE) [54].

Avoid. Straight percentage cuts or large investments with no near-term recovery [54].

2.5.10 Summary check

LCC scope. LCC = acquisition cost + O&S + disposal (true) [54].

Method selection. Early concept: analogy; preliminary design: parametric; detailed design: engineering; production run: actuals [54].

Learning curve. Applies to labor only (not material) [54].

Escalation. Convert constant-year to then-year using inflation and outlay rates [54].

2.5.11 Coursebook cue: cost-estimate review

Coursebook 3.1.6 emphasizes that a cost estimate is not complete just because the math is finished. The PMO prepares and defends the program estimate, the SYSCOM or component cost organization reviews the estimate for realism and method, and the CAPE role becomes central for major programs that require independent cost assessment before a milestone decision [54].

Table XXXII: Cost-estimate products and review roles.

Product	Typical owner	Board cue
CARD	Program office	Describes the program, technical baseline, schedule, quantities, and assumptions that estimators price.
POE	Program office	PMO estimate used to plan, budget, and defend the program.
CCE	Component cost agency	Independent component-level estimate or assessment used to challenge the program view.
ICE	Independent cost authority	Required when statute, policy, acquisition category, or decision authority requires an independent estimate.
CAPE review	OSD CAPE	Provides independent cost analysis for major decisions and helps the MDA judge affordability and realism.

Board cue. A defensible estimate has a technical baseline, documented assumptions, independent review, and a decision use. Say who owns the CARD, POE, component review, ICE, and CAPE role.

3 Contracting and Solicitation

3.1 Intro to Contracting Fundamentals

3.1.1 Why Contracts Are Needed

- Establish a legal relationship between the Government and the contractor.
- Define the rights and responsibilities of each party.
- Allow changes within the terms and conditions of the legal relationship.

Note

Contracts protect both the contractor and the Government throughout the acquisition process [69].

3.1.2 Essential Elements of a Contract

Mutual assent. A contract is formed when an offer is accepted by the offeree, creating a meeting of the minds on material terms such as price, quantity, and delivery [69].

Consideration. Each party provides something of value (e.g., performance, payment, schedule relief); courts consider sufficiency and adequacy in Government contracts [69].

Capacity. Parties must be legally competent and authorized to bind their organizations; incapacity or lack of authority undermines enforceability [69].

Lawful purpose. The objective must be legal and enforceable; contracts contrary to public policy or statute are void [69].

Note

Without all elements properly satisfied, a contract may not be valid in the eyes of the court [69].

3.1.3 Government Relationship to Contractors

Transactional. Contractors seek profit while the Government demands cost, schedule, and performance results; avoid unauthorized commitments [69].

Professional. Both sides must understand the contract and work cohesively to meet requirements [69].

Collaborative. Contractors bring technical expertise, while programmatic decision-making remains inherently governmental [69].

Constrained. Ethics rules and contractual obligations govern interactions; avoid conflicts of interest [69].

3.1.4 Federal Acquisition Regulation System

- The system establishes policies and procedures for contracting and procurement across executive agencies [69].
- It consists of:
  - The Federal Acquisition Regulation (FAR).
  - The Defense Federal Acquisition Regulation Supplement (DFARS) (DoW FAR Supplement), including DoW source selection procedures.
  - Agency supplements and acquisition regulations such as Navy Marine Corps Acquisition Regulation Supplement (NMCARS), SUPSHIP Operation Manual (SOM), the NAVSEA Source Selection Guide, and the NAVSEA Contracts Handbook.

Note

Contracting officers use the FAR and its supplements to carry out the contracting process [69].

3.1.5 Types of Contracting Officers

- PCO: Handles procurement from pre-solicitation through award and signs the contract on behalf of the Government [69].
- Administrative Contracting Officer (ACO): Performs contract administration functions as assigned by the PCO (see FAR 42.302(a)) [69].
- Termination Contracting Officer (TCO): After a termination notice, negotiates an equitable settlement with the contractor [69].

Note

A single Contracting Officer (KO) can serve in all three roles depending on organizational needs and contract type [69].

3.1.6 KO and PM Roles and Responsibilities

KO responsibilities. Ensure contract performance and compliance; serve as the contract advisor for the program [69].

KO major functions. Prepare solicitations and contracts; communicate with offerors and negotiate; ensure consistency with FAR/DFARS; award, administer, modify, and terminate contracts; legally bind the Government through warranted authority [69].

PM responsibilities. Manage the program and remain accountable to the MDA for cost, schedule, and performance in accordance with the DoW 5000 series [69].

> **Note**
>
> The KO, PM, and their teams must understand each other's responsibilities and governing regulations [69].

3.1.7 Differences Between the PM and the KO

Table XXXIII: Differences between the PM and the KO

Category	PM	KO
Authority	Tenure agreement	Warrant
Responsibility	Entire program	Contract
Background/Training	Technical	Business
Guiding directives	DoW 5000 series	FAR system and contract management
Organization	Program Office (IPT)	Matrix (IPT)

Source: *3.2.1. Intro to Contracting*, 2026 [69]

> **Note**
>
> PM authority is provided by the appropriate CAE in accordance with DoW 5000 and Department of War Instruction (DoWI) 5000.66; Head Contracting Activities (HCAs) provide KOs authority via a Certificate of Appointment (Standard Form 1402) warrant. Integrated Product Team (IPT) means Integrated Product Team [69].

3.1.8 Who's Responsible?

Table XXXIV: Contracting responsibilities by role

Action	Role
Decide which vendor will be awarded the contract	KO / SSA caveat
Conduct contract negotiations with the selected vendor	KO
Set the desired award date for the contract	PM
Set the maximum dollar amount to be awarded	PM
Monitor contractor performance	PM/KO
Make changes to the contract	KO

Source: *3.2.1. Intro to Contracting*, 2026 [69]

Note

The KO awards and signs the contract. In formal source selections, the Source Selection Authority (SSA) selects best value when appointed; under FAR 15.303, the contracting officer may serve as the SSA by default [69, 70].

3.1.9 Advantages of Competition

- Incentivizes lower prices for goods and services.
- Encourages innovation in transformational technologies.
- Promotes higher quality products and services.
- Enables best-value tradeoffs for performance improvements.
- Provides opportunities for capable small businesses.
- Strengthens the defense industrial base.

Note

Competition improves both value and capability outcomes [69].

3.1.10 Competition in Contracting Act (CICA)

Full and open competition. All responsible sources may compete for the requirement [69].

Exclusion of sources. Allows alternate-source strategies, small-business set-asides, and Small Business Administration (SBA) Section 8(a) actions [69].

Seven exceptions. FAR authorizes seven exceptions to full and open competition (e.g., sole source) [69].

Approval. Most other-than-full-and-open competition requires a written Justification and Approval (J&A) under FAR 6.303/6.304; the public-interest exception instead uses an agency-head Determination and Findings (D&F) with congressional notice. Required justifications are generally posted publicly after award under FAR 6.305, subject to timing exceptions [69, 70].

Seven Exceptions to Full and Open Competition

1. Only one responsible source will satisfy agency requirements.
2. Unusual and compelling urgency.
3. Industrial mobilization; engineering, developmental, or research capability; or expert services.
4. International agreement.

5. Authorized or required by statute.
6. National security.
7. Public interest.

Note

These exceptions align with FAR 6.302 and require documentation when used [69].

3.1.11 Contracting Terms

Sole source. A contract negotiated with only one source; other-than-full-and-open competition requires justification and approval (typically a J&A using NMCARS format) [69].

Responsiveness. For sealed bidding, awards are made without discussion; the bid must conform to all Invitation for Bids (IFB) requirements [69].

Responsibility. A prospective contractor must be determined responsible, including financial health, to receive an award (applies to IFB and Request for Proposal (RFP)) [69].

Note

Tip to remember: you are responsible for your own financial health [69].

3.1.12 Determining Responsibility

The prospective contractor (offeror) must:

- Have adequate financial resources or the ability to obtain them.
- Comply with the required or proposed delivery schedule.
- Maintain a satisfactory performance record.
- Maintain a satisfactory record of integrity and business ethics.
- Possess (or obtain) the necessary organization, experience, accounting and operational controls, and technical skills.
- Have the necessary equipment or the ability to obtain it.
- Be otherwise qualified and eligible under applicable laws and regulations.

[69]

3.1.13 Determining Responsiveness

- A bid must comply in all material respects with the IFB to be considered for award.
- Example: if the offered delivery date (including mailing or transmittal time) is later than required, the bid is nonresponsive and rejected.
- Example: if the solicitation requires copper but the bidder offers tin, the bid is nonresponsive.

[69]

3.1.14 Summary

- Contracts protect the contractor and the Government by establishing a binding legal relationship.
- Essential elements are mutual assent, consideration, capacity, and lawful purpose.
- Governing documents include the FAR, DFARS, and agency supplements.
- The three KO types are PCO, ACO, and TCO.
- PMs manage the program and are accountable to the MDA; KOs ensure contract performance and compliance.
- CICA implements competition requirements, and sole-source actions require approved justification.

[69]

3.2 Solicitation Preparation

3.2.1 Contracting Process Overview

- Pre-solicitation phase (determination of need, requirements package, method of contracting).
- Solicitation, evaluation, and award phase (solicitation of sources, response and evaluation, award).
- Post-award phase (contract administration, delivery, closeout).

[71]

3.2.2 Pre-Solicitation Phase

Determination of capability need. For systems with new capabilities, the contracting process starts only after a need is identified, non-materiel solutions are ruled out, and approval is granted to pursue a materiel solution [71].

Requirements package. The package establishes the baseline for solicitation development and includes [71]:

- Management reviews and approvals.
- Certification of funds availability.
- Requirements description (specifications, Statement of Work (SOW)/Statement of Objectives (SOO), purchase descriptions, contract data requirements).
- Special packaging and marking requirements.
- Recommended evaluation factors and criteria (team-developed).
- Inspection and acceptance requirements.
- Delivery or performance schedule.
- Special contract administration requirements.
- Special provisions or clauses (e.g., options, GFE).
- Recommended potential sources and market research results.
- Input for other-than-full-and-open competition approvals.
- Acquisition plan if the action exceeds $10M.

Note

A contract requirements package requires substantial effort across the team [71].

3.2.3 Statement of Objectives and Performance Work Statement

SOO. A Government-prepared, high-level objectives document that is provided in the solicitation in lieu of a Government-written Performance Work Statement (PWS); offerors use it to propose the PWS [71].

PWS. Describes work in terms of required results rather than how the work is performed or hours provided; may be Government-developed or derived from an SOO [71].

Note

Example: SOO = “maintain attractive, functional groundcover”; PWS = “maintain grass between 1 and 2 inches.” [71].

3.2.4 Pre-Solicitation Communication and Market Research

- Before soliciting offers, the requirement is advertised to gauge industry interest through market research, industry days, Request for Informations (RFIs), and draft RFPs [71].
- Robust communication improves the quality of the RFP and the source selection plan by clarifying Government needs and industry capabilities [71].

3.2.5 Market Research Purpose

- Conducted throughout the acquisition life cycle to inform acquisition strategy and planning [71].
- Evaluates commercial marketplace capability and identifies potential sources [71].
- Supports price analysis and scales in depth based on urgency, dollar value, complexity, and past experience [71].
- Conducted in an IPT environment [71].

3.2.6 IPT Roles in Market Research

PM. Owns overall responsibility, develops the acquisition strategy, and ensures transparency in sources-sought and market research activities [71].

KO. Negotiates, awards, administers, and terminates contracts; determines whether the requirement can be set aside for small business; supported by contract specialists, cost analysts, and other functional experts [71].

Technical IPT. Evaluates existing commercial products and non-developmental items for applicability [71].

Cost analysts. Provide input on appropriate contract pricing information [71].

3.2.7 Methods of Contracting

Sealed bidding. Used when award can be made on price, specifications are clear, discussions are unnecessary, and there is a reasonable expectation of more than one bid. Uses an IFB and awards to the lowest priced, responsive, responsible bidder [71].

Negotiation. Used when any sealed-bidding condition is lacking (time, price information, requirements definition, or competition). May be sole source or competitive using an RFP; awards are based on evaluation criteria and discussions as needed [71].

3.2.8 Contract Types Overview

- Contract type is a key pre-solicitation decision tied to risk, requirements clarity, and profit incentives [71].
- Two broad categories: fixed-price and cost-reimbursement (cost-plus) [71].

3.2.9 Fixed-Price Contracts

- Provide for a firm price or adjustable price constrained by a ceiling or target price [71].
- Used when scope is defined, cost estimates are reliable, and the Government wants to shift cost risk to the contractor [71].

FFP. Price not subject to adjustment; contractor assumes maximum risk and responsibility for costs and profit/loss [71].

FPEPA. Price adjusted up or down based on specified contingencies (often economic indices) [71].

FFP LOE. Fixed dollar amount for a specified level of effort over a stated period [71].

FPIF. Final price equals final cost plus adjusted profit based on a shareline formula; profit varies with cost performance and is bounded by a ceiling price [71].

Fixed-price award fee. Fixed-price with award fee based on evaluation criteria when performance cannot be measured objectively [71].

Fixed-price with prospective price redetermination. Firm fixed price for an initial period with future price redetermination for later periods [71].

Fixed-price with retroactive price redetermination. Fixed ceiling price with retroactive price redetermination after completion [71].

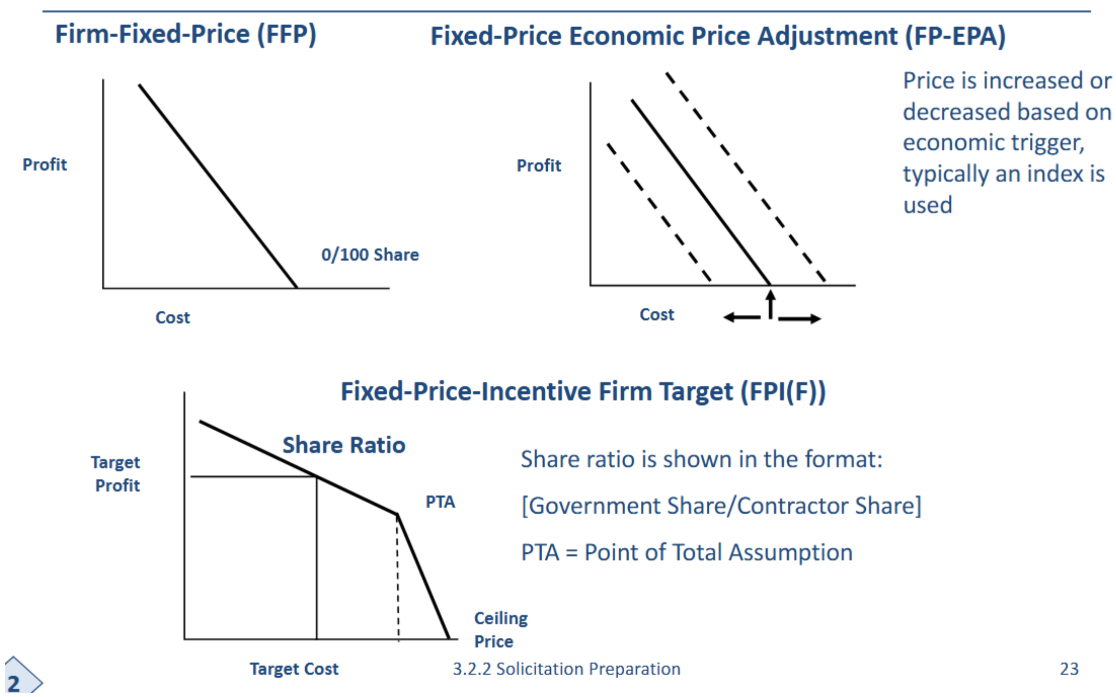

Figure 3.1. Fixed-price contract types and incentive geometry. Source: 3.2.2. Solicitation Preparation, *2026 [71].*

3.2.10 Fixed-Price Incentive (Firm Target) Example

- Given: target cost 100, target fee 10, target price 110, ceiling price 130, shareline 80/20 [71].
- Overrun/underrun = actual cost - target cost [71].
- Adjusted fee = target fee -/+ (0.2 x overrun/underrun) [71].
- Final price = actual cost + adjusted fee (if below the ceiling) [71].

Note

Point of Total Assumption (PTA) = (ceiling price - target price) / Government share + target cost [71].

3.2.11 Fixed-Price Incentive Practical Exercise

- Actual cost $90,000; target cost $100,000; target fee $10,000; target price $110,000.
- Shareline 80 Government / 20 contractor; ceiling price $130,000.
- Underrun = $10,000; adjusted fee = $10,000 + (0.2 x $10,000) = $12,000.
- Final price = $90,000 + $12,000 = $102,000.

[71]

3.2.12 Cost-Reimbursement Contracts

- Provide for payment of allowable incurred costs within a stated ceiling; the contractor may not exceed the ceiling at its own risk [71].
- Used when scope is uncertain, costs cannot be estimated accurately, an approved accounting system exists, and Government surveillance can ensure cost control [71].
- Place cost risk on the Government and can yield lower initial contract costs [71].

CPFF. Allowable costs plus a fixed fee; fee can change only if scope changes [71].

CPIF. Allowable costs plus incentive fee tied to a formula based on total costs versus target costs; includes min and max fee [71].

CPAF. Allowable costs plus base fee (possibly zero) and award fee [71].

Cost. Allowable costs only; no fee [71].

Cost sharing. Government pays an agreed portion of allowable costs; no fee [71].

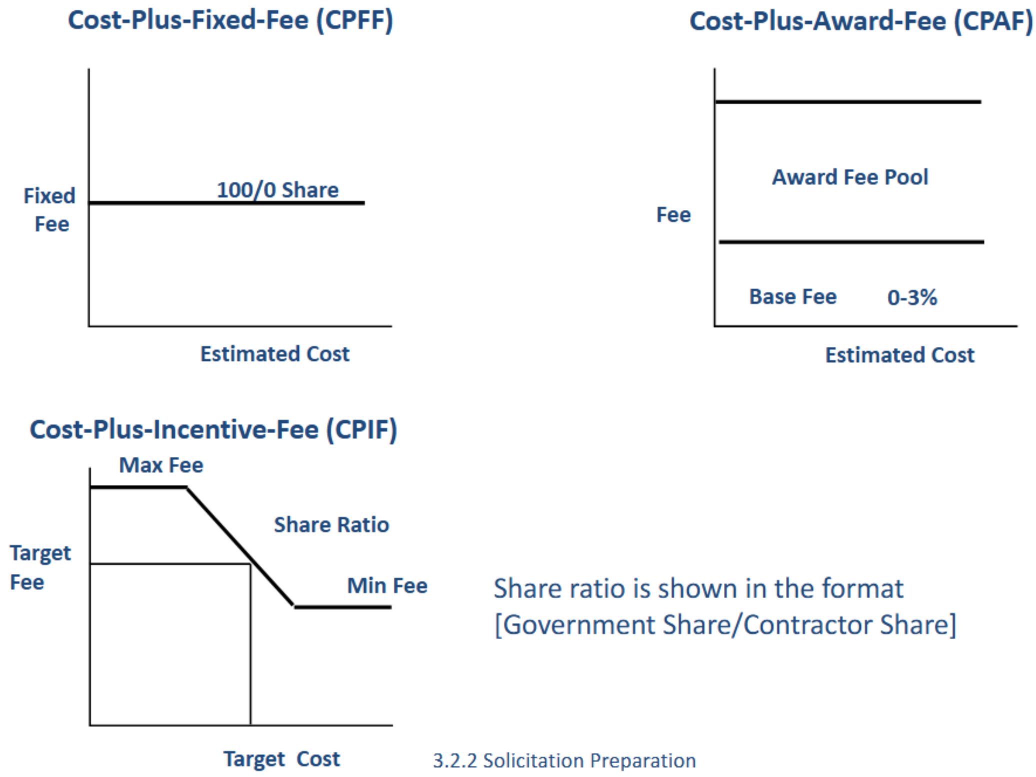

Figure 3.2. Cost-reimbursement contract types and fee structures. Source: 3.2.2. Solicitation Preparation, *2026 [71].*

3.2.13 Cost-Plus-Incentive-Fee Example

- Given: target cost 1000, target fee 80, max fee 100, min fee 60, shareline 80/20 [71].
- Overrun/underrun = actual cost – target cost [71].
- Adjusted fee = target fee –/+ (0.2 x overrun/underrun) [71].
- Final price = actual cost + adjusted fee [71].

3.2.14 Incentives and Award Fees

- Incentives are driven by the requirement and can target cost, performance, schedule, award fee, options, or mixed Contract Line Item Number (CLIN) structures [71].
- Incentive contracts allow higher profit for meeting cost, performance, or schedule objectives when targets are clear and attainable [71].
- Most incentive contracts rely on cost incentives implemented via a shareline formula [71].

Special incentives. Tailored focus areas such as schedule milestones, key events, or drawing release counts; common in construction or design contracts [71].

Award fee. Subjective criteria are used when performance cannot be measured objectively; award fee may be used in fixed-price or cost-plus contracts with base fee as applicable [71].

Note

Award fee criteria examples include systems engineering milestones, successful design reviews, test completion, and certifications [71].

3.2.15 Fixed-Price vs. Cost-Reimbursement Summary

Table XXXV: Fixed-price vs. cost-reimbursement comparison

Fixed-price	Cost-reimbursement
Promised outcome: acceptable goods or services	Promised outcome: best efforts
Payment timing: after delivery (progress payments possible)	Payment timing: as costs are incurred
Cost risk to contractor: high	Cost risk to contractor: low
Cost risk to Government: low	Cost risk to Government: high

Source: *3.2.2. Solicitation Preparation*, 2026 [71]

Note

A contract may include multiple contract types through the use of CLINs [71].

3.2.16 Contract Type by Acquisition Phase

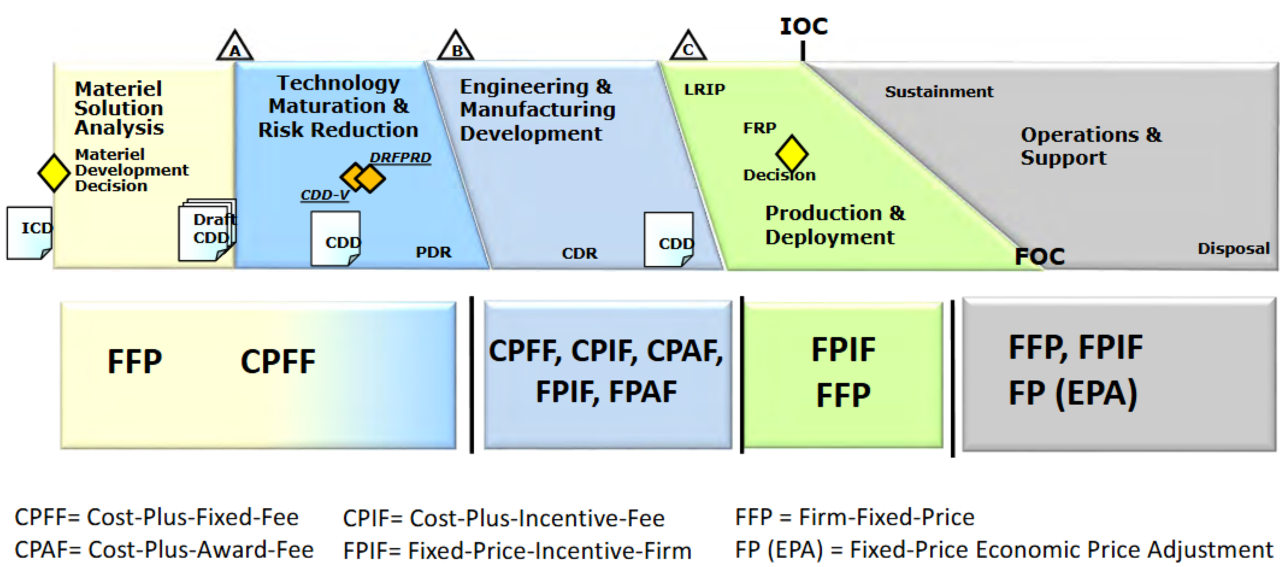

Figure 3.3. *Contract type by phase of the acquisition process (notional). Source:* 3.2.2. Solicitation Preparation, *2026 [71].*

Note

Contract type selection is influenced by program risk and requirements definition [71].

3.2.17 Risk and Contract Types

- Cost risk shifts from Government to contractor as requirements become better defined and technical risk declines [71].
- Cost-reimbursement vehicles (Cost-Plus-Fixed-Fee (CPFF), Cost-Plus-Incentive-Fee (CPIF), Cost-Plus-Award-Fee (CPAF)) align with higher technical uncertainty; incentive and fixed-price contracts align with defined requirements and determinable prices [71].

3.2.18 Practical Exercise: Contract Type Selection

- Long-standing contractor relationship with a clear, well-defined scope and strong cost confidence.
- First-of-class depot-level overhaul for a new submarine class with uncertain cost drivers.
- Consider the contract type you would choose as the Government and as the contractor, tying the decision to risk.

Source: *3.2.2. Solicitation Preparation*, 2026 [71]

3.2.19 Indefinite-Delivery Contracts

- Used when the Government has recurring needs but is uncertain about quantities or timing; reduces lead time for task or delivery orders [71].
- Three types: definite-quantity (Indefinite-Delivery, Definite-Quantity (IDDQ)), Indefinite-Delivery, Indefinite-Quantity (IDIQ) (single or multiple award), and requirements contracts [71].

3.2.20 Other Contract Types

T&M. Used when the extent or duration of work cannot be estimated with reasonable confidence; based on direct labor hours plus materials at cost [71].

LH. Similar to Time-and-Materials (T&M), but without material charges [71].

Letter contract (UCA). Preliminary instrument authorizing immediate performance with a price ceiling when definitive contract negotiations cannot be completed in time [71].

3.2.21 Socioeconomic Programs and Set-Asides

- Small business concerns are independently owned, not dominant in their field, and meet size standards [71].

- Socioeconomic programs include small business, woman-owned, veteran-owned, service-disabled veteran-owned, Historically Underutilized Business Zone (HUBZone), and SBA 8(a) certified firms [71].
- Set-asides reserve acquisitions exclusively for small businesses and are automatic for actions between $10K and $250K; for actions above $250K, set-asides are used when two or more responsible small businesses are expected and fair market prices are likely [71].
- Partial set-asides may be used for portions of multiple-award contracts, except construction [71].
- The SBA 8(a) program supports disadvantaged businesses with agency contracts and limited competition for up to nine years [71].

3.2.22 Small Business Innovation Research

- Small Business Innovation Research (SBIR) enables small businesses to develop innovative technologies for Government use and later commercialization [71].
- Large research and development organizations allocate approximately 3% of budgets to SBIR programs [71].
- Three phases: Phase I (6–12 months, $50K–$275K), Phase II (24 months, $750K–$1.8M), Phase III (commercialization with no SBIR funding) [71].

3.2.23 Other Transaction

- Other Transaction (OT) instruments are not standard contracts, grants, or cooperative agreements and provide flexible business arrangements [71].
- Used for research and development, prototypes, or models to evaluate feasibility or utility; often funded with RDT&E [71].
- Many Federal contract statutes do not apply, but procurement integrity rules and competitive practices still matter; OT decisions can be protested at the Government Accountability Office (GAO) (limited) and the Court of Federal Claims (COFC) [71].
- Prototype OT authority (10 U.S.C. § 4021) and follow-on production authority (10 U.S.C. § 4022) enable faster transitions when prototypes meet requirements [72, 73].
- Evidence should include problem statements, solution briefs, prototype demo results, and transition plans to support production decisions [74].
- GAO has noted that OT use has expanded faster than data collection and outcome tracking; program teams should capture transition metrics early to support oversight [75].

3.2.24 Data Rights Overview

- DoW policy emphasizes acquiring only the data rights essential to meet minimum Government needs [71].

- Main categories: unlimited rights, government purpose rights, limited rights, and restricted rights [71].

Unlimited rights. Government funded the entire development; Government may use, modify, reproduce, display, release, or disclose technical data for any purpose [71].

Limited rights. Developed exclusively at contractor expense; Government may use and disclose within the Government but cannot use for manufacturing or release to third parties without permission [71].

Government purpose rights. Mixed funding; Government may use for Government purposes, with rights typically converting to unlimited after a negotiated period (normally five years) [71].

Restricted rights. Applies to noncommercial computer software; Government use is limited (single machine use, minimum backup copies, transfer within Government, limited modification) [71].

3.2.25 Factors in Determining Data Rights

- Whether the item will be competitively acquired.
- Whether repair and overhaul will be contracted out or performed in-house.
- Whether parts are commercial items or require detailed drawings (build-to-spec or build-to-print).

Note

DoW policy requires procedures that are least intrusive on contractor economic interests as practicable [71].

3.2.26 Solicitation, Evaluation, and Award Phase

- Solicitations notify industry of a contracting need and its parameters [71].
- Method of contracting determines the solicitation type: IFB for sealed bidding, RFP for negotiated procurement [71].
- IFB and RFP use the Uniform Contract Format (UCF); submitted bids and proposals are considered offers [71].
- Solicitations and pre-solicitation opportunities are publicized on `https://sam.gov`; public notice is required for contract actions expected to exceed $25K [71].

3.2.27 Uniform Contract Format Essentials

- Part I (The Schedule) includes Sections A–H; Part II contains contract clauses (Section I); Part III lists documents and attachments (Section J); Part IV covers representations and instructions (Sections K–M) [71].
- Section B provides supplies or services and prices/costs, with each item identified by a CLIN [71].
- Section C provides descriptions, specifications, standards, and the SOW; it often references DD Form 1423 Contract Data Requirements List (CDRL) requirements [71].

- Section J lists attachments, including CDRL data requirements [71].
- Section L explains proposal submission requirements, organization, page limits, past performance data, and cost or pricing data [71].
- Section M states evaluation factors and their relative importance (e.g., price, past performance, technical, management, personnel, experience, schedule) [71].

Note

Data deliverables add cost; balance reporting needs and CDRL scope against program value [71].

3.2.28 Big-Picture Flow

- Requirements package → contracting officer → develop RFP or IFB → advertise on System for Award Management (SAM) → prospective offers [71].

3.2.29 Summary

- The two primary contract categories are fixed-price and cost-reimbursement.
- Fixed-price contracts transfer cost risk to the contractor.
- An IDIQ contract provides for indefinite quantities within stated limits during a fixed period.
- Pre-solicitation sequence: determination of need, requirements package, and method of contracting.
- IFB is used for sealed bidding when conditions for sealed bidding are met.

Source: *3.2.2. Solicitation Preparation*, 2026 [71]

3.3 Source Selection Process

3.3.1 Overview

- Formal source selection is used for high-dollar defense acquisitions to achieve best value in cost, quality, and timeliness [76].
- Source selection participants may be sequestered and required to sign a non-disclosure agreement [76].
- The process is guided by the Source Selection Plan (SSP) and the solicitation (RFP or IFB) [76].

3.3.2 Source Selection Plan

Purpose. The SSP details how the Government will solicit and evaluate proposals and execute source selection [76].

Origins and approval. It originates from the acquisition strategy, is prepared by the KO with the IPT, and is approved by the SSA [76].

Contents. Description of the procurement, source selection charter, team organization, best-value approach, evaluation factors and rating scheme, and milestone schedule [76].

3.3.3 Basis for Award and Best Value

- Sealed bidding focuses on responsiveness and responsibility [76].
- Negotiated acquisitions select best value, defined as the outcome that provides the greatest overall benefit in response to the requirement [76].
- Best-value policies include Lowest Price Technically Acceptable (LPTA) and tradeoff processes [76].

3.3.4 Commercial Solutions Opening and Other Transaction Prototypes

CSO focus. Commercial Solutions Openings (CSOs) are structured around problem statements and solution briefs, with emphasis on demonstrations and prototype results rather than FAR Part 15-style proposals [70, 74].

Prototype authority. OT prototype awards and follow-on production are supported under 10 U.S.C. § 4021 and § 4022, enabling faster transitions when prototypes succeed [72, 73].

Evidence discipline. Source selection documentation should capture problem-solution fit, demo results, and transition plans to support timely fielding decisions [74].

3.3.5 Lowest Price Technically Acceptable

- Best value results from the technically acceptable proposal with the lowest evaluated cost or price [76].
- Once technical acceptability is established, proposals are ranked on cost or price only [76].
- Tradeoffs are not permitted [76].

3.3.6 Tradeoff Process

- Best value is a balance of technical merit, performance risk, management capability, and price factors [76].
- Used when requirements are less defined, significant development work is needed, or performance risk is high [76].
- The solicitation must state evaluation factors and their relative importance, and the rationale for selection must be documented [76].

3.3.7 Source Selection Steps

Common start. Compare each proposal to the RFP requirements [76].

Tradeoff path. Perform comparative analysis, determine whether prices are fair and reasonable, and select the source for award [76].

LPTA path. Identify technically acceptable sources, then select the lowest-priced acceptable offer for award [76].

3.3.8 Source Selection Organization

- The SSA is the decision authority; the Source Selection Advisory Council (SSAC) and advisors compare proposals and provide recommendations [76].
- The Source Selection Evaluation Board (SSEB) evaluates proposals and is typically organized into technical, management, past performance, cost/pricing, and small business teams as needed [76].

3.3.9 Roles and Responsibilities

PCO/KO. Ensures compliance with procurement laws, reviews proposal completeness, manages communications, and awards the contract [76].

SSA. Oversees integrity of the process, appoints qualified personnel, and selects the best-value proposal [76].

SSAC. Reviews proposal analyses after SSEB evaluation and recommends the best-value source; functions may be assigned to the SSEB when a separate council is not used [76].

SSEB. Evaluates proposals against the solicitation's evaluation factors and rating standards, including Section M when the RFP uses the FAR uniform contract format, identifies strengths and weaknesses, documents deficiencies, and drafts discussion questions [70, 76].

3.3.10 Interactions with Contractors

Deficiencies. Material failures to meet minimum requirements or combinations of significant weaknesses that increase performance risk; must be corrected, and correction requires proposal revision [76].

Competitive range. The most highly rated offerors selected for further discussions [76].

Clarifications. Used when awarding without discussions; limited to minor or clerical errors and adverse past performance issues [76].

Communications. Used when inclusion in the competitive range is uncertain, to enhance Government understanding without proposal revision [76].

Discussions. Conducted after establishing the competitive range; offerors may revise proposals and negotiations may occur [76].

3.3.11 Guidance on Exchanges

- Do not favor one offeror over another or reveal another offeror's technical solution or price [76].
- Conduct meaningful discussions with all offerors in the competitive range, addressing all deficiencies and significant weaknesses [76].
- Allow reasonable time for proposal revision and document all discussions (oral or written) [76].

3.3.12 Fact-Finding and Analysis

- Fact-finding prepares the Government's pre-negotiation position and relies on Government research and controlled interactions that do not permit proposal revision [76].
- Typical analyses include technical analysis, cost analysis, price analysis, and cost-realism analysis (required for cost-reimbursement contracts) [76].
- External support can include Defense Contract Management Agency (DCMA), Defense Contract Audit Agency (DCAA), and field activities [76].

3.3.13 Preparing for Fact-Finding

1. Review the contracting situation and relevant legal and contractual terms [76].
2. Identify team members who will support analysis and negotiations [76].
3. List open issues, conduct research through evaluation teams, and develop the pre-negotiation position using cost and price techniques [76].
4. Draft questions and submit them in writing to the KO, who forwards them to offerors [76].

3.3.14 Negotiation Preparation and Objectives

- Be familiar with the solicitation and proposals, submit questions in advance, and align the team on acceptable trade-offs [76].
- Handle administrative details early (agenda, assignments, facilities) [76].
- The KO establishes pre-negotiation objectives using field pricing reports, DCAA audits, technical analysis, Independent Government Cost Estimate (IGCE) data, and price histories, with management review per policy [76].

3.3.15 Negotiation Skills

Do. Identify and prioritize discussion items, verify contractor authority, listen and probe for detail, draw estimate bases into the open, document action items, and keep thorough notes [76].

Do not. Reveal Government findings or specific numbers, coach the contractor, negotiate how to perform tasks, or let issues go unresolved [76].

3.3.16 Keep the End Goal in Mind

- The goal of negotiations is an agreement that enhances the quality of the product or service delivered to the Government [76].
- Ensure negotiation activities support that end state [76].

3.3.17 Contract Award and Protests

- The SSA signs the source selection decision document; the KO incorporates negotiated changes, obtains legal review, and signs the contract [76].
- Only the KO can award; the contract is binding only after signature and distribution, and the KO must ensure price reasonableness and protection of Government interests [76].
- Unsuccessful offerors are notified and may request a required post-award debriefing within three days after award notice; the Government should debrief within five days when practicable [70, 76].
- Protests are written objections to award, cancellation, or termination and may be filed with the agency, GAO, or the COFC; involve legal counsel immediately [70, 76].
- Timely GAO protests generally trigger a stay or suspension unless the HCA authorizes continued performance on best-interest or urgent-and-compelling grounds [70, 76].

3.3.18 Summary

- Guiding documents: SSP and the RFP or IFB [76].
- Formal steps: compare proposals to the RFP, perform comparative analysis, determine fair and reasonable price, and select the source [76].
- SSEB responsibilities: evaluate against solicitation evaluation factors, document strengths and weaknesses, and draft discussion questions [76].
- Fact-finding questions are communicated in writing through the KO [76].
- Successful negotiation steps: prepare, conduct fact-finding and analysis, set pre-negotiation objectives, ensure negotiating skills, and keep the end goal in mind [76].

3.4 Cost and Price Evaluation

3.4.1 Summary

Fair and reasonable price. The KO determines fair and reasonable price using price, cost, technical, field-pricing, and risk analyses along with sound business judgment [77].

Financial health signals. Balance sheets and income statements enable Return on Sales (ROS) and Return on Assets (ROA) calculations that inform responsibility and financial capacity decisions [77].

Direct vs. indirect. Direct costs tie to a single cost objective; indirect costs (overhead and G&A) support multiple objectives and must be allocated through defined bases [77].

Allowability tests. Billable costs must be reasonable, allocable, compliant with Cost Accounting Standards (CAS)/Generally Accepted Accounting Principles (GAAP), within contract terms, and within FAR 31.205 limitations [70, § 31.201-2] [77].

Rate discipline. Indirect rates, fully burdened rates, and forward pricing rate agreements sharpen negotiation positions and protect the prenegotiation objective [70, § 15.407-3] [77].

3.4.2 Fair and Reasonable Price Determination

KO responsibility. The KO evaluates price reasonableness using price, cost, technical, field-pricing support, and risk analysis inputs to reach a fair and reasonable determination [70, § 15.404-1(a)] [77].

Analysis toolkit. Price and cost analysis are primary techniques, supported by technical evaluations, field pricing support, and risk analysis as needed [77].

3.4.3 Cost or Pricing Data Requirements

Threshold and exceptions. The coursebook threshold is $2M; certified cost or pricing data are generally required above the threshold unless a FAR exception applies [70, § 15.403-1] [77].

Solicitation request. When required, the Government must request cost or pricing data in the solicitation and typically seeks a technical analysis of labor, material, and other proposal elements [77].

3.4.4 Cost and Price Analysis

Price analysis. Examination of the total proposed price without evaluating separate cost elements, using competition, historical prices, or market indicators [70, § 15.404-1(b)].

Cost analysis. Evaluation of individual cost elements and profit/fee to determine how well proposed costs represent what the contract should cost under reasonable economy and efficiency [70, § 15.404-1(c)] [77].

Cost vs. price lens. Cost analysis looks at direct labor, indirect rates, materials, and ODCs; price analysis looks at total proposed price with profit/fee embedded [77].

3.4.5 Field Pricing Support

Specialist support. Field pricing support includes audits and specialist assessments (accountants, auditors, cost/price analysts, negotiators, engineers, production specialists, and small and disadvantaged business utilization specialists) to assist the KO in determining reasonableness [77].

DCAA/DCMA thresholds. DCMA provides field assistance and DCAA supports audits for fixed-price proposals over $10M and cost-type proposals over $100M per the coursebook [77].

3.4.6 Contractor Financial Health

Balance sheet and income statement. Balance sheets show financial condition at a point in time; income statements summarize revenues and expenses over a period and enable profitability ratios [77].

ROS and ROA. ROS and ROA gauge profitability and efficiency; compare ratios to industry averages and investigate sustained shortfalls [77].

Responsibility risk. Debt, cash-on-hand, and cash-flow trends complement ratio checks to determine offeror responsibility [77].

3.4.7 Direct and Indirect Costs

Direct costs. Costs traceable to a single cost objective (direct labor, direct material, and ODCs) with clear identification to a specific requirement [77].

Indirect costs. Costs that benefit more than one objective and are allocated through overhead and G&A pools, such as management, legal, travel, utilities, and facility security [77].

The breakdown of the proposed price is shown in Figure 3.4.

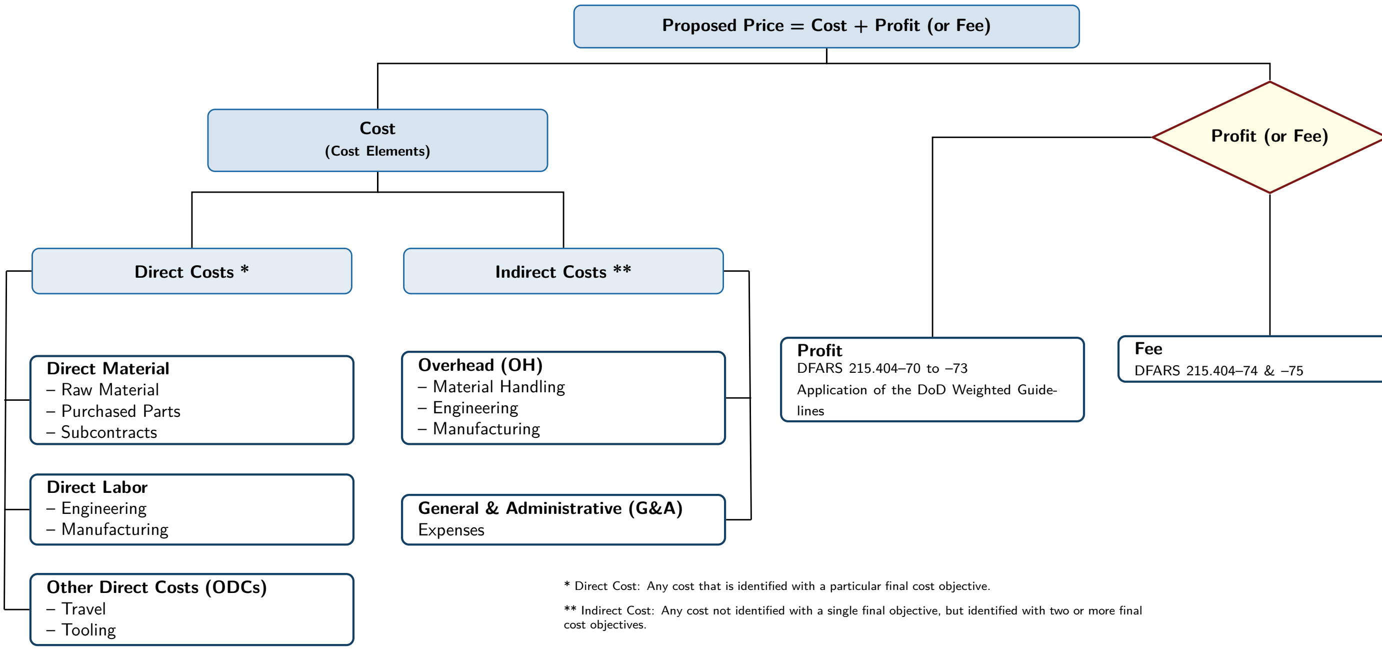

Figure 3.4. *Mapping direct vs. indirect contractor cost elements. Source:* 3.2.4. Cost and Price Evaluation, *2025 [77]*

3.4.8 Indirect Cost Pools, Bases, and Rates

Cost pools. Indirect cost pools group similar indirect costs (e.g., material handling, design support, building maintenance) that are allocated to cost objectives using a logical, causal-beneficial relationship [77].

Allocation bases. Bases include direct labor dollars or hours, material dollars, machine hours, or units produced and are used to compute indirect rates [77].

Apply the standard relationships from the coursebook when validating proposals; they are board favorites for quick-turn computations covering indirect pools, Total Cost Input (TCI), G&A, and profitability metrics such as ROS and ROA [77].

$$\text{Indirect Cost Rate} = \frac{\text{Indirect Cost Pool}}{\text{Allocation Base}}$$

$$\text{TCI} = \text{Direct Cost} + \text{Overhead Cost}$$

$$\text{G\&A Cost} = \text{G\&A Rate} \times \text{TCI}$$

$$\text{Fully Burdened Labor Rate} = \frac{\text{Direct Labor Cost} + \text{Indirect Costs} + \text{Profit/Fee}}{\text{Direct Labor Hours}}$$

$$\text{ROS} = \frac{\text{Operating Profit}}{\text{Net Sales}}$$

$$\text{ROA} = \frac{\text{Net Income}}{\text{Total Assets}}$$

3.4.9 Cost Allowability

Five bases. Reasonableness, allocability, compliance with CAS/GAAP, contract terms, and FAR 31.205 limitations govern allowability [70, § 31.201-2] [77].

Unallowable examples. Bad debts, contributions/donations, entertainment, and alcoholic beverages are unallowable [77].

Note

Slides may refer to "cash allowability"; FAR terminology is *cost* allowability.

3.4.10 Fully Burdened Rate

Definition. Fully burdened rates combine direct labor, indirect costs, and profit/fee to support comparisons across skill mixes and estimate labor-heavy work packages [77].

Example. A contractor with $1.0M direct labor, $0.5M manufacturing overhead, and $0.2M other costs across 50,000 labor hours has a fully burdened rate of $34 per hour [77].

3.4.11 Forward Pricing Rates

Purpose. Forward Pricing Rate Agreements (FPRAs) establish agreed indirect rates (typically one to three years) to speed negotiations and reduce repetitive audits by anchoring the prenegotiation objective [70, § 15.407-3] [77].

Process. Contractors submit a Forward Pricing Rate Proposal (FPRP); the Government reviews and negotiates rates into an FPRA, and the ACO monitors performance against the agreed rates [77].

3.4.12 Practitioner Steps

1. Confirm whether certified cost or pricing data are required and document the applicable exception if not; ensure the solicitation includes any required data requests [70, § 15.403-1] [77].

2. Execute price analysis first, then cost analysis when price analysis alone cannot support reasonableness; integrate technical analysis and field pricing support as needed [70, § 15.404-1] [77].

3. Validate direct and indirect cost elements, confirm allocation bases, and apply the five allowability tests to each significant cost element [70, § 31.201-2].

4. Analyze financial health using balance sheet and income statement ratios and note any responsibility concerns in the prenegotiation objective [77].

5. Capture DCAA and DCMA inputs, forward pricing rate agreements, and fully burdened rate calculations in the contract file to justify the final determination [77].

3.4.13 Checklist

- File price and cost analysis results (with technical inputs) showing how the fair-and-reasonable price determination was reached [70, § 15.404-1].
- Verify each significant cost element passes the five allowability tests and that allocation bases match current practice [70, § 31.201-2].
- Document any FPRA/FPRP used in the prenegotiation objective, or explain why legacy rates remain valid [70, § 15.407-3].
- Record ROS/ROA trends, cash flow, and DCAA/DCMA inputs that influenced the responsibility determination [77].

3.5 Source Selection Practical

3.5.1 Summary

SSP development. As an IPT member, build portions of the SSP that rank factors and subfactors and translate them into evaluation criteria that support best-value decisions [78].

Evaluation criteria. Criteria must discriminate among offerors, be tailored to the acquisition, and be communicated in the solicitation's evaluation factors, commonly RFP Section M when the uniform contract format is used, with clear statements of relative importance [70, 78].

Team roles. The SSA, SSAC, and SSEB execute a team-based source selection using standards defined in the SSP [78].

Best value trade-off. Apply evaluation standards to proposals and test results to select the best value contractor and compare outcomes to LPTA logic [78].

3.5.2 Request for Proposal Format

Table XXXVI: Standard RFP organization (Parts I–IV, Sections A–M).

Part	Section(s)	Purpose
Part I – Schedule	A–H	Solicitation/contract form; supplies or services and prices/costs; description/specifications/work statement; packaging/marking; inspection/acceptance; deliveries/performance; contract administration; special contract requirements.
Part II – Contract Clauses	I	Contract clauses.
Part III – List of Documents, Exhibits, and Attachments	J	Attachments.

Continued on next page

Table XXXVI: Standard RFP organization (Parts I–IV, Sections A–M). **(Continued)**

Part	Section(s)	Purpose
Part IV – Representations and Instructions	K–M	Representations/certifications; instructions/conditions/notices to offerors; evaluation factors for award.

Source: *3.2.5. Source Selection Practical*, 2025 [78]

Note

The uniform contract format has FAR 15.204 exceptions, and Part IV is not physically included in the resulting contract [70].

3.5.3 DoW Source Selection Process

Uniform procedures. DoW source selection procedures standardize rating criteria for technical and past performance factors and streamline DoW-wide reviews [78].

SSAC threshold. For competitive negotiated FAR Part 15 acquisitions, DoW source selection procedures require an SSAC for total estimated value of $100M or more unless waived or modified by agency procedure [78, 79].

3.5.4 Source Selection Plan

Purpose and content. The SSP defines team structure, the conduct of source selection, and proposal evaluation criteria to ensure a structured, fair, and impartial evaluation that supports competition and best-value selection [78].

Ownership. Prepared by the KO/IPT and approved by the SSA [78].

3.5.5 Program Manager Roles

- Ensure technical requirements are approved and stable.
- Establish technical specifications and develop the SOW, SOO, or PWS.
- Allocate resources to support the SSP and help form the source selection team.
- Assist in development of evaluation criteria and standards [78].

3.5.6 Evaluation Factors, Subfactors, and Standards

Good factors. Factors should discriminate among offerors, reflect the relative importance of the program's needs, and be tailored to the acquisition (examples: cost, technical, past performance, small business participation) [78].

Evaluation-factor requirement. The solicitation must state the relative importance of all factors and significant subfactors, and specify how non-cost factors compare to cost/price (significantly more important, approximately equal, or significantly less important); this is typically Section M when the RFP uses the uniform contract format [70, 78].

3.5.7 Evaluation Standards and Rating Scales

Standards. Standards are developed for each factor/subfactor, use words/numbers/colors, and must be clearly defined for the SSEB; they are not included in the RFP [78].

Sample factor. Reliability can be evaluated with metrics such as mean time between critical failure (Mean Time Between Critical Failure (MTBCF)); the course exercise applies a color scale (Blue, Purple, Green, Yellow, Red) to defined standards [78].

3.5.8 Exercise: CGW(X) Source Selection

Part 1: Background

- Scenario: CGW(X) ship program preparing for Milestone (MS) A approval.
- The team has developed trade-off opportunities, a Government preliminary contract design, and an RFP draft.
- A team is writing the SSP for full and open competition for the Detailed Design and Construction (DD&C) contract [78].

Part 1a: Subfactor Ranking

Rank the following technical subfactors by relative importance (use terms such as "significantly more important than" or "approximately equal to"):

- Sprint speed.
- Mission payload.
- Troop delivery.
- Aviation capability.
- Endurance (unrefueled range at 20 kts).
- Energy efficiency (fuel consumption at endurance speed) [78].

Part 1b: Thresholds and Objectives

Table XXXVII: Subfactor thresholds and objectives for the CGW(X) exercise.

Subfactor	Threshold	Objective
Sprint speed at 90 percent max continuous rating in sea state 3	30 kts	40 kts
Mission payload	600 ST	1000 ST
Troop delivery	Offload in 5 hours	Offload in 1 hour

Continued on next page

Table XXXVII: Subfactor thresholds and objectives for the CGW(X) exercise. (Continued)

Subfactor	Threshold	Objective
Aviation capability	Deck and hangar for 1 SH-60 (rotors folded)	Deck and hangar for 2 SH-60s (rotors folded)
Endurance (unrefueled range at endurance speed)	2500 NM	3500 NM
Energy efficiency (fuel consumption at endurance speed)	600 gallons/hour	400 gallons/hour

Source: *3.2.5. Source Selection Practical*, 2025 [78]

Part 2: Down-Select

- Shipyard A proposal: sprint speed 35 kts; mission payload 850 tons; troop delivery 4 hours; aviation capability deck for 2 SH-60s/hangar for 1 SH-60; endurance 3000 NM at 20 kts; efficiency 450 gallons/hour at 20 kts.
- Shipyard B proposal: sprint speed 42 kts; mission payload 600 tons; troop delivery 2 hours; aviation capability deck for 1 SH-60/hangar for 2 SH-60s; endurance 2500 NM at 20 kts; efficiency 500 gallons/hour at 20 kts.
- As the SSEB, rate each shipyard against the standards and develop an overall rating based on the relative importance of the subfactors.
- As the SSAC, recommend a contractor to the SSA using a best-value trade-off; note whether the decision would change under LPTA [78].

3.6 Contract Administration

3.6.1 Summary

Role in the process. Contract administration is a post-award function that protects Government interests, monitors contractor performance, manages changes, resolves disputes, and closes out contracts [80].

Key organizations. DCMA, DCAA, and Defense Finance and Accounting Service (DFAS) provide contract management, audit, and payment support to keep contracts compliant and funded [80].

Key personnel. The PCO, ACO, Contracting Officer's Representative (COR), and TCO coordinate with the PMO and program integrator to manage contract execution [80].

Changes and commitments. Distinguish change orders from supplemental agreements, track NECP/ECP actions, and avoid constructive changes and unauthorized commitments [80].

Disputes and termination. Use Alternative Dispute Resolution (ADR) and the disputes process, and apply termination for convenience, cause, or default based on commercial/noncommercial context [80].

3.6.2 Role in the Contracting Process

- Manage Government interests and monitor contractor quality, cost, and performance.
- Ensure contractor payment, administer contract modifications, and verify compliance with terms.
- Resolve disputes and execute contract close-out activities [80].

3.6.3 Administration Organizations

DCMA. Provides worldwide contract administration for non-shipbuilding contracts; functions include contract management, quality assurance, engineering/production oversight, property management, and program support [70, § 42.302] [80].

DCAA. Performs contract audit services and financial advisory support before and after award to inform negotiation, administration, and close-out actions [80].

DFAS. Executes contract payments and must meet Prompt Payment Act timelines to protect contractor cash flow, especially for small businesses [80].

Table XXXVIII: Contract administration organizations and services.

Organization	Program contract support	Financial support	Relationship to program
DCMA	Full spectrum of contract support	Transaction processing	Assistance
DCAA	Contract audit	Transaction processing	Audit
DFAS	Contract payments	Transaction processing	Functional

Source: *3.2.6. Contract Administration*, 2025 [80]

3.6.4 Key Personnel and Relationships

ACO and CAO. ACOs perform contract administration functions assigned by the PCO and operate within the Contract Administration Office (CAO); SUPSHIP and Regional Maintenance Center (RMC) equivalents perform ACO roles for shipbuilding and overhaul work [70, § 42.302] [80].

Program integrator. The program integrator serves as the CAO focal point for the program, leads the program support team, and provides status updates to the PMO and DCMA chain of command [80].

COR. The CO appoints qualified personnel as CORs with written authority; unauthorized acts can create personal liability [80].

TCO. After a termination notice, the TCO negotiates an equitable settlement with the contractor [80].

3.6.5 PMO and CAO Coordination

- Involve the CAO early, establish a Memorandum of Agreement (MOA), and conduct a combined post-award conference.
- Inform the CAO of program developments, reviews (e.g., Defense Acquisition Board (DAB)), and funding issues; notify the CAO of planned visits [70, § 42.402] [80].

3.6.6 Cash Flow

Timely Government payments drive contractor cash flow needed for materials, payroll, and subcontractor bills; late payments increase financing costs and can degrade performance, especially for small businesses [80].

3.6.7 Contract Changes and Modifications

Change proposals. Non-Engineering Change Proposal (NECP) and Engineering Change Proposal (ECP) proposals modify scope, cost, or configuration; value engineering changes save money and share savings with the contractor [80].

Deviations and waivers. Deviations authorize pre-manufacture nonconformance; waivers accept nonconformance during or after manufacture as suitable-as-is, often with reduced cost [80].

Modification requests. Field Modification Requests (FMRs) and Headquarters Modification Requests (HMRs) authorize ACO-approved contract modifications based on supporting NECP/ECP packages [80].

Contract changes. Supplemental agreements are bilateral modifications with equitable adjustments; change orders are unilateral and increase Government risk (undefinitized liabilities) while performance continues [80].

3.6.8 NECP/ECP Approval Levels

Level 1 (OPNAV). Changes to military characteristics, technical specifications exceeding the CNO ship cost adjustment, schedule slips beyond contract delivery, and changes with reclamas by the SPM or Configuration Control Board (CCB) sponsor.

Level 2 (SYSCOM). Changes affecting multiple program managers where a common decision cannot be reached.

Level 3 (SPM). Field-activity or contractor changes affecting dates, guarantees, incentives, hull strength or safety, interfaces, or significant cost impacts.

Level 4 (Field Activity). Changes not requiring higher approval [80].

3.6.9 Constructive Changes and Unauthorized Commitments

Constructive change. An authorized Government act (or failure to act) interpreted as a change order, requiring ratification and equitable adjustment when performance goes beyond minimum requirements [80].

Unauthorized commitment. An agreement made by a Government representative without authority; only the KO can bind the Government, and ratification is required [70, § 1.602-3] [80].

3.6.10 Privity of Contract

The Government has a direct contractual relationship with the prime contractor, not subcontractors; administration actions must respect this privity [80].

3.6.11 Disputes and Alternative Dispute Resolution

ADR. ADR promotes early, principled settlement of disputes and is preferred before formal processes [70, § 33.204] [80].

Disputes process. If unresolved at the contracting officer level, contractors may use the disputes process under the Contract Disputes Act [70, § 33.202] [80].

3.6.12 Termination Types and Notices

Convenience. Used when requirements change, funds are insufficient, quantities shrink, or Government needs evolve; it is a unique Government right [80].

Cause vs. default. Termination for cause applies to commercial contracts; termination for default applies to noncommercial contracts when performance fails [70, § 52.212-4] [80].

Notices. Cure notices provide 10 days to remedy failures; show-cause notices apply when time is insufficient or cure fails; termination notices explain consequences [80].

3.6.13 Commercial Item Definitions

Commercial products and services. FAR 2.101 definitions cover products, services, and non-developmental items available in the commercial marketplace, including customary or minor modifications [70, § 2.101] [80].

COTS. Commercial Off-the-Shelf (COTS) items are sold in substantial quantities in the commercial marketplace and offered to the Government without modification [70, § 2.101] [80].

3.6.14 Government Obligation to Pay

For cost-reimbursement contracts, the Government pays allowable costs incurred before termination, costs necessary after termination for default, approved subcontractor settlements, and prorated fees; excusable delays include epidemics, strikes, floods, Government actions, freight embargoes, fire, and unusually severe weather [80].

3.7 Earned Value Management (EVM)

3.7.1 What EVM is

Earned Value Management (EVM) integrates scope, schedule, and cost to quantify performance by comparing planned work to completed work and associated costs; it is widely used in government and defense programs to keep delivery on time and within budget.

3.7.2 Key Components of EVM

Core cost and schedule metrics establish the earned value performance baseline [81, 82].

Planned Value (PV) / Budgeted Cost of Work Scheduled (BCWS): The budgeted cost for the work scheduled to be completed by a specific date.

Earned Value (EV) / Budgeted Cost of Work Performed (BCWP): The budgeted cost for the work actually completed by a specific date.

Actual Cost (AC) / Actual Cost of Work Performed (ACWP): The actual cost incurred for the work completed by a specific date.

Budget at Completion (BAC): The total budget allocated for the project.

Estimate at Completion (EAC): The forecasted total cost of the project based on current performance.

Cost Performance Index (CPI): A measure of cost efficiency, calculated as $\mathrm{CPI} = \mathrm{EV}/\mathrm{AC}$. A CPI less than 1 indicates a cost overrun.

Schedule Performance Index (SPI): A measure of schedule efficiency, calculated as $\mathrm{SPI} = \mathrm{EV}/\mathrm{PV}$. A SPI less than 1 indicates a schedule delay.

Schedule Variance (SV): The difference between the earned value and the planned value, calculated as $\mathrm{SV} = \mathrm{EV} - \mathrm{PV}$. A negative SV indicates a schedule delay.

Cost Variance (CV): The difference between the earned value and the actual cost, calculated as $\mathrm{CV} = \mathrm{EV} - \mathrm{AC}$. A negative CV indicates a cost overrun.

Info

The first three are equivalent terms; this guide uses the PV, EV, AC forms.

3.7.3 When to Use EVM

EVM is mandated on cost or incentive contracts (and applicable subcontracts) when the program meets the policy thresholds and the effort is discretely measurable [81, 83]. Key decision points for the PM include:

$100 million and above: Require an American National Standards Institute (ANSI)/Electronic Industries Alliance (EIA)-748-compliant and acceptable Earned Value Management System (EVMS); when DoW is the cognizant Federal agency, DCMA determines system compliance.

$50 million to $99.9 million: Require an EVMS for cost or incentive contracts and subcontracts, but a formal compliance determination is not required unless separately directed.

Below $50 million: Use EVM only when a risk-based cost-benefit analysis shows that the management value justifies the reporting burden.

Additional policy considerations follow [81, 83, 84].

- Applicability determinations must confirm the work scope can be discretely measured before mandating EVM.
- EVM on Firm-Fixed-Price (FFP), time-and-materials, or Level of Effort (LOE) contracts is discouraged absent clear management benefit.
- Contracts primarily for software, including DoWI 5000.87 software-acquisition pathway contracts, are exempt from DFARS Subpart 234.2 under the active class deviation.
- Contracts that require EVM normally also require Integrated Program Manager's Data Analysis Report (IPMDAR) data and an Integrated Baseline Review (IBR) no later than 180 calendar days after contract award, exercise of a significant option, or major modification.

3.7.4 EVM Compliance

EVM clauses are flowed in the solicitation and award package. Under Class Deviation 2026-O0011, use DFARS 252.234-7998 and 252.234-7999 instead of the legacy 252.234-7001 and 252.234-7002 text [83]. Compliance expectations follow [81, 83, 84].

- The program office integrates EVM planning into the WBS, maintains the Performance Measurement Baseline (PMB), and ensures timely receipt and analysis of IPMDAR submissions.
- When DoW is the cognizant Federal agency, DCMA determines EVMS compliance against ANSI/EIA-748; the contracting officer approves or disapproves the system with input from functional specialists and auditors.
- Surveillance access is contractual: the applicable DFARS clause gives the Government access to contractor processes, data, and personnel needed to assess ongoing EVMS compliance.
- Service focal points such as Acquisition Data and Analytics and the DON Center for EVM adjudicate applicability determinations and coordinate policy updates.
- SUPSHIP can execute validation and surveillance responsibilities for shipbuilding programs on behalf of DCMA *(Appendix C.2: The May 2026 flag roster lists RDML Jonathan J. Jett-Parmer (United States Navy Reserve (USNR)) as Reserve Deputy, Naval Sea Systems Command; his Navy bio still reflects Deputy Commander, Supervisor of Shipbuilding, Conversion and Repair [12, 85].)*.

3.7.5 EVM Principal Players

Programmatic, contracting, and oversight roles synchronize to sustain disciplined performance management [81].

PMO: Implements EVM on the contract, ensures solicitations and awards contain the required clauses, and works with the contracting activity to tailor reporting and Integrated Master Schedule (IMS) requirements.

DCMA: When DoW is the cognizant Federal agency, determines contractor EVMS compliance, supports acceptance and surveillance, maintains the official acceptance roster, and analyzes IPMDAR and schedule submissions to focus government attention on emerging issues.

DCAA: Conducts accounting system audits, corroborates cost data, and supports DCMA and the contracting officer during surveillance events and reviews.

Acquisition Data and Analytics: Acts as the department-wide focal point for EVM policy, guidance, and competency management; issues interpretations to maintain consistent application across programs.

DON Center for EVM: Coordinates applicability determinations with Deputy Assistant Secretary of the Navy (Acquisition Policy and Budget) (DASN (APB)) and OSW, and serves as the Navy's central point of contact for EVM matters.

SUPSHIP: Performs many DCMA/DCAA surveillance roles for shipbuilding contracts when delegated by the cognizant program office. *(Appendix C.2: See Appendix C.2 for current SUPSHIP flag billet holders)*

3.7.6 PMB

Figure 3.5 shows the characteristic S-curve depicting cumulative PV/BCWS, while Figure 3.6 summarizes the progression from defining the SOW to final PMB adjustments. PMBs are:

- Scoped, scheduled, and budgeted plans for the authorized work
- Time-phased budgets that align to the master schedule
- The basis for cost and schedule performance management
- Effectively the PV for the entire project

Do not confuse the PMB with the Acquisition Program Baseline. The Acquisition Program Baseline is the program-level cost, schedule, and performance agreement with the Milestone Decision Authority; the PMB is the contract execution baseline used to measure earned value performance.

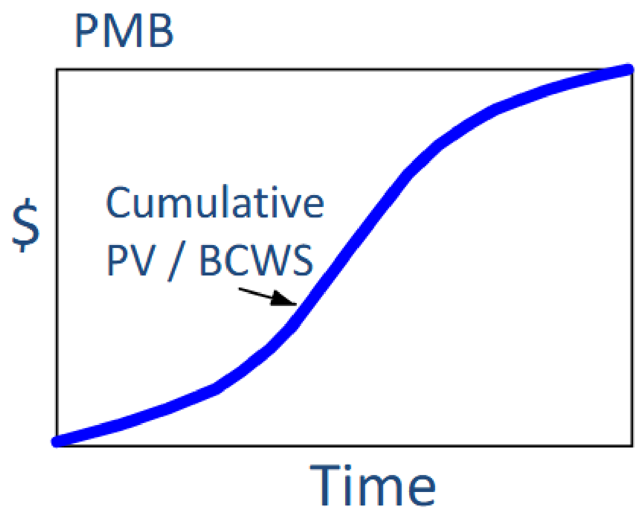

Figure 3.5. *PMB cost vs. time chart showing the "S"-curve. Source:* 3.6.1. Introduction to EVM, *2025 [81].*

Figure 3.6. *The development flow for PMB. Source:* 3.6.1. Introduction to EVM, *2025 [81].*

PMB Development

Step 1: *Define all work*. Decompose the effort using the WBS and align it with the organizational structure. Control accounts are the natural management points where Control Account Managers (CAMs) are assigned, and each control account contains work packages and (if needed) planning packages.

Step 2: *Schedule the work*. Develop the integrated schedule (often visualized with Gantt charts) that sequences the WBS elements, includes key milestones, and forms the backbone of the time-phased budget.

Step 3: *Budget the work*. Assign time-phased budgets to each work package, establish management reserve for in-scope known unknowns, and confirm that the sum of the control accounts equals the contract budget base.

Changes to the PMB must be formally controlled and documented. Reasons for changes include:

- Contract changes
- Internal replanning
- Formal reprogramming

3.7.7 EVM Reviews and Reports

Formal reviews validate the integrity of EVM data products before and after system acceptance [81].

Post-Acceptance Review

- Objective: ensure performance data remain accurate after system acceptance and identify any corrective actions required to reaffirm compliance.
- Timing: conducted as needed following system acceptance and prior to the next IBR.
- Led by the Review Director (typically DCMA) with membership tailored to the purpose of the review.
- Culminates in a formal report prepared by the Review Director.

Initial Compliance Evaluation

- Objective: validate that the Contractor's EVM system description matches actual practice and satisfies the EVMS criteria.
- Timing: executed prior to the initial IBR (and subsequently only if needed).
- Led by the DCMA Review Director with cross-functional support as required.
- Results documented in a report signed by the Review Director.

Integrated Baseline Review

- Participants: the contractor, PM and deputy, DCMA/DCAA/SUPSHIP personnel, and relevant technical staff.
- Purpose: joint assessment to verify SOW coverage, logical schedule sequencing, resource realism, and risks captured in the PMB.
- Timing: conducted no later than 180 calendar days after contract award, exercise of a significant option, or major modification in accordance with the DoW EVM Implementation Guide and service policy [81, 83].

Integrated Program Management Data Analysis Report

This contractually required report delivers cost, schedule, and technical status derived from the Contractor's EVM system, enabling the PMO to identify performance problem areas and emerging risks. When required, IPMDAR is normally delivered monthly; final reports are due no later than 16 Government business days after the contractor accounting period ends unless the contract tailors the deadline. The Rev. D IPMDAR Data Item Description components are [86]:

1. Contract Performance Dataset.
2. Schedule deliverables: the native IMS plus the Schedule Performance Dataset.
3. Performance Narrative Report.

A Contract Funds Status Report (CSFR) complements the IPMDAR by providing funding (price) information that reconciles to the cost-focused earned value data [81].

Problems

When using EVM, the following are indications that a problem exists [82].

- Use of management reserves
- Significant revisions to the PMB
- Zero variance
- Sudden change in trends
- Unreasonable Total Cost Performance Index (TCPI)

3.7.8 EVM Data Analysis

Figure 3.7 shows the S-curve representation of the EVM variables; the gaps between the curves are the cost and schedule variances that drive performance assessments [82].

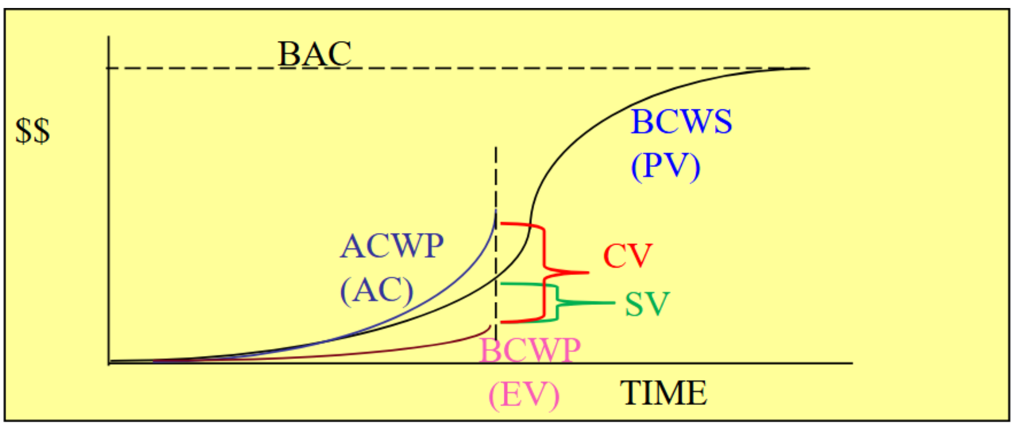

Figure 3.7. EVM chart showing the independent variables and variances. Source: 3.6.2. EVM Data Analysis*, 2025 [82]*

Data Analysis Steps

1. **Get current status**: capture the latest cost and schedule performance using CPI, SPI, and narrative explanations.
2. **Identify trends**: analyze indices and variance trajectories over time to highlight emerging risks and opportunities.
3. **Predict completion**: develop independent EAC and Variance at Complete (VAC) assessments to forecast final outcomes.
4. **Determine management actions**: decide where to apply resources, contract changes, or risk mitigations to recover performance [82].

Performance Measurement Techniques

Percent Complete: Applies to long-duration tasks lacking interim milestones; progress is reported as the earned value fraction of total budget.

Weighted Milestones: Uses milestone weights for long-duration tasks with clear waypoints, crediting earned value as milestones are completed.

Percent Start/Percent Finish: Suitable for short-duration tasks (less than two reporting periods) with credit apportioned at start and finish.

0/100 Method: Provides earned value only when the task is complete, offering a conservative measure for discrete, short tasks [82].

Equations

Standard cost and schedule performance calculations include [82].

$$\mathrm{CV} = \mathrm{EV} - \mathrm{AC}$$
$$\mathrm{SV} = \mathrm{EV} - \mathrm{PV}$$
$$\mathrm{CPI} = \frac{\mathrm{EV}}{\mathrm{AC}}$$
$$\mathrm{SPI} = \frac{\mathrm{EV}}{\mathrm{PV}}$$
$$\mathrm{CV\%} = \left(\frac{\mathrm{CV}}{\mathrm{EV}}\right) \times 100$$
$$\mathrm{SV\%} = \left(\frac{\mathrm{SV}}{\mathrm{PV}}\right) \times 100$$
$$\%\ \mathrm{Complete} = \left(\frac{\mathrm{EV}}{\mathrm{BAC}}\right) \times 100$$
$$\%\ \mathrm{Spent} = \left(\frac{\mathrm{AC}}{\mathrm{BAC}}\right) \times 100$$
$$\mathrm{EAC} = \frac{\mathrm{BAC}}{\mathrm{CPI}}$$
$$\mathrm{EAC} = \mathrm{AC} + \frac{\mathrm{BAC} - \mathrm{EV}}{\left(\frac{\mathrm{EV}}{\mathrm{AC}}\right)_{3\text{ months}}}$$
$$\mathrm{EAC} = \mathrm{AC} + \frac{\mathrm{BAC} - \mathrm{EV}}{\mathrm{CPI} \times \mathrm{SPI}}$$
$$\mathrm{EAC} = \mathrm{AC} + \frac{\mathrm{BAC} - \mathrm{EV}}{0.8\,\mathrm{CPI} + 0.2\,\mathrm{SPI}}$$
$$\mathrm{VAC} = \mathrm{BAC} - \mathrm{EAC}$$
$$\mathrm{TCPI} = \frac{\text{Work remaining}}{\text{Budget remaining}}$$
$$\mathrm{TCPI}_{\mathrm{EAC}} = \frac{\mathrm{BAC} - \mathrm{EV}}{\mathrm{EAC} - \mathrm{AC}}$$
$$\mathrm{TCPI}_{\mathrm{BAC}} = \frac{\mathrm{BAC} - \mathrm{EV}}{\mathrm{BAC} - \mathrm{AC}}$$

3.8 EVM Exercise

3.8.1 Summary

Purpose. Apply earned value calculations to a two-period data set to diagnose cost and schedule status [87].

Focus. Compute performance indicators, interpret trends, and project cost outcomes [87].

3.8.2 Exercise Inputs

- Two-period summary data for planned value, earned value, and actual cost [87].
- Program budget targets and any stated profit or funding assumptions [87].

3.8.3 Calculation Checklist

- SV and SPI, period and cumulative [87].
- CV and CPI, period and cumulative [87].
- TCPI to budget [87].
- EAC forecast and estimated overrun/underrun [87].
- Estimated contractor profit, government cost, and POM request impact [87].

3.8.4 Interpretation Prompts

Status. Explain the program's cost and schedule health using SV/CV and SPI/CPI trends [87].

Drivers. Identify the likely sources of variance and whether corrective action is required [87].

Projection. Describe the likely end-state cost using EAC and resulting profit or funding impacts [87].

4 Defense Acquisitions

4.1 Acquisition Policy and Players

4.1.1 Summary

Rules and regulations. Acquisition authority flows from acquisition-related law, Federal regulations (FAR, DFARS, and component supplements), and DoW/DON policy documents in the 5000 series and Secretary of the Navy Instruction (SECNAVINST) guidance [88].

Decision support systems. Joint Capabilities Integration and Development System (JCIDS) defines requirements, PPBE provides resources, and the DAS delivers materiel; effective program management aligns all three [88].

WAS emphasis. DoW redesignated the DAS as the Warfighting Acquisition System (WAS) to bias acquisition toward speed-to-capability, commercial-first approaches, and portfolio-level outcomes over process compliance [89, 90].

Risk and affordability. Cost, schedule, and performance risks define the trade space, while affordability keeps life-cycle cost within disciplined bounds [88].

Key acquisition players. Acquisition Program, Defense Acquisition Executive (DAE), CAE, MDA, PEO, and PM definitions establish who owns decisions and reporting [88].

Oversight and reporting. ACAT categories scale oversight by cost and special interest; reporting relies on Research, Development, and Acquisition Information System (RDAIS), Defense Acquisition Executive Summary (DAES), Selected Acquisition Report (SAR), Unit Cost Report (UCR), and Defense Acquisition Visibility Environment (DAVE) [88].

4.1.2 Rules and Regulations

1. Acquisition-related laws.
2. Federal Government regulations.
3. DoW and DON acquisition policy documents [88].

4.1.3 Acquisition-Related Law

Statutory authority from Congress provides the legal basis for systems acquisition. Title 10, U.S.C. governs the organization, structure, and operation of the Armed Forces, and the annual NDAA is a primary mechanism for modifying acquisition authorities and reporting requirements [88].

- Armed Services Procurement Act (1947), as amended.
- Small Business Act (1963), as amended.
- Office of Federal Procurement Policy Act (1983), as amended.
- Competition in Contracting Act (1984).
- Department of War Procurement Reform Act (1985).
- Department of War Reorganization Act (Goldwater-Nichols) (1986).
- Defense Acquisition Workforce Improvement Act (1990).
- Federal Acquisition Streamlining Act (1994).
- Clinger-Cohen Act (1996).
- Weapon Systems Acquisition Reform Act (2009) [88].

4.1.4 Federal Government Regulations

- The FAR is the primary regulation for all Federal agencies acquiring supplies and services with appropriated funds; it guides acquisition planning, competition, contract award, and warranty policy.
- The DFARS supplements the FAR with DoW-specific procedures.
- Component-unique supplements (for example, NMCARS for the Navy) tailor policy at the Service level.

The FAR is the highest-level acquisition regulation [88].

4.1.5 Revolutionary FAR Overhaul Class Deviations

Presidential action established FAR-overhaul policy direction through "Restoring Common Sense to Federal Procurement" [91]. OMB Memo M-25-26 provides implementation direction for agencies and the FAR Council during execution [92]. FAR Council deviation guidance then supports interim operations while part-by-part text is revised and final rulemaking proceeds [93]. DoD class deviations posted by Defense Pricing, Contracting, and Acquisition Policy (DPCAP) currently cover Part 6 (competition), Part 10 (market research), Part 12 (commercial products and services), and Part 18 (emergency acquisitions), effective Feb 1, 2026 [94, 95, 96, 97]. The Acquisition.gov deviation guide is the live part-by-part status tracker for agency implementation [98, 99]. For source authority, use the EO, OMB memo, and FAR Council guidance as controlling implementation references; White House fact-sheet and DAU materials are informative context [100, 101].

4.1.6 Acquisition Policy Documents

- Department of War Directive (DoWD) 5000.01 (Defense Acquisition System) sets policy and assigns senior acquisition responsibilities.

- DoWI 5000.02 (Operation of the Adaptive Acquisition Framework) establishes six pathways to tailor acquisition strategies.
- Acquisition Guidebooks provide non-mandatory guidance, lessons learned, and best practices.
- SECNAVINST 5000.2 and the SECNAV M 5000.2 DoN Acquisition and Capabilities Guidebook provide DON-specific policy and tailoring guidance [88].

4.1.7 Adaptive Acquisition Framework

The Adaptive Acquisition Framework (AAF) supports the National Defense Strategy by providing tailored pathways for delivering capability. Pathway instructions include:

- Major Capability Acquisition (Major Capability Acquisition (MCA)) – DoWI 5000.85.
- Urgent Capability Acquisition – DoWI 5000.81.
- Middle Tier of Acquisition (Middle Tier of Acquisition (MTA)) – DoWI 5000.80.
- Software Acquisition – DoWI 5000.87.
- Defense Business Systems – DoWI 5000.75.
- Defense Acquisition of Services – DoWI 5000.74.

Functional area guidance includes engineering, test and evaluation, cybersecurity, Analysis of Alternatives (AoA), cost estimating, intellectual property, protection, human systems integration, intelligence, and information technology [88].

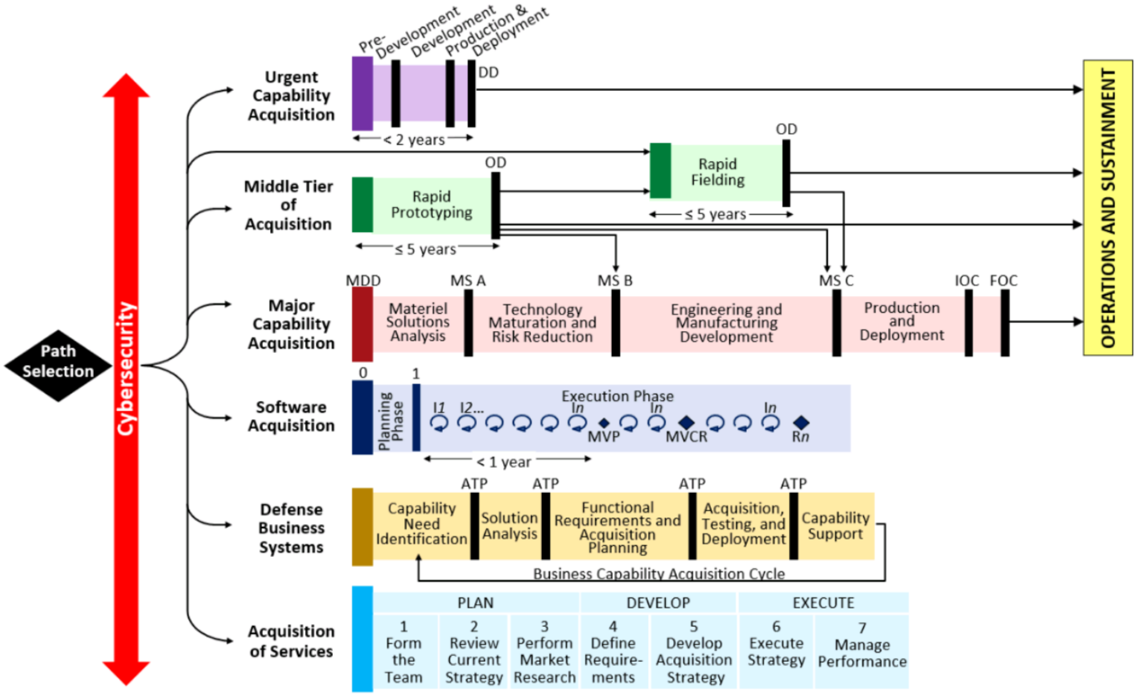

Figure 4.1. Adaptive Acquisition Framework. Source: 3.3.1. Acquisition Policy and Players*, 2025 [88].*

4.1.8 Defense Acquisition System Focus

The DAS manages the acquisition of weapon systems and other materiel. Major Capability Acquisition is the primary pathway for this course and is event-driven with five phases, four major decision points, and three milestones [88].

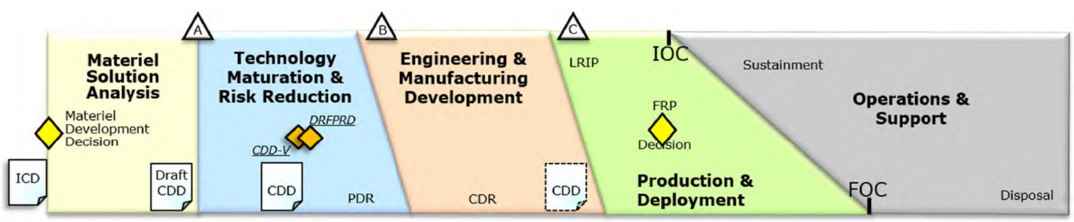

Figure 4.2. Major Capability Acquisition phases, milestones, and decision points. Source: 3.3.1. Acquisition Policy and Players*, 2025 [88].*

4.1.9 Warfighting Acquisition System Emphasis

Redesignation. The DAS is now referred to as the WAS to reinforce acquisition as a warfighting function focused on rapid capability delivery [89].

Speed and outcomes. Portfolio leaders are expected to trade cost, schedule, and performance for faster fielding when warranted, using only the documentation needed to manage risk and statutory compliance [89, 90].

Commercial-first tools. WAS guidance elevates commercial-first sourcing and flexible acquisition tools, including CSO solicitations and OT prototype pathways, to shorten sourcing timelines and widen access to non-traditional vendors [72, 74, 89].

4.1.10 Key Acquisition Reform Initiatives

- Streamlined FAR/DFARS updates and continued modernization of the 5000 series.
- Mil-Spec review for cost effectiveness and open standards.
- Advanced Concept Technology Demonstrations (Advanced Concept Technology Demonstration (ACTD)) to reduce technical risk.
- Control of TOC in addition to procurement cost.
- Integrated Product and Process Development (IPPD) and IPT structures to integrate disciplines early.
- Cost as an Independent Variable (CAIV) to trade performance and cost.
- Single Process Initiative to reduce process duplication.
- Open Systems Architecture (Open Systems Architecture (OSA)) and simulation-based acquisition to shorten time and cost.
- Better Buying Power (Better Buying Power (BBP)) and Back to Basics (Back to Basics (BtB)) workforce initiatives [88].

4.1.11 Risk Factor Impact

Cost. Drivers include material price changes, labor rates, and regulatory compliance such as Environmental, Safety, and Occupational Health (ESOH) requirements.

Schedule. Late deliveries, political pressure, and changing capability needs can drive schedule risk, as can late compliance planning.

Performance. New materials or technologies and unproven operational concepts can increase performance risk.

Managing risk by balancing cost, schedule, and performance is central to successful acquisition [88].

Figure 4.3. *Cost, schedule, and performance trade space. Source:* 3.3.1. Acquisition Policy and Players, *2025 [88].*

4.1.12 Affordability and Trade Space

Affordability keeps life-cycle cost within an established range while using schedule and performance as the trade space. It informs AoA work, requirements, and engineering decisions throughout the life cycle [88].

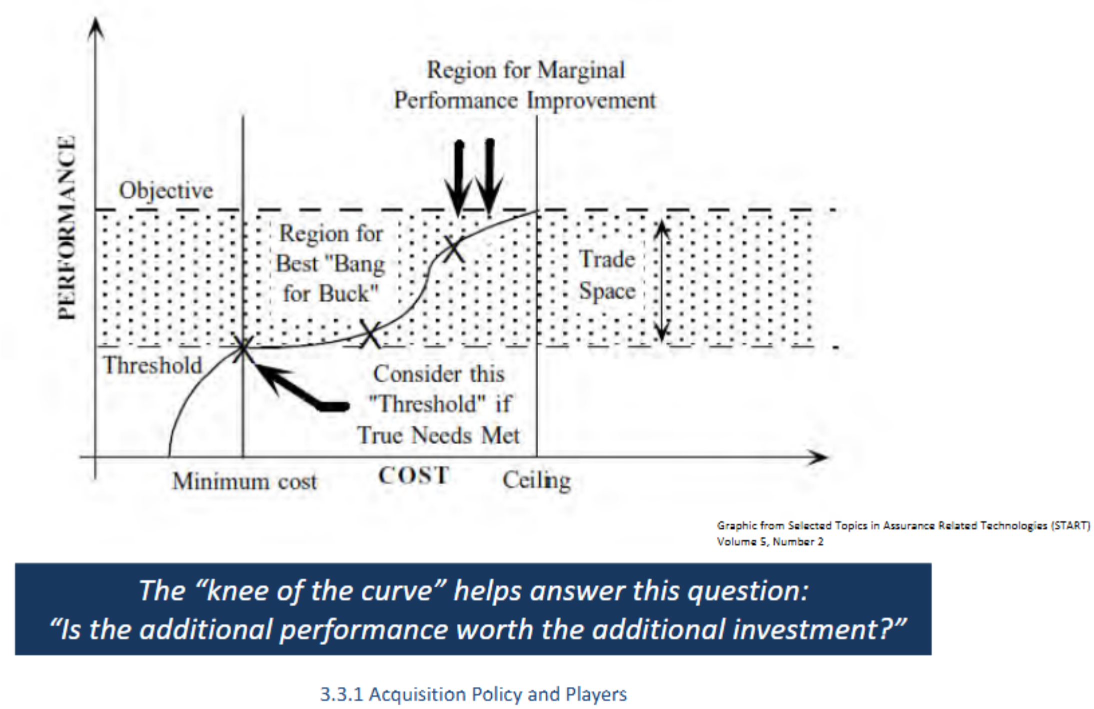

Figure 4.4. *Trade space of performance vs. cost showing the knee of the curve. Source:* 3.3.1. Acquisition Policy and Players, *2025 [88].*

4.1.13 Decision Support Systems

Three decision support systems must work together for acquisition success [88].

JCIDS. Requirements generation system that identifies, assesses, and prioritizes warfighter capability needs.

PPBE. Resource allocation process that builds the DoW budget for acquisitions.

DAS. Management process for developing and buying weapon systems; implemented via the AAF.

Program Managers must understand all three systems and synchronize them [88].

4.1.14 Major Institutions of Defense Acquisition

At the national level, defense acquisition is shaped by the Executive Branch, Congress, and defense industry. Program Managers execute day-to-day coordination and reporting across these institutions [88].

4.1.15 Defense Industry Landscape

Public vs. private companies. Public companies issue shares and disclose financial results; private companies are closely held and generally have fewer reporting obligations.

Large/medium businesses. Establish divisions, team with other large firms, and pursue large RDT&E contracts to gain experience.

Small businesses. Often centralize staff functions, outsource support services, and rely on relationships with KOs, the SBA, and the DoW Office of Small Business Programs; the coursebook threshold is 500 employees or $7.5M in revenue.

Commercial vs. government markets. Commercial markets set prices freely; government markets are regulated by the FAR and profit caps but are generally less risky [88].

4.1.16 Organizational Structures

Common organizational models include functional, divisional, and matrix structures. Small firms typically centralize functions, while large firms create multiple divisions aligned to product lines or markets. A sound structure clarifies authority and enables coordination; a poor structure blurs accountability and slows decision-making [88].

4.1.17 Defense Trends

Vulnerabilities. Economic downturns, technology breakthroughs, workforce availability, energy issues, raw materials, and wartime demand can stress suppliers and schedules.

Opportunities. Shifting threats create opportunities in cyber, counter-terrorism, missile defense, space, and science and technology; creative partnerships can leverage commercial markets [88].

4.1.18 Acquisition Players (DoWD 5000.01 Definitions)

Acquisition Program. A directed, funded effort to deliver a new, improved, or continuing materiel, weapon, information system, or service capability in response to an approved need.

DAE. The top-level official for defense acquisitions; acts as MDA for designated MDAPs and MAISs and may delegate authority to Component leaders.

CAE. Senior official responsible for acquisition functions within a DoW Component; authority may be re-delegated.

MDA. Designated official with overall program responsibility; approves phase entry and is accountable for cost, schedule, and performance reporting to higher authority.

PEO. Directs multiple acquisition programs and may be delegated MDA for ACAT II/III/IV programs.

PM. Responsible for achieving program objectives and accountable for cost, schedule, and performance reporting to the MDA [88].

4.1.19 Office of the Secretary of War Decision Makers

Key decision makers within OSW include Under Secretary of War for Acquisition and Sustainment (USW(A&S)), Under Secretary of War for Research and Engineering (USW(R&E)), the DoW Chief Information Officer (CIO), and Director, Operational Test and Evaluation (DOT&E). CAPE provides cost and program evaluation support [88].

USW(C). Controls the budget and release of funds; aligned to PPBE budgeting and execution.

USW(P). Aligns DoW plans with national security objectives and supports the planning phase of PPBE.

USW(A&S). Principal advisor to SECWAR and Deputy Secretary on acquisition and sustainment; sets workforce policy and makes milestone decisions for designated MDAPs.

DOT&E. Provides independent operational test oversight, approves operational test plans and live fire strategies, and reports directly to SECWAR and Congress.

CAPE. Oversees AoA guidance and evaluates program costs, schedules, and impacts across the DoW portfolio.

USW(R&E). Chief technology officer for DoW; provides technical risk assessments and oversees innovation organizations such as () and Defense Advanced Research Projects Agency (DARPA).

DoW CIO. Oversees IT and cyber acquisition and may serve as MDA for MAIS programs.

The DAB is the senior forum advising USW(A&S) on critical acquisition decisions and includes senior DoW leadership from policy, financial management, test, intelligence, and Service Secretaries [88].

4.1.20 Intelligence Community Touchpoints

Service and national intelligence organizations provide threat assessments and updated intelligence throughout the acquisition life cycle. Program review involvement typically includes Service intelligence, Defense Intelligence Agency (DIA), and Director of National Intelligence (DNI) depending on program impact [88].

4.1.21 Department of the Navy Decision Makers

ASN(RD&A) serves as the DON CAE and oversees SYSCOM commanders, PEOs, and Deputy Assistant Secretary of the Navys (DASNs). ASN(RD&A) represents the DON to USW(A&S) and Congress on acquisition policy and program execution [88].

4.1.22 Acquisition Categories

Programs are grouped into ACAT categories primarily by cost and level of interest to scale oversight and statutory compliance. Factors include cost, location in the acquisition framework, and special interest [88].

ACAT I

ACAT I programs are MDAPs requiring expenditure of at least $525M RDT&E or $3.065B procurement (FY 2020 constant dollars), or those designated by USW(A&S). ACAT ID programs are managed at the DAE level; ACAT IC programs are managed by the CAE. Examples include DDG-1000 and the Navy Multiband Terminal [88].

ACAT II

ACAT II major systems do not meet ACAT I criteria and are estimated to exceed $200M RDT&E or $920M procurement (FY 2020 constant dollars). The CAE (or designee) is the MDA. Automated information systems do not use ACAT II [88].

ACAT III and ACAT IV

ACAT III programs do not meet ACAT I, ACAT IA, or ACAT II criteria and, in DON policy, generally cover programs estimated above $75M but below $200M RDT&E or above $250M but below $920M procurement (FY 2020 constant dollars). ACAT III programs are assigned to the lowest appropriate MDA, normally the cognizant PEO, DRPM, SYSCOM commander, or designee. ACAT IIIA covers less-than-major automated information systems. ACAT IV programs are DON-unique categories for efforts not designated ACAT III, and are split into IVT (test) and IVM (monitor) depending on operational test requirements [9, 88].

4.1.23 Program Reporting Requirements

RDAIS. Authoritative DON database tracking program cost, schedule, and performance with monthly and quarterly updates.

DAES. Early warning reports from DON to USW(A&S) for ACAT I programs between milestones, using RDAIS inputs.

DAVE. Tracks MTA efforts with status, issues, cost, and performance updates.

SAR. Annual report from SECWAR to Congress for ACAT I programs; quarterly reporting is required for schedule slips of six months or more or unit cost increases of 15% over the current APB objective (or 30% over the original baseline).

UCR. Quarterly report for ACAT I programs submitting SAR to ensure unit costs are visible and not masked by quantity changes [88].

4.1.24 Current Navy Acquisition Issues

- China, culture, climate change, and supply chain issues.
- "More than 350" fleet objective from Force Design 2045 (2022) for manned and unmanned platforms.
- Make cybersecurity part of program DNA from requirements through sustainment.

- "Get Real, Get Better" as a leadership imperative for continuous improvement.
- Ongoing acquisition reform [88].

4.1.25 Summary Highlights

- Decision support systems: JCIDS, PPBE, and DAS.
- Rules and regulations: FAR, DoWI 5000.02, and SECNAVINST 5000.2.
- Risk focus: cost, schedule, and performance.
- Key players: DAE, CAE, MDA, PEO, and PM.
- ACAT categories exist to scale oversight and tailor reviews [88].

4.1.26 Coursebook cue: systems acquisition management

Coursebook 3.3.1 uses "Systems Acquisition Management" to describe the organized process of translating military need into fielded, supportable capability through requirements, resourcing, contracting, engineering, test, production, sustainment, and disposal. The EDO answer should connect the management system to three outcomes: deliver required capability, keep the program executable in cost and schedule, and preserve competitive advantage by using government contracting and technical authority to buy what the fleet actually needs rather than what a vendor happens to offer [88].

Board cue. Systems acquisition management is the disciplined conversion of need into fielded capability. Tie the answer to requirements, resources, contracting, engineering, test, production, sustainment, and competitive advantage.

4.2 Acquisition Milestones and Phases

4.2.1 Summary

DAS structure. The DAS uses Milestones A/B/C, five phases, and decision points (Materiel Development Decision (MDD), Capabilities Development Document Validation (CDD-V), Development Request for Proposal Release Decision (DRFPRD), Full-Rate Production (FRP)) to pace major capability acquisitions [102].

Baseline discipline. The APB sets cost, schedule, and performance thresholds/objectives that define the program trade space and trigger deviation actions [102].

Pre-systems acquisition. The MDD, Materiel Solution Analysis (MSA), and AoA converge on a validated materiel solution, initial acquisition strategy, and draft requirements [102].

Systems acquisition. Technology Maturation and Risk Reduction (TMRR) and Engineering and Manufacturing Development (EMD) reduce risk and mature designs before Production and Deployment Phase (P&D); MS B/C decisions commit the program to development and production [102].

WAS decision evidence. WAS emphasis adds time-to-capability, producibility, and production scalability as first-order decision criteria alongside cost, schedule, and performance [89, 90].

Navy overlays. Ships, SCN timing, and the Navy 2-pass/7-gate review shape milestone execution and resourcing expectations [102].

4.2.2 Introduction: DAS and Urgent Needs

The DAS supports the National Defense Strategy through the AAF, tailoring acquisition to program urgency and risk [102].

UON. Urgent Operational Need (UON) addresses capability gaps impacting ongoing or anticipated contingency operations; when validated by a single DoW Component it is a Component UON [102].

JEON. Joint Emergent Operational Need (JEON) is a joint requirement identified by CCMD, CJCS, or Vice-Chairman of the Joint Chiefs of Staff (VCJCS) for an anticipated contingency [102].

JUON. Joint Urgent Operational Need (JUON) is a joint requirement identified by CCMD, CJCS, or VCJCS for an ongoing contingency [102].

4.2.3 DAS Structure: Milestones, Phases, Decision Points

- Milestones: Milestone A initiates technology maturation and risk reduction; Milestone B initiates engineering and manufacturing development; Milestone C initiates production and deployment [102].
- Phases: MSA, TMRR, EMD, P&D, and O&S [102].
- Decision points: MDD, CDD-V, DRFPRD, and FRP [102].

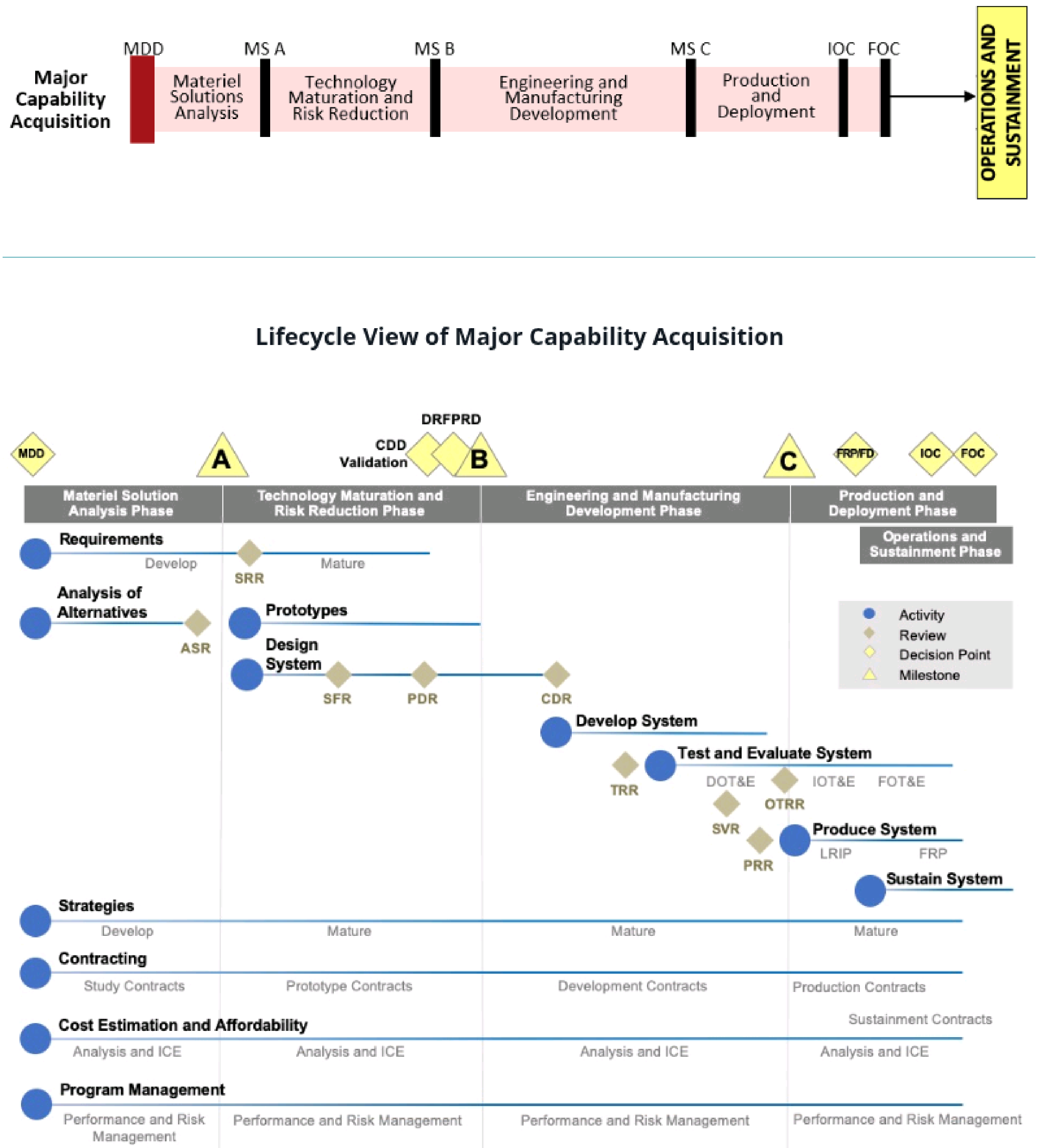

Figure 4.5. *Major Capability Acquisition phases and decision points. Source:* 3.3.2. Acquisition Milestones and Phases, *2025 [102].*

4.2.4 Milestone Review Information Requirements

Information for milestone reviews is tailored by the MDA and must align with DoWI 5000.85 requirements; it should cover program progress/risk, affordability, trade-offs, acquisition strategy updates, and exit criteria for the next phase [102].

4.2.5 Warfighting Acquisition System Evidence Priorities

Time-to-capability. Demonstrate how the plan accelerates fielding, including rapid prototypes, iterative delivery, and production ramp assumptions [89, 90].

Producibility and scalability. Provide evidence of manufacturing readiness, supplier depth, and surge capacity to move from prototype to production [89, 90].

Operational evidence. Favor demo results, test data, and transition plans over paper-only artifacts when assessing phase exit criteria [89].

4.2.6 Entrance and Exit Criteria

Entrance criteria. Minimum accomplishments required before entering a phase; documented in DoWI 5000.85 and required for all programs [102].

Exit criteria. Program-specific technical, schedule, or management risk areas proposed by the PM and approved by the MDA; recorded in the Acquisition Decision Memorandum (ADM) and must be specific and demonstrable [102].

Risk linkage. Programs must meet exit criteria for the current phase and entrance criteria for the next phase before transition; the MDA uses exit criteria to confirm risk reduction [102].

4.2.7 Acquisition Program Baseline (APB) and Trade Space

Purpose. The APB documents cost, schedule, and performance thresholds/objectives and defines the trade space for the program [102].

Thresholds. Minimum acceptable values; exceeding schedule, cost, or performance thresholds constitutes a deviation and requires MDA involvement [102].

Objectives. Desired values that represent meaningful improvements beyond thresholds and guide PM trade decisions [102].

Default thresholds. Unless specified, performance threshold equals objective, cost threshold equals objective plus 10%, and schedule threshold equals objective plus 6 months [102].

4.2.8 Program Deviations

Deviation definition. A program deviation occurs when the current estimate exceeds APB thresholds for cost, schedule, or performance; early indicators are set by the MDA (e.g., 10% schedule slip) [102].

Nunn–McCurdy. Significant breaches are 15% growth from the current baseline or 30% from the original baseline; critical breaches are 25% current or 50% original, and are subject to termination review; significant breaches require congressional notification within 45 days after the Program Deviation Report [102].

4.2.9 Program Deviation Actions

- Immediately notify the MDA through appropriate management channels [102].
- Within 30 days, submit a Program Deviation Report describing the cause and corrective actions [102].
- Within 90 days, either return the program within APB parameters or submit information to the Overarching IPT for a MDA decision on APB revision [102].

4.2.10 Requirements Relationship to the DAS

Legacy JCIDS teaching describes requirements generation as identifying warfighter capability gaps, developing joint concepts, and producing validated requirements that feed the acquisition framework [102]. Current Joint Staff guidance rescinds and replaces JCIDS with the Joint Force Requirements Process: Service and Component requirements are approved first, then Joint Capability Integration, Joint Capabilities Board (JCB), or JROC endorsement or approval applies when the Joint Staffing Designator (JSD) and joint impact warrant it [103].

4.2.11 Materiel Development Decision (MDD)

Purpose. Confirms a materiel solution is required and triggers the AoA [102].

MDA inputs. Validated Initial Capabilities Document (ICD), evidence of technical foundation, AoA study guidance and plan, and a Capstone Threat Assessment [102].

MDA decisions. Designate the lead DoW Component and the phase entry point; entry may occur at any point that meets entrance criteria and statutory requirements [102].

4.2.12 Materiel Solution Analysis (MSA)

Purpose. Assess candidate materiel solutions and prepare for Milestone A [102].

Guided by. Validated ICD, AoA study guidance, and AoA plan [102].

Major activities. Establish PM/PMO, conduct AoA, develop initial Acquisition Strategy, TEMP, Systems Engineering Plan (SEP), Life-Cycle Sustainment Plan (LCSP), and cybersecurity strategy; the user writes a draft Capabilities Development Document (CDD) [102].

Minimum funding. Funding supports all phase activities and the Milestone A decision [102].

Exit. MDA approves the materiel solution and Acquisition Strategy [102].

4.2.13 Analysis of Alternatives (AoA)

Focus. Materiel solution, mission effectiveness, key cost/capability trades, schedule, Concept of Operations (CONOPS), risk, producibility, and total life-cycle cost [102].

Innovation/competition. Emphasizes competition and encourages industry to surface candidate solutions to sponsors and stakeholders [102].

Hierarchy of alternatives. Commercial solutions first, then existing U.S./allied systems, international cooperative development, joint/government agency development, and finally DoW-unique development [102].

Approval. CAPE approves the AoA; results inform the Acquisition Strategy and industry competition [102].

4.2.14 Acquisition Strategy

Requirement. The PM prepares and the MDA approves an Acquisition Strategy at Milestone A [102].

What it does. Tailors the acquisition framework, sets the master schedule, and provides the basis for plans such as TEMP, contract awards, and SEP [102].

Affordability. The strategy must maximize affordability from initiation through post-production support [102].

4.2.15 International Acquisition and Exportability

Drivers. U.S. strategy and DoWI 5000.02 require international cooperation, sales, and exportability to be considered in acquisition strategies [102].

IA&E objectives. Operational interoperability, economic cost-sharing, technical access, political alliance strength, and industrial base health; additional considerations include security risk and foreign dependency [102].

Common forms. International cooperative programs, FMS, building partner capacity initiatives, and Direct Commercial Sales (DCS) arrangements (do not confuse with program ownership and contract privity roles) [102].

4.2.16 Evolutionary vs. Single-Step Acquisition

Evolutionary. Time-phased increments that define the current increment and subsequent evolutions to meet full capability needs [102].

Single-step. Full capability delivered in one step before Milestone C; major modifications of sufficient cost/complexity are managed as separate efforts [102].

Preference. Evolutionary acquisition is preferred when rapidly fielding mature technology [102].

4.2.17 Key Performance Parameters

Definition. Key Performance Parameters (KPPs) are system attributes critical to delivering effective capability; at Milestone B, KPPs from the validated CDD are listed verbatim in the APB [84, 102].

Validation. Under legacy JCIDS teaching, JROC, JCB, or DoW Component validation depended on program category and joint interest. Under current JFRP governance, do not assume universal JROC validation; Component approval normally comes first, with Joint Staff endorsement or approval driven by JSD and joint impact [102, 103].

Mandatory. Force Protection, System Survivability, Sustainment, and Energy; Net-Ready remains required but can be treated as a KPP or performance attribute [102].

4.2.18 Milestone A

PM inputs. Proposed materiel solution, Acquisition Strategy, SEP/TEMP, CDD/CONOPS, LCSP/ICE, risk assessment, and should-cost targets [102].

Component inputs. Affordability analysis, cost estimate, and evidence the program is fully funded in the FYDP [102].

MDA decisions. Approves materiel solution, authorizes TMRR RFP release, and sets TMRR exit criteria in the ADM [102].

Certification. 10 U.S.C. 2366a requires ICD alignment, AoA-validated solution, cost estimate with CAPE concurrence, JROC priority, and DIA-validated Validated Online Life-cycle Threat (VOLT) report [102].

Milestone A Entry Criteria (Pre-A Decision)

You must demonstrate problem clarity and viable concept space before the Milestone A decision [84, 102]:

Validated capability need. Approved ICD confirming the capability gap and required operational capability [102].

Analysis of Alternatives (AoA). Completed or sufficiently mature AoA with preferred solution(s) identified [102].

Affordability and alignment. Initial cost targets and affordability caps established [102].

Threat and operational context. Validated threat baseline and CONOPS [102].

Technology maturity (early). Critical technologies identified (typically Technology Readiness Level (TRL) 3–5) [102].

Acquisition strategy (draft). Includes prototyping approach and competition strategy [102].

Milestone A Exit Criteria (Milestone A Decision Approval)

Approval allows entry into TMRR [84, 102]:

Approved Acquisition Strategy. MDA-approved strategy tailoring the acquisition framework [102].

Approved AoA results. CAPE-approved AoA with preferred solution [102].

Initial Systems Engineering Plan (SEP). Technical approach defined [102].

Initial Test & Evaluation Strategy (TES). Test approach outlined [102].

Funding aligned in FYDP. Program fully funded through TMRR [102].

Risk reduction plan defined. Path to reduce technical risk before EMD commitment [102].

Feasibility judgment. Program judged feasible to mature technology [102].

Key judgment: Is there a credible path to reduce technical risk before committing to EMD? [84, 102]

4.2.19 Technology Maturation and Risk Reduction (TMRR)

Purpose. Reduce technology, engineering, integration, and life-cycle cost risks; demonstrate critical technologies; complete preliminary design [102].

Guided by. ICD, Acquisition Strategy, draft CDD, and SEP [102].

Basis for entry. MDA-approved materiel solution and Acquisition Strategy [102].

Activities. Competitive prototyping, Preliminary Design Review (PDR), CDD-V, sustainment planning, DRFPRD, Technology Readiness Assessment (TRA), and Reliability, Availability, and Maintainability Cost Rationale (RAM-C) report [102].

Exit criteria. Affordable increment identified, technology demonstrated in relevant environment (TRL 6), PDR completed before Milestone B (unless waived), validated requirements, full FYDP funding, and affordability compliance for production/sustainment [102].

TRLs are a technology-maturity overlay, not a substitute for the MDA's phase-entry, exit, affordability, requirements, and test judgments. For board purposes, use TMRR as the bridge from promising concepts toward TRL 6 evidence that critical technologies can support Milestone B and EMD entry [84, 102, 104].

Table XXXIX: Technology Readiness Levels mapped to MCA phases

TRL	Technology maturity meaning	Best-fit MCA phase	Board-prep interpretation
1	Basic principles observed and reported	Pre-MCA or early MSA	Science exists, but there is not yet an acquisition-ready materiel concept.
2	Technology concept or application formulated	Pre-MCA or MSA	A possible military application is described, but feasibility remains mostly conceptual.
3	Analytical and experimental proof of concept	MSA	Early proof supports alternatives analysis and helps decide whether a materiel solution is worth maturing.

Continued on next page

Table XXXIX: Technology Readiness Levels mapped to MCA phases **(Continued)**

TRL	Technology maturity meaning	Best-fit MCA phase	Board-prep interpretation
4	Component or breadboard validated in a laboratory environment	Late MSA or early TMRR	The technology works in controlled conditions, but relevant-environment risk is still high.
5	Component or breadboard validated in a relevant environment	TMRR	The program is retiring technical risk under more realistic conditions before committing to development.
6	System or subsystem model or prototype demonstrated in a relevant environment	Late TMRR, before or around Milestone B	Critical technologies should be mature enough to support EMD entry and detailed design.
7	System prototype demonstrated in an operational environment	EMD	An integrated prototype has been shown in an operationally realistic setting, often supporting operational assessment evidence.
8	Actual system completed and qualified through test and demonstration	Late EMD or P&D	The system is qualified and ready to support production, deployment, or fielding decisions.
9	Actual system proven through successful mission operations	O&S	The fielded system has demonstrated performance in real operational use.

Source: *DoWI 5000.85, 3.3.2. Acquisition Milestones and Phases, Technology Readiness Assessment Guide*, 2020, 2025, 2020 [84, 102, 104]

4.2.20 Capability Development Document Validation

Authority. The applicable requirements validation authority validates the CDD; JROC or JCB action applies when joint impact and the JSD require Joint Staff review, while Component-approved requirements can support acquisition execution when no joint endorsement or approval is required [102, 103].

Focus. Confirms cost-performance trades, affordability, testability, and alignment with user priorities; KPPs/Key System Attributes (KSAs) guide PDR and DRFPRD [102].

4.2.21 Configuration Steering Boards

Authority. Chaired by the CAE with USW(A&S) and Joint Staff participation; formed after CDD-V [102].

Responsibilities. Review requirement changes for ACAT I/IA programs, propose adjustments to meet affordability, and evaluate cost/schedule impacts; changes require identified funds and mitigated schedule impacts [102].

Cadence. Meets at least annually and recommends de-scope options to the MDA [102].

4.2.22 Development RFP Release Decision

The DRFPRD authorizes release of the RFP for EMD (often with Low-Rate Initial Production (LRIP) options), confirms affordability/executability, and is the last point for major changes without program disruption; the MDA determines preliminary LRIP quantities [102].

4.2.23 Milestone B

Purpose. Development decision authorizing entry into EMD and formal program initiation [102].

MDA approvals. EMD entry, Acquisition Strategy, APB, LRIP quantities, contract type, exit criteria, and Milestone B certification/determination with ADM issuance [102].

Shipbuilding. Milestone B triggers SCN funding for shipbuilding programs [102].

Milestone B Entry Criteria (Pre-B Decision)

This is the most demanding milestone [84, 102]:

Requirements stabilized. Approved CDD with validated capability requirements sufficient to support EMD entry [84, 102].

Technology maturity. Critical technologies at TRL 6 (demonstrated in relevant environment) [102].

Preliminary Design Review (PDR). Completed with acceptable design maturity [102].

Systems Engineering maturity. SEP, architecture, and technical baseline established [102].

Cost, schedule, performance baseline. Independent cost estimate (ICE) completed [102].

Test planning. Approved Test & Evaluation Master Plan (TEMP) [102].

Industrial base. Manufacturing feasibility assessed [102].

Cybersecurity and sustainment planning. Initial sustainment strategy and cyber posture defined [102].

Milestone B Exit Criteria (Milestone B Approval)

Approval allows entry into EMD and is typically program initiation [84, 102]:

Approved Acquisition Program Baseline (APB). Cost, schedule, and performance thresholds/objectives established [102].

Full funding certification. Program fully funded in FYDP [102].

Affordable and executable program. Program judged affordable and executable [102].

Design stability. Design stable enough to proceed [102].

Risk manageable. Risk reduced to manageable levels [102].

Ready to build. Program ready to build and integrate system prototypes [102].

Key judgment: Can the program execute development within cost/schedule/performance constraints? [84, 102]

4.2.24 Engineering and Manufacturing Development

Purpose. Develop, build, and test the product to verify all requirements and support production decisions [102].

Guided by. Acquisition Strategy, CDD, TEMP, and SEP [102].

Activities. Complete design, retire risks, build/test prototypes, establish product baselines, conduct Critical Design Review (CDR), and perform operational assessments [102].

Exit. Design stability, requirements met via DT and early Operational Test (OT), and all EMD exit criteria met before Milestone C [102].

4.2.25 Milestone C

At Milestone C the MDA reviews cost estimates, validated life-cycle threat reporting, environmental issues, design stability, Operational Assessment (OA) results, software maturity, manufacturing risk, validated CDD, supportability, interoperability, affordability, and phased production ramp-up before committing to production [102].

Milestone C Entry Criteria (Pre-C Decision)

Design maturity. Critical Design Review (CDR) completed [102].

System performance demonstrated. Developmental testing shows requirements are met [102].

Operational testing readiness. Initial Operational Test & Evaluation (IOT&E) readiness [102].

Production readiness. Manufacturing processes stable (often Manufacturing Readiness Level (MRL) 8–9) [102].

Sustainment readiness. Logistics, training, and support systems in place [102].

Affordability confirmed. Updated cost estimates and funding in place [102].

Milestone C Exit Criteria (Milestone C Approval)

Approval allows entry into P&D, normally LRIP, initial production, or limited deployment; FRP and full deployment follow later when the tailored decision criteria and statutory IOT&E/Live Fire Test and Evaluation (LFT&E) gates are satisfied [84, 102, 105, 106]:

Production decision authorized. MDA authorizes entry into production and deployment, including approved LRIP quantities or a tailored limited deployment decision [84, 102].

System meets threshold performance. Threshold performance requirements met [102].

Test results support acceptable production risk. Developmental test, early operational test or assessment, and certification evidence support production-representative articles and planned IOT&E/LFT&E; final operational effectiveness, suitability, and survivability findings normally support the later beyond-LRIP/FRP decision [84, 102, 105, 106].

Industrial base capable. Industrial base capable of production [102].

Sustainment system executable. Sustainment system is executable [102].

Key judgment: Is the system ready to be produced and fielded with acceptable risk? [84, 102]

4.2.26 Cross-Milestone Comparison

Table XL: Cross-Milestone Comparison

Area	Milestone A	Milestone B	Milestone C
Phase Entry	TMRR	EMD	Production
Requirements	ICD	CDD	Stable/validated
Tech Maturity	TRL 3–5	TRL 6	Fully integrated
Design Review	–	PDR	CDR
Cost Baseline	Rough order	APB established	Refined
Key Output	Feasible approach	Program start	Production decision

Source: *DoWI 5000.85, 3.3.2. Acquisition Milestones and Phases*, 2020, 2025 [84, 102]

4.2.27 Navy-Specific Nuances

Gate rigor varies. Rigor varies by ACAT level and Milestone Decision Authority (MDA) (PEO vs ASN(RDA) vs OSD) [102].

ASN(RDA) emphasis. Assistant Secretary of the Navy for Research, Development and Acquisition often enforces stronger emphasis on affordability caps and early prototyping evidence before Milestone B [102].

Tailored criteria. Programs may use tailored entrance/exit criteria under the AAF; these are not strictly one-size-fits-all [102].

Integrated Baseline Review. IBR timing is tied to contract award, significant options, or major modifications; it is often seen during EMD startup, but it is not a milestone-tied review [83, 102].

4.2.28 Bottom Line

Milestone A. Should we pursue this? [84, 102]

Milestone B. Can we build it? (formal commitment) [84, 102]

Milestone C. Are we ready to produce and field it? [84, 102]

4.2.29 Production and Deployment

Purpose. Produce and deliver requirements-compliant systems [102].

Guided by. Acquisition Strategy, TEMP, CDD, SEP, and LCSP [102].

Activities. LRIP, OT in realistic threat conditions, Beyond LRIP report, Full-Rate Production Decision Review (FRPDR), fielding, Initial Operational Capability (IOC), and Full Operational Capability (FOC) [102].

Exit. IOC achieved and deployment in full swing [102].

4.2.30 Operations and Support

Purpose. Execute the support strategy, sustain readiness, and plan disposal [102].

Guided by. Product Support Strategy and Programmatic Environmental, Safety, and Occupational Health Evaluation (PESHE) [102].

Activities. Life-cycle sustainment resourcing, technical data/tooling planning, and disposal in accordance with safety, security, and environmental requirements; phase completes when assets are disposed [102].

4.2.31 Ships Are Different

- The PM is designated as early as possible, often before the MDD [102].
- Initial Acquisition Strategy and TEMP set the long-term context and gain early leadership alignment [102].
- Between Milestone A and Milestone B/C, RFP and specification development is for the ship, not an engineering development model [102].

4.2.32 Navy 2-Pass / 7-Gate Review

Gates 1–4 (requirements). Led by the CNO or CMC; starts before MDD and leads to approved ICD, AoA guidance, a selected AoA alternative, approved CDD, CONOPS, and a System Design Specification development plan [102].

Gates 5–7 (acquisition/sustainment). Led by ASN(RD&A); starts after Gate 4 and continues through Milestone B; approves System Design Specification, RFP release, and production readiness (including EVMS and IBR checks), with Gate 6 pre- / post-Milestone C and FRPDR follow-on [102].

4.2.33 Scenario Drills (Entry-Point Decisions)

Use the following vignettes to select the most appropriate entry point into the acquisition framework [102].

1. **Situation 1.** Joint mid-course interceptor capability; validated ICD; multiple materiel approaches; AoA guidance and plan approved; MDA wants a lead DoW Component and an evolutionary strategy. *Question:* best entry point?

2. **Situation 2.** Validated CDD for a new side-arm with proven commercial options; field evaluations meet thresholds; full procurement funding. *Question:* best entry point?

3. **Situation 3.** Validated ICD; AoA and Acquisition Strategy complete; promising but immature technology; significant performance risk; draft CDD. *Question:* best entry point?

4. **Situation 4.** Mature lab eye-protection technology; validated ICD/CDD; full funding offsets identified; technology not yet tested outside the lab and not integrated with helmet systems. *Question:* best entry point?

4.2.34 Quick Review

- Which entry point requires a validated ICD, AoA study guidance, and AoA plan [102]?
- Which milestone authorizes entry into P&D [102]?
- Which milestone requires all sources of program risk to be adequately mitigated [102]?
- In MSA, which two artifacts are approved by the MDA [102]?

4.2.35 Coursebook cue: lifecycle-entry scenario rule

Coursebook 3.3.2 uses milestone scenarios to test whether the student can place a program into the right acquisition phase. Use the following decision rule before naming the milestone: ask what evidence exists, what decision is being requested, and what work the program wants authority to begin [102].

Table XLI: Lifecycle-entry scenario answer key.

Scenario evidence	Likely answer
Validated capability need, alternatives analysis, and request to mature concepts	Enter Materiel Solution Analysis or prepare for Milestone A.
Technology has matured enough to begin preliminary design and risk reduction	Milestone A authorizes Technology Maturation and Risk Reduction.
Stable requirements, preliminary design evidence, and request to start detailed design	Milestone B authorizes Engineering and Manufacturing Development.
Design, test, manufacturing, and support evidence show readiness to build for use	Milestone C authorizes production, deployment, or limited fielding as appropriate.
Urgent operational need needs faster fielding with tailored documentation	Use the urgent or rapid pathway, but still identify requirement, funding, test, sustainment, and authority.

5 Requirements

5.1 Joint Capabilities Integration & Development System

5.1.1 Summary

Decision support triad. JCIDS (requirements), PPBE (funding), and the DAS (materiel) operate continuously and must stay aligned to enable timely decisions [107].

Current JFRP note. Current Joint Staff guidance cancels the 2021 JCIDS manual and rescinds/replaces the JCIDS process with the Joint Force Requirements Process (JFRP); this section preserves JCIDS terminology for legacy EDO lecture context and calls out current JFRP governance where it changes the answer [103].

Purpose. JCIDS ensures the joint warfighter receives required capabilities, institutionalizes joint needs, and initiates the acquisition management process [107].

CBA foundation. Capability-Based Assessments (CBAs) identify capability gaps and recommend materiel or non-materiel solutions based on operational risk and feasibility [107].

Capability documents. ICD and CDD (with CDD updates) align to acquisition phases and provide validated requirements for program execution [107].

Key players. CJCS, JROC, JCB, Functional Capabilities Boards (FCBs), the Joint Staff Gatekeeper (J-8), and Joint Staff J-6 drive validation, prioritization, and interoperability decisions [107].

5.1.2 Decision Support Systems

Requirements. JCIDS captures warfighter capability needs and requirements [107].

Funding. PPBE provides the resource decisions to execute validated priorities [107].

Materiel. The DAS executes acquisition strategies that deliver validated capabilities [107].

Synchronization. These systems must interface continuously to keep requirements, funding, and execution aligned [107].

5.1.3 Capabilities-Based Assessment (CBA)

Objective. A CBA analyzes mission needs and capability gaps, assesses risk, and recommends materiel and non-materiel solutions [107].

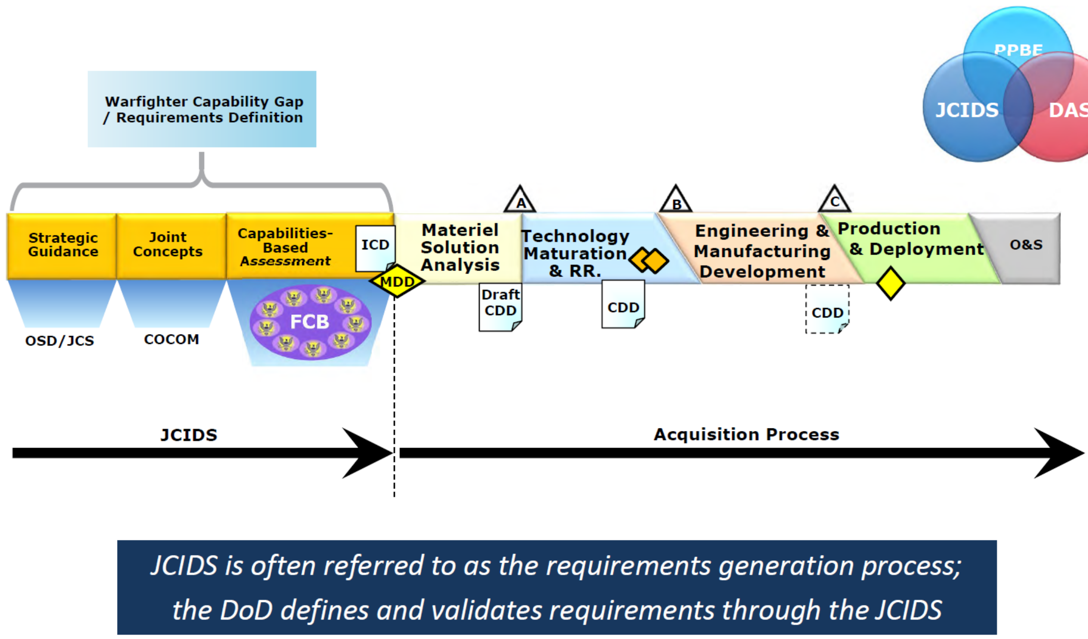

Figure 5.1. *Relationship between JCIDS and the DAS. Source:* 3.3.3. Joint Capabilities Integration Development System, *2025 [107].*

Initiation. A CBA can start from any DoW organization, a joint concept, a CONOPS, or an operational need [107].

Outputs. The CBA identifies mission and capability needs, gaps and risks, validates non-materiel feasibility, and recommends a path forward [107].

5.1.4 CBA Phases

Study definition. What does the mission require [107]?

Needs assessment. How well do current capabilities meet those needs; where are the gaps [107]?

Solutions recommendation. What materiel or non-materiel options should close the gaps [107]?

5.1.5 CBA Recommended Solutions

Materiel. New or modified systems acquired through the DAS; guided by validated JCIDS documents [107].

Non-materiel. Changes in doctrine, organization, training, leadership, personnel, facilities, or policy; often faster and less costly than new systems [107].

Document path. Non-materiel solutions lead to a DOTMLPF-P Change Recommendation (DCR); materiel solutions lead to an ICD [107].

5.1.6 DOTMLPF-P Elements

Doctrine. Fundamental guiding principles [107].

Organization. How the DoW organizes to fight [107].

Training. How the force prepares to fight tactically [107].

Materiel. Existing equipment and systems [107].

Leadership and education. Leader development requirements [107].

Personnel. People required to accomplish the mission [107].

Facilities. Property, installations, and industrial facilities supporting forces [107].

Policy. Policies that impact military operations [107].

5.1.7 Capability Documents

Purpose. Capability documents communicate operator needs to acquisition, test, and resourcing communities [107].

Primary documents. ICD (approved prior to MDD) and CDD (approved prior to Milestone B); CDD updates support production [107].

AoA support. AoA is performed during the MSA phase as part of the requirements-to-acquisition handoff [107].

5.1.8 Initial Capabilities Document (ICD)

Purpose. Summarizes the CBA and justifies requirements to resolve capability gaps within a timeframe, including possible solutions for MSA and TMRR [107].

Inputs. Summarizes Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, Facilities, and Policy (DOTMLPF-P) analysis and capability gaps [107].

Supports. MDD, AoA, Acquisition Strategy, and Milestone A [107].

5.1.9 Capability Development Document (CDD)

Purpose. Defines authoritative, measurable, and testable capabilities for EMD and the program technical baseline [107].

Affordability. Captures information needed to deliver an affordable and supportable capability across increments [107].

Performance. Specifies performance attributes, including KPPs, needed to design systems that close the gaps in the ICD [107].

Updates. Can be updated near end of development to refine production attributes, quantities, and subsequent increments [107].

5.1.10 CDD Update and Production Requirements

Policy change. The 2018 JCIDS manual removed the requirement for a separate Capability Production Document (CPD); a CDD update supports production needs [107].

Legacy CPD intent. Defined capabilities for Production and Deployment, refined CDD thresholds/objectives, and was approved before Milestone C after CDR [107].

Legacy production role. A legacy CPD defined the production baseline requirements used for test and evaluation, establishing the threshold performance attributes that had to be demonstrated before Milestone C [84, 107].

CDD update function. Modern CDD updates incorporate production attributes, quantities, and supportability considerations that traditionally appeared in separate CPD documents [107].

5.1.11 JCIDS Within the Acquisition Framework

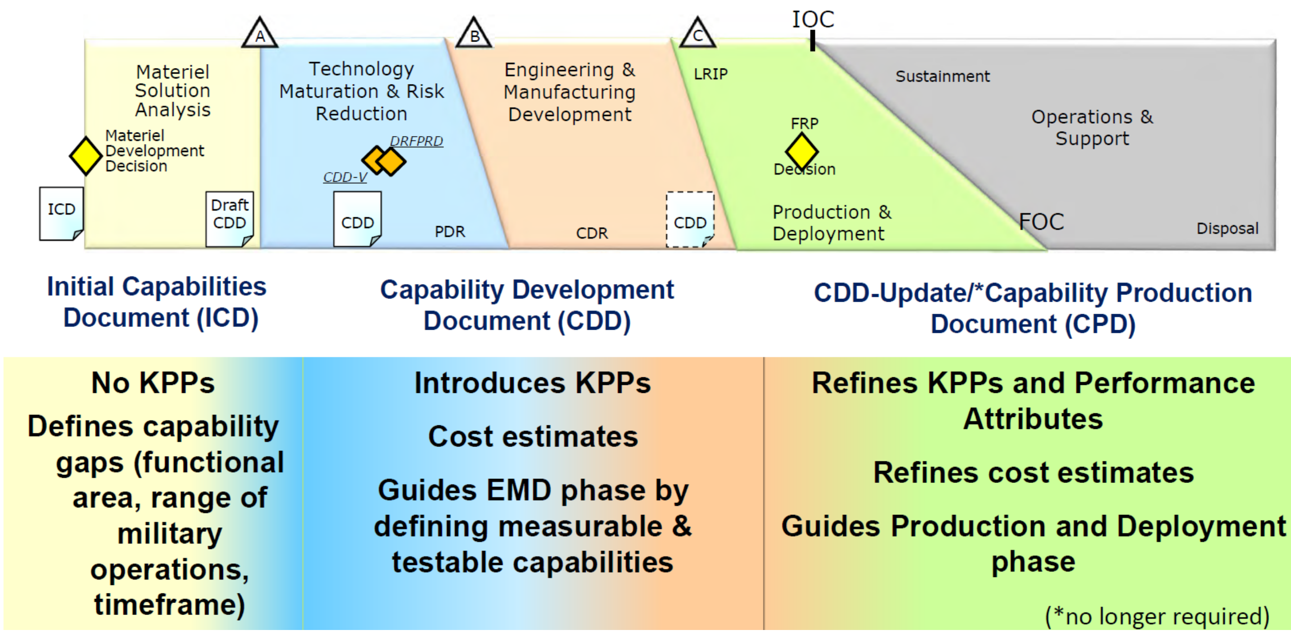

Figure 5.2. Placement of JCIDS documents within MCA phases. Source: 3.3.3. Joint Capabilities Integration Development System, *2025 [107].*

ICD. Defines capability gaps and supports MDD, MSA, and TMRR (no KPPs) [107].

CDD. Introduces KPPs, provides cost estimates, and guides EMD with measurable, testable capability definitions [107].

CDD update. Refines KPPs and performance attributes, updates cost estimates, and guides Production and Deployment [107].

Team effort. Requirements development must involve the warfighter and all stakeholders to produce executable technical requirements [107].

5.1.12 Key Players

Gatekeeper (J-8/DDR). Manages document flow, assigns JSD, and designates lead/supporting FCBs for staffing and validation [107].

Chairman of the Joint Chiefs of Staff. The CJCS is the principal military adviser to the President, SECWAR, and the National Security Council (NSC); sets the agenda and presides over the Joint Chiefs of Staff [107].

Joint Requirements Oversight Council. The JROC is the process owner; validates requirements, prioritizes joint capabilities, and validates KPPs unless delegated [107].

Joint Capabilities Board. The JCB supports the JROC, reviews documents, adjudicates issues, and validates KPPs for JCB-interest documents [107].

Functional Capabilities Boards. The FCBs are permanent bodies that analyze, prioritize, and integrate stakeholder views across functional areas; include CCMD participation [107].

5.1.13 Joint Staff J-6 (C4/Cyber)

Net-ready validation. Validates the Net-Ready performance attribute (KPP or KSA) for JROC/JCB-interest documents [107].

Interoperability certification. Performs IT and national security system interoperability and supportability certifications; Joint Interoperability Test Command (JITC) issues certification before full-rate production [107].

5.1.14 Key Performance Parameters and Performance Attributes

Definition. KPPs and related performance attributes are system attributes critical to delivering effective capability and are documented in the CDD and APB [84, 102].

Current governance. Under the current JFRP, the JROC remains the senior joint requirements body, but Joint Staff review is focused on Joint Force Design, Joint Capability Integration, Combatant Command requirements, and by-exception reviews; Service and Component requirements are not automatically held for legacy JCIDS validation before acquisition execution [103].

Traceability. Programs still need clean traceability from validated capability needs to KPPs, KSAs, other performance attributes, verification methods, and decision evidence [84, 107].

5.1.15 KPP vs KSA vs APA Comparison

Table XLII: KPP vs KSA vs APA Comparison

Parameter Type	Definition	Governance Level	Impact if Unmet	Navy Example
Key Performance Parameter (KPP)	Critical capability attribute tied to the validated capability need and acquisition baseline	JROC/JFRP review when tied to joint-force requirements or by-exception joint impact; Component validation for Service-specific programs	Program stoppage; statutory reporting required; Milestone decisions blocked	Illustrative Aegis Weapon System example: defended-area detection coverage sufficient to support raid engagement timelines
Key System Attribute (KSA)	Important performance attribute that supports KPP achievement and operational suitability	Joint Staff or functional review where the attribute has joint/interoperability impact; Component validation for program-specific attributes	Interoperability certification delays; reduced operational effectiveness; potential re-baseline	Illustrative Aegis Weapon System example: identification data exchange responsive enough to support engagement coordination
Additional Performance Attribute (APA)	Performance attribute required for full system capability but not critical enough for KPP	Component validation; documented in CDD/APB	Reduced test priority; capability gaps if not met; schedule impacts	Illustrative Aegis Weapon System example: electronic-protection behavior that preserves mission effectiveness in representative threat environments

Source: *DoWI 5000.85*, *CJCSM 5123.01*, *3.3.3. Joint Capabilities Integration Development System*, 2020, 2026, 2025 [84, 103, 107]

5.1.16 IT Acquisition Requirements

Net-Ready attribute. Documented as a KPP or KSA to ensure IT supports operations, is manageable on networks, and exchanges information effectively [107].

Architecture compliance. Systems must align to Department of War Architecture Framework (DoWAF); architecture views are mandatory parts of the CDD and must be consistent with Department of War Information Technology Standards Registry (DISR) standards [107].

5.1.17 Importance of Interoperability

Definition. Ability of systems, units, or forces to provide and accept services and operate effectively with U.S. and coalition partners [107].

Operational risk. Poor interoperability increases risk to people and equipment and reduces operational effectiveness [107].

Cost discipline. Stovepiped systems are costly and hinder joint operations; interoperability is a core JCIDS validation priority [107].

5.1.18 Validation Authority

Joint Requirements Oversight Council. Under the JFRP, the JROC remains the senior joint requirements body and reviews prioritized Joint Force and Combatant Command requirement issues, but it is not the default validator for every Service-generated acquisition requirement document [103].

Joint Capabilities Board. The JCB supports JROC deliberations and adjudicates staffing issues for Joint Capability Integration and by-exception reviews directed by the Joint Staff [103].

Component sponsor. Services and Components generate and validate their own requirements; Joint Capability Integration occurs after Component validation and in parallel with acquisition execution when joint impact warrants it [103].

5.1.19 Document Review and Staffing

Submission. Under legacy JCIDS teaching flow, sponsors submit documents to Joint Staff J-8 for staffing; under current JFRP, this routing is selective and focused on joint-force integration priorities rather than universal staffing of Service documents [103, 107].

JSD assignment. J-8 assigns staffing and review mechanisms according to joint impact and JFRP governance decisions [103].

FCB review. Lead FCB analysis still informs joint integration decisions when a requirement has cross-functional or joint implications [103].

Validation. Board answer: state Component validation first, then explain Joint Staff/JROC integration review as required by scope and joint impact [103].

5.1.20 Quick Review

- What system institutionalizes a joint capability need and sponsors further evaluation of a materiel solution [107]? **Answer:** JCIDS.
- Which acquisition phases use the ICD and CDD [107]? **Answer:** MSA/TMRR, EMD, and Production and Deployment.
- Which document defines capabilities and is transformed into a program start [107]? **Answer:** CDD, with Milestone B approval by the MDA.
- JROC and the Joint Staff ensure new capabilities are conceived and developed in what context [107]? **Answer:** Joint warfighting context.
- Warfighter gap identification is driven by Combatant Commander participation in what body [107]? **Answer:** FCBs.

5.1.21 Coursebook cue: legacy JCIDS language

Coursebook 3.3.3 uses JCIDS-era vocabulary to teach capability gaps, requirements documents, and validation. For a board answer, state the coursebook concept first, then add the current-policy caveat when relevant: legacy JCIDS terms remain useful for explaining need, gap, validation, and traceability, but current Joint Staff requirements reform may change document names, routing, and review mechanics [107].

5.2 Research & Development

5.2.1 Summary

Technology posture. The U.S. relies heavily on industry research and commercial technologies while Navy labs retain a strong S&T role in acquisition decisions [108].

S&T role. S&T reduces integration risk by maturing technologies, conducting assessments, and advising requirements and acquisition communities [108].

Continuum and readiness. The S&T continuum runs from basic research to advanced technology; TRL maturity is central to insertion timing and risk [108].

Transition mechanisms. Advanced Technology Demonstration (ATD) and Joint Concept Technology Demonstration (JCTD) accelerate technology transition with distinct objectives, oversight, and user involvement [108].

Support partners. Federally Funded Research and Development Centers (FFRDCs) and University Affiliated Research Centers (UARCs) provide objective, long-term research and systems acquisition support [108].

5.2.2 R&D Perspective and Posture

Near term (today's Navy). Focused on current readiness and fleet needs led by the CNO, CMC, CCMDs, Fleet, and the United States Marine Corps (USMC) [108].

Mid term (next Navy). Programs and resources shaped by PEOs, SYSCOMs, NRO, and OPNAV [108].

Long term (Navy the Nation needs). DON S&T, the Naval War College, and research centers such as Naval Information Warfare Center (NIWC), NSWC, the Naval Air Warfare Center, NUWC, Naval Research Laboratory (NRL), ONR, , and DARPA [108].

5.2.3 Using Commercial Technology

Industry leverage. DoW S&T budgets drive greater reliance on commercial investments while Navy labs retain an advisory role [108].

Common pathways. OSA, Dual Use Applications Program (DUAP), Independent Research and Development (IR&D), Manufacturing Technology (ManTech), Technology Transfer (T2), and S&T affordability initiatives bring commercial technologies into defense systems [108].

5.2.4 R&D in the Acquisition Framework

Technology opportunities. State-of-the-art assessments, experiments, and demonstrations inform program decisions [108].

Insertion timing. New technology can enter in any phase, but is most likely inserted prior to Milestone B [108].

5.2.5 Role of Science & Technology

Component S&T executives. Evaluate battlefield deficiencies, establish S&T projects, mature technologies, conduct independent assessments, and advise requirements and acquisition communities [108].

Transition mechanisms. ATD, JCTD, experiments, and projects with explicit transition requirements move technology toward acquisition [108].

5.2.6 FFRDCs and UARCs

FFRDCs. Government-sponsored centers that perform and manage basic/applied research and development; provide innovation, systems acquisition support, and policy guidance (e.g., JPL, LLNL, LANL, CNA, NDRI, RAND) [108].

UARCs. A specialized FFRDC model leveraging university capabilities; valued for adaptability, objectivity, conflict-free advice, rapid response, and continuity (e.g., USC, Georgia Tech, MIT, JHU, Penn State) [108].

5.2.7 Technology Readiness Levels

- TRL 1: Basic principles observed and reported [108].
- TRL 2: Technology concept or application formulated [108].
- TRL 3: Analytical and experimental proof of concept [108].
- TRL 4: Component or breadboard validation in laboratory environment [108].
- TRL 5: Component or breadboard validation in relevant environment [108].
- TRL 6: System/subsystem model or prototype demonstrated in relevant environment (desired for Milestone B) [108].
- TRL 7: System prototype demonstrated in operational environment (desired for Milestone C) [108].
- TRL 8: Actual system completed and qualified through test and demonstration [108].
- TRL 9: Actual system proven through successful mission operations [108].

5.2.8 RDT&E Budget Activities (0601–0607)

6.1 Basic research. Foundational scientific study and experimentation [108].

6.2 Applied research. Translating basic research into potential solutions [108].

6.3 Advanced technology development. Demonstrations and prototypes to mature technologies [108].

6.4 Advanced component development and prototypes. Integrating and evaluating prototype components [108].

6.5 System development and demonstration. System-level development and risk reduction [108].

6.6 RDT&E management support. Test ranges, support activities, and infrastructure [108].

6.7 Operational system development. Upgrades and development of fielded systems [108].

5.2.9 IRL and SRL

Integration maturity. Integration Readiness Level (IRL) and System Readiness Level (SRL) complement TRL by measuring how well technologies integrate across interfaces [108].

Nine-level scale. Like TRL, IRL/SRL use a 1–9 scale to communicate integration readiness and mission support maturity [108].

5.2.10 Advanced Technology Demonstrations (ATDs)

Purpose. Hardware/software prototypes that refine basic or applied research and demonstrate feasibility at relatively low cost [108].

Funding and timing. Often ONR-funded (0603) and usually pre-Milestone B [108].

User involvement. Limited or no user involvement; no leave-behind capability objective [108].

5.2.11 Joint Concept Technology Demonstrations (JCTDs)

Purpose. Evaluate military utility, develop CONOPS and doctrine, and establish realistic requirements [108].

MUA. Produces a Military Utility Assessment (MUA); can substitute for an ICD for Milestone A [108].

Transition focus. Aims for early transition to acquisition or sustainment; may leave a go-to-war interim capability when appropriate [108].

5.2.12 Small Business Innovation Research

Purpose. SBIR stimulates innovation, meets federal R&D needs, expands participation, and commercializes federally funded innovations [108].

Phase I. Proof of concept (6–12 months, roughly $50K–$275K) [108].

Phase II. Technology development (about 24 months, roughly $750K–$1.8M) [108].

Phase III. Commercialization with no SBIR funding [108].

5.2.13 Quick Review

- What is the U.S. technology posture in defense acquisition [108]? **Answer:** Heavy reliance on industry research and commercial technology.
- What is the role of S&T in defense acquisition [108]? **Answer:** Mature technology and reduce integration risk through research and assessment.
- When are new technologies most likely introduced in the life cycle [108]? **Answer:** Prior to Milestone B.
- What are key characteristics of ATD [108]? **Answer:** Feasibility/maturity demonstrations and risk reduction.
- What are key characteristics of JCTD [108]? **Answer:** Military utility evaluation, CONOPS/doctrine development, and potential go-to-war capability.

5.3 Risk Management & Problem Solving

5.3.1 Summary

Definition. Acquisition risk is the inability to achieve cost, schedule, and performance objectives; risk is defined by probability, consequence, and root cause [109].

Sources. Risks arise from external drivers (threat, funding, contractor, politics) and internal drivers (technology, design, manufacturing, support, management) [109].

APB linkage. The APB sets the baseline used to plan, control, and manage risk across cost, schedule, and performance goals [109].

Process. Risk management is a continuous cycle: plan, identify, analyze, handle, and monitor [109].

Problem solving. Trade studies, Multi-Attribute Utility Theory (MAUT), fishbone diagrams, the 5 Whys, and multi-voting support objective decisions and root-cause discovery [109].

5.3.2 What Is Acquisition Risk?

Risk management. Identifies, evaluates, and prioritizes risks, then applies resources to minimize, monitor, and control probability or impact [109].

Program management. Risk management is integral to program management and systems engineering [109].

Risk components. Probability of occurrence, consequence (impact), and root cause define each risk [109].

5.3.3 General Sources of Acquisition Risk

External. Threat and requirements, funding, contractor capability, and politics [109].

Internal. Technology, design/engineering, manufacturing, support/logistics, cost/schedule, and management [109].

5.3.4 APB and Risk Management

Baseline. The APB documents cost, schedule, and performance thresholds/objectives agreed by the PM and MDA [109].

Planning anchor. The APB provides the baseline used to plan, control, and manage risk to attain program goals [109].

Ground rules. Analyze risks with realistic timeframes, consider event timing, and identify risks at the lowest feasible WBS level [109].

5.3.5 Risk Management Model

Planning. Define the program approach, roles, resources, and battle rhythm; document in SEP and Acquisition Strategy; often combined in a Risk, Issue, and Opportunity (RIO) plan or Risk and Opportunity Management Board (ROMB) [109].

Identification. Determine what can go wrong and document root causes using program data, test results, and trends [109].

Analysis. Size risk by probability and consequence, use risk matrices, and isolate root causes (the 5 Whys) [109].

Handling. Define responses, resources, ownership, and reporting requirements [109].

Monitoring. Track burn-down plans, update risk status, and adjust actions based on performance metrics [109].

5.3.6 Risk Identification Inputs

- Current/proposed staffing, resources and dependencies, test failures, negative data trends, and performance shortfalls [109].
- Technical Performance Measure (TPM), life-cycle cost information, WBS, IMS, EVM, and technical baseline maturity [109].

5.3.7 Risk Mitigation Techniques

- Prototyping, modeling and simulation, and technology demonstrations with decision points [109].
- Intelligence analyses, data dependencies, and threat projections [109].
- Multiple design approaches, alternative designs, and phased activities to address high-risk areas early [109].
- Manufacturability, industrial base availability, supply-chain analysis, and independent risk assessments [109].
- Schedule and funding margins sized to identified risks [109].

5.3.8 Risk Handling Options

Control. Reduce probability/impact (Pre-Planned Product Improvement (P3I), reuse software, parallel design, additional testing) [109].

Avoid. Change course to eliminate the risk (redesign, reduce scope, extend schedule, use COTS) [109].

Accept. Continue the plan while identifying resources or schedule to respond to consequences [109].

Transfer. Shift responsibility to another entity (warranties, fixed-price contracts, more capable organization, or safety reviews through Weapons Safety and Explosives Review Board (WSERB)/Naval Ordnance Safety and Security Activity (NOSSA)) [109].

5.3.9 Risk Monitoring

Status and communication. Share risk status with stakeholders and review burn-down plans through periodic IPT meetings and IBRs [109].

Tracking. Update the risk matrix, alert leadership when plans need adjustment, and evaluate impacts across milestones [109].

5.3.10 Risk Management Terminology

Concern. A newly identified item before Risk Management Working Group (RMWG) review [109].

Candidate. A concern accepted for analysis or a segment risk elevated to program level and presented to the Enterprise Risk Management Board (ERMB) [109].

Active. Risk managed under a formal handling plan in planning or tracking sub-phases [109].

Watch. Low-level risk monitored without a handling plan; elevated if risk increases [109].

Retired/Rejected. Retired risks are mitigated or no longer relevant; rejected risks are removed due to low concern [109].

Issue. A realized risk or problem whose consequence has occurred or is certain [109].

5.3.11 Risk Management Software Tools

Standards. Tools often align to Project Management Institute, National Institute of Standards and Technology, or International Standards Organization guidance [109].

Capabilities. Risk registers, modeling/forecasting, Monte Carlo analysis, Office compatibility, and web access [109].

5.3.12 Opportunity Management

Definition. Opportunities are future uncertainties that can exceed performance, accelerate schedules, or reduce cost [109].

Parallel process. Uses the same organization and process as risk management; handling options are ignore, transfer, or pursue [109].

5.3.13 Issue Management

Definition. Issues are realized risks or problems that already occurred or are certain to occur [109].

Handling. Similar to risk management but without likelihood assessment; options are accept, transfer, or resolve/control [109].

5.3.14 Trade Study Fundamentals

Purpose. Evaluate viable alternatives and justify the preferred choice for requirements, design, or processes [109].

Core trade. Always includes analysis of cost versus performance [109].

5.3.15 Generic Problem Solving Model

1. Define the problem using facts, flowcharts, and cause-effect diagrams [109].
2. Generate alternatives through IPT brainstorming [109].
3. Evaluate and select an alternative [109].
4. Implement the solution and follow up [109].

5.3.16 Cause and Effect (Fishbone) Diagram

Purpose. Identifies root causes using an Ishikawa diagram [109].

Six cause categories. Materials, machines, methods, measurement, manpower, and milieu (environment) [109].

5.3.17 The 5 Whys

Method. Iterative questioning to reach the underlying root cause; the number five is a guideline [109].

Best practices. Use an accurate problem statement, unbiased answers, and focus on root causes not symptoms [109].

5.3.18 Multi-Voting

Purpose. Narrows long lists of options using group judgment and weighted voting [109].

How. Select top items individually, tally votes (weighted if needed), and repeat until priorities are clear [109].

5.3.19 Multi-Attribute Utility Theory (MAUT)

1. Identify alternatives [109].
2. Choose attributes and define a numerical grading scale [109].
3. Apply weights to attributes [109].
4. Score each alternative on each attribute [109].
5. Multiply scores by weights and sum [109].
6. Compare results [109].

5.3.20 Quick Review

- Risk is comprised of which three variables [109]? **Answer:** Root cause, probability of occurrence, and consequence.
- Risk mitigation option described as lowering event probability through multiple tests is which [109]? **Answer:** Control.
- “Use of proven technologies in commercial and non-developmental items eliminates technical risk” [109]? **Answer:** False.

5.4 Environmental Requirements and Responsibilities

5.4.1 Summary

NEPA foundation. National Environmental Policy Act (NEPA) is a procedural statute requiring the DoW to consider environmental impacts for major federal actions, involve the public, and document decisions [110].

2025 update. Executive Order 14154 directed Council on Environmental Quality (CEQ) to rescind 40 CFR 1500; CEQ rescinded its regulations effective April 11, 2025 and the DoW issued new NEPA procedures on June 30, 2025 [110].

Review levels. Categorical Exclusion (CATEX), Environmental Assessment (EA), and Environmental Impact Statement (EIS) drive outcomes such as Finding of No Significant Impact (FONSI) or Record of Decision (ROD), with public involvement when appropriate [110].

OCONUS planning. NEPA does not apply Outside the Continental United States (OCONUS); Executive Order 12114 and 32 CFR 187 require a NEPA-like analysis for significant overseas impacts [110].

ESOH integration. ESOH compliance is embedded in acquisition; PESHE documentation is required for all ACAT milestone reviews and covers compliance, safety, hazards, pollution prevention, and NEPA status [110].

5.4.2 National Environmental Policy Act (NEPA)

Procedural statute. For major federal actions affecting the human environment, NEPA requires the DoW to consider environmental impacts, involve the public, seek less damaging alternatives, and document decisions [110].

PM responsibility. Program managers should design systems to minimize negative impacts on human health and the environment [110].

Compliance schedule. DoW policy requires a NEPA compliance schedule documented in the PESHE and integrated into the program schedule [110].

5.4.3 What Is New with NEPA

Executive Order 14154. Directed CEQ to rescind 40 CFR 1500 and provide new implementation guidance; CEQ rescinded its regulations effective April 11, 2025 [110].

Agency actions. Federal agencies were instructed to repeal existing CFR NEPA rules and develop agency-specific procedures [110].

DoD implementation. The DoW released new NEPA procedures on June 30, 2025 to standardize compliance across components and maintain continuity in the review process [110].

Component rules. As of June 30, 2025, component NEPA regulations in the CFR were repealed and replaced by enterprise procedures; DON will issue supplemental guidance [110].

DoN statement. DON reaffirmed its commitment to NEPA compliance, transparency, and stakeholder communication under the new procedures [110].

5.4.4 When NEPA Review Is Not Required

- No final agency action, statutory exemption, or legal conflict with other requirements [110].
- Congress limits DoW discretion by statute or another law satisfies NEPA requirements [110].
- Actions that are not major federal actions or have minimal federal involvement or funding [110].
- Issuance of the DoW NEPA procedures themselves [110].

5.4.5 Determining the Level of NEPA Review

CATEX. Check for existing CATEX and apply DoW or other agency CATEX if applicable; consider revising or creating a CATEX if needed [110].

Environmental Assessment. If reasonably foreseeable effects are not likely significant or are unknown, prepare an EA [110].

Environmental Impact Statement. If impacts are likely significant, prepare an EIS [110].

5.4.6 DoW as a Single Federal Agency for CATEX

DoW and its components are treated as a single federal agency for CATEX purposes, regardless of which component established the exclusion [110].

5.4.7 Other Laws, Statutes, and Executive Orders Integrated with NEPA

- Water and air statutes: Rivers and Harbors Act of 1899, Wild and Scenic Rivers Act, Safe Drinking Water Act, Coastal Zone Management Act, Clean Water Act (CWA), and Clean Air Act (CAA) [110].
- Species and wildlife protections: Magnuson-Stevens Fishery Conservation and Management Act, Endangered Species Act (ESA), Bald and Golden Eagle Protection Act, and Migratory Bird Treaty Act [110].
- Land and agriculture: Executive Order 11990 (Wetlands), Executive Order 11988 (Floodplain Management), and the Farmland Protection Policy Act [110].
- Cultural and tribal resources: National Historic Preservation Act (NHPA), Archaeological Resources Protection Act, Native American Graves Protection and Repatriation Act, and American Indian Religious Freedom Act [110].
- Chemical and hazardous waste: Toxic Substances Control Act (TSCA), Resource Conservation and Recovery Act (RCRA), and Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) [110].
- Human health: Executive Order 13045 (Protection of Children from Environmental Health and Safety Risks) [110].

5.4.8 NEPA Process Outcomes

CATEX record. A categorical exclusion may produce a record of CATEX [110].

EA outcome. An EA may result in a FONSI when impacts are not significant; public involvement occurs as appropriate [110].

EIS outcome. An EIS requires a notice of intent, scoping, draft and final EIS publication, public and agency comment, and concludes with a ROD and mitigation monitoring [110].

5.4.9 Environmental Planning Outside the United States

OCONUS. NEPA does not apply outside U.S. territories and territorial seas; Executive Order 12114 governs environmental effects abroad [110].

Implementation. DoW implementing regulation is 32 CFR 187; actions require a NEPA-like analysis when significant impacts are possible [110].

5.4.10 Environment, Safety, and Occupational Health (ESOH)

Scope. ESOH integrates environmental compliance, system safety, occupational safety and health, hazardous materials management, and pollution prevention [110].

Acquisition responsibility. Program managers must consider ESOH requirements throughout the life cycle when making design decisions [110].

5.4.11 ESOH Requirements

Compliance process. Poor ESOH planning can lead to civil or criminal penalties and program delays [110].

Execution. Compliance responsibility is assigned primarily to facility and installation managers and maintainers [110].

PM focus. Program managers must integrate ESOH impacts as part of acquisition systems engineering [110].

5.4.12 ESOH Law Summary

- Air pollution control: CAA [110].
- Water pollution control: CWA [110].
- Hazardous waste: RCRA and CERCLA [110].
- Chemical production and tracking: TSCA and Emergency Planning and Community Right-to-Know Act (EPCRA) [110].
- Endangered species and marine mammals: ESA and Marine Mammal Protection Act (MMPA) [110].
- Workplace safety: Occupational Safety and Health Act (OSHA) [110].
- Environmental policy and planning: NEPA [110].
- Cultural resources: NHPA [110].

5.4.13 Clean Air Act (CAA)

Standards. Establishes national air quality standards for public health; amended in 1990 [110].

State implementation. States execute State Implementation Plans with permits, reporting, and operational controls for stationary and mobile sources [110].

Industrial controls. The Environmental Protection Agency (EPA) sets emission standards for aerospace manufacturing and shipbuilding and repair, emphasizing compliant materials and control devices [110].

5.4.14 Clean Water Act (CWA)

Scope. Major federal legislation to control water pollution [110].

Permit program. Requires a National Pollutant Discharge Elimination System permit program for point and non-point sources discharging to waters of the United States [110].

Examples. Point sources include ships and industrial facilities; non-point sources include highways and parking lots, and are the biggest challenge [110].

5.4.15 Clean Water Act Jurisdiction Update

Wetlands test. The Supreme Court limited CWA jurisdiction to wetlands with a continuous surface connection that makes them practically indistinguishable from waters of the United States [110].

5.4.16 Resource Conservation and Recovery Act (RCRA)

Purpose. Major federal legislation on hazardous waste management enacted in 1976 to protect health, conserve resources, reduce waste, and ensure sound disposal [110].

Cradle-to-grave control. Regulates generation, management, storage, transport, and disposal of hazardous waste with manifests and permits; applies to shore activities [110].

Aberdeen Three. Federal employees were convicted for improper storage, treatment, and disposal of hazardous waste under RCRA [110].

5.4.17 Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA)

Cleanup authority. Authorizes federal response to the release or threat of release of hazardous substances and focuses on old or inactive sites; costs are assigned to responsible parties [110].

Reporting. Releases of CERCLA hazardous substances must be reported to the National Response Center [110].

Limitations. Petroleum, oils, and lubricants are covered by other programs, not CERCLA [110].

Common name. CERCLA is commonly known as Superfund [110].

5.4.18 Toxic Substances Control Act (TSCA)

EPA authority. Enables the EPA to track industrial chemicals and require reporting or testing; 2016 amendments streamlined listing and bans based on risk [110].

DoN concerns. Regulates polychlorinated biphenyls, lead-based paint, asbestos, and radon [110].

5.4.19 Endangered Species Act (ESA)

Purpose. Protects threatened and endangered species and prohibits "taking" listed species [110].

Federal duty. Requires federal agencies to consult and affirmatively conserve ecosystems supporting protected species [110].

5.4.20 Marine Mammal Protection Act (MMPA)

Purpose. Maintains the health and stability of the marine ecosystem and limits "take" or harassment of marine mammals [110].

Authorizations. Activities such as sonar, sea trials, and weapons firings may require authorization from the National Oceanic and Atmospheric Administration and NEPA analysis [110].

5.4.21 Cultural Resources

NHPA compliance. NHPA requires federal agencies to assess impacts on historic property and consult with states and Native American tribes when actions affect historic characteristics [110].

5.4.22 Emergency Planning and Community Right-to-Know Act (EPCRA)

Notifications. Requires immediate notification of state and local emergency planners and the facility's major claimant for releases over threshold levels [110].

Public information. Provides information on hazards associated with toxic chemical releases and encourages local emergency planning [110].

Fence line owner. The base commander files EPCRA reports for the entire facility; releases that expose only personnel within facility boundaries do not require state or local notification [110].

5.4.23 Workplace and Employee Safety

OSH Act. The Occupational Safety and Health Act requires safe and healthful workplaces; OSHA is the common reference [110].

Federal compliance. Federal agencies must comply with OSHA standards unless the Secretary of Labor approves an alternate standard; Executive Order 12196 applies [110].

Scope. Uniquely military equipment and operations are not covered, but DoW workplaces comparable to private industry are covered under 29 CFR 1960 [110].

5.4.24 DoW Policy

DoD 5000 series. Requires acquisition programs to prevent, mitigate, or remediate environmental damage; address issues early; and evaluate alternatives to reduce impacts across design, test, operation, and disposal [110].

Compliance duty. All DoW personnel, tenants, and contractors must comply with applicable federal, state, and local environmental requirements [110].

5.4.25 Defense Acquisition Program Requirements

Lifecycle compliance. Program managers must ensure systems can be produced, tested, fielded, operated, trained, maintained, and disposed of in compliance with ESOH requirements [110].

Outcome focus. The goal is safeguarding the environment, reducing accidents, and protecting human health [110].

5.4.26 Programmatic ESOH Evaluation (PESHE)

PM documentation. The PESHE documents the program process, the schedule for completing NEPA documentation, and ESOH risk status [110].

Requirement. Required for all ACAT programs at milestone review and serves as a management and reporting tool for the PM [110].

Analysis areas. Environmental compliance, system safety and health, hazardous materials management, pollution prevention management, and NEPA [110].

5.4.27 Quick Review

- What environmental-related document must be updated before every milestone review [110]? **Answer:** PESHE.
- Which environmental law allows for discharge into the waters of the United States [110]? **Answer:** CWA.
- What procedural law established the review process for major federal actions [110]? **Answer:** NEPA.

5.5 Software Acquisition Environment

5.5.1 Summary

Computer fundamentals. A computer receives input, processes data, stores it, and produces outputs; software provides the instructions that make the system operate [111].

Software importance. Software drives risk for many programs, requires continuous updates, and continues to grow in size and complexity across modern weapon systems [111].

Software-intensive systems. A system is software-intensive when software is a major risk driver for development cost, process, time, or functionality [111].

System domains. DoW software domains include weapon systems, C4ISR systems, and Defense Business System (DBS) systems, each with distinct architecture, data latency, and mission focus [111].

Policy environment. The Clinger-Cohen Act and DoWI 5000.02, DoWI 5000.75, and DoWI 5000.87 set expectations for software acquisition, modular delivery, and interoperability [111].

5.5.2 Computer Basics

Computer. A device that receives data through inputs, processes it, stores it, and provides outputs; inputs include sensors such as pressure, temperature, radar, or electronic warfare sensors [111].

Software. Instructions that allow computer components to operate and communicate; includes system software (operating systems) and application software written in programming languages [111].

5.5.3 Programming Language Categories

Machine language. Binary code (0s and 1s) that computers directly execute [111].

Assembly language. Mnemonics translated to machine code by an assembler [111].

Higher-order languages. Languages such as C++ or Java compiled into machine code, enabling programmers to work without detailed hardware knowledge [111].

Fourth-generation languages. Higher-level languages closer to human language (for example, SQL or LabVIEW) [111].

5.5.4 The Importance of Software

Mission dependence. Software is a crucial and growing part of weapon systems and the national security mission [111].

Risk driver. Software challenges drive risk across a large share of acquisition programs [111].

Continuous updates. Software does not end at delivery; it requires continuous updates and sustainment [111].

Growing complexity. Source lines of code have grown dramatically across successive aircraft generations, highlighting the scale of software growth [111].

5.5.5 Software-Intensive System Definition

Definition. A software-intensive system is one where software has essential influence on system design, construction, deployment, and evolution [111].

DoW threshold. In DoW, software is considered intensive when it is a major risk driver for development cost, process, time, or system functionality [111].

5.5.6 Classifications of Software-Intensive Systems

Weapon systems. Purpose-built software, embedded hardware, unique or proprietary architecture, safety as the primary design feature, and real-time data measured in nanoseconds [111].

C4ISR systems. Mix of proprietary and COTS architecture with security as the primary design feature and data lifespans measured in seconds (near real-time) [111].

DBS systems. Routine administrative systems built on COTS architecture with privacy as the primary design feature and data lifespans measured in hours or longer [111].

5.5.7 Software-Intensive System Summary

Weapons. Safety and performance driven, real-time sensor-to-shooter environment [111].

C4ISR. Security and decision support driven, near real-time tactical and strategic command environments [111].

DBS. Privacy and administrative driven, data-intensive business environments with non-real-time data [111].

5.5.8 Software Management Policies and Practices

- Use robust systems engineering, open architectures, and COTS products where practicable [111].
- Exploit software reuse opportunities and select contractors with relevant domain experience and mature development processes [111].
- Plan early for transition to support activities, including documentation, host systems, test beds, and tools needed for life-cycle support [111].

5.5.9 Clinger-Cohen Act (CCA) of 1996

Structure. Combines the Federal Acquisition Reform Act and the Information Technology Management Reform Act [111].

ITMRA focus. Sets DoW policy for software acquisition with life-cycle management and accountability through CIOs [111].

Key themes. Capital planning and investment control via OMB, performance-based management, incremental and modular acquisitions, and COTS preference [111].

National security systems. Defines national security systems as intelligence, cryptologic, command and control systems, or software integral to weapon systems [111].

Overall impact. The Clinger-Cohen Act governs acquisition of software-intensive systems across the federal government [111].

5.5.10 Clinger-Cohen Act Requirements

- Conduct an analysis of alternatives and an economic analysis (or life-cycle cost estimate for non-automated information systems) [111].
- Verify the acquisition supports core DoW functions and no better source exists in the private or public sector [111].
- Establish clear accountability, outcome-based performance measures, and program metrics linked to strategic goals [111].
- Ensure consistency with DoW information enterprise architecture and standards, including a cybersecurity strategy aligned to policy and architecture [111].
- Use modular contracting and phased, successive increments that deliver measurable benefit and redesign processes to maximize COTS use [111].

- Register mission-critical and mission-essential systems with the DoW CIO [111].

5.5.11 Commercial Off-The-Shelf Software

Benefits. Can save design and development cost, reduce risk, enable rapid prototyping, and leverage industry expertise [111].

Tradeoffs. Still requires integration and qualification, has configuration-management challenges, and introduces vendor-driven release schedules and licensing costs [111].

Security. Security characteristics of COTS solutions may be poorly understood and require additional testing [111].

5.5.12 DoW Instruction Provisions

DoWI 5000.02. Covers acquisition categories, systems engineering, program management, test and evaluation, acquisition of IT, human systems integration, and cybersecurity [111].

DoWI 5000.75. Implements Clinger-Cohen requirements for business systems, minimizes COTS customization, and uses a tailorable Business Capability Acquisition Cycle (BCAC) with authority-to-proceed decision points [111].

DoWI 5000.87. Establishes the Software Acquisition Pathway, exempts software programs from MDAP treatment and JCIDS, streamlines requirements and budgeting, and integrates Agile, DevSecOps, Lean, and user-centered design [111].

5.5.13 Software Acquisition Pathway Execution

Iterative delivery. The pathway treats software as continuously delivered capability with frequent releases, user feedback loops, and rapid defect remediation [111].

Risk management. Cybersecurity and Risk Management Framework (RMF) requirements are integrated into release planning and verification, avoiding one-time compliance gates that stall delivery [111].

Artifacts and metrics. Programs emphasize release cadence, operational use metrics, and sustainment pipelines in place of a single monolithic baseline snapshot [111].

5.5.14 Business Capability Acquisition Cycle

- Streamlines activities to align with rapid, frequent, and measurable deliveries [111].
- Brings the user closer to the acquisition community [111].
- Delivers meaningful capability early and often [111].

5.5.15 Software Items

Building blocks. Software items are smaller software building blocks used to manage requirements, test qualification, control interfaces, ensure configuration management, and mitigate risk [111].

Decomposition. Software items can be broken into smaller software units for programming, and lack of control increases risk and vulnerability [111].

5.5.16 Interoperability and Architecture

Net-ready. DoW systems must be Net-Ready (NR) and interoperable in compliance with enterprise architecture; net-ready certification follows DoWI 8330.01 [111].

Architecture framework. DoWAF provides the overarching framework for interoperability and architecture descriptions, with tailored views selected by the sponsor [111].

5.5.17 Quick Review

- A software-intensive system is one in which software is a major risk driver for development cost, time, process, or functionality [111].
- Software-intensive system classifications include weapon systems, C4ISR systems, and DBS systems [111].
- Which law drives DoW software acquisition policy [111]? **Answer:** The Clinger-Cohen Act.

5.6 Software Acquisition Management

5.6.1 Summary

Common trouble sources. Software-intensive programs struggle when requirements are weak, users are not engaged, architectures and interfaces are unclear, hardware deficiencies are overlooked, or integrated teams are not maintained [112].

Best practices. Manage risk continuously, estimate cost and schedule empirically, implement strong configuration control, and plan early for life-cycle support [112].

Life-cycle costs. Software costs peak late in the life cycle while early decisions drive those costs, making up-front planning decisive [112].

Execution focus. Execution emphasizes rapid, iterative design, development, testing, and delivery supported by a product roadmap and program backlog [112].

Support share. Post-deployment software support dominates total life-cycle cost and must be planned early [112].

5.6.2 Common Issues in Software-Intensive Systems

Root causes. Poor requirements, limited user involvement, unclear architecture and interfaces, overlooked hardware deficiencies, and weak integrated teams drive software problems [112].

Mission impacts. Software delays can delay ship or boat delivery, and upgrade delays can trigger stop-work when software access is blocked [112].

5.6.3 Best Practices

Risk. Identify and manage risk continuously, use metrics to monitor risk, and address risks from reusing existing software (commercial or non-developmental) [112].

Cost. Estimate cost and schedule empirically and track earned value [112].

Configuration control. Implement configuration management and formally inspect requirements and design products [112].

Life-cycle planning. Plan early for software changes and enhancements; establish support concepts and acquire Post-Deployment Software Support (PDSS) resources [112].

Other focus areas. Trace requirements to the lowest level, ensure data interoperability, conduct continuous testing, and account for end-of-service and licensing considerations [112].

5.6.4 Software Cost Distribution

Software costs peak late in the program life cycle, while early decisions drive those eventual costs [112].

5.6.5 Planning Phase

Purpose. Understand user and system needs and plan the approach to deliver capabilities [112].

Entry information. Requires an ADM and a draft capability needs statement that captures mission deficiencies, interoperability needs, and legacy interfaces [112].

Artifacts before execution. Capability needs statement, user agreement, cost estimates and Cost Analysis Requirements Description (CARD), acquisition strategy, test strategy, cybersecurity strategy, contracting strategy, intellectual property strategy, and program support strategy [112].

5.6.6 Acquisition Strategy Planning

- Address data rights, security, system integration, open systems architecture, software reuse, logistics support, software support and licensing, and software modification for emerging missions [112].
- Intellectual property strategy and information support planning address these issues [112].

5.6.7 Contracting and Solicitation Planning

Software quality. The program office works with end users to generate requirements [112].

Development environment. Require an automated software life-cycle development and support environment for the contractor [112].

Data rights. Determine the amount of data rights to be purchased [112].

Contracting approach. Support small, frequent releases that are responsive to change (modular contracting) [112].

5.6.8 Cost Estimating

Timing and updates. Estimates are required before execution and updated annually [112].

Attributes. Estimates should be timely, iterative, informative, flexible, and credible [112].

Approaches. Analogy, parametric, engineering build-up, and actuals are common methods; teams also select top-down, bottom-up, or Delphi approaches depending on data and program context [112].

5.6.9 Execution Phase

Purpose. Rapidly and iteratively design, develop, integrate, test, deliver, and operate resilient software capabilities that meet user priorities [112].

Artifacts. Periodic updates to planning artifacts, systems architecture, product roadmap, program backlog, value assessment, Clinger-Cohen compliance, test plans, and program metrics [112].

5.6.10 Systems Engineering Linkage

Definition. Systems engineering transforms operational needs into system performance parameters and a preferred system configuration [112].

Policy. DoW policy requires PM use of systems engineering principles in software design and development [112].

5.6.11 Software Development Activities

System requirements and design. Develop top-level specifications and determine which requirements are best performed through software [112].

Requirements analysis. Select Software Items (SIs) and define detailed requirements for each [112].

Software design. Build detailed designs for each SI and verify requirements are met (not applicable to Agile development) [112].

Coding and implementation. Developers code each software unit and test units stand-alone; each SI may include multiple software units [112].

Unit integration and testing. Progressively integrate software units and test internal interfaces [112].

Software item qualification. Qualify each SI as an entity, documented in the Software Requirements Specification (SRS) [112].

Software item integration. Integrate SIs (and hardware items) into subsystems [112].

Subsystem qualification. Integrate, test, and qualify subsystems [112].

5.6.12 Software Development Models

Selection. The model depends on acquisition strategy and system type [112].

Preferred approach. DoW prefers empirical (Agile) development because it supports continuous inspection and adaptation to emerging threats [112].

5.6.13 Agile Overview

Agile definition. Requirements and solutions evolve through collaboration between cross-functional teams and users, enabling adaptive planning, early delivery, and continuous improvement [112].

Agile vs traditional. Empirical Agile emphasizes frequent inspect-adapt points and delivery of priority features; traditional approaches plan and manage to full requirements up front [112].

5.6.14 Project Management Methodologies

Scrum. Framework for complex projects using cross-functional teams and time-boxed sprints (typically 1–4 weeks) [112].

Kanban. Visual workflow management with to-do, in-progress, and completed tasks [112].

Minimum viable product. An Minimum Viable Product (MVP) is an early version of software provided to users for feedback [112].

Minimum viable capability release. An Minimum Viable Capability Release (MVCR) is an initial set of features suitable to be fielded as a completed product [112].

Lean. Tools that help teams improve processes [112].

5.6.15 Software Process and Maturity

Process. Activities, methods, and practices used to develop and maintain software products (requirements, code, test cases, manuals) [112].

Process maturity. The extent a process is documented, managed, measured, controlled, and continually improved [112].

5.6.16 Selecting a Software Developer

Key discriminators. Organization and resource management, software and systems engineering process management, tools and techniques, development expertise, domain experience, and Capability Maturity Model Integration (CMMI) accreditation [112].

CMMI appraisal. Programs can evaluate capability using Standard CMMI Appraisal Method for Process Improvement (SCAMPI) even if a contractor is not CMMI-rated [112].

Common practices. Use measurement to plan and track development, use mature development processes, develop architectures that support open systems, exploit commercial products, and train personnel on tools and environments [112].

5.6.17 CMMI Maturity Levels

- Level 5 (Optimizing): focus on process improvement [112].
- Level 4 (Quantitatively Managed): process measured and controlled [112].
- Level 3 (Defined): process characterized for the organization; projects tailor from organizational standards [112].
- Level 2 (Managed): process characterized for projects and often reactive [112].
- Level 1 (Initial): processes unpredictable, poorly controlled, and reactive [112].

5.6.18 Defense Innovation Board Ten Commandments of Software

1. Make computing, storage, and bandwidth abundant to DoW developers and users [112].
2. Start software procurement programs small, iterate, and build on success or terminate quickly [112].
3. Ensure acquisition processes support the full iterative life cycle of software [112].
4. Adopt a DevSecOps culture for software systems [112].
5. Automate testing to deploy critical updates in days to weeks, not months [112].
6. Include source code as a deliverable for purpose-built DoW software systems [112].
7. Ensure local DoW software experts can modify or extend software through source code or Application Programming Interface (API) access [112].
8. Only run operating systems receiving and using security updates for new vulnerabilities [112].
9. Make security a first-order consideration and encrypt data unless it is part of active computation [112].
10. Store, mine, and make available all data generated by DoW systems for machine learning [112].

5.6.19 Metrics Selection Considerations

- Identify the risks being tracked, required measures, data definitions, collection costs and cadence, and system architecture [112].
- Use a minimal set of metrics that are understandable, economical, field tested, leveraged, useful at multiple levels, and timely [112].

- Tailor metrics to management needs, collect across the life cycle, and use metrics to identify risks, improve process, and evaluate practices [112].
- Use automated collection where possible [112].

5.6.20 Software Measurement Metrics Categories

Process. Maturity and robustness of organizational processes, including productivity, rework, and technology impacts [112].

Quality. Attributes affecting performance, user satisfaction, supportability, and ease of change [112].

Management. Compare actual progress against plans to detect trouble early and adjust plans [112].

5.6.21 Software Support

Definition. Software support includes activities, resources, and infrastructure to keep software performing its intended role and to modify design for new or changed requirements [112].

Maintenance. Modification after delivery to correct faults, improve performance, or adapt to a changed environment [112].

5.6.22 Post-Deployment Software Support

Dominant activities. Improve performance, adapt to new hardware, add features, improve efficiency and reliability, and correct errors [112].

Cost share. PDSS accounts for approximately 67 percent of total software life-cycle cost [112].

5.6.23 PDSS Best Practices

- Plan early for changes and enhancements and account for cost, schedule, and performance [112].
- Establish software support concepts and acquire resources, including support for COTS systems [112].
- Include PDSS planning in program documentation [112].

5.6.24 Quick Review

- The engineering build-up cost estimating approach is used when a detailed analysis of work packages is required [112].
- The three categories of software metrics are process, quality, and management [112].
- The preferred software development model is empirical (Agile) [112].
- PDSS is approximately 67 percent of total software life-cycle cost [112].

5.7 Cybersecurity Awareness

5.7.1 Summary

Definition. DoWI 8500.01 defines cybersecurity as preventing, protecting against, and restoring damage to communications and information systems to preserve confidentiality, integrity, availability, non-repudiation, and authentication [113].

Threat evolution. Events from Solar Sunrise through ransomware campaigns and major breaches show escalating sophistication and impact [113].

Acquisition integration. The PM owns cybersecurity across the life cycle; the cybersecurity strategy annex is part of the Program Protection Plan (PPP) starting at Milestone A and updated each milestone [113].

RMF backbone. The RMF is a six-step process that bakes cybersecurity into acquisition for Information System (IS) and Platform Information Technology (PIT) systems and drives continuous monitoring [113].

Navy frameworks. CYBERSAFE and Defense-in-Depth Functional Implementation Architecture (DFIA) standardize defense-in-depth across the enterprise and prioritize mission assurance [113].

5.7.2 What Is Cybersecurity?

Confidentiality. Protect information from unauthorized disclosure [113].

Integrity. Protect information from unauthorized modification or destruction [113].

Availability. Ensure information and systems are accessible when needed [113].

Non-repudiation. Prevent parties from denying they sent or received information [113].

Authentication. Verify the identity of users or systems [113].

5.7.3 Consequences of Cybersecurity Failure

Loss of confidentiality. Unauthorized disclosure of Personally Identifiable Information (PII) or classified data (for example, exfiltration from Secret Internet Protocol Router Network (SIPRNet)) [113].

Loss of integrity. Unauthorized modification of Global Positioning System (GPS) navigation data or destruction of PII on a DBS [113].

Loss of availability. Denial of Service (DoS) attacks on Nuclear Command, Control, and Communications (NC3) networks, logistics ordering systems, or unmanned system C2 links [113].

5.7.4 Driving Emphasis for Cybersecurity

- 1998: Solar Sunrise network intrusion on DoW networks [113].

- 2006: State Department networks hacked by unknown foreign intruders [113].
- 2008: Agent.btz Universal Serial Bus (USB) intrusion on DoW computers [113].
- 2008: Supply chain attack on China-made credit card readers [113].
- 2009: Conficker worm infects military databases and grounds French naval aircraft [113].
- 2013: Operation Rolling Tide in response to adversary intrusions on Navy networks [113].
- 2015: Office of Personnel Management (OPM) data breach [113].
- 2017: Equifax data breach [113].
- 2018–2020: Ransomware attacks on U.S. cities and hospitals [113].

5.7.5 Cyber-Related Organizations

DoN CIO. Provides policy and guidance, evaluates security of IT systems, and shapes network architecture and interoperability [113].

USCYBERCOM. Combatant command for global cyber operations; sets enterprise policy [113].

Fleet Cyber Command/U.S. 10th Fleet. Navy component commander to United States Cyber Command (USCYBERCOM); responsible for operations, maintenance, and defense of Navy networks [113].

NCDOC. Cyber Security Service Provider (CSSP) performing defense, incident response, and forensics; threat focused [113].

NAVNETWARCOM. Tactical command and control of Navy networks; directs, operates, maintains, and secures communications and networks [113].

NAVIFOR. Cyber TYCOM that organizes, mans, trains, and equips the cyber workforce [113].

NAVWAR. Navy cybersecurity technical authority; sets standards, tools, and protocols and warrants cybersecurity technical warrant holders within SYSCOMs [113].

5.7.6 Intelligence Community Support

Threat inputs. PMs must use cyber threat information from the Intelligence Community to shape cybersecurity strategy and risk assessments [113].

Products. VOLT reports, Capstone Threat Assessment (CTA), and technology targeting risk assessments support acquisition programs [113].

5.7.7 Cybersecurity in the Acquisition Process

PM accountability. The PM is responsible for cybersecurity requirements throughout the program life cycle [113].

Program protection. The cybersecurity strategy annex is documented in the PPP and describes how the system operates in a cyber-contested environment and the consequences of breach [113].

RMF requirement. The RMF is required for all acquisitions containing IT and is updated each milestone starting at Milestone A [113].

5.7.8 Risk Management Framework Overview

Purpose. Integrates security and risk management into the acquisition life cycle and emphasizes continuous monitoring and timely correction [113].

Applicability. Applies to IS, PIT, and DoW partnered systems [113].

Goals. Perform risk assessment throughout the life cycle, bake in cybersecurity, and improve understanding of risks to Navy systems and missions [113].

5.7.9 RMF Step 1: Categorize the System

Categorization. Identify information types, select confidentiality, integrity, and availability objectives, and assign impact levels (low, moderate, high) [113].

Documentation. Record the security category in the PPP, initiate the security plan, register the system with the component cybersecurity program, and assign qualified personnel [113].

5.7.10 RMF Step 2: Select Security Controls

Control selection. Select controls based on categorization and Committee on National Security Systems Instruction (CNSSI) 1253, then tailor or supplement based on system risk [113].

Monitoring plan. Develop the system-level continuous monitoring strategy and approve the security plan [113].

5.7.11 RMF Step 3: Implement Security

Implementation. Implement controls consistent with component cybersecurity architectures and integrate them into system design specifications [113].

Documentation. Record control implementation in the security plan [113].

5.7.12 RMF Step 4: Assess Security Controls

Assessment plan. Develop and approve the security assessment plan and assess controls for proper implementation [113].

Assessment outputs. The Security Control Assessor (SCA) prepares a SAR and recommends cybersecurity risk acceptance or denial; conduct initial remediation [113].

5.7.13 RMF Step 5: Authorize the System

Authorization package. Prepare Plan of Action and Milestones (POA&M) based on vulnerabilities and submit the security plan, SAR, and POA&M to the Authorizing Official (AO) [113].

Decision. The AO issues an Authorization to Operate (ATO), Interim Authorization to Test (IATT), or Denial of Authorization to Operate (DATO) based on risk determination [113].

5.7.14 RMF Step 6: Monitor Security Controls

Continuous monitoring. Assess controls annually, conduct remediation, update the security plan, SAR, and POA&M, and report status to the AO [113].

Lifecycle closeout. Implement a system decommissioning strategy when required [113].

5.7.15 Cyber Hygiene

- Install and maintain firewalls and review logs for signs of intrusion [113].
- Perform routine scans for malware and known vulnerabilities; mitigate findings quickly [113].
- Use whitelisting to allow only authorized programs to run [113].
- Maintain approved baseline images and follow configuration management policies [113].

5.7.16 EDO Role in Cybersecurity

PM responsibility. PMs are responsible for cybersecurity across programs and life-cycle stages [113].

EDO leadership. EDOs must be accountable leaders who embed cybersecurity into programmatic, engineering, and acquisition processes and involve cyber professionals [113].

Workforce duty. Cybersecurity responsibility extends to all members of the acquisition workforce [113].

5.7.17 CYBERSAFE Program

Purpose. Navy Cyber Safety Program (CYBERSAFE) hardens mission-critical systems and components to provide mission assurance and resiliency [113].

Approach. Assesses mission risk across system-of-systems and applies elevated controls to a subset of systems and components [113].

Certifications. System design certification, enclave design certification, and platform certification [113].

5.7.18 Defense-in-Depth Functional Implementation Architecture

Definition. DFIA provides a Navy defense-in-depth Information Assurance (IA) approach across SYSCOM boundaries, enabling inheritance and improving interoperability and cyber situational awareness [113].

Goals. Provide a uniform security framework to support interoperability, reduce cost, improve cybersecurity, and eliminate duplication [113].

Integration. DFIA standardizes implementation of security controls through RMF, CYBERSAFE, and acquisition processes [113].

5.7.19 Quick Review

- The cyber TYCOM is Naval Information Forces Command (NAVIFOR) [113].
- The Navy component commander to USCYBERCOM is Fleet Cyber Command/U.S. 10th Fleet [113].
- The process used to bake cybersecurity into acquisition programs is the RMF [113].

5.8 Systems Engineering Overview

5.8.1 Summary

SE definition. Systems engineering is an interdisciplinary management process that evolves and verifies balanced, integrated system solutions to satisfy customer needs across the life cycle [114].

DoW policy. Defense Acquisition Guidebook (DAG) and DoWI 5000.88 position systems engineering as the technical framework that balances cost, schedule, performance, and risk for delivery of capability [114].

Good design traits. Supportability, mission performance, producibility, testability, cost efficiency, integrated software, and technology transition define good design [114].

SE process model. Eight technical processes and eight technical management processes guide decomposition, realization, and management across the life cycle [114].

SEP anchor. The SEP documents the program's technical approach and is required at each milestone review starting at Milestone A [114].

5.8.2 Introduction to Systems Engineering

Purpose. Systems engineering provides the technical framework for delivering materiel capability to the warfighter and supports program success [114].

Approach. A methodical, disciplined approach to specification, design, development, realization, management, operations, and retirement [114].

Lifecycle relevance. Systems engineering applies continuously and iteratively across the entire product life cycle [114].

5.8.3 DoW Policy on Systems Engineering

Defense Acquisition Guidebook. DAG Chapter 3 frames systems engineering as foundational to balanced cost, schedule, performance, and risk across the program life cycle [114].

DoDI 5000.88. Emphasizes a structured, multidisciplinary approach that balances cost, schedule, and performance while maintaining acceptable risk [114].

Best practices. Military services, PMs, and engineering leaders implement concept exploration, mission engineering, technical baselines, reviews, test and evaluation, risk and configuration management, and secure technical decisions [114].

Documentation. The systems engineering approach and processes are documented in the SEP and updated as programs evolve [114].

5.8.4 Characteristics of Good Design

- Supportability: hardware and software can be sustained across the life cycle [114].
- Mission performance: meets required performance thresholds [114].
- Producibility: can be manufactured efficiently [114].
- Testability: requirements can be verified [114].
- Cost efficiency: reduces total ownership cost [114].
- Integrated software: software and hardware integration is planned and verified [114].
- Technology transition: incorporates technology refresh and transition [114].

5.8.5 Integrated Product and Process Development

Definition. IPPD integrates acquisition activities using multidisciplinary teams to optimize design, manufacturing, business, and supportability [114].

Role of IPTs. IPTs plan, execute, and implement life-cycle decisions [114].

Key tenets. Customer focus, concurrent product and process development, early life-cycle planning, flexible optimization, robust design, event-driven scheduling, multidisciplinary teamwork, empowerment, and proactive risk management [114].

5.8.6 Systems Engineering Disciplines

Core disciplines. Systems engineering, test and evaluation, life-cycle logistics, science and technology, production and manufacturing, hardware/software engineering, and cybersecurity [114].

EDO roles. EDOs perform jobs across all of these disciplines [114].

5.8.7 System Engineering Plan

Purpose. Program office engineers develop the SEP to communicate and manage the technical approach guiding all program technical activities [114].

Content. Describes technical risks, processes, resources, metrics, products, organizations, and design considerations; updated as the program evolves [114].

Scope. Defines the who, what, where, when, why, and how of the systems engineering approach, including review criteria and tools [114].

Approval. Submitted at each milestone review beginning at Milestone A and approved by the MDA [114].

5.8.8 Systems Engineering Process Model

The systems engineering process model includes eight technical management processes and eight technical processes (three decomposition and five realization processes) [114].

5.8.9 Systems Engineering ”V” Model

Iterative application. Processes are applied iteratively, recursively, and in parallel across the life cycle [114].

Left side. Decomposition processes translate stakeholder needs into requirements and architecture [114].

Right side. Realization processes build, integrate, verify, validate, and transition the system [114].

5.8.10 Eight Technical Processes

Stakeholder requirements definition. Translate stakeholder needs into technical requirements [114].

Requirements analysis. Decompose user needs into clear, achievable, verifiable requirements and system functions [114].

Architecture design. Develop alternative design solutions and establish candidate architectures [114].

Implementation. Produce detailed designs and fabrication or software build procedures [114].

Integration. Assemble lower-level elements into higher-level system elements [114].

Verification. Provide evidence the system meets performance requirements; involves developmental testing [114].

Validation. Provide evidence the system meets stakeholder performance in the intended environment (effectiveness, suitability, sustainability, survivability) [114].

Transition. Move system elements to the next level; for end-items, field the system to the user [114].

5.8.11 Eight Technical Management Processes

Technical planning. Define scope of technical effort and provide inputs to program planning and life-cycle cost [114].

Decision analysis. Convert decision opportunities into traceable, actionable plans [114].

Technical assessment. Compare results to criteria to assess technical maturity, status, and risk [114].

Requirements management. Maintain traceability from user needs to performance specs and delivered capability [114].

Risk management. Identify, analyze, mitigate, and monitor risks to meet cost, schedule, and performance goals [114].

Configuration management. Establish and control baselines and manage technical change across reviews and audits [114].

Technical data management. Identify and manage technical data and software needed to support the system [114].

Interface management. Ensure interface definition and compliance among system elements and external systems [114].

5.8.12 Systems Engineering Process Inputs and Outputs

Inputs. Customer needs (missions, KPPs, Measure of Effectivenesss (MOEs), environments, constraints), technology base, prior SE outputs, program decisions, and commercial standards [114].

Outputs. Decision database, WBS, system and configuration item architectures, operational and technical architectures, and program-unique specifications and baselines [114].

5.8.13 Requirements Traceability Flow

Document flow. CBA → ICD → CDD → CDD update (legacy CPD function) → SEP → TEMP → Developmental Test and Evaluation (DT&E) → IOT&E → Milestone C [84, 107].

Artifact ownership. CBA by Joint Capability Developer (JCD)/CCMD; ICD by JCD/FCB; CDD and CDD updates by the requirements validation authority; SEP by PM; TEMP by PM; DT&E/IOT&E by Chief Developmental Tester (CDT)/Lead Developmental Test Organization (LDTO) [84, 107].

Decision support. Each artifact informs specific acquisition decisions: ICD supports MDD; CDD supports Milestone B; the CDD update or legacy CPD supports Milestone C; SEP and TEMP support technical reviews; DT&E/IOT&E support operational readiness decisions [84, 107].

5.8.14 Standards and Specifications

Standard. A description of a process and recommended practices to achieve objectives with sufficient quality [114].

Specification. A technical description of a product or service that documents design and production details [114].

5.8.15 Program-Unique Specifications

System specification. Describes what the entire system will do; derived from CDD KPPs and includes verification approach [114].

Item performance specification. Defines performance requirements for major subsystems (design-to requirements) [114].

Item detail specification. Defines specific design parameters or build-to requirements [114].

Item process specification. Defines fabrication processes such as welding, soldering, or bonding [114].

Item material specification. Defines raw or semi-fabricated materials such as copper pipe or aluminum wire [114].

5.8.16 Government and Contractor Roles

Government. Manage total program, set user requirements, ensure responsible design/testing/manufacturing, manage technical risks, and verify solutions meet customer needs [114].

Contractor. Use SE processes for planning, design, and internal testing to meet contractual requirements [114].

Shared. Translate, define, monitor, ensure quality of, and test the design together [114].

5.8.17 Quick Review

- Systems engineering is a translation and feedback process from operational needs to system requirements [114].
- The SE process includes eight technical processes and eight technical management processes [114].
- The SEP documents the program technical approach and is approved at each milestone review [114].

5.9 SE Analysis & Control: Work Breakdown Structure

5.9.1 Summary

Product-oriented hierarchy. The WBS is a product-oriented hierarchy and a core output of the systems engineering process [115].

Program management tool. The WBS supports planning, scheduling, cost estimating, logistics, progress reporting, and EVM baselines across the life cycle [115].

SE linkage. The WBS supports IPT organization, system specifications, technical reviews, ECPs, and interface and risk management [115].

Standards. MIL-STD-881 provides the DoW format and guidance for standardized WBS development [115].

Types. Program WBS is owned by the PMO; contract WBS is owned by the contractor and extends the program structure to lower levels [115].

5.9.2 Introduction

Purpose. The WBS breaks large projects into manageable parts using a DoW-standard format [115].

Lifecycle use. The WBS is used across life-cycle phases and disciplines including program management, contracting, logistics, finance, and budgeting [115].

5.9.3 Basic Role of the WBS

- Establishes the foundation for program and technical plans through the systems engineering process [115].
- Supports acquisition strategy, contracting documents, schedules, and cost and budget formulation [115].
- Supports logistics planning, progress tracking, and problem analysis [115].
- Enables PMB establishment for EVM [115].

5.9.4 WBS as a Management Tool

PM focus. PMs use the WBS as a roadmap for integrated product and process development [115].

Tradeoffs. The WBS supports monitoring cost, schedule, performance, manufacturing, life-cycle support, testing, and risk [115].

5.9.5 WBS and the SE Process

SE outputs. The WBS is an output of the systems engineering process, alongside decision databases, architectures, and baselines [115].

SE support. The WBS helps organize IPTs, group specifications, structure technical reviews, analyze ECPs, manage interfaces, and focus risk management on high-impact items [115].

5.9.6 WBS Hierarchy

Product orientation. A hierarchical, product-oriented structure that defines the product or service to be developed and relates elements of work to the end product [115].

System elements. Represent product parts of the WBS (subsystems, components, assemblies, parts) and align to Configuration Items (CIs) [115].

Enabling system elements. Common elements that provide the means to realize the product (processes, equipment, facilities), shared across programs [115].

5.9.7 WBS Guidance

MIL-STD-881. Provides the principal guidance and standardized formats for WBS development across system types [115].

System types. Formats include aircraft, electronic, missile, ordnance, sea, space, surface vehicle, unmanned air, unmanned maritime, launch vehicle, and automated information systems [115].

5.9.8 WBS Types

Program WBS. Controlled and maintained by the PMO, encompasses the entire program, and provides the basis for the contract WBS [115].

Contract WBS. Controlled and maintained by the contractor; extends the program WBS to lower levels to manage cost and schedule data [115].

5.9.9 Program WBS

Scope. Encompasses the entire program, including contract WBS and government elements such as operations, manpower, government-furnished equipment, and testing [115].

Ownership. Prepared and maintained by the program office and tailored from MIL-STD-881 [115].

Levels. Typically includes Levels 1–3 and serves as the framework for program objectives, technical reviews, and cost and schedule performance [115].

5.9.10 Contract WBS

Purpose. Defines the portion produced by a contractor and is the basis for collecting contract cost and schedule data [115].

Structure. Extends the program WBS below Level 3 (beginning at Level 4) and includes all elements for hardware, software, data, or services [115].

Responsibility. Prepared by the prime contractor and should include subcontractor inputs [115].

5.9.11 Quick Review

- The WBS is a product-oriented hierarchy and an output of the systems engineering process [115].
- MIL-STD-881 provides guidance for developing a WBS [115].
- Program WBS includes Levels 1–3; contract WBS extends the program WBS to Levels 4 and below [115].
- Program WBS is maintained by the Government; contract WBS is maintained by the contractor [115].

5.10 SE Analysis & Control: Configuration Management & Technical Reviews

5.10.1 Summary

CM purpose. Configuration management establishes and maintains consistency between requirements, the product, and configuration information across the life cycle [116].

CI control. Designating CIs enables separate control, documentation, reviews, and qualification testing [116].

Baselines. Functional, allocated, and product baselines connect specifications to technical reviews and change control [116].

Interfaces and data. Interface control and technical data management preserve configuration discipline across government and contractor boundaries [116].

Technical reviews. Reviews assess design maturity, readiness, compliance, and risk at key points in the acquisition life cycle [116].

5.10.2 Configuration Management

Definition. A technical and management process that maintains consistency between product requirements, the product, and configuration information across baselines [116].

Implementation. Technical and administrative direction to identify and document CIs, control changes, and record change status to provide an audit trail [116].

5.10.3 Why CM Matters

- Provides measurable performance parameters and a common basis for acquisition and use [116].
- Ensures correct, current information for change decisions and minimizes avoidable cost [116].
- Correlates products with requirements, design, and product information to avoid guesswork [116].
- Verifies configuration against required attributes and increases confidence in product information [116].

5.10.4 Configuration Items

CI definition. A hardware/software aggregation treated as a single entity for configuration management and uniquely identified at a reference point [116].

Designation impacts. Separate control, documentation, formal approval of changes, individual design reviews, and qualification tests [116].

5.10.5 Considerations for Designating CIs

- Qualification level, integration of tested/untested components, and interfaces with other controlled CIs [116].
- Need to know exact configuration during the life cycle, high-risk or safety concerns, and new technology [116].
- Each development contract should designate at least one CI; cost alone is not a valid consideration [116].

5.10.6 CM Functional Components

Identification. Identify and document functional and physical characteristics of CIs [116].

Status accounting. Record and report all configuration changes [116].

Control. Manage CIs and documentation throughout the life cycle [116].

Verification and audits. Verify conformance to requirements, specifications, drawings, and interface documents [116].

5.10.7 IPPD and CM

Baseline linkage. CM ties requirements and specifications to baselines and supports IPPD across disciplines [116].

Cross-discipline impact. Configuration changes affect test and evaluation, manufacturing, logistics, software development, and technical data management [116].

5.10.8 Commercial Item Impacts

Government control limits. Design details and full Technical Data Package (TDP) may not be available, complicating manuals and sustainment [116].

Interfaces and support. Interfaces must be carefully defined, and support strategies must address commercial repair and updates [116].

Test strategy. Tests must ensure performance at both item and integrated levels [116].

5.10.9 Configuration Control

Engineering change. Modifies, adds, deletes, or supersedes original parts [116].

Deviation. Written authorization before manufacture to depart from requirements for specific units or time [116].

Waiver. Written authorization during or after manufacture to accept a nonconforming item as-is or after rework [116].

Approved configuration. Approved baseline plus approved changes [116].

5.10.10 Configuration Baselines

Functional baseline. Defines overall system performance and functional characteristics [116].

Allocated baseline. Defines performance requirements for CIs that make up a system [116].

Product baseline. Defines functional and physical characteristics of CIs [116].

5.10.11 Specifications, CM Planning, and Baselines

System specification. Establishes mission performance requirements and interfaces tied to the functional baseline [116].

Performance specification. Defines form, fit, and function of CIs and supports the allocated baseline [116].

Detail/process/material specifications. Define build-to requirements and fabrication processes that support the product baseline [116].

5.10.12 Interface Control

Interfaces. Boundaries between CIs or between contractor and GFE requiring mechanical, electrical, operational, or software alignment [116].

ICWG. The Interface Control Working Group (ICWG) identifies, documents, and controls interface characteristics and produces interface control documents [116].

Government role. Government sets interface standards; contractors design to meet interface specifications [116].

5.10.13 Technical Data Management

Purpose. Processes to plan, acquire, manage, and use technical data to support the system life cycle [116].

Data requirements. Documented in the CDRL; only minimum required data should be procured in contractor format [116].

TDP support. TDPs support technical reviews, including Functional Configuration Audit (FCA) and Physical Configuration Audit (PCA) [116].

5.10.14 Configuration Management Strategy

Government-controlled. Functional baseline, system-level specs, external interfaces, and system or major subsystem reviews [116].

Contractor-controlled. Allocated and product baselines, item performance/detail specs, internal interfaces, and component/vendor reviews [116].

Benefits. Promotes design flexibility, reduces government burden for low-level ECPs, and supports smaller program offices [116].

5.10.15 Technical Reviews Purpose

- Check design maturity and readiness for the next development level [116].
- Assess compliance with requirements and review technical risk [116].
- Provide visibility into contractor implementation of the statement of work [116].

5.10.16 MCA Technical Reviews by Phase

MCA technical reviews are event-driven maturity points, not calendar ceremonies. Unless waived through the SEP approval process, DoWI 5000.88 requires the PM to conduct system-level reviews or equivalents for requirements/functionality, preliminary design, critical design, verification/configuration audit, production readiness, and physical configuration audit; Alternative Systems Review (ASR), Test Readiness Review (TRR), and Operational Test Readiness Review (OTRR) remain important best-practice or test-readiness events that are tailored to the program and test strategy [117, 118, 119].

Table XLIII: Major Capability Acquisition technical reviews by phase

Review	Typical MCA phase	Primary baseline/readiness focus	Board-prep purpose
ASR	MSA	Preferred material solution	Confirms the preferred material solution is technically feasible, affordable, and understood well enough to support a draft performance specification.
SRR	TMRR	System requirements	Confirms the developer and government share a clear, testable, traceable understanding of system performance requirements.
SFR	TMRR	Functional baseline	Confirms the functional architecture satisfies performance requirements and supports preliminary design with acceptable risk.
PDR	TMRR, normally before Milestone B	Allocated baseline	Confirms the preliminary design and architecture are mature enough to proceed to detailed design and support the Milestone B decision.

Continued on next page

Table XLIII: Major Capability Acquisition technical reviews by phase **(Continued)**

Review	Typical MCA phase	Primary baseline/readiness focus	Board-prep purpose
CDR	EMD	Initial product baseline	Confirms detailed design stability, expected performance, affordability posture, and readiness to start coding, fabrication, integration, and test.
TRR	EMD, before major test events	Test readiness	Confirms the test article, procedures, instrumentation, resources, environment, safety posture, and success criteria are ready for the planned test.
SVR/FCA	EMD	Verification against functional or allocated baseline	Verifies the system or configuration item satisfies approved requirements and that discrepancies are understood before production or operational test decisions.
PRR	EMD before LRIP; repeated in P&D when needed	Production readiness	Confirms producibility, quality, manufacturing planning, supply chain, facilities, tooling, and production risk are acceptable for LRIP or FRP.
OTRR	P&D, before IOT&E, OA, or FOT&E	Operational test readiness	Certifies that the production-representative configuration, reliability, maintainability, supportability, hazards, resources, and test planning are ready for operational test.
PCA	P&D, after successful OT&E and before FRP decision	Final product baseline	Audits the as-built production-representative item against its technical documentation, resolves discrepancies, and establishes the final product baseline.

Source: *DoWI 5000.88*, *Engineering of Defense Systems Guidebook*, *Test and Evaluation Enterprise Guidebook*, 2020, 2024, 2022 [117, 118, 119]

Please see Figure 4.5 for a visual showing when each of the technical reviews are held.

Note

For a board answer, lead with the baseline progression: requirements/function first, allocated design at PDR, product design at CDR, verified configuration at System Verification Review (SVR)/FCA, production readiness at Production Readiness Review (PRR), operational test readiness at OTRR, and final as-built product baseline at PCA.

5.10.17 Specifications, Baselines, and Reviews

System specifications. Link to the functional baseline and System Requirements Review (SRR)/System Functional Review (SFR) [116].

Performance specifications. Link to the allocated baseline and PDR [116].

Detail/process/material specifications. Link to the product baseline and CDR/FCA/PCA [116].

5.10.18 General Principles for Reviews

Why. Assess design maturity and program risk; not a check-the-box event [116].

When. Driven by the SEP, baseline completion, operational test prep, and upcoming milestone reviews [116].

Who. Led by a Technical Review Board (TRB) with PM representation, engineers, independent reviewers, logistics, cost, legal, contracting, sponsors, and users [116].

5.10.19 Steps to Effective Design Reviews

- Plan the review and scope [116].
- Familiarize participants with data and requirements [116].
- Conduct individual reviews before team review [116].
- Hold team review and resolve issues [116].
- Follow up to close actions [116].

5.10.20 Quick Review

- CM avoids cost, reduces risk, and controls design changes [116].
- CM functions are identification, status accounting, control, and verification/audits [116].
- Baselines are functional, allocated, and product; changes are engineering changes, deviations, and waivers [116].
- Commercial items reduce Government control of design details and TDP access, increasing interface and support planning needs [116].
- In MCA, technical reviews should be explained as risk-reduction gates tied to phase maturity: SRR/SFR in TMRR, PDR before EMD, CDR and verification reviews in EMD, and PRR/OTRR/PCA around production and operational test [118, 119].

5.11 Acquisition Logistics

5.11.1 Summary

Discipline. Acquisition logistics is a multi-functional technical management discipline spanning design through sustainment and modernization to ensure supportability across the life cycle [120].

Supportability focus. Acquisition logisticians ensure the system is designed for supportability and that the support infrastructure is ready when the system is fielded [120].

Readiness metrics. Readiness is measured through Reliability, Availability, and Maintainability (RAM) parameters; early RAM focus improves operational effectiveness and reduces cost [120].

Cost impact. Operations and Support (O&S) costs often account for roughly 70–80% of LCC, so early design decisions drive long-term affordability [120].

Product support. Product support strategy, Performance-Based Logistics (PBL), LCSP, and the 12 Integrated Product Support (IPS) elements integrate sustainment across the life cycle [120].

5.11.2 Defining Acquisition Logistics

Scope. A discipline associated with design, development, testing, production, fielding, sustainment, and modernization [120].

Life-cycle lens. Technical and management activities ensure supportability implications are considered early and throughout acquisition [120].

Goal. Field cost-effective systems that achieve peacetime and wartime readiness needs [120].

5.11.3 Role of Acquisition Logistics

- Ensure the system is designed for supportability [120].
- Design the support infrastructure and support structure required at fielding [120].
- Measure supportability using RAM readiness parameters [120].

5.11.4 Acquisition Logistics Policy

PM accountability. DoW policy makes the PM accountable for life-cycle management, including early O&S planning and balancing supportability with cost, schedule, and performance [120].

Design fundamentals. PMs implement design and manufacturing fundamentals that yield reliable and maintainable systems across the service life [120].

Product support strategy. PMs design an enduring, affordable Product Support Strategy (PSS) with metrics and the best mix of public and private capabilities [120].

5.11.5 Policy Implementation: PSM and LCSP

PSM requirement. MDAP programs require a Product Support Manager (PSM) per 10 U.S.C. 4324 [120].

DoW guidance. DoWI 5000.91 states the PM is the single point of accountability and the PSM supports a performance-based PSS for all covered systems [120].

Strategy delivery. The PSM develops, plans, and implements a comprehensive PSS; the LCSP is required to capture it [120].

Business model. The Product Support Business Model (PSBM) and 12-step product support strategy process are operationalized through the LCSP [120].

5.11.6 RAM Definitions

Readiness is measured through RAM parameters [120].

Reliability. Ability to perform the mission without failure, degradation, or demand on the support system [120].

Availability. Degree to which an item is operable and committable at mission start for an unknown time [120].

Maintainability. Ability to retain or restore an item to required condition with specified skills, procedures, and resources at prescribed maintenance levels [120].

5.11.7 Acquisition Logistics and RAM

Early objectives. PMs establish RAM objectives early and treat them as design parameters throughout acquisition [120].

Requirement basis. RAM system requirements are derived from capability documents and TOC considerations, expressed in quantifiable operational terms and measured in test [120].

System-wide scope. RAM requirements include support and training equipment, manuals, spares, and tools [120].

5.11.8 RAM Metrics

Sustainment KPP. For ACAT I programs, the sustainment KPP includes Material Availability (Am) and Operational Availability (Ao) [120].

Supporting KSAs. Reliability (e.g., Mean Time Between Failures (MTBF)), maintainability (e.g., Mean Time To Repair (MTTR)), and TOC are mandatory sustainment KSAs for ACAT I programs [120].

Lower ACAT. For ACAT II and below, the sponsor determines sustainment KPP/KSA applicability [120].

5.11.9 RAM Techniques

- Design for human engineering, accessibility, visibility, testability, standardization, simplicity, and minimized failures [120].
- Consider Naval Supply Systems Command (NAVSUP) support, Reliability-Centered Maintenance (RCM) analysis, skill requirements, maintenance levels, tools, and test equipment [120].

5.11.10 Design to Minimize Failures

- Address mission, environmental, and life-profile demands [120].

- Use stress and worst-case analysis, Failure Mode, Effects and Criticality Analysis (FMECA), and Fault Tree Analysis (FTA) [120].
- Apply allocation, part/material selection, derating, simplification, and design reviews [120].

5.11.11 Support Costs and Life-Cycle Cost

Cost share. O&S costs are about 70–80% of total LCC for typical programs [120].

Ship O&S examples. Cost categories include mission personnel, unit-level consumption, intermediate maintenance, depot maintenance, contractor support, sustaining support, and indirect support [120].

Early influence. Early analyses and decisions lock in most life-cycle cost outcomes [120].

5.11.12 Logistics-Related Concepts

- IPPD/IPTs integration and continuous LCC consideration [120].
- COTS/Non-Developmental Item (NDI) reduce development risk but can increase long-term support risk [120].
- Maintain organic core depot capability and avoid overuse of Contractor Logistics Support (CLS) [120].
- RAM as a force effectiveness multiplier and open systems to enable upgrades [120].
- Cradle-to-grave program management for sustainment outcomes [120].

5.11.13 Performance-Based Logistics

Definition. PBL is outcome-based, performance-focused product support through performance-based arrangements [120].

Not transactional. Transactional support purchases parts or repairs as needed; PBL focuses on outcomes across the life cycle [120].

Preferred approach. PBL is the preferred DoW approach for product support [120].

5.11.14 PBL Metrics and Contracting Strategy

Metric design. Metrics should be defined so outcomes can be tracked, measured, and revalidated; they should drive improvement, efficiency, and innovation [120].

Contract attributes. Effective PBL contracts specify requirements, parameters, incentives, and risk mitigation; successful contracts are often fixed-price, multiyear, and align with FAR Part 12 [120].

Funding strategy. Funding stability, time, scope, and appropriation category shape PBL contract funding [120].

5.11.15 Supportability Analysis

Definition. Product Support Analysis (PSA) is a body of procedures used to define system performance and supportability goals [120].

Methods. Common analyses include FMECA, FTA, RCM, Level of Repair Analysis (LORA), and maintenance task analysis [120].

SE integration. Supportability analysis is conducted as part of the systems engineering process to ensure supportability is a system performance requirement [120].

Timing. Must occur early in the life cycle to influence design; assessment continues throughout the life cycle [120].

5.11.16 Product Support Analysis

Requirement. PSA is required to create the product support package and is guided by MIL-HDBK-502A [120].

System-level start. Analysis begins at the system level to shape design and operational concepts [120].

Cost leverage. Early analysis is less costly because it can influence design decisions up front [120].

5.11.17 Supportability and the Acquisition Life Cycle

Milestone A. Ensure support considerations are integral to system design requirements and reduce LCC early [120].

Milestone B. Identify, develop, and acquire infrastructure for initial fielding and operational support [120].

Milestone C. Ensure the system can be cost-effectively supported across its life cycle [120].

5.11.18 Supportability Issues and Impact

- Operating requirements, readiness metrics (Ao, MTBF, MTTR), and manpower/skill requirements including Navy Enlisted Classifications (NECs) [120].
- Maintenance requirements, repair-vs.-discard decisions, self-diagnostics, and battle damage reparability [120].
- Environment, safety, health, and transportability constraints [120].
- Failure to resolve supportability issues increases LCC and degrades readiness; reliability and maintainability must be contract performance parameters [120].

5.11.19 Product Support Business Model

Framework. PSBM defines how product support is accomplished across components, subsystems, and platforms to balance availability and affordability [120].

Roles. Total Life-Cycle Systems Management (TLCSM) responsibility lies with the PM; life-cycle product support management lies with the PSM [120].

Integration. Product Support Providers (PSPs) provide support and are integrated by one or more Product Support Integrators (PSIs) [120].

5.11.20 12-Step Product Support Strategy Process

1. Integrate warfighter requirements and support [120].
2. Form the product support management IPT [120].
3. Baseline the system [120].
4. Identify and refine performance outcomes [120].
5. Conduct business case analysis [120].
6. Perform product support value analysis [120].
7. Determine support acquisition method(s) [120].
8. Designate product support integrator(s) [120].
9. Designate product support provider(s) [120].
10. Identify and refine financial enablers [120].
11. Establish and refine product support agreements [120].
12. Implement and provide oversight [120].

5.11.21 Life-Cycle Sustainment Plan

Purpose. The LCSP documents how the program will implement its product support strategy across the life cycle [120].

Management tool. The LCSP is a living document that communicates the sustainment approach, resources, and roles [120].

Not a checklist. The LCSP is not a static rehash of policy or a milestone-only artifact [120].

5.11.22 LCSP Development by Milestone

Milestone A. Focus on sustainment metrics and actions prior to Milestone B to reduce future O&S costs [120].

DRFPRD and Milestone B. Finalize sustainment metrics, integrate sustainment with design activities, and refine execution plans for acquisition, fielding, and sustainment competition [120].

Milestone C. Ensure operational supportability and include a comprehensive product support package description, competition plan, and fielding plan [120].

5.11.23 12 Integrated Product Support Elements

- Product support management [120].
- Design interface [120].
- Sustaining engineering [120].
- Supply support [120].
- Maintenance planning and management [120].
- Packaging, Handling, Storage, and Transportation (PHS&T) [120].
- Technical data [120].
- Support equipment [120].
- Training and training support [120].
- Manpower and personnel [120].
- Facilities and infrastructure [120].
- Computer resources [120].

5.11.24 IPS Element Highlights

Product support management. Integrates all IPS activities to achieve KPPs and KSAs and begins at Milestone A [120].

Design interface. Ensures RAM parameters are incorporated early and aligned with operational effectiveness and suitability [120].

Sustaining engineering. Minimizes downtime, lowers risk, and drives design and infrastructure changes when needed [120].

Supply support. Establish provisioning and replenishment, including Allowance Parts List (APL) and Coordinated Shipboard Allowance List (COSAL) development [120].

Maintenance planning. Establishes maintenance concept, repair levels, Planned Maintenance System (PMS) interface, tools, and facilities [120].

Packaging and transport. Defines resources and methods for preservation, storage, and shipment, including environmental and PHS&T considerations [120].

Technical data. Identify minimum technical data needs, documented through the CDRL, with special care for software-intensive systems [120].

Support equipment. Identify required test and support equipment and develop the Allowance Equipage List (AEL) [120].

Training. Define individual and crew training, develop the Navy Training Plan (NTP), and select delivery methods [120].

Manpower. Identify military and civilian skills and minimize manning impacts on LCC [120].

Facilities. Identify facility requirements early because major projects have long lead times [120].

Computer resources. Define software life-cycle support plans, configuration management, support agents, and licensing [120].

5.11.25 Logistics and Program Planning

Start point. Supportability planning begins in MSA [120].

Common issues. Training and technical data, manpower, supply support, facilities, and PHS&T gaps can derail fielding [120].

5.11.26 Fielding and Deployment

Deployment planning. Coordinated by the PMO in support of operational commanders; milestones include trained users, deployed teams, spares, and technical data [120].

Materiel release. Certification that logistics deficiencies from OT&E are corrected and initial fielding resources are available [120].

IOC linkage. Initial fielding activities culminate in materiel release and IOC [120].

5.11.27 Post-Production Support

Activities. CLS, component redesign, P3I, data management, multiple sources, open systems, life-of-type buys, spares modernization, training infrastructure, and funded ISEA support [120].

Challenges. Long life spans, uneconomic order quantities, high spares usage, Diminishing Manufacturing Sources and Material Shortages (DMSMS), proprietary technical data, and technology obsolescence [120].

5.11.28 Quick Review

- Acquisition logistics ensures supportability across the life cycle and is measured by RAM readiness parameters [120].
- O&S costs dominate LCC; early analysis and design decisions shape affordability [120].
- PSA, PSBM, PBL, and LCSP integrate product support planning and execution [120].
- Fielding without infrastructure degrades readiness; post-production support requires proactive risk management [120].

5.12 Manufacturing and Producibility

5.12.1 Why manufacturing belongs in acquisition decisions

Manufacturing is not a post-design activity. A program can meet technical requirements on paper and still fail because the design cannot be produced, tested, funded, or sustained at the required rate. Coursebook 3.7.1 frames the manufacturing decision as a cost, schedule, and performance control problem: the program manager must use producibility thinking early enough to influence system design, supplier choices, manufacturing planning, and quality management before production commitments become expensive to reverse [121].

5.12.2 Nine industrial process elements

Table XLIV: Industrial process elements to check before production commitment.

Element	Program-manager leverage	Board cue
Design	Direct influence	Push producibility, testability, modularity, tolerances, and design margins into the design baseline.
Materials	Direct influence	Check critical materials, alternate sources, lead times, and obsolescence risk before rate decisions.
Cost and funding	Direct influence	Align design choices, production rate, and quality controls with executable funding.
Process capability and control	Direct influence	Ask whether the process can repeatedly hold the required tolerances at the planned rate.
Quality management	Direct influence	Treat prevention and appraisal as cheaper than repeated inspection, rework, and field failure.
Manufacturing planning, scheduling, and control	Direct influence	Ensure work instructions, tooling, labor, suppliers, and schedule logic support the required flow.
Technology and industrial base	Indirect influence	Assess fragile vendors, unique tooling, special processes, foreign dependencies, and surge capacity.
Facilities	Indirect influence	Identify facility constraints, environmental permits, capital equipment, and layout limits.
Personnel	Indirect influence	Check whether the supplier has trained labor, certified operators, supervision, and quality culture.

5.12.3 Manufacturing objectives for design

The manufacturing objective is to make the design producible, affordable, and testable without losing the operational capability that justified the program. The board answer should connect three levers [121]:

Design. Develop a producible and testable design rather than a design that only works in engineering demonstration.

Process. Design and mature manufacturing processes that can reliably build the item.

Process control. Establish process controls that satisfy requirements while minimizing avoidable cost.

5.12.4 Producibility trade space

Producibility is a trade, not a slogan. The program manager should be ready to explain how operational performance, reliability, availability, supportability, technology constraints, ownership cost, schedule, and producibility interact. A design that maximizes one attribute can damage another; for example, a very tight tolerance may improve a performance margin while increasing supplier yield loss and delaying delivery [121].

Board cue. If asked why an EDO cares about manufacturing, answer: production is where design, funding, industrial base, quality, and schedule converge. Producibility must be designed in before production locks in cost and risk.

5.13 Production, Quality, and Manufacturing

5.13.1 Production system variables

Coursebook 3.7.2 ties production performance to the variables that create process variation. The standard board framing is the “5Ms”: material, machinery, method, manpower, and measurement. Some quality tools add environment, or milieu, as a sixth source of variation [122].

Table XLV: Production variables and the EDO question to ask.

Variable	EDO question
Material	Are incoming materials, parts, and supplier processes consistent enough for the required product?
Machinery	Are equipment, tooling, fixtures, and calibration capable at the planned rate?
Method	Are work instructions, routing, sequencing, and process controls mature and repeatable?
Manpower	Are operators trained, certified, supervised, and available?
Measurement	Can inspections and tests detect the variation that matters?
Environment	Do temperature, humidity, cleanliness, safety, or other workplace conditions affect output?

5.13.2 Product and process structures

Production structures move along a variety-rate continuum. Job shops support high variety and low volume; batch production groups similar work; assembly lines increase flow for more stable products; continuous-flow processes support very high volume with highly standardized products. The acquisition question is whether the product design, production rate, and industrial base match the selected process structure [122].

5.13.3 Variation, capability, and statistical control

Process proofing and variation reduction precede confidence in rate production. The program manager should distinguish tolerance from capability: a drawing tolerance says what is acceptable, while process capability shows whether the production process can repeatedly produce within that tolerance. Statistical process control is useful only when measurements are tied to the critical process characteristics and acted on before defects escape [122].

5.13.4 Lean, ISO 9001, and cost of quality

Lean. Lean manufacturing is a people-oriented business philosophy that focuses on value, value stream, flow, pull, and perfection.

ISO 9001. DoD can accept an ISO 9001 quality system as evidence of disciplined quality management for complex or critical systems, but the program manager should not treat registration as the only possible quality approach.

Cost of quality. Prevention and appraisal cost are deliberate investments to avoid internal failure, external failure, rework, warranty, and fleet-impact costs.

These coursebook quality concepts are linked because they all push the program manager toward prevention, stable process control, and supplier-quality evidence instead of late defect discovery [122].

5.13.5 Problem-solving tools

Use "5 Whys" only after the problem statement is accurate and the answers are complete, unbiased, and honest. Use multivoting to reduce a long list of candidate causes or improvements to the few priorities the team will actually work. Use cause-and-effect, or fishbone, diagrams to organize possible causes by production variable before choosing corrective action [122].

Board cue. Quality is built into the process, not inspected into the product. Tie every production-quality answer to variation, process capability, control, and cost of quality.

5.14 Continuous Process Improvement

5.14.1 Purpose

Continuous Process Improvement (CPI) is the disciplined use of improvement methods to remove waste, reduce variation, elevate constraints, and improve fleet outcomes. Coursebook 3.7.3 links CPI to the EDO tool set: Lean, Theory of Constraints, and Six Sigma are practical methods for improving engineering, maintenance, production, and administrative processes [123].

5.14.2 Lean thinking

Table XLVI: Five Lean principles.

Principle	Meaning for an EDO process
Value	Define value from the customer or fleet perspective.
Value stream	Map every step that creates, supports, delays, or obscures value.
Flow	Remove queues, rework, handoff delays, and avoidable batching.
Pull	Let demand signal the next step instead of pushing excess work into the system.
Perfection	Continue improving after the first visible bottleneck is removed.

Lean waste is any activity that does not add value for the customer. Common waste categories include waiting, excess inventory, unnecessary motion, transportation, overproduction, overprocessing, defects, and unused talent [123].

5.14.3 Theory of Constraints

Theory of Constraints starts from the premise that every system has a limiting constraint. The five focusing steps are: identify the constraint, exploit the constraint, subordinate other work to the constraint, elevate the constraint, and repeat when the constraint moves [123]. A bottleneck is a visible queue or capacity limit; a constraint is the system limit that controls total throughput. They may be the same, but the EDO should verify rather than assume.

5.14.4 Six Sigma and DMAIC

Six Sigma focuses on reducing variation and defects. The common performance shorthand is 3.4 defects per million opportunities. The DMAIC improvement cycle is define, measure, analyze, improve, and control [123].

Table XLVII: DMAIC board cue.

Step	Question
Define	What problem, customer, process, and outcome are in scope?
Measure	What data prove the current baseline and variation?
Analyze	What root causes explain the performance gap?
Improve	What countermeasures change the process?
Control	What controls prevent regression after the event ends?

Board cue. Do not answer CPI as a slogan. Name the method, state the constraint or waste, show the metric, and explain how the control plan prevents backsliding.

5.15 Integrated Case Study

5.15.1 Purpose and scope

The Integrated Case Study (ICS) is an EDO School application exercise, not an industrial-control-system topic. Coursebook 3.8.1-4 uses the case study to make students apply program-office organization, requirements, funding, technical authority, systems engineering, and stakeholder communication to an integrated acquisition problem [124].

5.15.2 Expected products

Table XLVIII: Integrated Case Study products and board value.

Product	Purpose
Team charter	Defines team purpose, roles, norms, decision process, and delivery expectations.
Stakeholder map	Identifies who owns requirements, funding, technical authority, fleet need, and execution risk.
Reporting chain	Shows how the issue moves through SYSCOM, PEO, program office, resource sponsor, and fleet channels.
Requirements trace	Connects the proposed action to capability need, acquisition phase, and technical baseline.
Funding trace	Identifies the appropriation, fiscal year, color-of-money risk, and execution constraint.
Presentation package	Forces the team to defend the recommendation with evidence, assumptions, risks, and alternatives.

5.15.3 Team and IPT behavior

The case study is graded as both technical reasoning and team performance. A strong team makes assumptions explicit, separates facts from judgment, assigns owners, controls scope, and resolves disagreements through evidence. Common leadership barriers include unclear decision authority, unspoken assumptions, failure to include the right stakeholders, and treating a symptom as the root problem [124].

5.15.4 Program-office application checklist

Use this checklist to translate the case-study ground rules and stakeholder requirements into a defensible program-office recommendation [124].

- State the operational problem and requirement source.
- Identify the owning program office, resource sponsor, technical authority, and fleet stakeholder.
- Define the acquisition phase, decision point, and required artifact.

- Trace funding to appropriation, fiscal year, and execution constraint.
- Identify technical, schedule, cost, test, logistics, cybersecurity, and contracting risks.
- Present alternatives, recommendation, assumptions, and residual risk.

Board cue. If a board question sounds like an integrated scenario, answer as a program-office problem: requirement, authority, funding, technical baseline, stakeholders, risks, alternatives, and recommendation.

5.16 Test & Evaluation

5.16.1 Board Prep Summary

T&E provides the objective evidence that a system can achieve the warfighter outcomes promised in requirements documents before obligating the fleet to fielding, sustainment, and congressional reporting commitments [84, 125]. A test is the controlled activity (lab, range, digital thread, fleet event) that produces data, while an evaluation is the analytic judgment that compares that data to thresholds, suitability expectations, and risk acceptance criteria [125]. Effective T&E integrates developmental, operational, and live-fire evidence so that each MCA milestone is backed by verified design data and validated mission performance insights rather than optimistic modeling alone [126]. Programs that synchronize T&E with systems engineering and program control expose performance gaps early, recycle fixes through the technical baselines, and enter OTRR with a defendable TEMP, resource plan, and deficiency scrub.

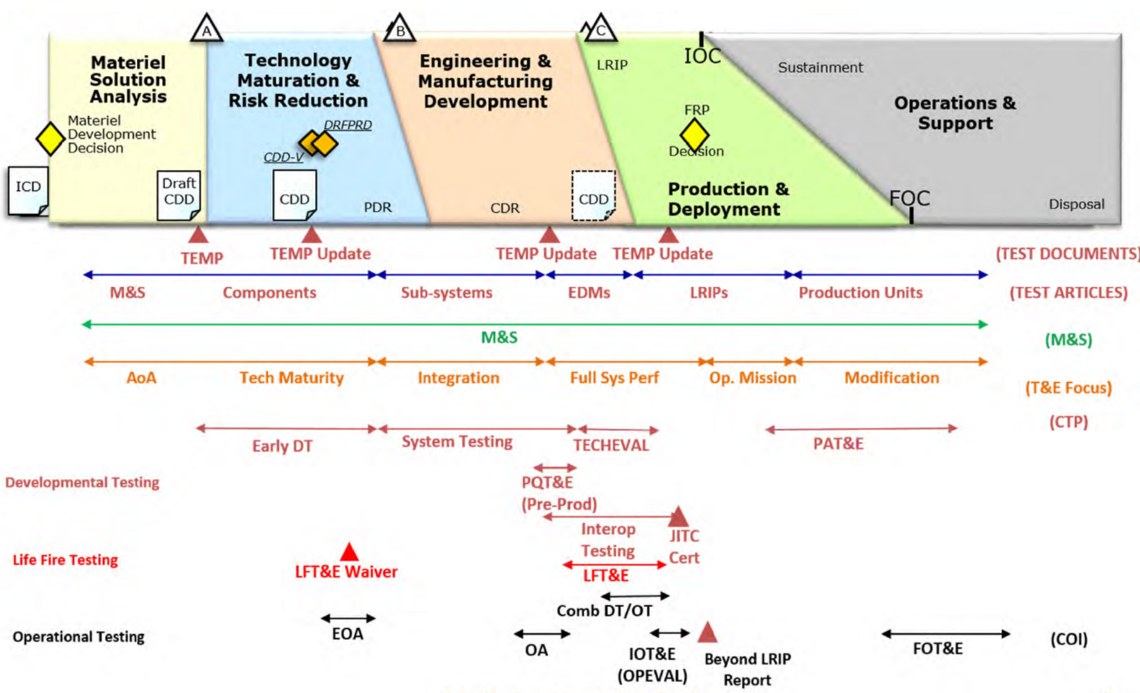

Figure 5.3. Key MCA test and evaluation touchpoints that frame DT, OT, and certification decisions. Source: 3.5.5. Test and Evaluation Part I, *2025 [125].*

5.16.2 Definition and Relationship to Verification and Validation

DT&E verifies "did we build the system right" (verification) by showing the technical baseline meets allocated requirements, standards, and interface controls, while Operational Test and Evaluation (OT&E) validates "did we build the right capability" (validation) by proving Sailors and Marines can accomplish missions under representative threat, environment, and sustainment conditions [127]. In board-prep terms:

- **How do we ensure we are building the right thing?** Through *Validation* (operational testing), which evaluates the system in a realistic environment with representative operators against capability requirements (KPPs/KSAs).
- **How do we ensure we are building the thing right?** Through *Verification* (developmental testing), which evaluates the system against documented technical specifications, drawings, and baseline standards.

Verification relies on repeatable measurements (e.g., propulsion plant efficiency in a land-based engineering site) and feeds configuration control; validation uses operational scenarios, tactics, and logistics supportability to demonstrate mission effectiveness and suitability. DoWI 5000.88 further directs programs to execute frequent, iterative end-user validation of features and usability so mission thread owners can accept the delivered capability before major decisions [117]. Verification and Validation (V&V) and Verification, Validation and Accreditation (VV&A) activities must be planned in the TEMP and SEP so that models, simulations, hardware-in-the-loop facilities, and live events share common data assumptions and confidence levels before milestone reviews [126]. *Verification* therefore proves compliance with the documented specification (analysis, inspection, test, demonstration), whereas *validation* proves the warfighter accepts the delivered capability and can close the mission threads; both sides of the V&V matrix must be satisfied before the MDA can certify readiness for production or fleet release [126].

5.16.3 Operational vs. Developmental Testing

Purpose.

DT&E is an engineering feedback loop: find design, interface, reliability, cybersecurity, safety, and manufacturability problems early enough to fix the baseline. OT&E is a readiness judgment: determine whether a production-representative system is operationally effective, suitable, survivable, and supportable for Fleet use [126, 128].

Primary question.

Developmental testers ask whether the program built the system correctly against allocated technical requirements. Operational testers ask whether operators can accomplish representative missions with the system under realistic threat, environment, training, logistics, and command-and-control conditions [127].

Timing and environment.

DT&E starts early and repeats through component, subsystem, integration, qualification, and certification events in labs, digital environments, ranges, and prototypes. OT&E occurs when the configuration, support package, operators, training, and threat-representative scenarios are mature enough to support an operational adequacy decision before LRIP, FRP, Fleet release, or major follow-on fielding [84, 105].

Who leads.

Program managers, the CDT, LDTO, warfare centers, contractors, and certification authorities normally lead DT&E. Independent Operational Test Agencies such as Commander, Operational Test and Evaluation Force (COMOPTEVFOR), with DOT&E oversight for covered programs, lead OT&E so the Fleet and Congress receive an evaluation independent of program advocacy [126, 129].

Evidence and failure meaning.

DT&E failures are expected risk-discovery events that drive redesign, deficiency correction, reliability growth, or requirements trades. OT&E failures indicate the system may not be ready for the next operational decision; they can delay fielding, restrict Fleet release, require corrective action, or prevent a beyond-LRIP/FRP decision [105, 128].

Note

Board answer: DT&E builds technical confidence while OT&E proves operational fitness. Combined testing may collect shared data, but it does not erase the independence, realism, or adequacy expectations of operational evaluation [126, 127].

5.16.4 Statutory Requirements and Key Roles

DOT&E (10 U.S.C. § 139). Establishes the DOT&E as the independent statutory advisor to the SECWAR and Congress. The DOT&E is responsible for:

- Approving operational test plans and live-fire test strategies for all covered programs.
- Monitoring operational and live-fire test execution.
- Submitting independent adequacy and operational capability assessment reports directly to Congress, the SECWAR, and the MDA before a program can proceed beyond LRIP or enter full-rate production.
- Ensuring programs are realistically stressed under operational threat environments.

10 U.S.C. § 4171. Requires covered programs to complete adequate operational test and evaluation before proceeding beyond LRIP; DOT&E reports independent operational evaluation results to defense leadership and congressional defense committees [105].

10 U.S.C. § 4172. Requires survivability and lethality testing, or an approved waiver, for covered systems and major munitions programs before full-scale production decisions; it also directs reporting of vulnerability and lethality findings to defense committees [106].

DASD(T&E). Serves as the OSW focal point for test resources, designates CDTs for ACAT I programs, charters LDTOs, and co-chairs Defense-level T&E Working Integrated Product Teams [126].

CDT and LDTO. The CDT orchestrates the integrated test schedule, data requirements, and deficiency resolution cadence, while the LDTO (often a warfare center or system command test directorate) furnishes instrumentation, modeling, and range execution authority for DT&E [126].

Operational Test Agencies. Plan and execute independent OT&E to evaluate operational effectiveness, suitability, survivability, and supportability, providing independent evidence for the milestone record. The Navy's agency is the Commander, Operational Test and Evaluation Force (COMOPTEVFOR).

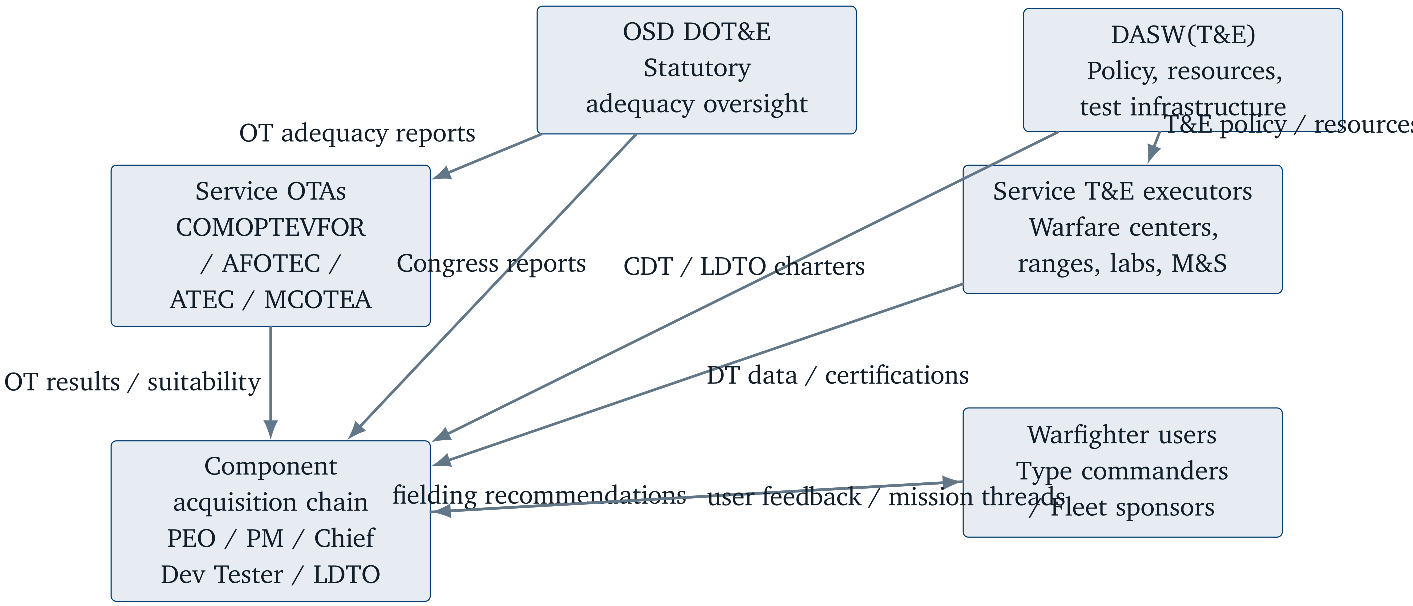

Figure 5.4. *DoW test and evaluation governance chain linking OSW oversight, Service Operational Test Agencies, warfare centers/ranges, and program-level test leadership.*

5.16.5 Developmental Test Activities

Component and qualification testing. Bench, lab, and land-based engineering site events that verify materials, software builds, cyber hardening, interoperability, and safety factors before integration; data feeds configuration audits and reliability growth curves [125].

Subsystem and integration testing. Hardware-in-the-loop, systems integration labs, and shore-based platform integration tests that expose interface defects and mission thread timing issues before at-sea trials [125].

System-level DT&E. Full-up prototypes or first-of-class hulls execute combined developmental/operational scenarios, cybersecurity assessments, and logistics demonstrations to validate readiness for TRR and Technical Evaluation (TECHEVAL) [84].

Reliability growth and maintainability demonstrations. Planned stress events and failure-mode tracking prove that Mean Time Between Operational Mission Failure (MTBOMF) and corrective maintenance timelines align with sustainment strategies before Operational Test Readiness [126].

Certification testing. Airworthiness, safety, information assurance, weapon certification, and electromagnetic environmental effects tests that deliver statutory certifications needed for fielding [125].

Note

Edge case: Software pathway efforts may collapse TECHEVAL/Operational Evaluation (OPEVAL) into continuous DevSecOps assessments, but the program must still show how combined testing and user evaluations satisfy Title 10 adequacy requirements before fielding [127].

5.16.6 Live Fire Test and Evaluation

When required.

Title 10 demands LFT&E for every new or significantly upgraded crew-occupied platform, combat vehicle, major munitions program, or missile defense component that will face threat lethality; full-up system-level shots must be realistically stressed prior to requesting Full-Rate Production approval [106, 126].

When waived.

Only the SECWAR can waive full-up LFT&E when the system is unmanned, when destructive testing would be unreasonably expensive or impractical, or when alternative testing (component shots, validated modeling, allied data) provides equivalent insights; Congress must be notified of the rationale and the alternative plan [106, 126].

Execution and use.

Programs plan vulnerability and lethality events in the TEMP, employ full-up articles plus critical subsystems and surrogates, and collect battle damage, crew survivability, and cascading-effect data to inform design hardening, tactics, and survivability equipment upgrades; the results become an annex to the DOT&E report before the FRPDR [84, 126].

5.16.7 Operational Test Phases

EOA. A qualitative review, often using simulations or limited field events, that informs CDD refinement and acquisition strategy decisions before Milestone B [128].

OA. Focused operational events (sometimes embedded in Fleet Battle Problems or Fleet Exercises) that characterize mission effectiveness, training load, and logistics demands prior to LRIP or dedicated IOT&E [128].

IOT&RE. Navy programs often combine IOT&E and Readiness Evaluation to prove both mission performance and Fleet introduction readiness; Title 10 requires that the IOT&E portion be adequate, realistic, and reported to Congress before FRP [105].

FOT&E. Conducted after FRP or major configuration changes to verify corrective actions, CDD update or legacy CPD thresholds, cybersecurity resilience, and software drops prior to wide deployment [126].

5.16.8 Fundamental Test and Evaluation Terms

COI. The mission-outcome questions (“Can the ship defend itself against a raid?”) that structure OT&E plans and reports [128].

MOE & MOS. Measures of Effectiveness and Suitability quantify warfighting value (probability of raid annihilation) and fleet readiness attributes (reliability, maintainability, training burden) tied to Critical Operational Issues (COIs) [128].

MOP. Measures of Performance (e.g., radar detection range) show lower-level technical behaviors that roll up to MOEs/Measure of Suitability (MOS) [125].

Deficiency categories. Category I deficiencies are mission-critical failures that could cause death, severe injury, or major mission loss; Category II items are significant but less critical issues requiring tracking and correction [128].

Combined testing. Planned events that collect developmental and operational data simultaneously to save schedule and align datasets for evaluation teams [127].

5.16.9 Critical Technical Parameters and Technical Performance Measures

Critical Technical Parameters (CTPs) are the small subset of design attributes that must be controlled to assure a KPP or KSA remains achievable; they are derived from the ICD/CDD and baselined inside the APB [84]. Each CTP maps to one or more TPMs that engineers track continuously using lab data, digital models, or DT&E results; TPMs trends highlight when technical debt threatens TEMP and SEP assumptions so leadership can adjust scope, schedule, or funding. Programs are expected to show a clean trace from COI → KPP/KSA → CTP → TPM → planned test points in the TEMP, ensuring every test objective underwrites an operational outcome and every requirement has an observable verification method [125, 126].

5.16.10 Test and Evaluation Master Plan

The TEMP captures the integrated T&E strategy, schedule, resources, and data management approach across the lifecycle; it harmonizes SEP, LCSP, and APB so that every milestone package shows how risk will be retired [130]. Part I summarizes program context and key performance requirements, Part II defines the integrated test program (DT, OT, certification, modeling, and analysis), Part III outlines resource needs (ranges, threat surrogates, workforce, instrumentation), and annexes address data rights, stats, and cyber accreditation [130]. ACAT I(D) TEMPs require DOT&E approval; other TEMPs are approved by Component Acquisition Executive. Updates are mandatory when requirements change, critical deficiencies emerge, or significant schedule/funding revisions alter test sequencing [84, 130].

5.16.11 How CDD Updates Drive Test and Engineering

TEMP linkage. TEMP validates KPPs and KSAs through test events derived from CDD update production requirements, or from legacy CPD baselines; ensures test methods directly verify production-representative capabilities [84, 107].

SEP linkage. SEP ensures design meets CDD update or legacy CPD requirements through verification methods and engineering allocations; provides technical baseline for production testing [114, 117].

KPP traceability example. KPP (radar detection range) → engineering allocation (antenna gain, processor power) → test method (instrumented range test) → TEMP test event → Milestone C evidence [84, 107].

Integrated verification. SEP and TEMP must align verification methods to ensure CDD update or legacy CPD requirements are consistently validated through both engineering analysis and test events [117, 126].

5.16.12 Modeling and Simulation

Modeling and simulation provide the only practical way to explore the mission design space before hardware exists, rehearse high-risk scenarios (e.g., hypersonic raid defense), and extend sparse live-test data across the operational envelope [125]. Programs must document model purpose, pedigree, uncertainty, and VV&A status in the TEMP; accredited models can bound live-test needs, size instrumentation, and support decision-quality evaluations when safety, cost, or treaties limit destructive testing [126]. Mission engineering digital threads (system-of-systems simulations, campaign models) allow combined DT/OT teams to choreograph events, align data management plans, and feed statistical confidence calculations before and after live events.

5.16.13 MCA Testing Lifecycle

MCA programs must align T&E activities with milestone decision points to provide defensible evidence for phase transitions [84]. DoDI 5000.98, effective December 9, 2024, consolidates OT&E and LFT&E policy previously in DoDI 5000.89; DoDM 5000.100 governs TEMP and T&E strategy content and timing; and DoDI 5000.89 remains the primary source for integrated T&E and DT&E policy [126, 130, 131]. Cyber DT&E policy is specified in DoDM 5000.103, effective February 25, 2026 [132].

Table XLIX: MCA Testing Lifecycle by Phase

Phase / Decision	Primary Test Emphasis	Evidence Needed	Decision Supported	Key Players
Pre-Milestone A / MSA	Technology readiness, feasibility testing, early EOA	TRL 6 evidence, AoA data, initial TEMP outline	MDD approval, AoA guidance	PM, CDT, warfare centers, LDTO
TMRR	Component and subsystem DT&E, competitive prototyping, cyber DT&E planning	PDR completion, technology demonstration, CDD-V inputs, TEMP Part I	Milestone A / DRFPRD	PM, CDT, LDTO, contractor, DASD(T&E)
Milestone B / EMD entry	System-level DT&E, integration testing, early OA, cyber DT&E execution	CDR completion, TEMP approval, APB baselines, OTRR readiness	EMD authorization	MDA, PM, CDT, OTA, DOT&E (covered programs)
EMD	Full-up system DT&E, LFT&E planning, OA, cyber DT&E and OT&E integration	Design stability, TRR success, OA results, approved TEMP/T&E strategy, LFT&E strategy, cyber accreditation	Milestone C / LRIP decision	PM, CDT, OTA, DOT&E, LDTO, warfare centers
Milestone C / LRIP	LRIP start, production-representative articles, IOT&E/LFT&E readiness, cyber OT&E preparation, interoperability certification planning	LRIP quantity approval, production readiness, stable configuration, approved operational test planning, required certifications on track	P&D entry / LRIP / limited deployment	MDA, PM, CDT, OTA, DOT&E, JITC

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Table XLIX: MCA Testing Lifecycle by Phase **(Continued)**

Phase / Decision	Primary Test Emphasis	Evidence Needed	Decision Supported	Key Players
IOT&E / LFT&E	Independent operational evaluation, realistic threat testing, cyber adversarial assessment	Operational effectiveness/suitability, survivability/lethality, cyber mission assurance	Beyond-LRIP / FRP	DOT&E, OTA, COMOPTEVFOR, PM, MDA
FRP / Full Deployment	FOT&E, software increment testing, cyber update validation, sustainment verification	Corrective action validation, CDD update or legacy CPD compliance, cyber resilience, supportability	Full deployment	MDA, OTA, PM, CDT
Post-fielding / FOT&E	FOT&E, software increments, cyber updates, reliability growth	Ongoing effectiveness, deficiency correction, cyber posture, fleet feedback	Sustainment decisions	PM, OTA, CDT, fleet users

Source: *DoWI 5000.85*, *DoWI 5000.89*, *DoWM 5000.100*, *DoWI 5000.98*, *DoWM 5000.103*, 2020, 2020, 2024, 2024, 2026 [84, 126, 130, 131, 132]

5.16.14 Test Type Matrix

Different test types answer distinct questions across the MCA lifecycle [126, 131, 132].

Table L: Test Type Matrix for MCA Programs

Test Type	Primary Question Answered	Typical Timing	Requirements / Planning Considerations	Major Players	MCA Decision Relevance
Contractor Test & Evaluation (CT&E)	Does the design meet contract specifications?	Continuous through EMD	Contract requirements, quality assurance, data rights	Contractor, PM	Design maturity, risk reduction
Developmental Test & Evaluation (DT&E)	Did we build the system right against technical requirements?	MSA through P&D	TEMP Part II, SEP, CTP/TPM traceability	PM, CDT, LDTO, warfare centers	TRR, TECHEVAL, OTRR readiness
Integrated Testing	Can combined DT/OT data support multiple objectives efficiently?	EMD and P&D	TEMP integrated test plan, data sharing agreements, independence safeguards	CDT, OTA, PM	Schedule efficiency, data alignment
Early Operational Assessment (EOA)	Is the operational concept viable for requirements refinement?	MSA / early TMRR	EOA plan, user involvement, representative scenarios	OTA, PM, users	CDD refinement, acquisition strategy

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Table L: Test Type Matrix for MCA Programs **(Continued)**

Test Type	Primary Question Answered	Typical Timing	Requirements / Planning Considerations	Major Players	MCA Decision Relevance
Operational Assessment (OA)	Is the system operationally effective enough to proceed?	Late EMD / pre-LRIP	OA plan, threat-representative environment, logistics support	OTA, PM, fleet	Milestone C / LRIP input
Operational Test & Evaluation (OT&E)	Can warfighters accomplish missions with the system under realistic conditions?	P&D / O&S	TEMP OT annex, DOT&E approval (covered), operational realism	OTA, DOT&E, COMOPTEVFOR, PM	FRP decision, Title 10 adequacy
Initial Operational Test & Evaluation (IOT&E)	Is the production-representative system ready for full deployment?	LRIP to FRP	IOT&E plan, DOT&E oversight, statistical confidence	OTA, DOT&E, COMOPTEVFOR	Beyond-LRIP / FRP statutory gate
Follow-on Operational Test & Evaluation (FOT&E)	Do corrective actions, software drops, or updates maintain operational effectiveness?	Post-FRP / major updates	FOT&E plan, deficiency validation, cyber update testing	OTA, PM, fleet	Sustainment decisions, fielding authorization
Live Fire Test & Evaluation (LFT&E)	Is the system survivable and lethal under realistic combat threats?	EMD through FRP	LFT&E strategy, Title 10 waiver process, threat surrogates	DOT&E, PM, test ranges	FRP statutory requirement (covered systems)
Cybersecurity DT&E	Is the system cyber-resilient against mission-focused threats throughout development?	Continuous DT&E phases	DoDM 5000.103, CVPA, RMF, cyber TEMP annex	PM, cyber certifiers, CDT	TRR, OTRR, ATO/CATO
Cybersecurity OT&E	Can the system maintain mission assurance under cyber attack in operational conditions?	IOT&E and FOT&E	Cyber OT&E plan, adversarial assessment, mission thread testing	OTA, cyber certifiers, PM	FRP, fielding, cyber accreditation
Interoperability T&E / JITC Certification	Can the system exchange and use information with joint/coalition/external systems?	EMD through FRP	Interface Control Documents, JITC test plan, certification timeline	JITC, PM, interface owners	FRP, fielding, coalition operations
Modeling & Simulation / VV&A	Can accredited models supplement or extend live-test evidence?	Lifecycle-wide, especially early and for LFT&E	DoDM 5000.102, VV&A plan, pedigree documentation	PM, CDT, DOT&E, model sponsors	LFT&E waivers, OT&E augmentation, cost savings

Source: *DoWI 5000.89*, *DoWI 5000.98*, *DoWM 5000.103*, *DoWM 5000.96*, 2020, 2024, 2026, 2024 [126, 131, 132, 133]

5.16.15 ACAT-Level Differences

ACAT level affects oversight intensity, MDA level, required documentation, approval authority, and reporting pathways [84]. Non-DOT&E oversight does not eliminate the requirement for adequate tailored T&E; statutory, cyber, interoperability, and LFT&E requirements apply regardless of ACAT [105, 106, 134].

Table LI: ACAT-Level Differences for Test and Evaluation

ACAT Level	Oversight Intensity	MDA Level	Required Documentation	Approval Authority	Reporting
ACAT ID	Highest; DOT&E statutory oversight	DAE (USW(A&S))	Full TEMP, DOT&E reports, SAR, UCR	DOT&E approves TEMP and OT plans	Quarterly DAES, SAR, UCR, DOT&E reports to Congress
ACAT IC	High; DOT&E oversight for covered programs	CAE (ASN(RD&A))	Full TEMP, DOT&E reports (if covered), SAR, UCR	CAE approves TEMP; DOT&E approves OT (covered)	Quarterly DAES, SAR, UCR, DOT&E reports (if covered)
ACAT IB	Moderate; tailored DOT&E oversight	CAE or designee	Tailored TEMP, DAES	CAE or designee approves TEMP	DAES (if designated)
ACAT II	Moderate; Service/component oversight	CAE or PEO	Tailored TEMP	Component Acquisition Executive or PEO	Component reporting
ACAT III / IV	Low; tailored oversight	CAE, PEO, or SYSCOM	Tailored TEMP or T&E Strategy	Designated MDA	Minimal reporting

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Table LI: ACAT-Level Differences for Test and Evaluation **(Continued)**

ACAT Level	Oversight Intensity	MDA Level	Required Documentation	Approval Authority	Reporting
Non-DOT	E oversight	Tailored; Service/component channels	Component MDA	Tailored TEMP or T&E Strategy	Component Acquisition Executive or designee	Component reporting; statutory requirements still apply

Source: *DoWI 5000.85*, *DoWI 5000.89*, *DoWI 5000.98*, 2020, 2020, 2024 [84, 126, 131]

> **Note**
>
> Board trap: Non-DOT&E oversight does not mean no formal T&E. It means T&E is tailored and approved through Service/component channels unless statutory, cyber, interoperability, LFT&E, or other oversight requirements apply [84, 134].

5.16.16 Major Players and Responsibilities

T&E success depends on clear role delineation and collaboration across the acquisition enterprise [126, 131].

Table LII: Major Test and Evaluation Players and Responsibilities

Player	Role	Controls / Approves	Does Not Control
Program Manager (PM)	Overall program responsibility; integrates T&E into acquisition strategy	TEMP, test resources, contractor test plans, deficiency resolution	Independent OT judgments, DOT&E reports, statutory certification decisions
Milestone Decision Authority (MDA)	Approves phase entry, TEMP, and acquisition strategy	Milestone decisions, APB, TEMP approval, ACAT designation	Test execution details, independent OT findings, technical baseline content

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Table LII: Major Test and Evaluation Players and Responsibilities **(Continued)**

Player	Role	Controls / Approves	Does Not Control	
Chief Developmental Tester (CDT)	Orchestrates integrated DT&E schedule, data requirements, and deficiency resolution	Integrated test plan, DT&E resource allocation, deficiency tracking	OT plans, OT execution, DOT&E oversight, certification authority	
T&E WIPT / Integrated Test Team	Coordinates T&E planning and execution across stakeholders	Test schedules, data sharing, cross-discipline coordination	Milestone decisions, budget authority, independent OT independence	
DOT	E	Independent statutory oversight for covered programs; reports to SECWAR and Congress	OT plan approval, LFT&E strategy approval, OT adequacy assessment	DT planning, DT&E execution, certification decisions, program acquisition strategy
USD(R&E) / DTE&A	Sets T&E policy, designates CDTs, charters LDTOs	T&E policy, CDT designations, LDTO charters, test infrastructure	Individual program TEMP content, OT execution, milestone decisions	

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Table LII: Major Test and Evaluation Players and Responsibilities (Continued)

Player	Role	Controls / Approves	Does Not Control
Operational Test Agency (OTA)	Plans and executes independent OT&E	OT plan, OT execution, OT report, OTRR certification	DT planning, contractor testing, program acquisition strategy, certification decisions
OPTEVFOR (Navy)	Navy independent operational test agency for maritime systems	Navy OT plans, OT execution, OT reports, OTRR certification	DT planning, SYSCOM engineering decisions, certification decisions beyond OT
JITC	Joint interoperability certification for information systems	Interoperability test plans, certification decisions	OT execution, DT planning, program acquisition strategy
Requirements Sponsor / User Community / Fleet	Defines operational context, validates mission effectiveness	ICD/CDD, CONOPS, operational scenarios, user feedback	Test planning, technical specifications, certification details

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Table LII: Major Test and Evaluation Players and Responsibilities **(Continued)**

Player	Role	Controls / Approves	Does Not Control
SYSCOM Developmental Test Organizations	Executes DT&E, provides labs/ranges/test infrastructure	DT execution, range scheduling, instrumentation, certification testing	OT planning, OT execution, acquisition strategy
Contractor	Executes CT&E, provides test articles, supports DT&E	Contractor test plans, test article delivery, CT&E data	OT plans, OT execution, independent OT judgments, DOT&E oversight
Threat / Intelligence Community	Provides threat data, intelligence support, and threat-representative targets	Threat assessments, threat surrogates, intelligence products	Test planning, test execution, certification decisions

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Table LII: Major Test and Evaluation Players and Responsibilities (Continued)

Player	Role	Controls / Approves	Does Not Control
Certification Authorities (Cyber, Interoperability, Safety, Airworthiness, Spectrum, Nuclear Surety)	Issue statutory or regulatory certifications	Certification criteria, certification decisions, ATO/CATO	Test planning, test execution, acquisition strategy

Source: *DoWI 5000.89*, *DoWI 5000.98*, *DoWM 5000.103*, *COMOPTEVFOR Official Website*, 2020, 2024, 2026, 2025 [126, 131, 132, 135]

5.16.17 Navy-Specific Test and Evaluation Context

Navy T&E execution reflects maritime operational requirements, shipbuilding constraints, and the distributed nature of naval forces [128, 135].

COMOPTEVFOR as Navy OTA. COMOPTEVFOR reports directly to the CNO, ensuring operational independence from program managers, PEOs, and SYSCOMs. Key responsibilities include:

- Planning, executing, and reporting independent OT&E (including OAs, IOT&E, and Follow-on Operational Test and Evaluation (FOT&E)) for Navy and Marine Corps systems.
- Determining if a system is operationally effective (meets mission needs) and operationally suitable (reliability, training, supportability, and interoperability criteria).
- Serving as the sole certification authority for the Operational Test Readiness Review (OTRR) to certify that a program is ready for OPEVAL.
- Recommending restricted Fleet release to the CNO and MDA if critical Category I deficiencies remain.
- Designing realistic test scenarios and operational test plans, independent of contractor or program office influence.

SYSCOM Support for DT. NAVSEA, NAVWAR, and NAVAIR SYSCOMs support DT&E through warfare centers, engineering sites, labs, ranges, and test infrastructure [125]. SYSCOMs provide technical expertise, instrumentation, certification testing (airworthiness, safety, information assurance, weapon certification), and integration support. SYSCOM developmental test organizations execute DT&E under CDT and LDTO direction but do not replace independent OT.

OPNAV / Resource Sponsors. OPNAV N9 warfare-systems sponsors help define requirements, operational context, and portfolio tradeoffs but do not replace independent OT [128]. N94 is the Test & Evaluation office within the N9 warfare-systems sponsor family, but N94 is not the sole owner of all Navy testing. N94 coordinates T&E policy and resource alignment across the Navy T&E enterprise.

Interoperability Coordination. For systems requiring interoperability with joint or coalition partners, programs must coordinate early with JITC and interface owners to ensure certification timelines align with FRP decisions [136]. Navy-specific interface control working groups (ICWGs) support this coordination.

Cyber Testing Integration. For cyber-heavy systems, cyber DT&E and cyber OT&E must be integrated early in the lifecycle, not deferred until IOT&E [132, 134]. Navy cyber testing leverages NIWC Atlantic and Pacific, FCC, and NCDOC for cyber DT&E and OT&E support.

5.16.18 Common Board Traps

Avoid these high-yield misconceptions when discussing MCA testing [84, 126, 131, 132].

Table LIII: Common Test and Evaluation Board Traps

Trap	Correct Understanding
DT&E proves technical performance; OT&E proves operational mission performance	DT&E answers "did we build the system right" against technical requirements; OT&E answers "did we build the right capability" by validating mission effectiveness and suitability under realistic conditions [126, 131]
Integrated testing erases IOT&E	Combined testing can collect shared data and improve efficiency, but it does not eliminate the need for independent IOT&E with operational realism, adequate sample size, and statistical confidence before FRP [131]
Contractor testing substitutes for independent operational testing	Contractor Test & Evaluation (CT&E) is essential for design maturity but is not independent OT; OTA must conduct IOT&E with operational users and threat-representative scenarios [126]
ACAT level changes the need for adequate evidence	ACAT level changes oversight intensity and approval pathways, but all programs require adequate T&E evidence for milestone decisions; statutory, cyber, interoperability, and LFT&E requirements apply regardless of ACAT [84, 134]
Cyber testing is a late checklist	Cyber DT&E and cyber OT&E must be integrated throughout the lifecycle, starting in MSA and continuing through EMD, IOT&E, and FOT&E; DoDM 5000.103 mandates lifecycle-wide cyber testing [132, 134]

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Table LIII: Common Test and Evaluation Board Traps (Continued)

Trap	Correct Understanding
Interoperability testing is just paperwork certification	Interoperability must be demonstrated in representative operational context through JITC testing, not just documented through interface specifications; certification requires operational validation [136]
LFT&E applies only to specific systems	Title 10 requires LFT&E for covered systems (crew-occupied platforms, combat vehicles, major munitions, missile defense components) before beyond-LRIP decisions; waivers require SECWAR approval and congressional notification [106, 131]
Modeling and simulation requires no validation	Models used for decision-quality evaluations require appropriate VV&A per DoDM 5000.102; unvalidated models cannot substitute for required live-test evidence without appropriate accreditation [137]

Source: *DoWI 5000.85*, *DoWI 5000.89*, *DoWI 5000.98*, *DoWM 5000.103*, 2020, 2020, 2024, 2026 [84, 126, 131, 132]

5.16.19 Navy Overlays and Integration

Office of the Chief of Naval Operations Instruction (OPNAVINST) 3960-series policy (captured in the coursebook) requires Navy programs to plan TECHEVAL (led by the engineering agent) before OPEVAL, to obtain OPNAV N9 concurrence on test objectives that bear on Fleet introduction risk, and to route OTRR packages through COMOPTEVFOR for certification that logistics, training, and safety enablers are in place [128]. First-of-class ships also execute Builder's and Acceptance Trials, Board of Inspection and Survey (INSURV) events, and dedicated combat system Ship Qualification Trials that integrate with TECHEVAL/OPEVAL data packages. Navy Commander, Operational Test and Evaluation Force (COTF) retains authority to recommend restricted Fleet release if Category I deficiencies remain; those actions feed the Key Roles appendix update for Deputy Assistant Secretary of War for Test and Evaluation (DASD(T&E)), CDT, and COTF responsibilities.

5.16.20 Fiscal and PPBE Touchpoints

RDT&E appropriations fund most DT&E events, test articles, range time, and contractor support, while procurement colors such as SCN or Weapons Procurement, Navy buy dedicated OT assets when production-representative hardware is required for statutory adequacy [125, 138]. Operations and Maintenance funds cover Fleet participation, temporary duty, and sustainment of government test ranges. Because OT adequacy is a precondition for FRP and sustainment decisions, Program Objective Memorandum issue papers must protect test range modernization, threat-representative targets, and Modeling & Simulation accreditation costs or risk a PPBE disconnect when COMOPTEVFOR identifies unfunded test objectives.

5.16.21 Assumptions and Uncertainties

- Coursebook Modules 3.5.5 and 3.5.6 (March 2025 editions) remain the latest DON-specific T&E references.
- COMOPTEVFOR manual updates post-2024 were not available offline; Navy-specific terminology (e.g., OPEVAL taxonomy) is based on coursebook guidance.
- Upcoming NDAA changes to Title 10 T&E statutes have not been incorporated pending official publication.

5.17 Ship Trials

5.17.1 Summary

Purpose. Ship trials verify readiness, safety, and contractual compliance before delivery and Fleet introduction [139].

Major trial types. Surface ship trials include Builder's Dock Trials (BDT), Builder's Sea Trials (BST), Acceptance Trials (AT), Integrated Acceptance Trials (IAT), Post Delivery Test and Trials (PDT&T), Final Contract Trials (FCT), Shock Trials (ST), and post-shakedown availability events [139].

Key players. INSURV, OPNAV, NAVSEA, PEO, TYCOM, SUPSHIP, and ship's force each hold defined roles in trials and acceptance [139].

Submarine emphasis. Submarine trials include fast cruise, ALPHA, BRAVO, INSURV combined trials, and Guarantee Material Inspection (GMI), with SUBSAFE certification and Unrestricted Operations (URO) authorization [139].

Certification. Post-sea trial certification requires completion messages, deficiency tracking, and readiness determinations for Fleet operations [139].

5.17.2 Ship Trial Requirements

Title 10. 10 U.S.C. 7304 directs SECNAV to designate a board of officers to examine Navy vessels and report material condition to CNO [139].

OPNAV policy. OPNAVINST 4700.8 series governs trials, acceptance, commissioning, fitting out, and post-shakedown availability for ships under construction or conversion [139].

Mission capable prior to delivery. Ships should be fully mission capable before delivery, except for crew certification or range requirements not possible until after delivery [139].

5.17.3 Key Terms

Accepting authority. Officer designated by the CNO to accept a vessel for the Navy, normally the cognizant PEO [139].

Presenting authority. Officer designated to present the ship to the trial board and certify readiness for trials [139].

OWLD. Obligations for construction and post-delivery work must occur before the Obligation Work Limiting Date (OWLD), typically 11 months after completion of fitting out [139].

Delivery and guarantee period. Delivery is the date the Navy accepts the ship; the shipbuilder remains responsible for correcting defects during the guarantee period [139].

5.17.4 Key Players and Responsibilities

- INSURV provides independent verification of readiness for acceptance and Fleet introduction [139].
- OPNAV makes final determinations of readiness for service; NAVSEA provides formal delivery recommendations to CNO via ASN(RD&A) [139].
- PEO acts as accepting authority for new construction and conversion ships; SUPSHIP prepares, presents, and documents discrepancies [139].
- TYCOM represents the warfighter; Fleet commanders provide trial services; ship's force executes post-delivery trials [139].
- The Naval Supervising Authority assumes custody of the ship's material condition during industrial periods [139].

5.17.5 Surface Ship Construction Trials

BDT. Pier-side tests to determine readiness to safely conduct sea trials and rehearse the AT schedule of events; presented by shipbuilder to SUPSHIP [139].

BST. Conducted after BDT to demonstrate seaworthiness and readiness for AT; presented by shipbuilder to SUPSHIP with NAVSEA and ship's force observers [139].

AT. Conducted after BST to assess contract compliance and operational readiness; INSURV documents discrepancies and recommends acceptance or additional trials [139].

IAT. Consolidates BST and AT before delivery to accelerate schedules for mature production lines [139].

5.17.6 Post Delivery Test and Trials

When. Conducted prior to the shakedown period [139].

Purpose. Executes warfare-area trials such as acoustic trials, propulsion plant examinations, CSSQT (surface) or weapons system accuracy trials (submarines), and ASW/weapon accuracy events [139].

Responsibility. Conducted by the ship's commanding officer and ship's force, with special assistance teams as needed [139].

5.17.7 Final Contract Trials

When. After AT deficiency corrections and prior to post-shakedown availability; before guarantee period expiration [139].

Purpose. Final check for shipbuilder-related defects and operational readiness; similar in scope to AT but executed by Navy personnel [139].

Responsibility. Presented by the shipbuilding program manager to INSURV and observed by NAVSEA, SUPSHIP, and the shipbuilder [139].

5.17.8 Shock Trials

When. Conducted prior to post-shakedown availability for first-of-class ships [139].

Purpose. Demonstrate the ability to fight the ship and satisfy Congressional LFT&E requirements [139].

Responsibility. Shipbuilding program manager oversees; NSWC Carderock designs and instruments trials; ship's CO serves as Officer in Tactical Command [139].

5.17.9 Post-Shakedown Availability

When. After shakedown and prior to expiration of SCN funding and the OWLD [139].

Purpose. Correct new construction deficiencies and accomplish authorized improvements [139].

Responsibility. Supervising authority and ship's CO determine test scope; ship's force operates systems and the shipyard remains responsible for completion [139].

5.17.10 Special Trials

NAVSEA trials. Conducted on representative ships after delivery to determine class characteristics or for ship-specific needs [139].

Examples. Standardization, tactical, acoustical, vibration, fuel economy, heat balance, emergency recovery, and stability/control trials [139].

5.17.11 Submarine Trials Timeline (New Construction)

Fast cruise. TYCOM-directed evolutions for 96 hours with required crew rest to support certification [139].

ALPHA trials. Propulsion plant full-power trials, test depth, ship control, and initial tightness checks [139].

BRAVO trials. Combat systems, navigation, communications, and acoustic performance to prepare for INSURV trials [139].

BRAVO work period. Correct discrepancies and conduct post-dive checks and inspections [139].

INSURV trials and delivery. INSURV conducts combined acceptance and final contract trials; delivery is followed by GMI [139].

5.17.12 Post-Industrial Availability Trials

Purpose. Conducted after repairs to verify readiness for service; scope depends on the work package [139].

Full power. Required after each regular overhaul; other repairs use trial scopes agreed by oversight and the ship's CO [139].

5.17.13 Post-Sea Trials Certification

CNO availabilities. Naval supervising authority, lead maintenance activity, and ship's force conduct a departure conference and issue an availability completion message detailing unresolved work and configuration changes [139].

Submarine requirements. NAVSEA certifies SUBSAFE material condition and recommends URO; TYCOM authorizes URO to test depth [139].

5.17.14 Submarine New Construction vs. Repair

NAVSEA certification. New construction certifies SUBSAFE material condition for the entire submarine; repair certifies only the work performed [139].

Testing. New construction uses multiple at-sea trial periods; repair typically uses one trial period to complete all testing [139].

5.17.15 Ship Program OT&E

Dedicated OT phases. Per COMOPTEVFORINST 3980.2, ships use EOA, OA, Initial Operational Test and Readiness Evaluation (IOT&RE), VCD, and FOT&E to assess operational suitability and effectiveness [139].

Program coordination. MDA and DOT&E align on test requirements; DT&E and OT&E may be combined with SECNAV concurrence, but TECHEVAL and IOT&E are not combined [139].

Pre-Milestone B testing. DT&E and OT&E before Milestone B address individual systems, land-based testing, or validated modeling and simulation when necessary [139].

5.17.16 Quick Review

- Ship trials are mandated by Title 10 and OPNAVINST policy to certify readiness prior to delivery [139].
- BDT, BST, AT, and IAT are the core surface-ship construction trials; PDT&T and FCT follow delivery [139].
- Submarine trials culminate in INSURV combined trials, GMI, and SUBSAFE certification with URO authorization [139].
- Post-sea trial certification and availability completion messages confirm readiness and track deficiencies [139].

5.17.17 Coursebook cue: ship-trial types and roles

Coursebook 3.5.7 expects more than the names of trials. The board answer should connect timing, lead organization, certification purpose, and platform distinctions [139].

Table LIV: Ship-trial board cue.

Trial cue	What to say
Builder's trials	Contractor-led event used to prove readiness before government acceptance events.
Acceptance trials	Government-witnessed event used to support delivery or acceptance decisions.
Final contract trials	Post-delivery event used to verify correction of deficiencies and contract compliance.
Special trials	Focused events for unique systems, mission areas, or certification needs.
Role split	Builder, supervisor of shipbuilding, INSURV, program office, TYCOM, and technical authorities each have distinct interests.
Submarine distinction	Nuclear, SUBSAFE, deep-submergence, and certification constraints can change trial sequencing and evidence requirements.

6 Maintenance

6.1 Fleet Maintenance Assets

6.1.1 Summary

Levels. Fleet maintenance is organized into organizational (O), intermediate (I), and depot (D) levels, each matched to increasing complexity, equipment, and workforce specialization [140].

Activities. O-level is ship's force; I-level includes tenders and shore intermediate facilities; D-level includes public and private shipyards, regional maintenance centers, and ship repair facilities [140].

SUPSHIP and RMCs. SUPSHIPs administer new construction and modernization contracts, while RMCs manage depot projects and provide I-level support and training [140].

Chains and funding. Reporting chains and funding streams differ across organizations and must be aligned for effective maintenance execution [140].

Workforce. Maintenance activities rely on blended military, civilian, and contractor workforces with structured training pipelines and significant early-career demographics at shipyards [140].

6.1.2 Organizational (O) Level Maintenance

Definition. Planned maintenance from the PMS and limited repair performed by ship's force [140].

Typical work. Tasks range from simple lubrication to modular change-outs and valve overhauls [140].

Purpose. Leverages operator experience to keep the ship as self-sufficient as possible [140].

6.1.3 Intermediate (I) Level Maintenance

Definition. Planned and corrective maintenance on shipboard hull, mechanical, and electrical systems and components [140].

Typical work. Routine calibration, gas turbine change-outs, pump and engine overhauls, corrosion control, and rigging/weight testing [140].

Workforce. Sailors and civilians at Intermediate Maintenance Activity (IMA)/Intermediate Maintenance Facility (IMF) sites gain NEC qualifications through OJT-driven repair work [140].

6.1.4 Depot (D) Level Maintenance

Definition. Major industrial work requiring specialized equipment, machinery, and skill sets [140].

Typical work. Docking and underwater hull work, shaft repairs, engine rebuilds, modernization and ship alteration installation, and array resurfacing [140].

Workforce. Executes with skilled civilian and private-sector workforces beyond I-level capability or capacity [140].

6.1.5 Maintenance Trends

- O-level tasking shifted ashore for some platforms (e.g., LCS and DDG-1000), with crews regaining some checks over time [140].
- Regional consolidation of I- and D-level work led to integrated shipyard and intermediate facilities at Puget Sound and Pearl Harbor [140].
- A 2020 GAO review highlighted workforce capacity, unplanned work, planning adherence, facility condition, and IT constraints in surface maintenance [140].

6.1.6 Maintenance Organizations

O-level. Ship's force [140].

I-level. Submarine tenders, Strike Force Intermediate Maintenance Activity (SFIMA), Naval Submarine Support Facility (NSSF), and Trident Refit Facility (TRF) [140].

D/I-level. Naval shipyards and IMFs, U.S. Naval Ship Repair Facility, RMCs, SUPSHIP (minor repair role), private shipyards, and some warfare center divisions [140].

6.1.7 Afloat Maintenance Activities

Submarine tenders. Provide deployable I-level maintenance to submarines, conduct casualty repair, and support Indo-Asia-Pacific sustainment (e.g., USS Emory S. Land and USS Frank Cable in Guam) [140].

SFIMA. Afloat IMA capability using carrier and amphibious platforms to maximize battle force self-sustainment while deployed [140].

6.1.8 Tender Repair Organization (Typical)

Repair officer. Leads the tender repair organization; an EDO O-5 billet in many organizations [140].

Divisions. Hull (R-1), Mechanical (R-2), Electrical (R-3), Electronic (R-4), Radiological Controls (R-5), Repair Services (R-6), Planning (R-7), Outside Machinery (R-8), Quality Assurance (R-9), and Tender Maintenance (R-T) [140].

Support. Repair administration, ship superintendent, and technical library functions align under repair leadership [140].

6.1.9 Submarine Support Facilities

NSSF. Provides I-level maintenance to fast-attack submarines and support craft; works closely with Regional Support Group (RSG) and Nuclear Regional Maintenance Department (NRMD); based in Groton, CT [140].

TRF. Provides industrial support for incremental overhaul and refits of ballistic- and guided-missile submarines; conducts I-level maintenance during refit cycles in Kings Bay, GA and Bangor, WA [140].

TRIPER. TRF supports equipment replacement using a rotatable pool concept under Trident Planned Equipment Replacement (TRIPER) [140].

6.1.10 Naval Shipyards

Mission. Provide affordable, timely, and quality depot-level maintenance, modernization, inactivation, disposal, and emergency repair of carriers and submarines [140].

Capabilities. Maintain dry docks, piers, and production shops; sustain a skilled workforce; provide organic nuclear and conventional repair capacity [140].

Product lines. Submarine availabilities/defueling/refueling; CVN availabilities; and combined inactivation, reactor compartment disposal, and hull recycling at specific yards [140].

Locations and Specializations. The Navy operates four public Naval Shipyards:

- **Portsmouth Naval Shipyard (PNSY)** (Kittery, ME): Specializes in nuclear-powered fast attack submarine (Attack Submarine, Nuclear (SSN)) maintenance, refueling, and modernization.
- **Norfolk Naval Shipyard (NNSY)** (Portsmouth, VA): Specializes in aircraft carrier (CVN) and submarine (SSN/SSBN) maintenance, modernization, and inactivation.
- **Puget Sound Naval Shipyard & IMF (PSNS & IMF)** (Bremerton, WA): Specializes in CVN and submarine (SSN/SSBN/Guided Missile Submarine, Nuclear (SSGN)) maintenance, defueling, inactivation, reactor compartment disposal, and hull recycling. It is the only shipyard that recycles nuclear-powered ships.
- **Pearl Harbor Naval Shipyard & IMF (PHNSY & IMF)** (Pearl Harbor, HI): Specializes in submarine (SSN) maintenance and modernization, and emergent aircraft carrier (CVN) support for the Pacific Fleet.

6.1.11 Ship Repair Facility (Japan)

SRF/JRMC. A non-nuclear depot facility that executes I- and D-level availabilities and provides fleet technical assistance and assessments for Seventh Fleet support [140].

Services. Planning/execution of availabilities by organic workforce (Yokosuka) and contract oversight (Sasebo), plus dive locker and underwater husbandry services [140].

Product lines. Homeported ship support in Yokosuka and Sasebo, plus emergent repair brokered by TYCOM [140].

6.1.12 Supervisor of Shipbuilding (SUPSHIP)

Primary mission. Administer Navy new construction, nuclear repair, and modernization contracts [140].

Key roles. Acts as ACO for shipbuilding/conversion contracts, manages schedules, conducts technical oversight, administers funds, and coordinates logistics support [140].

RCOH. Conducts CVN RCOH at Newport News Shipbuilding and is expanding repair responsibilities at nuclear SUPSHIPs [140].

6.1.13 Major Platform Private Shipyards

- Huntington Ingalls Industries (Newport News Shipbuilding, Ingalls Shipbuilding) [140].
- General Dynamics (NASSCO, Bath Iron Works, Electric Boat) [140].
- Fincantieri Marinette Marine, Austal USA [140].

6.1.14 Regional Maintenance Centers

Mission. Provide D-level project/contract management, I-level repairs and training, fleet technical assistance, Total Ship Readiness Assessment (TSRA), underwater ship husbandry, regional calibration, and logistics support [140].

Organization. RMCs organize around contracts, logistics, financial, corporate operations, environmental safety, engineering and planning, waterfront operations, quality assurance, and production resources [140].

Locations. Commander, Navy Regional Maintenance Center (Commander, Navy Regional Maintenance Center (CNRMC)) oversees six RMCs:

- **Mid-Atlantic Regional Maintenance Center (MARMC)** (Norfolk, VA).
- **Southeast Regional Maintenance Center (SERMC)** (Mayport, FL).
- **Southwest Regional Maintenance Center (SWRMC)** (San Diego, CA).
- **Forward Deployed Regional Maintenance Center (FDRMC)** (Naples, Italy, with detachments in Rota, Spain and Manama, Bahrain).
- **Northwest Regional Maintenance Center (NWRMC)** (Bremerton, WA) – co-located and integrated with PSNS & IMF.
- **Hawaii Regional Maintenance Center (HRMC)** (Pearl Harbor, HI) – co-located and integrated with PHNSY & IMF.

Cooperative Support. Public Naval Shipyards are government-owned and focused on nuclear-powered vessels (submarines and carriers). RMCs manage conventional surface ship maintenance and modernization, executing depot repairs primarily through private contractors (using Multi-Ship, Multi-Option contracts) while providing intermediate-level (I-level) military repair capacity and fleet technical assistance.

EDO Support roles. EDOs provide technical, business, and leadership expertise at these activities:

- At shipyards, EDOs serve as Shipyard Commanders (COs), Production Officers, Project Superintendents, Quality Managers, and Chief Test Engineers.
- At RMCs, EDOs serve as Commanding Officers, Waterfront Chief Engineers (WFCHENGs), Project Managers, and Contracting Officer's Representatives.
- At SUPSHIPs, EDOs serve as Supervisors, Deputy Supervisors, and Administrative Contracting Officers (ACOs) administering new construction and repair contracts at private yards.

6.1.15 Funding and Reporting Chains

Modernization funds. Maintenance and modernization use separate funding streams, including D-Alt and K-Alt modernization accounts; an OPN pilot supports some private contracted availabilities [140].

Operational chain. CNO delegates readiness through fleet commanders and TYCOMs who authorize work and manage maintenance budgets [140].

Technical chain. NAVSEA provides engineering standards and technical authority for maintenance execution at RMCs and NSYs [140].

6.1.16 Responsibilities by Organization

OPNAV N83B. Establishes ship maintenance policy and procedures, allocates shipyard man-days, and publishes depot availability schedules [140].

Fleet commanders. Assess material readiness and resource maintenance funding through Fleet Maintenance Officers [140].

TYCOM. Acts as fleet commander agent for assigned platforms; authorizes work and screens maintenance requirements to RMCs and depots [140].

NAVSEA 04/05. Develop technical requirements and engineering standards, approve maintenance strategies, and set industrial activity policy [140].

SEA 21 and CNRMC. Integrate maintenance strategies, modernization plans, training, and in-service life-cycle management for surface ships [140].

6.1.17 Workforce Composition and Training

Workforce mix. Maintenance activities employ military, civilian, and contractor workforces, with IMA and IMF pipelines providing OJT-based skill development [140].

Shipyard demographics. FY17 Q1 data show a large early-career workforce: Portsmouth 40.7%, Norfolk 46.2%, Puget Sound 43.3%, and Pearl Harbor 35.8% with five years or less experience [140].

6.1.18 Quick Review

- Maintenance levels are O, I, and D; each level increases in complexity and industrial capability [140].
- I-level activities include tenders, SFIMA, NSSF, and TRF; D-level includes shipyards, Ship Repair Facility (SRF), RMCs, and private yards [140].
- SUPSHIPs administer new construction and modernization contracts and hold a minor repair role [140].

6.2 Navy Modernization Program

6.2.1 Summary

Purpose. Navy Modernization Program (NMP) manages ship modernization so alterations are properly engineered, approved, funded, and installed with configuration control and logistics support [141].

Goals. Modernization improves capability, safety, reliability, and fleet readiness; maintenance sustains current performance; repair returns equipment to baseline performance [141].

Governance. NMP provides the authoritative process and data environment for ship change approval, prioritization, and scheduling [141].

Documentation. The Ship Change Document is the single authorized record for ship changes under the NDE entitled process [141].

Policy. Public Law 102-396 limits modernization for ships scheduled for retirement within five years, with national security waiver provisions [141].

6.2.2 Background and Need for Modernization

- NMP began in 2002 to transform maintenance and modernization planning into a more efficient, enterprise-aligned process [141].
- The program addresses emerging threats and new technology while ensuring engineering rigor, configuration control, interoperability, and logistics support [141].
- NMP supports an Optimized Fleet Response Plan (OFRP)-ready fleet by aligning modernization with regional maintenance integration [141].

6.2.3 Goals of Modernization and Definitions

Modernization goals. Improve capability and material condition through approved alterations; improve safety, reliability, maintainability, habitability, and environmental compliance; and increase fleet readiness through timely warfighting support [141].

Modernization. Improves capability above the initial configuration [141].

Maintenance. Allows a ship to continue operating to its current design reliability and performance characteristics [141].

Repair. Returns a ship to its current design reliability and performance characteristics [141].

6.2.4 Goals and Attributes of the NMP

Program goals. Maintain modernization policy and guidance, provide SME support, enable ship change review and approval, and ensure authoritative alteration scheduling [141].

Process discipline. NMP structures identification, planning, design, approval, funding, and installation while maintaining configuration control and supportability [141].

Risk control. Prevents unauthorized, non-supported ship changes and selects funded changes based on technical, readiness, and cost benefits [141].

6.2.5 Modernization Funding

- Modernization funding is divided into D-Alt and K-Alt accounts [141].
- A pilot program allows limited OPN use for ship maintenance, repair, and modernization in private contracted availabilities (started FY20) [141].

6.2.6 Legacy Alterations and SHIPALT

Legacy alteration types. Ship Alteration (SHIPALT) titles, Alteration Equivalent to Repair (AER), equipment alterations (ordnance, machinery, engineering changes, software deliveries, field changes, nuclear alterations, Trident alterations, stand-alone removals, crossdeck alterations), and temporary alterations [141].

Temporary alterations. Temporary ship and equipment alterations require a corresponding removal alteration [141].

SHIPALT definition. Any change in hull, machinery, equipment, or fittings that changes design, materials, number, location, or relationships of component parts; modernization is any change from approved class plans or configuration [141].

6.2.7 Legacy SHIPALT Funding (FMP)

K- and KP-alts. Funded by SYSCOM; require centrally procured material; installed at depot or by ship/AIT depending on title [141].

D- and F-alts. Funded by TYCOMs; D-alts may require centrally procured material and depot or IMA installation; F-alts are installed by ship or IMA without centrally procured material [141].

6.2.8 Alteration Equivalent to Repair (AER)

Approval. Technical alteration approved by NAVSEA [141].

Characteristics. Uses improved materials from standard stock, replaces obsolete or damaged parts with efficient designs, strengthens parts to improve reliability, or implements minor modifications to prevent recurrence of deficiencies [141].

6.2.9 Legacy Documentation

Justification/Cost Form. Prepared by NAVSEA 05 or SYSCOM engineering; approved by the SPM; defines purpose, applicable ships, design/installation/integrated logistics support requirements, and cost [141].

SHIPALT Record. Typically prepared by the planning yard and approved by the SPM; provides detailed materials, test equipment, design/installation/ILS requirements, and cost [141].

6.2.10 Planning Yards

Role. Provide engineering for alteration development, conduct ship checks, prepare drawings and installation data, validate alteration packages, and provide on-site availability support [141].

Relationship. Responsible to the program manager for development engineering and to the installing activity during execution [141].

Examples. Planning yards include Bath Iron Works (DDG-51/1000), Electric Boat (SSBN/SSGN/SSN classes), Huntington Ingalls (LCS, amphibious classes), Newport News (CVN/SSN classes), Naval Shipyards, MSC, and MARAD for specific hull families [141].

6.2.11 NMP Enhancements

Two change categories. Program (K-Alt) and Fleet (D-Alt) alterations replace dozens of legacy types; funding source determines category [141].

Single database. Navy Data Environment (NDE) provides the authoritative ship change database with prioritized change lists [141].

Decision hierarchy. O-6, 1–2 star, and 3-star boards approve changes by threshold and priority [141].

Installation alignment. Installation activities are determined by complexity and assigned to O-, I-, or D-level organizations [141].

6.2.12 NDE Entitled Process and Ship Change Document

NDE-EP. The entitled process supports NMP by collapsing alteration categories, providing a workflow, and enforcing hierarchical decision-making [141].

SCD purpose. The Ship Change Document (SCD) replaces the Justification/Cost Form, SHIPALT Record, and ECP; it is the single authorized document for ship changes in NDE [141].

Authorization. An SCD becomes a ship change after procurement and installation authorization at Decision Point 3 [141].

6.2.13 SCD Initiation and Submission

Initiators. Any user with an NDE account may initiate an idea, including program offices, port engineers, ISEAs, group staffs, TYCOMs, RMCs, and government planning yards [141].

Authorized submitters. Only authorized submitters can forward ideas for board review (about 130 submitters), designated across stakeholder organizations and trained in NMP/SCD processes [141].

6.2.14 Approval Thresholds

Program alterations. Approved based on ACAT designation and monetary thresholds [141].

Fleet alterations. O-6 board approves changes under $50M; 1–2 star board approves $50M–$200M; 3-star board sets priorities, approves Surface Ship Modernization Plan POM submissions, and approves changes over $200M [141].

6.2.15 NDE-NM Capabilities

- Tracks alteration development by category, ship class, and hull; planning and budgeting for programmed alterations; and estimated/actual man-days and costs [141].
- Tracks logistics status, alteration maturity, Ship Installation Drawing status, execution schedules, and which ships have which alterations [141].
- Serves as the authoritative source for all CNO availability dates [141].

6.2.16 Public Law 102-396

Restriction. Prohibits modernization funding for ships planned for retirement within five years of the scheduled completion of the modification [141].

Exceptions. Safety modifications are exempt; SECNAV may waive in the national security interest with written notification to congressional defense committees [141].

6.2.17 Quick Review

- Modernization improves capability and readiness; maintenance sustains baseline performance; repair returns to baseline performance [141].
- Legacy FMP funding: SYSCOM funds K/KP; TYCOMs fund D/F; NMP uses Program vs Fleet categories [141].

- The SCD replaces JCF, SHIPALT Record, and ECP as the single authorized ship change document [141].

6.3 Fleet Maintenance Policy and Implementation

6.3.1 Summary

Policy baseline. OPNAVINST 4700.7 sets ship maintenance policy and requires RCM for new ships, systems, and equipment [142].

Maintenance strategies. RCM underpins Condition-Based Maintenance (CBM) and Time Directed Maintenance (TDM) strategies; the Navy uses CBM to the maximum extent available [142].

Key documents. Class Maintenance Plan (CMP), Integrated Class Maintenance Plan (ICMP), Office of the Chief of Naval Operations Notice (OPNAVNOTE) 4700, and Navy Data Environment - Navy Modernization (NDE-NM) define requirements, intervals, and schedules for depot maintenance [142].

Scheduling governance. Fleet Availability Scheduling Team (FAST) integrates operational requirements with OPNAVNOTE 4700 and recommends schedules to the Fleet Maintenance Board of Directors (FMBOD) for approval [142].

Availability types. CNO scheduled and Fleet scheduled availabilities, plus class-specific maintenance strategies, structure depot maintenance execution across the fleet [142].

6.3.2 Fleet Maintenance Policy

Readiness and service life. Ships are maintained to the highest practical readiness, to meet required operational availability, in a safe material condition, to meet habitability standards, and to achieve expected service life [142].

Right level of maintenance. Work is performed by the maintenance level best suited to accomplish the task, balancing law, urgency, capability, capacity, crew impact, and total cost [142].

RCM requirement. New ships and systems use RCM methodology to define maintenance requirements [142].

6.3.3 Influences on Maintenance Policy

- Force structure: size of fleet and mix of ship classes [142].
- Operational transformation: surge and pulse demands, minimal manning approaches, and increased forward presence [142].
- OFRP alignment, industrial capacity, and the balance of public and private sector capability [142].
- Proprietary and Original Equipment Manufacturer (OEM) information, available funding, and readiness drivers such as continuous maintenance and CBM [142].

6.3.4 Maintenance Principles

RCM. An engineering discipline that determines maintenance requirements based on likely functional failures and their impact on safety, operations, and life-cycle cost [142].

CBM. Uses objective evidence of need (e.g., sensors, inspections, diagnostics) to schedule maintenance and sustain safety, reliability, availability, and cost targets [142].

TDM. Time-directed tasks restore resistance to failure on a fixed interval regardless of condition when no effective condition-directed task exists [142].

Failure-finding tasks. Tests identify hidden failures in protective devices, such as smoke detectors, to ensure readiness [142].

Task effectiveness. Tasks must be applicable and effective to preserve inherent reliability at minimum cost [142].

6.3.5 Maintenance Philosophy

- RCM is the core maintenance philosophy; CBM and TDM are the primary strategies derived from its analysis [142].
- Navy maintenance policy applies CBM to the maximum extent available using RCM rules [142].

6.3.6 Depot Maintenance Framework

OPNAVINST 4700.7. Establishes overall policy, broad O/I/D maintenance concepts, CMP requirements, and organizational responsibilities [142].

CMP. Defines maintenance strategy, availability cycle, and tasks by scope, level, and periodicity; developed by the SPM and technical community based on Technical Foundation Paper (TFP) guidance [142].

TFP. Captures notional class maintenance requirements across expected service life and informs programming and budgeting [142].

Ship sheets. Detail maintenance requirements by hull across the FYDP and support the annual POM process [142].

ICMP. Transitions class plans to a CBM-based, equipment-oriented approach with standard tasks and consistent planning data [142].

OPNAVNOTE 4700. Consolidates class maintenance schedules and publishes notional intervals, durations, and man-days by ship class [142].

NDE-NM. Provides the current and future schedule of depot maintenance by hull and balances operational schedules with industrial workload [142].

6.3.7 CNO Availability Scheduling and Approval

FAST. Integrates OPNAVNOTE 4700 with operational requirements and proposes workload allocations by shipyard capacity, homeport, and private sector availability for FMBOD approval [142].

FAST membership. Includes OPNAV N83, N2/N6, N95, N96, fleet N43 representatives, CNRMC/SEA 04, TYCOMs, NAVWAR, and other PMOs as required [142].

FMBOD. Fleet Maintenance Officers chair the board with NAVSEA 04/08, OPNAV N43, TYCOM N43, and CNRMC/SEA 21 membership [142].

Scheduling artifacts. Workload and Resource Report (WARR) products from naval shipyards and port loading charts from RMCs inform workload and resource decisions [142].

6.3.8 Public-Private Split and Assignment Considerations

50/50 rule. Title 10 U.S.C. Section 2466 requires at least 50 percent of DoW depot maintenance funding to be executed in public depots; no more than 50 percent may go to private industry [142].

Exclusions. CVN refueling and contracted elements of public shipyard work packages are excluded from the 50/50 calculation [142].

NSY assignments. Nuclear ship work, urgent operational commitments, homeport skills, workload capacity, and drydock availability drive public shipyard assignments [142].

Private competition. Conventional surface ships and large deck dry-docking are competed with homeport or coast-wide bids based on availability length and contractor capacity [142].

6.3.9 Budget and Special Cases

- Availabilities often have multiple customers; each contributes maintenance and modernization funding as appropriate [142].
- FYDP maintenance estimates include notional work and emergent growth, while modernization estimates are based on planned alterations [142].
- President's Budget updates refine estimates as execution dates approach [142].
- Urgent repair, emergent dry docking, and collision or battle damage follow the same approval process, with stakeholders informed on scheduling impacts [142].

6.3.10 Types of Availabilities

CNO scheduled. Regular Overhaul (ROH), Complex Overhaul (COH), Engineered Refueling Overhaul (ERO), Phased Maintenance Availability (PMA), Depot Modernization Period (DMP), Planned Incremental Availability (PIA), Docking Planned Incremental Availability (DPIA), Selected Restricted Availability (SRA), Docking Selected Restricted Availability (DSRA), and Continuous Incremental Availability (CIA) are among the scheduled depot availability categories [142].

Fleet scheduled. Technical Availability (TAV), Restricted Availability (RAV), Fleet Maintenance Availability (FMAV), Continuous Maintenance Period (CMAV), and Voyage Repair (VR) are Fleet-scheduled availability types [142].

6.3.11 Maintenance Strategies

EOC. A structured approach to sustain or increase operational availability through periodic inspections, planned maintenance, scheduled availabilities, and modernization; submarine examples include DMP at 120 months and Engineered Overhaul (EOH) at 240 months for 688-class boats [142].

PROG. Supports reduced manning ship classes with limited organizational maintenance, forward deployment, major modernization every 10 years, a rotatable pool, and a dedicated repair facility [142].

Phased maintenance. Depot work is executed through short, frequent PMAs, with repairs authorized by port engineers or maintenance managers and performed in homeport [142].

IMP. Keeps CVN 68 and 78 classes in acceptable material condition using CIAs, PIAs, and DPIAs, with mid-life RCOH still required [142].

Continuous maintenance. Depot-level work flows continuously outside scheduled CNO availabilities, shifting to a decentralized, condition-based process that validates CSMP and maintenance cost knowledge in real time [142].

6.3.12 Optimized Fleet Response Plan

O-FRP objectives. OFRP aligns maintenance, training, evaluations, and a single eight-month deployment within a 36-month cycle to increase readiness and predictability [142].

Availability alignment. O-FRP integrates maintenance and modernization planning for carrier strike groups, amphibious ready groups, and submarines, supporting surge readiness and port loading adjustments [142].

Modernization integration. Modernization improvements are synchronized with integrated maintenance, including Shipboard Operational Verification Test (SOVT) and supporting system verification [142].

6.3.13 Quick Review

- Overall ship maintenance policy is defined by OPNAVINST 4700.7 [142].
- CMP defines maintenance policy for a ship class and is developed by the SPM [142].
- Progressive Maintenance (PROG) is the maintenance philosophy centered on major modernization every 10 years and dedicated repair facilities [142].
- Incremental Maintenance Plan (IMP) applies to CVN 68 and 78 classes [142].

6.4 AIT Fundamentals and Lessons Learned

6.4.1 Summary

Definition. An Alteration Installation Team (AIT) is a trained team under an AIT manager responsible for installing and completing ship changes and alterations on specified ships [143].

Why use AITs. AITs apply specialized skills, leverage lessons learned across installations, and execute large numbers of ship changes efficiently [143].

Governance. AIT alteration development and execution follow the Navy Modernization Process Management and Operations Manual and TS9090-310 procedures aligned with the Joint Fleet Maintenance Manual (JFMM) [143].

Key roles. Sponsors, Regional Maintenance and Modernization Coordination Office (RMMCO), AIT managers, On-Site Installation Coordinator (OSIC), Lead Maintenance Activity (LMA), and Naval Supervising Activity (NSA) coordinate planning, execution, and certification during availabilities [143].

MOA discipline. A MOA is required to align stakeholder expectations and support services before installation begins [143].

6.4.2 Alteration Installation Team

Definition. An AIT is a unit (military, government activity, and/or contractor) trained and equipped to accomplish specific ship changes or alterations on specified ships [143].

Responsibility. AITs are responsible for the installation, performance, and completion of the assigned ship change or alteration [143].

Composition. Teams may be government-led, contractor-led, or a combination of government and contractor personnel [143].

6.4.3 Purpose and Advantages

- Use AITs when technical complexity or specialized oversight is required [143].
- Reuse the same team to capture lessons learned and improve installation quality [143].
- Apply AITs to large numbers of ships when repeatable alterations are required [143].
- Example AIT alterations include CANES, Aegis Weapon System (AWS), and GPS-based Positioning, Navigation, and Timing Service (GPNTS) [143].

6.4.4 Roles and Responsibilities

AIT sponsor. Assigns tasks and funding to the AIT manager; sponsors may include SYSCOM, PEO, PARM, SPM, fleet commanders, TYCOM, or CNO organizations [143].

AIT manager. Assigns and manages the AIT, designates the OSIC, and plans and oversees alteration accomplishment in accordance with modernization policy and procedures [143].

AIT OSIC. Acts with the authority of the AIT manager, manages installation conduct, coordinates with the NSA, and resolves issues and quality discrepancies [143].

AIT. Executes the alteration under the direction of the AIT manager or designated agent [143].

NSA. Integrates and verifies work accomplished during an availability, controls AIT access, ensures authorized work and compliance with technical specifications, and executes quality assurance monitoring [143].

LMA. Integrates all maintenance and modernization activities, coordinates testing controls, and reports work status to the NSA [143].

RMMCO. Serves as the initial point of entry and gatekeeper for waterfront AIT installations, ensuring compliance with TS9090-310 requirements and delivery of logistics products for fleet self-sufficiency [143].

6.4.5 Need for RMMCO

- Rapid 1990s installation of command, control, communications, computers, intelligence, combat system, and hull, mechanical, and electrical improvements created coordination, interoperability, configuration control, and supportability challenges [143].
- Many alterations lacked approved drawings and adequate integrated logistics support products [143].
- Fleet leadership halted uncoordinated installations in 1998 and established RMMCOs in 1999 to enforce integration [143].

6.4.6 Requirements

AIT. The OSIC coordinates with the NSA, installs on a not-to-interfere basis, allows Ship's Force quality monitoring under the JFMM, and follows an approved quality plan [143].

NSA. Certifies all work in the availability, integrates AIT work with other activity work, and coordinates with the LMA for quality monitoring [143].

Ship's Force. Maintains overall responsibility for the safety and security of AIT work [143].

6.4.7 Planning

- Generate the alteration list using ship program manager letters of authorization, TYCOM authorizations, and inputs from PARM, NAVWAR, and NAVSEA 08 as required [143].
- Review alteration scope using the SCD and validate impact on sequencing and availability integration [143].
- Validate AIT quality systems; NAVSEA maintains the list of approved contractor and government quality systems [143].

- The AIT manager coordinates early with the NSA and provides production and test schedules before the availability starts [143].
- AIT managers submit detailed support service requirements to the NSA to integrate work, especially during private availabilities [143].
- RMMCO check-in occurs 30 days prior to installation and requires authorization, integrated logistics support certification or risk assessment, test procedures, approved installation drawings, SCDs, and NDE-NM scheduling confirmation [143].
- Conduct an AIT in-brief to reiterate requirements, deconflict support services, and confirm MOA expectations [143].

6.4.8 Memorandum of Agreement

Purpose. AITs are not fully self-sufficient; the MOA aligns LMA, AIT managers, Ship's Force, and NSA expectations and responsibilities [143].

Required content. Occupational safety, work control and tag-out, work hours, configuration management, quality assurance, cleanliness, support services, clearances, training, and fire safety protocols [143].

Timing. The MOA is signed out by A-1 to confirm stakeholder alignment before availability execution [143].

6.4.9 Execution

Workmanship. The OSIC enforces NAVSEA standard items and maintenance standards; planning yards provide oversight when tasked; housekeeping is required [143].

Testing. SOVT demonstrates operational capability and is included in integrated test plans, including Submarine Safety Program (SUBSAFE) and security requirements [143].

Logistics turnover. Training, documentation, drawings, and allowance parts lists are turned over to the ship, along with spares and special test equipment [143].

Out-brief. The AIT conducts an out-brief with Ship's Force, NSA, RMMCO, and TYCOM; the completion report records status, deficiencies, and command comments [143].

6.4.10 Product Line Integration Walkthrough

The path for a new equipment product line from production to integration into a ship or submarine involves the following standard steps:

1. **System Production**: The manufacturer produces and packages the hardware and software components.
2. **Factory Acceptance Testing (FAT)**: Verifies that the manufactured system meets design specifications and quality standards before shipment.

3. **Staging and Delivery**: The system is shipped to a dedicated staging facility, warehouse, or directly to the ISEA or installation site.

4. **NMP and SCD Approval**: Under the Navy Modernization Process (NMP), a Ship Change Document (SCD) must be formally approved. The SCD details the physical change, weight and moment impacts, and power/cooling requirements, and is signed by the Program Manager and Technical Warrant Holder.

5. **Installation Package Development**: The ISEA or Planning Yard develops the Ship Installation Drawings (SIDs) and drafts the SOVT procedures.

6. **RMMCO Gatekeeping**: The AIT Manager checks in with the Regional Maintenance Modernization Coordinating Office (RMMCO) (typically 30 days prior to installation) to verify the approved SCD, final SIDs, SOVT plan, and certified Integrated Logistics Support (ILS) products.

7. **Waterfront Coordination**: The AIT coordinates with the NSA and LMA to gain ship access. A Memorandum of Agreement (MOA) is signed by A-1 to align support services.

8. **Physical Installation**: The AIT performs the physical installation on board (e.g., running cables, mounting racks, structural modifications).

9. **System Testing**: The AIT and Ship's Force execute the SOVT (including cold and hot checks) to demonstrate operational capability.

10. **Quality Assurance and Certification**: The NSA and AIT OSIC conduct final quality inspections and sign the certification/completion paperwork.

11. **Logistics Handover**: The AIT turns over technical manuals, spares, training, and Allowance Parts Lists (APLs) to the ship's force.

12. **Configuration Update**: The ship's configuration database (CDMD-OA) is updated within 30 days of completion to reflect the install.

6.4.11 Modernization Differences by Class

Modernization processes differ significantly between ship classes depending on their nuclear or safety boundaries:

Conventional Surface Ships. Managed under the standard NMP. Changes are driven by SCDs and gated on the waterfront by the local RMMCO to ensure logistics readiness before install.

Submarines and Nuclear Vessels. Experience extremely tight configuration control due to SUBSAFE, Fly-By-Wire, and Nuclear Power (NAVSEA 08) boundaries:

- **Boundary Control**: Any change within a certified boundary (e.g., SUBSAFE or Fly-By-Wire) requires formal engineering authorization, execution by certified activities (e.g., public shipyards), and rigorous quality audits.

- **Nuclear Systems**: Alterations affecting the reactor plant or reactor support systems require direct, explicit NAVSEA 08 review and authorization.
- **Temporary Alterations (TempAlts)**: Submarine TempAlts are strictly time-bound, logged in active status lists, and must be completely uninstalled before the approved expiration date.

ISEA Involvement. The ISEA serves as the lifecycle engineering agent for the system, responsible for developing the engineering changes, verifying SIDs, authoring SOVT plans, ensuring ILS product availability, and often providing the onsite AIT Manager or technical oversight.

6.4.12 Lessons Learned

- Integration issues arise when AITs, Ship's Force, and industrial activities compete for the same space, systems, or services [143].
- Validate the AIT quality system, ensure authorization is complete, and keep support service requirements current throughout execution [143].
- Provide AIT production schedules early, sign the MOA before the availability, and conduct in-brief and out-brief sessions [143].

6.4.13 Quick Review

- AITs are used when specialized work warrants, installations must be repeated, or many ships require the same change [143].
- AIT installations may be performed by government teams, contractors, or a combination [143].
- RMMCO serves as the gatekeeper for waterfront alteration installations [143].
- The MOA promotes stakeholder discussion and alignment during planning [143].

6.5 Waterfront Organizations, Work Controls, and Safety

6.5.1 Summary

Waterfront focus. Most SYSCOM products are installed during new construction or overhaul periods, so program managers must understand waterfront safety and work control rules [144].

Key nuclear players. Naval Nuclear Propulsion Program (Naval Reactors), Naval Reactors Representative Office (NRRO), Reactor Plant Contractor's Office (RPCO), and Joint Test Groups enforce nuclear safety and testing discipline at public shipyards [144].

Safety programs. The loss of United States Ship (USS) THRESHER drove SUBSAFE requirements, and the loss of USS GRAYBACK established the Deep Submergence Systems (DSS) program [144].

Industrial safety manuals. The GUITARRO accident drove the 6010 manual for submarine ship safety, while the MIAMI fire drove the 8010 manual for fire prevention and response [144].

Work control. All work requires a valid Technical Work Document (TWD) and authorization, with tag-out requirements defined by the Tag-out Users Manual (TUM) [144].

6.5.2 Waterfront Context

- SYSCOM products are installed at the waterfront during new construction, overhaul, and repair periods [144].
- Waterfront culture emphasizes safety of people, equipment, and facilities; program managers may be held accountable for compliance [144].

6.5.3 Naval Nuclear Propulsion Program

Program basis. The Atomic Energy Act of 1954 established the Naval Nuclear Propulsion Program as a joint Department of Energy and Navy effort, with Executive Order 12344 and Public Law 98-525 defining responsibilities, and the FY2000 NDAA relocating Naval Reactors under the National Nuclear Security Administration [144].

NRRO. Ensures reactor safeguards are not compromised by enforcing standards, providing independent assessment, and keeping Naval Reactors informed [144].

RPCO. Represents prime reactor contractors, supports joint test and refueling groups, and can stop work or testing if unsafe conditions exist [144].

Joint Test Groups. Coordinate nuclear and non-nuclear testing with the NSA chief test engineer, NRRO oversight, Ship's Force participation, and reactor contractor technical advice [144].

6.5.4 SUBSAFE and Deep Submergence Systems

Origins. The loss of USS THRESHER led to the SUBSAFE program, and the loss of USS GRAYBACK led to the DSS program [144].

SUBSAFE purpose. The SUBSAFE program provides maximum reasonable assurance of hull integrity and operability of critical systems needed to control and recover from flooding casualties [144].

Fundamentals. Work discipline, material control, documentation (design products and Objective Quality Evidence (OQE)), and compliance verification through inspections, reviews, audits, and certification [144].

General requirements. SUBSAFE certification is maintained through life via re-entry control, the URO program, and certification and functional audits; initial certification is coordinated through the cognizant SUPSHIP [144].

Level I material. Level I material requires testing, traceability to original test reports, and receipt inspection by NAVSEA-approved certifying activities [144].

DSS scope. Covers manned noncombatant submersibles, submarine-oriented DSS including diving systems, and man-rated handling systems, applicable when Navy personnel use any DSS or any personnel use a Navy-owned DSS [144].

Fly-by-wire program. Provides maximum reasonable assurance that submarine fly-by-wire controls will not cause or impede recovery from a jam dive casualty; requires re-entry controls within the flight critical boundary with ISEA support [144].

6.5.5 Industrial Ship Safety Manuals

6010 manual (GUITARRO). The USS GUITARRO sinking led to the 6010 manual, which controls work and testing that affect ship conditions during construction and availabilities, especially for buoyancy, list, trim, watertight integrity, and stability [144].

6010 requirements. Establishes ship safety and ship test organizations, controlling documents (safety plan of the day, daily test schedule, and prerequisite lists), and flooding and fire prevention procedures [144].

8010 manual (MIAMI). The USS MIAMI fire led to the 8010 manual for fire prevention and response in the industrial environment and highlighted the need for early detection, rapid response, and extended response capability [144].

8010 applicability. Applies to all ship repair and construction activities and ship availabilities, requiring MOA agreements on fire prevention, detection, and response roles [144].

Fire safety governance. Fire safety councils coordinate local approvals for hull cuts, fire zones, access routes, alarms, and firefighting systems, and oversee compliance and risk mitigations [144].

6.5.6 Work Authorization and Control

Work control triangle. Training, procedures, supervision, and audit or OQE scale with the complexity and criticality of the work [144].

Unplanned events. Team learning sessions (submarines) or critiques (surface) drive root cause analysis, corrective actions, and adjustment of controls when severity is unacceptable [144].

Minimum requirements. All work requires a valid TWD and authorization to perform work [144].

TWD types. Maintenance procedures, formal work packages, and controlled work packages are selected based on task complexity, worker knowledge, and the need for OQE [144].

Authorization process. All outside activity work requires formal authorization via the Work Authorization Form (WAF) to ensure safety standards in public and private availabilities [144].

6.5.7 Tag-Outs in Maintenance

TUM guidance. The TUM covers repair activity responsibilities, barrier criteria, divers' tags, and maintenance-period tag-out situations, with specific appendices for amplifications and shift operations [144].

New construction. Contractors use occupational safety and health isolation processes until the pre-commissioning crew embarks or, for nuclear work, when the crew takes control of a system [144].

Repair activity responsibilities. Ensure qualified personnel, verify tag-out accuracy, sign tag-outs as the repair activity representative, and remove tags when no longer needed [144].

Divers' tags. Diving activity representatives review tags, brief supervisors on isolations, and coordinate with WAFs when multiple tag-out logs are affected [144].

Double barrier protection. Required for high temperature, high pressure, sea-connected systems, hull penetrations below the waterline, low flash point fluids, oxygen, and hazardous or toxic vapors [144].

6.5.8 Quick Review

- Naval Reactors' primary mission is reactor safety; RPCO reports to Department of Energy Naval Reactors [144].
- THRESHER led to SUBSAFE requirements, while DSS protects occupants of deep submergence systems [144].
- GUITARRO drove the 6010 manual; MIAMI drove the 8010 manual for industrial fire safety [144].
- Minimum work requirements are a valid TWD and authorization; double-barrier conditions follow TUM guidance [144].

6.5.9 Coursebook cue: RPCO and Deep Submergence Systems

The RPCO is a Naval Nuclear Laboratory waterfront presence that oversees technical aspects of naval nuclear work and evolutions. The Resident Manager, RPCO has authority to stop work, operations, or testing when conditions are unsafe, and the RPCO can provide technical advice to nuclear test groups even when not acting as a voting member [144].

The Deep Submergence Systems program grew from the need to protect occupants of deep-submergence systems from material or procedural failures. It applies to Navy-owned or Navy-used deep-submergence systems, including manned noncombatant submersibles, submarine-oriented diving systems, and man-rated portions of handling systems. The board cue is that DSS is a certification and safety-assurance regime, not a generic diving label [144].

6.6 Battle Damage Assessment and Repair

6.6.1 Summary

Why it matters. Battle Damage Assessment and Repair (BDAR) familiarization is critical for EDO career development and wartime readiness [145].

Repair continuum. Phases span damage control, afloat salvage, damage assessment, expeditionary repair, and repair at dedicated facilities [145].

Key enablers. Military Sealift Command (MSC), Mobile Diving and Salvage Unit (MDSU), and Supervisor of Salvage and Diving (SUPSALV) enable salvage; RMC and NSY teams support assessment; Forward Deployed Ship Repair Team (FDSRT), Expeditionary Maintenance and Repair Facility (EMRF), and Shop-in-a-Box (SIB) enable expeditionary repair [145].

Command and control. C2 architecture aligns supported and supporting commands to move, task, and integrate repair forces [145].

Contracting and funding. Contingency acquisition flexibilities and emergent maintenance funding enable rapid response, with fleet commanders responsible for budgeting and salvage [145].

6.6.2 Need for BDA/R

- Renewed great power competition with the People's Republic of China (PRC) and Russia, plus Democratic People's Republic of Korea (DPRK) provocations, increases the likelihood of battle damage and contested logistics [145].
- BDAR planning supports readiness for high-end conflict and distributed operations [145].

6.6.3 Guiding Documents and WARP

Guidance. Strategic and operational guidance includes national security and defense strategy, SECWAR sustainment frameworks, SECNAV policy, CNO maritime sustainment strategy, and fleet planning orders [145].

WARP. The Wartime Acquisition Response Plan calls on every SYSCOM and PEO to establish capacity, capability, data-driven actions, contingency execution, and workforce training for wartime pivot [145].

6.6.4 BDA/R Continuum

Damage control. Ship's Force combats fires, flooding, and casualties to keep the ship operable until disposition is determined [145].

Afloat salvage. Rescue and assistance, towing, emergency repair, and cleanup are provided by MSC, MDSU, and SUPSALV resources [145].

Damage assessment. Ship's Force, salvage forces, and ashore teams (RMC, NSY, NAVSEA) assess integrity and capability to recommend disposition [145].

Expeditionary repair. Forward teams such as FDSRT execute in-theater repairs to restore partial or full mission capability without returning to a depot [145].

Repair. Major repairs occur at public or private shipyards under NSA oversight, with options for complete repair, partial repair, or dismantling for parts [145].

6.6.5 Disposition Factors

- Threat level and danger to rescue and repair personnel [145].
- Operational timelines and the minimum capabilities required to return to the fight [145].
- Extent of damage, mission-critical impacts, and availability of nearby facilities, escorts, and resources [145].

6.6.6 Maritime Support Plan Force Packages

MSP. Maritime Support Plan (MSP) directs NAVSEA to provide maintenance and repair force packages across the spectrum of conflict and serve as lead integrator [145].

Force packages. Afloat augmentation, afloat independent, ashore robust, ashore lean, and ashore austere packages combine personnel and equipment to meet repair requirements [145].

6.6.7 Expeditionary Repair Capabilities

FDSRT. Provides expeditionary repair, BDAR teams, contract management oversight, and fleet technical assist using broad-skill teams sourced from shipyards, RMCs, and warfare centers [145].

SIB. Tiered SIB containers provide tools and materiel for afloat salvage and damage assessment (Tier 1), expeditionary repair (Tier 2), and major expeditionary repair (Tier 3) [145].

EMRF. Mobile expeditionary repair facilities provide containerized capabilities for hull, structural, piping, and combat system repairs [145].

6.6.8 Command and Control

Supported command. Fleet or task force commanders provide movement, lay-down, and tasking for repair forces [145].

Supporting command. NAVSEA provides force packages and maintenance capabilities and retains administrative control while operational tasking flows through the supported commander [145].

Exercises. Ship wartime repair and maintenance exercises, BDAR exercises, expeditionary repair availabilities, salvage exercises, and voyage repair events rehearse repair integration [145].

6.6.9 Contracting and Funding

Contracting. Contingency contracting uses FAR/DFARS emergency flexibilities; letter contracts authorize immediate performance as Undefinitized Contract Actions (UCAs) when definitive contracts cannot be negotiated in time [145].

Ship repair restrictions. U.S.-based ships must be repaired by U.S. companies, with foreign voyage repair limited to restoring return capability; forward deployed naval forces are the exception [145].

Funding. Fleet commanders budget and execute maintenance and modernization, fund emergent repairs under restricted or technical availability programs with OMN, and finance salvage or recovery for assets under their cognizance [145].

6.6.10 Quick Review

- BDAR spans damage control, salvage, assessment, expeditionary repair, and depot repair [145].
- MSP force packages and FDSRT/EMRF capabilities enable forward repair options [145].
- Contingency contracting and emergent funding sources enable rapid repair action in wartime [145].

6.6.11 Coursebook cue: EDO role in BDAR

For BDAR, the EDO role is to translate damage information into safe, executable technical options under time pressure. That requires understanding the ship's mission priority, residual material condition, casualty-control boundaries, technical authority, available parts and skills, and the commander's acceptable risk. The right answer is rarely "restore everything"; it is the repair or workaround that preserves mission, safety, watertight integrity, combat-system effectiveness, mobility, and survivability with the resources available [145].

6.6.12 Peacetime and wartime technical authority

In peacetime, technical authority normally moves through the formal engineering and certification chain before a configuration change. In wartime or contested conditions, the same engineering principles still matter, but the decision may be compressed into rapid risk advice to the operational commander. The board answer should make the trade explicit: what is known, what is unknown, what limit is being accepted, who has authority, and what follow-on repair or certification is required [145].

6.6.13 Contested logistics and sustainment system

BDAR depends on people, parts, and process. People provide damage assessment, engineering judgment, and skilled repair. Parts include onboard allowance, cross-deck material, expeditionary stocks, additive or improvised repair options, and supply-chain reachback. Process includes reporting, triage, technical reachback, configuration control, work authorization, and operational risk acceptance. Contested logistics changes the answer because parts, connectivity, transportation, and depot access may not be available when needed [145].

BDAR course-of-action checklist. State the mission need, safety boundary, casualty data, technical authority, parts and skills available, temporary repair limits, operational risk, and follow-on permanent repair.

6.7 CNO Availability Planning

6.7.1 Summary

Planning balance. Availability planning balances flexible requirements with stable early planning to execute within schedule and resources [146].

AWP sources. Availability Work Package (AWP) work candidates come from CMP/ICMP, NAVSEA-authorized alterations, TYCOM-authorized alterations/repairs/PMRs/baseline AWPs, pre-availability tests and inspections, CSMP, and Fleet Technical Support records [146, 147].

AWP timeline. Public JFMM stages are Baseline, Preliminary, Proposed, Approved, and Completed; current public text supports a Preliminary AWP about 12–14 months before start, Proposed within two months after WDC/PRC, Approved at PAC, and Completed within six months after completion [146, 147].

Work definition. Public shipyards use job summaries, component unit phases, and task group instructions; surface availabilities use 4E work item specifications [146].

Planning guidance. Baseline Project Management Plan (BPMP) and Advanced Industrial Management (AIM) processes guide baseline project management and industrial planning [146].

Pre-start goals. Long-Lead Time Material (LLTM), tooling, early start requirements, critical path, logistics, training, and on-time contract award or definitization must be ready before A-0 [146].

6.7.2 Sources of AWP Work Candidates

- CMP/ICMP routine maintenance and inspections [146, 147].
- NAVSEA-authorized alterations and TYCOM-authorized alterations, repairs, PMRs, and baseline AWPs [146, 147].
- Pre-availability tests and inspections (TSRA, Pre-Overhaul Test (POTS)/Pre-Availability Test (PATS)) [146, 147].
- CSMP items and Fleet Technical Support records [146, 147].

6.7.3 Pre-Start Planning Goals

- Order LLTM, have special tooling on-site, and complete early start requirements [146].
- Define critical path and key events in the integrated schedule [146].
- Resolve logistics and infrastructure needs (power, berthing, galley, force protection, LAN migration) [146].
- Train the crew on shipyard procedures and secure on-time contract award or definitization [146].

The most important planning output is a fully integrated schedule of all availability work [146].

6.7.4 Public Shipyard AWP Development

Baseline AWP. Planning requirements from SUBMEPP (submarines) or Program Manager, Ships (PMS) 312C (aircraft carriers); the A-36 cue is retained as coursebook planning shorthand [146, 147].

Preliminary AWP. Consolidates baseline AWP, authorized alterations, and repairs; public JFMM text places this about 12–14 months before start [146, 147].

Proposed AWP. Issued within two months after WDC/PRC; coursebook cues include TYCOM review of pre-availability tests and inspections, Initial Planning Meeting (IPM) for submarines, and A-12 carrier planning [146, 147].

Approved AWP. Approved at PAC; the A-3/post-final cue is coursebook shorthand for final planning updates [146, 147].

Completed AWP. Issued within six months of availability completion as requirements gain fidelity over time [146, 147].

6.7.5 RMC AWP Development

Baseline AWP (BAWP). Developed by Surface Maintenance Engineering Planning Program (SURFMEPP) using CMP, TFP, and ship sheets [146].

Availability Work Package. Refines Baseline Availability Work Package (BAWP) at A-360 with assessment results, CSMP, and modernization to balance life-cycle maintenance and readiness [146].

6.7.6 Public Shipyard Work Definition

Job Summary (JS). Groups work into logical evolutions with task lists, material, trades, and man-hour estimates [146].

Component Unit Phase (CU Phase). A specific phase of work for a specific Component Unit (CU) [146].

Task Group Instruction (TGI). Detailed instructions for tasks within a CU phase [146].

Component Unit (CU). The physical unit of work control [146].

CU phases. A (repair onboard), C (calibrate/align), D (open/inspect), E (restore prior to test), F (fabricate), H (repair off-ship), I (inspect), K (clean/prepare), M (maintain), P (paint/preserve), R (reinstall/reassemble), S (stage/prepare), T (test), U (unship/remove) [146].

6.7.7 Integrated Schedule (Public Shipyard)

1. Define the authorized AWP and establish key events and milestones [146].
2. Develop job summaries and sequence CU phases [146].
3. Build the network diagram (PERT), integrate Ship's Force/contractor/AIT work, and estimate resources, time, and cost [146].
4. Determine the critical path and produce the integrated schedule [146].

6.7.8 Critical and Controlling Paths

Critical path. Longest duration sequence of work with the least float that sets minimum project length [146].

Controlling path. Near-critical activities with slightly more float that can still drive key events due to scope, material, or complexity [146].

6.7.9 RMC Work Definition and Schedule

4E work item specifications. Contractual statements of work defining repair or alteration requirements in standard format for Master Ship Repair Agreement or Agreement for Boat Repair [146].

Integrated schedule. Defines AWP, phases, key events, milestones, network sequence, and critical path; LMA generates the schedule, contracted through NAVSEA Standard Item 009-60, with RMC review and acceptance [146].

6.7.10 Platform Differences

Baseline planning owners. SUBMEPP (submarines), PMS 312C (aircraft carriers), and SURFMEPP (surface) maintain the baseline AWP [146, 147].

Planning organizations. Submarines and carriers rely on public/private shipyard planning products; surface ships use contractor or third-party Specification Development and Availability Execution Support (SDAES) planning with 4E specifications [146].

Common sources. All platforms use the same work candidate sources, but planning timelines differ across platforms [146].

6.7.11 Planning Team Roles

BSPO. Customer interface and funding authority; processes new work and funding [146].

Project superintendent. Leads cost, schedule, and performance; oversees project planning and sequencing [146].

PEPM. Principal planner who directs Job Summary (JS) boundaries, planning products, and schedule development [146].

Project engineers/test engineers. Support planning, sequencing, and technical oversight [146].

Port engineer. TYCOM representative who screens work to the proper maintenance organization [146].

RMC project manager. Leads the RMC project team, ensures funding for planning, and screens work to contractor execution [146].

LMA project manager. Contractor LMA point of contact who leads integrated schedule development and coordinates with the RMC [146].

SMMO. Ship's Force point of contact for maintenance; assists with screening and 2K development [146].

6.7.12 RMC Contracting Strategy (MACMO)

MAC-MO. Firm-fixed-price surface ship repair contracts that separate planning from execution (SDAES or organic planning), maximize competition, and balance risk between Navy and industry [146].

Award approach. CNO availabilities of 10+ months compete coast-wide; under 10 months award in homeport; continuous maintenance uses homeport competition; emergent work uses existing contractor, competition if time allows, or rotational IDIQ awards if immediate [146].

6.7.13 Quick Review

- Core AWP source buckets: CMP/ICMP, authorized alterations and repairs, pre-availability tests and inspections, CSMP, and Fleet Technical Support records [146, 147].
- Baseline AWP producers: SUBMEPP (subs), PMS 312C (aircraft carriers), SURFMEPP (surface) [146, 147].
- The most important planning product is the integrated schedule [146].
- Multiple Award Contract, Multi-Order (MAC-MO) uses fixed-price competition to drive cost down and improve accountability [146].

6.7.14 Coursebook cue: Team One comparison and planning products

Coursebook 4.2.1 compares submarine, carrier, and surface planning to show that the planning logic is common even when the owning organizations and products differ [146].

Table LV: Availability planning comparison.

Community	Baseline AWP owner	Planning cue
Submarine	SUBMEPP	Public or private shipyard plans and sequences work using job summaries and task group instructions.
Carrier	PMS 312C	Carrier Team One focuses on aircraft-carrier maintenance performance and uses similar work-package logic.
Surface	SURFMEPP	Surface Team One focuses on surface-ship service-life performance; planning may use contractor or third-party Specification Development and Availability Execution Support with 4E work-item specifications.

Table LVI: Availability planning products.

Product	Use
Availability Work Package	Authoritative list of planned work for the availability.
Job Summary	Planning record containing the information needed to plan and schedule a work evolution.
Control Unit phase	Phase of work performed on a specific control unit.
Task Group Instruction	Detailed work instructions and technical information for a control-unit phase.
4E work-item specification	Surface-ship statement of work for the planned repair or modernization item.
Integrated schedule	Networked schedule that links authorized work, key events, critical path, and execution constraints.

6.8 CNO Availability Execution

6.8.1 Summary

Project teams. Public shipyard and RMC teams define roles from superintendent and manager through zone execution, with Ship's Force and contractor coordination embedded in the team [148].

Execution flow. CNO availability execution moves through nesting/ship arrival, rip-out and repair, open and inspect, progress assessment, and testing, trials, and certification [148].

Schedule language. Key events, milestones, Task Group Instructions (TGIs), and work items frame schedule progress against the critical path [148].

NSA/LMA. The NSA certifies completion while the LMA integrates execution and work control, including WAFs, tag-outs, and test sequencing for CNO availabilities [148].

Certification discipline. NSA certification depends on executing activity qualifications, integrated test planning, and OQE review and approval [148].

Momentum. Second-half performance hinges on documentation quality, staffing continuity, and coordinated testing and trials [148].

6.8.2 Public Shipyard Project Management Team

Project superintendent. Equivalent to the CO in the chain of command, with total responsibility for cost, schedule, safety, and quality; assembles the project team, develops the project management plan and execution strategy, and serves as the single shipyard point of contact with the ship CO [148].

Deputy project superintendent (DPS). The Deputy Project Superintendent (DPS) is equivalent to the Executive Officer (XO); assigned for large, complex projects or those with significant nuclear work, and performs duties assigned by the project superintendent [148].

Assistant project superintendents (APS). The Assistant Project Superintendent (APS) role is equivalent to a department head; APSs are experienced in all phases of their assigned area with delegated authority for non-technical decisions and recommendations. The Ship's Force representative on the project team may be titled an APS or overhaul coordinator; a nuclear-trained limited duty officer may fill the role for nuclear availabilities [148].

Project engineering and planning manager. The Project Engineering and Planning Manager (PEPM) reports to the project superintendent; directs project planning and management, job planning, and job sequencing/scheduling, develops the planning budget, and defines JS boundaries [148].

Business and strategic planning office. The Business and Strategic Planning Office (BSPO) reports to the project superintendent; obtains project funding, identifies authorized work, authorizes the expenditure of funds, and works directly with the customer on cost and schedule [148].

Zone managers (ZM). Zone Managers (ZMs) are equivalent to division officers; they report to the appropriate APS, control execution within their zone, supervise first-line supervisors, and develop daily production schedules [148].

6.8.3 RMC Project Management Team

Port engineer. Maintenance team leader, TYCOM representative, and resident expert in ship maintenance [148].

Project manager. RMC organization lead responsible for the entire availability; ensures work is certified after completion, approves changes to the availability contract with the KO, screens work to the LMA contractor, and manages funding [148].

Ship superintendent. Manages RMC production work at the intermediate level and assists in work sequencing and deconfliction [148].

Ship's commanding officer. Responsible for overall ship material condition, the ship's annual maintenance budget, and coordination of the maintenance team [148].

Ship's maintenance and material officer. The Ship's Maintenance and Material Officer (SMMO) is the Ship's Force point of contact for maintenance items; liaison between ship, maintenance activity, and RMC; assists in screening work and work package development; coordinates with the project manager [148].

Contractor program manager. Point of contact between the maintenance team and contractor workforce; assists in work sequencing and deconfliction and coordinates with the project manager [148].

Support specialists. Project officer (if assigned), shipbuilding specialists, project support engineer, administrative contract specialist, quality assurance specialist, financial and material specialists, government property manager, integrated test engineer, alteration coordination engineer, and AWP manager provide focused support and may be shared across projects [148].

6.8.4 Execution Phases

Nesting/ship arrival. Welcome Ship's Force as a member of the project team; train the crew on fire watch, tag-outs, work control (JFMM), and shipyard procedures; explain access and temporary services plans; coordinate berthing barge arrangements; execute availability agreements (MOA) and the 8010 fire drill; and begin system turnover from Ship's Force to the shipyard. The ship becomes an industrial site (public shipyard codes 246 and 2340, safety office code 106), so training, temporary services, and safety dominate early execution [148].

Rip-out & repair. Start as soon as possible; expect access issues, unexpected hazardous material, and tag-outs as common roadblocks. Push equipment to shops early to uncover surprises, recognize that "last off ship" often means "first back on ship," and manage labor intensity so activity translates into progress [148].

Open & inspect. Conduct inspections onboard, inside shops, and on Ship's Force work items; NAVSEA Standard Item 009-001 requires reports that could change or add work by the first 20% of the availability unless approved otherwise, and Standard Item 009-060 supports integrated production schedule planning of O&I reports by the first 20% plus 25% review impact analysis. Use inspection reports to authorize work, capture new and growth work, adjust anticipated repair quantities, and report schedule and cost impacts to financial customers (TYCOM/fleet). Treat the A+60% growth/new-work-on-contract rule as a coursebook or local planning cue unless the local contract invokes it [148, 149, 150].

Progress assessment. Monitor the critical path continuously because slippage or new work can change it at any point; assess progress at the key event, milestone, TGI (public shipyards), and work item/task levels [148].

Testing, trials, & certification. Enter the testing phase only after meeting the Production Completion Date (PCD); for AIT work, production completion does not substitute for checkout, because completed/signed alteration completion records plus completed SOVT support final acceptance. Coordinate tests and trials with Ship's Force, the shipyard, and the customer so test scope aligns to the work performed and certification requirements [148, 151].

6.8.5 New Work, Growth Work, and Anticipated Repairs

New work. Additional work identified or authorized after contract award that is not related to a work item in the original contract; may be discovered when equipment fails or deficiencies are found post-award [148].

Growth work. Additional work identified or authorized after contract award that expands a work item already in the original contract; often expected where work is difficult to scope fully before inspection (for example, tanks) [148].

Anticipated repairs. Work not fully quantifiable before the availability starts but expected based on historical lessons and planned for in budget, resources, and schedule before inspection [148].

New and growth work are common sources of schedule delay and must be reviewed and agreed to by the maintenance/project team; in some cases the TYCOM (or other government representative) must authorize work before it begins based on requirements, risk assessment, and schedule impact [148].

6.8.6 Schedule Terminology

Key event. High-level breakdown of principal activities with hard dates set for completion [148].

Milestone. Lower-level event that supports higher-level key events [148].

Task group instruction (TGI). TGI are used in public shipyards to manage discrete work packages and execution control [148].

Work item or task. The smallest execution unit used to track progress and assess schedule performance [148].

6.8.7 Testing, Trials, and Certification Details

PCD definitions vary. Clarify the project definition of PCD early because it must be met before the testing phase begins [148].

Testing governance. Public shipyard testing is supervised by the chief test engineer; tests are grouped for efficiency, scope depends on the work performed, and early testing mitigates schedule impacts from deficiencies. Ship's Force operates equipment during shipyard-administered tests, requiring tight coordination [148].

Trials and certification. The need for sea trials, crew certification, Light-Off Assessment (LOA), CSSQT, and (preliminary) operational reactor safeguards exams depends on the scope and length of work and the platform; the exact plan is determined ahead of time with Ship's Force, the shipyard, and the customer [148].

6.8.8 NSA and LMA Roles

Naval Supervising Activity. The NSA is responsible for certification of work accomplished during any type of availability and may be a NSY, SUPSHIP, or RMC. The NSA ensures work is authorized and completed in accordance with applicable technical requirements and maintenance policy and must possess a NAVSEA technical warrant [148].

Lead Maintenance Activity. The LMA is responsible for work being accomplished during any type of availability; coordinates work and testing controls including WAFs, tag-outs, and test sequencing (CNO availabilities only), integrates the work of all activities, tracks progress of maintenance execution, and provides the sea trials agenda for CNO availabilities [148].

CNO availability pairing. For CNO public shipyard availabilities, the public NSY is both NSA and LMA; for CNO private-sector availabilities, the RMC or SUPSHIP is NSA and the contractor is LMA [148].

6.8.9 Availability Work Certification

Certification plan. The NSA certification plan verifies work is completed and technically correct by confirming executing activity qualifications, NSA approval of mandatory technical requirements, an integrated test plan, and adequate NSA oversight of all availability work [148].

OQE. Each executing activity must provide OQE showing technically correct completion of work [148].

Waivers and deviations. NSA review and approval of waivers and deviations ensures exceptions do not adversely impact key event completion, providing maximum reasonable assurance the availability is complete and technically correct [148].

6.8.10 Lessons Learned

Keep momentum. First-half enthusiasm fades as undocking approaches, equipment installation peaks, and new work emerges; the test program accelerates at the same time [148].

Second-half obstacles. Documentation deficiencies, daily critical path changes, SOVT test deficiencies, Ship's Force manning shortfalls and turnover, inspection team scheduling, and material failures during waterborne testing can derail execution [148].

Second-half helpers. Maintain strong communication, build project team links across engineering/planning, planning yard, test engineering, supply, and the ship's Chief Engineer (CHENG), keep Ship's Force trained for testing, ensure habitability is ready for move aboard, keep supply informed on material needs, track completion risks, address fast cruise discrepancies early, and maximize sea trial effectiveness [148].

6.8.11 Quick Review

- The PEPM directs project planning and management, job planning, and job sequencing and scheduling for public shipyard availabilities [148].
- The BSPO obtains funding, identifies authorized work, and works directly with the customer on cost and schedule [148].
- The port engineer serves as the RMC maintenance team leader [148].
- Typical execution phases are nesting/arrival, rip-out and repair, open and inspect, progress assessment, and testing, trials, and certification [148].

6.9 Availability Work Package Requirements and Development Stages

The current JFMM landing page identifies COMUSFLTFORCOMINST 4790.3 Rev E dated 2026-04-27; detailed public text available for this board-prep summary was checked against recent Rev D volume changes, so use chapter-level paragraph details with that caveat. The JFMM defines the AWP as the consolidated industrial work plan for a scheduled maintenance availability, combining maintenance, modernization, and repair requirements drawn from the CMP/ICMP, NAVSEA-authorized alterations, TYCOM-authorized alterations/repairs/PMRs/baseline AWPs, pre-availability tests and inspections, CSMP, and Fleet Technical Support records. [146, 147] The AWP progresses through sequential stages (Baseline, Preliminary, Proposed, Approved, and Completed), with public text supporting Preliminary AWP development about 12–14 months before start, Proposed AWP within two months after WDC/PRC, Approved AWP at PAC, and Completed AWP within six months after availability completion. [146, 147] Baseline AWP owners are SUBMEPP for submarines, SURFMEPP for surface ships, and PMS 312C for aircraft carriers. [146, 147]

6.10 Planning Milestones and Availability Type Differences

Planned milestones for SRA and other CNO-scheduled availabilities establish early schedule alignment, work package integration, and contracting actions, with key gates at roughly A-720 (availability schedule), A-360 (letter of authorization), A-220 (work package integration conference), A-165 (100 percent work package lock letter), A-120 (contract award), and A-45 to A-30 (work package execution review) before A-0 start of availability. [146, 147, 148] By contrast, CMAV planning compresses the cycle, with planning beginning around A-180, integration and ship checks closer to A-60 to A-20, and a shortened execution review near A-10 before A-0 start. [146, 147, 148] Both paths use progress conferences during execution and culminate in completion reporting and lessons learned capture after availability closeout. [148]

6.11 System Operational Verification Test Definition and Significance

The SOVT is the structured verification that an alteration or ship change performs as intended following maintenance, and AIT or ISEA teams must conduct these tests under conditions that simulate normal service; current authoritative TS access may be CAC-controlled, so this summary is based on the available TS9090-310G/public-coursework alignment. [151] The alteration is not complete until the SOVT or stage testing is successfully executed and witnessed by Ship's Force, and test reports must be provided to Ship's Force, the NSA, and the LMA. [151] SOVT completion is distinct from the PCD: production completion supports readiness for checkout, while a completed/signed alteration completion record plus completed SOVT supports AIT checkout and closeout. System integration testing is complete only after successful testing with Ship's Force witnessing and signoff. [151]

6.12 Signed and Unsigned System Operational Verification Tests

A signed SOVT is a completed test plan that has been executed, witnessed, and recorded with results and discrepancies, enabling formal acceptance and closeout of the alteration. [151] An unsigned SOVT indicates the test was not fully executed or witnessed, or that discrepancies remain unresolved, which delays acceptance and introduces schedule risk through retesting or deferred closeout actions. [151]

Current as of 2026-05-27.

7 Leadership and Professional Development

7.1 Naval Justice and Standards of Conduct

7.1.1 Summary

Legal resources. Command leaders rely on Commander's Quick Reference Legal Handbook (QUICKMAN), Staff Judge Advocates (SJAs), legal officers/clerks, and regional legal support to spot issues early and obtain advice before acting [152].

Purpose of military law. Military law promotes justice, good order and discipline, and national security while reinforcing the obligation to follow the Law of Armed Conflict (LOAC) under stress [152].

Investigations. Commands use preliminary inquiries and command investigations to gather facts, while certain events must be referred to Naval Criminal Investigative Service (NCIS) [152].

Discipline tools. Non-punitive measures, Non-Judicial Punishment (NJP), and Administrative Separation (ADSEP) provide graduated responses to misconduct, with defined rights and procedures [152].

Ethics baseline. Standards of conduct, gift rules, political activity limits, and fundraising constraints protect public trust and individual careers [152].

7.1.2 Legal Resources and Roles

Staff Judge Advocate. The SJA advises the commander on legal risk, considerations, and options; commanders make the final decision [152].

Regional legal support. Regional Legal Service Office (RLSO)/Legal Services Support Section (LSSS) command services attorneys provide command advice, while RLSO/LSSS legal assistance attorneys counsel individuals on civil matters (wills, family law, landlord-tenant, and similar issues) [152].

Defense and victim counsel. The Defense Service Office (DSO) advises personnel facing disciplinary action, while Victims' Legal Counsel (VLC) support qualifying victims and coordinate through victim advocates or Sexual Assault Response Coordinator (SARC) channels [152].

Command legal enablers. Legal officers/clerks and legalmen are unit-level resources, and the QUICKMAN helps spot legal issues but does not replace SJA advice [152].

7.1.3 Legal Authorities and Purpose of Military Law

Hierarchy of law. The Constitution is supreme, Congress enacts statutes, the Executive issues orders and regulations, and military orders derive authority from higher law and leaders (including SECNAV and CNO) [152].

Purpose. Military law promotes justice, deters misconduct, and maintains good order and discipline to strengthen national security and enable LOAC compliance in combat stress [152].

7.1.4 Investigations and Referrals

Administrative investigations. Common types include preliminary inquiries, command investigations, litigation-report investigations, admiralty letter reports, dual purpose investigations, and courts/boards of inquiry [152].

Preliminary inquiry. A rapid fact-gathering step (typically within three days) led by the convening authority or a designated officer, used to decide on next actions (no action, further investigation, consult a judge advocate, or disciplinary action) [152].

Command investigation. A formal, written investigation led by an investigating officer, usually completed within 30 days, used for significant events (loss/destruction, collisions, some line-of-duty cases, and some deaths) with review by the General Court-Martial Convening Authority (GCMCA) unless the matter is internal to the unit [152].

Mandatory NCIS referral. Refer major criminal offenses, sexual assault reports (when not restricted), non-combat deaths not due to natural causes, missing members where foul play is possible, loss/compromise of classified information, and fires or explosions of unknown origin [152].

7.1.5 Rights Advisement and Statements

Article 31(b) rights. Uniform Code of Military Justice (UCMJ) Article 31(b) includes the right to remain silent and advisement on the consequences of waiver; the right to counsel applies to custodial interrogation, and commands should use the standard rights advisement forms [152].

Cleansing warnings. When a prior statement may be inadmissible, a cleansing warning is used; while rules of evidence do not apply to NJP or ADSEP, rights violations can affect court-martial admissibility and appeals [152].

7.1.6 Search Authorization, Inspections, and Inventories

Search basics. A search is a government quest for evidence where a reasonable expectation of privacy exists [152].

Consent vs. command authorization. Best practice is to request consent first (Permissive Authorized Search and Seizure (PASS)), with Command Authorized Search and Seizure (CASS) used when consent is questionable or revoked; obtain authorization in writing when possible [152].

Probable cause. Command-authorized searches require probable cause that a crime occurred and evidence is located in the place to be searched; commanders must not rely solely on another's determination [152].

Inspections. Inspections and inventories are not searches when primarily for military purpose (security, fitness, good order, readiness). Unscheduled or targeted inspections can be improper and risk suppression of evidence [152].

7.1.7 Line of Duty Determinations

When required. A line-of-duty determination is required for injuries or disease that may result in permanent disability, inability to perform duties for more than 24 hours, or death; a preliminary inquiry is the minimum starting point [152].

7.1.8 Non-Punitive Measures and Non-Judicial Punishment

Non-punitive measures. Extension of working hours, liberty risk, and extra military instruction are tools for correcting minor misconduct before formal punishment [152].

NJP authority. COs may award NJP to all command members (including those on temporary additional duty); Officer in Charges (OICs) may award NJP to enlisted members only, and authority cannot be directed but may be withheld for specific offenses or categories [152].

DRB and XOI. The Disciplinary Review Board (DRB) and Executive Officer Inquiry (XOI) provide fact-finding and corrective-action screening before NJP [152].

Right to refuse. Sailors may refuse NJP unless attached to or embarked on an operational vessel; the right can be exercised until punishment is awarded and does not create a demand for court-martial [152].

Rights at NJP. The accused may be present, remain silent, have a personal representative, examine evidence, present extenuation/mitigation, and request an open hearing (public is not the same as all-hands mast) [152].

Appeals. NJP appeals are normally submitted within five working days; members have a right to consult defense counsel, and appeals are limited to unjust or disproportionate punishment reviewed by the GCMCA for Navy cases [152].

7.1.9 Administrative Separation

Purpose. ADSEP separates enlisted members who do not meet performance, conduct, or discipline standards after reasonable efforts at counseling, retraining, and rehabilitation [152].

Reasons. Bases include misconduct, unsatisfactory performance, drug and alcohol rehabilitation failure, defective enlistments, security concerns, convenience of the government, and separation in lieu of trial by court-martial [152].

Common misconduct bases. Drug abuse, commission of a serious offense, and a pattern of misconduct (two or more incidents in the same enlistment) are common bases for processing [152].

Mandatory processing. Mandatory bases include sexual misconduct, certain sexual harassment, violent misconduct with severe outcomes, drug paraphernalia or abuse, extremist conduct, family advocacy program failure, alcohol rehabilitation failure, second DUI, and nonconsensual distribution of intimate images [152].

Procedures and boards. Notification procedures apply when the least favorable characterization is honorable or general; board procedures apply when the least favorable characterization is other than honorable and include a formal board unless waived [152].

Board findings. Boards determine whether a basis is met, whether to separate or retain, and the characterization (honorable, general, or other than honorable) [152].

7.1.10 Ethics and Standards of Conduct

Ethics foundation. The standards of ethical conduct (5 C.F.R. Part 2635) and the Joint Ethics Regulation (JER) are punitive and enforce the principle that public service is a public trust [152].

Gifts. Employees may not solicit or accept gifts from prohibited sources or because of official position; prohibited sources include entities doing or seeking business with the Department or whose interests are affected by official duties [152].

Gift exceptions. Exceptions include the 20/50 rule, personal relationships, awards, spouse employment, informational materials, and generally available discounts; low-value items like coffee, plaques, or contests are not gifts [152].

Gifts between employees. Subordinates generally should not give gifts to superiors, and employees may not accept gifts from subordinates, with exceptions for modest occasional gifts, shared refreshments, personal hospitality, and special infrequent occasions (marriage, condolence, birth/adoption, or termination of the relationship) [152].

Group gifts. Group gifts to superiors require voluntary contributions, no more than $10 per contributor, and a total value cap of $480 (raised from $300 on 30 Mar 2023) [152].

7.1.11 Fundraising and Political Activity

Commercial dealings. Personnel should not sell goods to subordinates or their families except in limited personal or off-duty retail contexts without coercion [152].

Fundraising. Do not use uniform, title, or authority to fundraise or fundraise in the workplace, with limited exceptions for Combined Federal Campaign (CFC), Navy-Marine Corps Relief Society (NMCRS), and similar official campaigns; consult legal counsel before fundraising [152].

Political activity. Members may vote, register, donate, attend rallies as spectators, and express personal opinions, but may not campaign, fundraise, hold public office, distribute partisan material, post political signs in government housing, or wear uniforms to political events [152].

7.1.12 Operational Law

Operational discipline. The purpose of military law and training is to strengthen readiness and enable compliance with the LOAC during operations [152].

7.1.13 Quick Review

- QUICKMAN is a spotting tool; SJAs remain the authoritative source of legal advice [152].
- Preliminary inquiries are fast fact-finding tools; command investigations are formal written products for significant events [152].
- NJP rights include presence, silence, a representative, evidence review, and mitigation opportunities [152].
- ADSEP procedures and characterization thresholds determine whether notification or board processing applies [152].
- Ethics rules prohibit accepting gifts from prohibited sources and restrict political activity and fundraising [152].

7.2 Ethics and Acquisition

7.2.1 Summary

Ethical leadership. Ethical conduct is foundational to mission success and is the most important component of naval leadership [153].

Program success. Successful acquisition is judged differently by stakeholders (program managers, contractors, users, and Congress), so ethical choices balance competing perspectives [153].

Decision models. Ethical models (Golden Rule, Kant, utilitarianism, altruism) and the principled decision-making model provide structured ways to resolve dilemmas [153].

Influences. Ethical decisions are shaped by personal factors, organizational pressures, and professional context; recognizing these influences reduces risk [153].

7.2.2 Ethics and Successful Leadership

Why it matters. Personal conduct is the most significant threat to a leader's career, so leaders must develop and sustain their own character while developing the character of their teams [153].

Misconduct patterns. Senior official misconduct often involves personal/ethical violations, improper personnel actions, misuse of government resources, travel violations, or procurement/security issues [153].

Causal factors. Common drivers include failure to consider consequences, misjudged privacy, alcohol abuse, loss of command stature, poor risk controls, weak moral compass, and temptation amplified by position and stress [153].

7.2.3 Ethical Decision-Making Foundations

Decision types. Simple decisions follow established rules, while complex or gray-area decisions require deeper ethical analysis and deliberation [153].

Influences. Instinct, religion, education, gender, life experience, hierarchy, mental health, laws and regulations, society, peers, family, professional ethos, job position, and physical condition all influence ethical choices [153].

7.2.4 Ethical Decision-Making Models

Golden Rule. Treat others the way you want to be treated; its limitation is that others may not want the same treatment [153].

Kant's absolute moral philosophy. Ethical principles always take precedence, with a duty to tell the truth; it lacks a method to choose among competing ethical values [153].

Consequentialism/Utilitarianism. Weigh outcomes to maximize overall good; this model can be manipulated to justify ends-over-means thinking [153].

Altruism. Act to increase others' welfare; it struggles when stakeholders have conflicting interests and offers limited guidance on trade-offs [153].

7.2.5 Principled Decision-Making Model

Six pillars of character. Trustworthiness, respect, responsibility, justice/fairness, caring, and civic virtue/citizenship take precedence over non-ethical values [153].

Decision tests. The Washington Post test (public scrutiny), generalization test (precedent for others), and mirror test (personal conscience) validate choices [153].

1. **Stop and think.** Create time and space before deciding [153].
2. **Clarify goals.** Balance short-term and long-term aims [153].
3. **Determine facts.** Identify what you know and what you need to know [153].
4. **Develop options.** Generate options and consult trusted advisors [153].
5. **Consider consequences.** Assess stakeholder impacts and the six pillars [153].
6. **Choose.** Make the decision and frame it with the decision tests [153].
7. **Monitor and modify.** Reassess outcomes and adjust if needed [153].

7.2.6 Acquisition Program Success Perspectives

Program manager view. Success means delivering required capability on time and within budget [153].

Contractor view. Success means profitability and positive cash flow [153].

User view. Success means fielding an effective and available capability [153].

Congressional view. Success balances social, environmental, and defense needs while ensuring appropriated funds are used for their intended purpose [153].

7.2.7 Acquisition Ethical Dilemmas

Budget insight. Informal requests for RDT&E impact data (e.g., a 30% cut) require caution, transparency, and alignment with ethics rules and program leadership [153].

Program pressures. Acquisition dilemmas (e.g., cost/schedule/performance problems with a prime contractor and congressional oversight) should be resolved using the principled model and the six pillars of character [153].

7.2.8 Procurement Integrity and OCI Board Prep

Procurement integrity. Before award, do not knowingly disclose or obtain contractor bid/proposal information or source-selection information. FAR 3.104 also covers employment-contact reporting, disqualification, and one-year compensation restrictions for specified officials on procurements over $10M [70].

OCI. Under FAR Subpart 9.5, the KO must identify and evaluate Organizational Conflict of Interest (OCI) issues as early as possible, then avoid, neutralize, or mitigate significant conflicts. Board answer: focus on biased judgment and unfair competitive advantage, and bring legal and contracting into the discussion early [70].

7.2.9 Quick Review

- Ethical leadership is essential to mission success and career longevity [153].
- The principled decision-making model provides seven steps anchored in the six pillars of character [153].
- Ethical models (Golden Rule, Kant, utilitarianism, altruism) help diagnose dilemmas but each has limits [153].
- Procurement integrity and OCI controls protect competition, source-selection fairness, and public trust [70].
- Acquisition success criteria vary by stakeholder, so ethical choices must balance competing priorities [153].

7.3 Presentation Techniques

7.3.1 Summary

Preparation. Effective briefs begin with audience analysis, understanding the deliverable, and clarifying the purpose (inform, inspire, motivate, persuade, or decide) [154].

Structure. A clear opening, focused body, and decisive closing, with deliberate slide limits and time for questions, drives a coherent message [154].

Delivery. Strong verbal, vocal, and visual execution, plus disciplined Q&A handling, determines whether the audience trusts the message [154].

7.3.2 Student Presentation Requirements

Scenario. Teams collaborate with IPTs to brief a professional decision brief aligned to the Integrated Case Study objectives [154].

Execution. Team briefs are limited to 20 minutes; each student briefs at least five slides and up to 10 minutes, with team grades for content and individual grades for presentation technique [154].

Follow-up. Briefs are recorded for self-evaluation and debrief with the director of training [154].

7.3.3 Preparation and Design

Audience analysis. Identify the audience's background, expected questions, and desired outcomes to tailor depth and framing [154].

Purpose. Define the purpose early and keep it concise; most Integrated Case Study briefs are decision briefs that seek buy-in [154].

Structure. For briefs under 30 minutes, target roughly 15% opening, 75% body, 10% closing, limit slide count, and plan three to five minutes per slide including questions [154].

Opening. Gain attention, establish relevance and credibility, preview the topic, state purpose, and set two to three audience expectations tied to closing takeaways [154].

Body. Present evidence for the purpose, use logical transitions, and structure content with chronological, problem/solution, or cause/effect patterns [154].

Closing. Reinforce takeaways, make the final case for the recommended option, and ask for a decision without recapping the entire brief [154].

7.3.4 Recommended Courses of Action

Option framing. Present the recommended option first or last, describe all options objectively, and explain assumptions, risks, and impacts if funding is not received [154].

Supporting material. Use charts, tables, and graphics sparingly, label everything, keep units consistent, and leverage backup slides for detail [154].

7.3.5 Delivery Components

Three components. Presentation effectiveness depends on verbal content, vocal delivery, and visual presence; all must be prepared together [154].

Visual design. Use white space, minimal animation, fragments instead of full sentences, left-justified text, consistent colors with legends, readable fonts (about 16–24), and numbered slides; spell out acronyms and test visuals on the actual screen [154].

Body language. Stand with balanced posture, move purposefully, make eye contact across the room, and avoid nervous gestures or overusing the pointer [154].

Verbal and vocal habits. Speak clearly and concisely, avoid filler words and reading slides, use examples, command attention with authentic energy, and keep humor controlled [154].

7.3.6 Q&A and Tough Questions

Q&A discipline. Relax, pause, repeat questions if needed, start with the conclusion, then explain and stop when answered; ensure the audience understands and build in time for questions [154].

Tough questions. Do not bluff; provide facts, identify missing data, offer qualified opinions, and keep the brief on track without letting non-principals derail the session [154].

7.3.7 Virtual Presentation Tips

Virtual delivery. Use the webcam, ensure bandwidth, minimize distractions, engage with polls or direct questions, and adjust visuals (larger fonts, lighting, professional background) to compensate for reduced presence [154].

7.3.8 Common Failure Modes

Death by PowerPoint. The failure is not the tool, but poor connection to the audience, unclear expectations, weak delivery, inaccurate content, or slides that are too dense or disorganized [154].

7.3.9 Checklist and Written Communication

- Ensure smooth slide transitions, clear expectations and takeaways, accurate numbers, labeled charts, consistent formatting, spelled-out acronyms, tested equipment, and time for questions [154].
- Written communication should be clear and concise, tailored to the receiver, fact-checked, professional, and explicit about response timelines [154].

7.3.10 Quick Review

- Analyze the audience, understand the deliverable, and define the purpose before building slides [154].
- Opening sets expectations, the body supports them, and the closing drives the decision or call to action [154].
- Verbal, vocal, and visual execution must align to earn trust and drive outcomes [154].

7.4 Team Building

7.4.1 Summary

IPT focus. IPPD and IPTs are central to acquisition execution; empowered cross-functional teams improve integration and decision quality [155].

Team dynamics. Effective teams share clear goals, empowered members, balanced productivity and satisfaction, and constructive conflict management [155].

Leadership impact. IPT leaders act as supervisors, facilitators, and coaches while managing stakeholder relationships and resources [155].

Cohesion stages. Teams progress through forming, storming, norming, performing, and adjourning; leaders must adapt at each stage [155].

7.4.2 IPPD and IPT Foundations

IPPD definition. IPPD integrates essential acquisition activities with multidisciplinary teams to optimize design, manufacturing, and supportability across the life cycle [155].

IPT definition. An IPT is a multidisciplinary team collectively responsible for delivering a defined product or process, composed of empowered representatives across functions [155].

IPT purpose. IPTs build successful programs, identify and resolve issues, and make sound, timely recommendations to enable decisions [155].

7.4.3 IPT Types and Roles

Oversight IPTs. Overarching Integrated Product Teams (OIPTs) assist the MDA with investment decisions and ensure programs are structured and resourced to succeed [155].

Working-level IPTs. Working-level Integrated Product Teams (WIPTs) are formed by the program manager with OSW staff and stakeholders to focus on topics like cost/performance, program baselines, acquisition strategy, test and evaluation, and contracting [155].

Program IPTs. Program Integrated Product Teams (PIPTs) represent functional areas and organizations involved in designing, developing, testing, manufacturing, deploying, and supporting products or services [155].

7.4.4 Team Characteristics

Effective teams. Clear purpose and goals, empowered participation, top management support with bottom-up leadership, effective communication, balanced productivity and satisfaction, cohesiveness, and objective process reviews [155].

Ineffective teams. Unclear goals, lack of management support, unempowered members, weak planning, limited training, poor access to lessons learned, unequal participation, inadequate resources, and unrealistic schedules [155].

7.4.5 IPT Leadership Requirements

Leadership roles. IPT leaders may serve as supervisors, facilitators, and coaches while interfacing with upper management and serving on higher-level teams [155].

Core skills. Leaders must manage disputes, orchestrate communication, obtain resources, build stakeholder relationships, and apply group process skills with flexibility and empowerment [155].

7.4.6 Group Decision Making

Consensus focus. Collaboration and consensus improve commitment and solution quality, but can be time-consuming and vulnerable to difficult personalities [155].

Decision commitment. The goal is a shared decision that members understand, discussed, and can support [155].

7.4.7 Virtual Teaming

Inclusion. Actively include remote members, establish core hours, and use multiple communication channels to maintain cohesion [155].

Cohesion practices. Recognize remote contributions through awards, celebrations, and shared events to reinforce team identity [155].

7.4.8 Team Development Stages

Forming. Members learn roles, expectations, tasks, and ground rules while deferring to authority and establishing mission and charter [155].

Storming. Members challenge ideas, form alliances, and raise personal issues; leaders must work through conflict or the team never gels [155].

Norming. Members build trust, establish boundaries, and express issues constructively; norms re-emerge after storming [155].

Performing. Teams collaborate with open communications, manage risk while innovating, and handle conflict constructively [155].

Adjourning. Members close out tasks, capture lessons learned, and transition focus to follow-on work [155].

7.4.9 Building Effective Teams

Start-up actions. Make introductions, clarify purpose, define operating agreements, set roles and responsibilities, and open communication channels [155].

Direction. Draft mission and vision statements, produce a team charter, and identify critical success factors to sustain a positive atmosphere [155].

Leadership posture. Use participative leadership, manage conflict, and reinforce balanced participation [155].

7.4.10 Quick Review

- IPPD relies on empowered, multidisciplinary IPTs to integrate life-cycle decisions [155].
- Effective IPTs combine clear goals, empowerment, communication, and balanced productivity [155].
- Teams progress through forming, storming, norming, performing, and adjourning [155].
- Strong team charters and participative leadership enable sustained performance [155].

7.5 Leading People

7.5.1 Summary

Leadership core. Leading people is dynamic and shaped by personal experience, but effective leaders consistently model integrity, accountability, and a commitment to developing others [156].

Responsibility triad. Responsibility, accountability, and authority must be aligned for mission success and healthy command climates [156].

Intent-based leadership. Clear purpose, shared understanding, and competence enable decentralized initiative in complex environments [156].

Get Real, Get Better. The Navy-wide culture shift emphasizes transparency, problem solving, and learning teams to deliver warfighting advantage [156].

7.5.2 Leadership Styles

- Vision, coaching, affiliative, democratic, pace-setting, and commanding styles may be situationally appropriate depending on mission and team needs [156].

7.5.3 Responsibility, Accountability, and Authority

Responsibility. The moral, legal, or mental obligation to perform duties and own outcomes [156].

Accountability. The willingness to answer for actions and decisions [156].

Authority. The power to influence or command thought, opinion, or behavior [156].

7.5.4 Leadership Expectations

CNO 34 Day One Message. Focus on Foundry (industrial base and shipbuilding), Fleet (integrated force readiness), and Fight (lethality and survivability), with measures of success tied to on-time delivery, readiness, ordnance production, repair parts, and mastery-level training [156].

Charge of Command. Lead with integrity, courage, and humility; invest in Sailors; own risk; prepare, stress plans, learn, and innovate; and honor the privilege and gravity of command [156].

7.5.5 Get Real, Get Better

Culture shift. Get Real, Get Better is the Navy standard for leadership and problem solving, building teams that self-assess, self-correct, and learn continuously [156].

Get Real behaviors. Align on standards, act transparently, and embrace the red to expose performance gaps [156].

Get Better behaviors. Focus on what matters, use proven problem-solving methods, and fix or elevate barriers with urgency and accountability [156].

Learning teams. Build trust, respect, and clear ownership while reinforcing collaboration and transparency [156].

7.5.6 Intent-Based Leadership

Core idea. Intent-based leadership empowers people at all levels to make decisions aligned with mission intent rather than relying on centralized direction [156].

Building blocks. Teams need shared understanding of purpose, clarity on each member's role, and technical competence to make good decisions [156].

7.5.7 Team of Teams

Initiative culture. Historical examples (Nelson's "touch") show that winning complex fights requires a culture that rewards initiative and critical thinking beyond strict execution [156].

7.5.8 Where to Start

- Commit to continuous personal and professional development and expand leadership knowledge through reading, podcasts, and guided learning resources [156].
- Use Navy learning resources (e.g., DoD MWR Library, Libby App, and leadership networks) to build a sustainable learning habit [156].

7.5.9 Quick Review

- Leadership relies on aligned responsibility, accountability, and authority [156].

- Get Real, Get Better emphasizes transparency, problem solving, and learning teams [156].
- Intent-based leadership depends on shared purpose and competence [156].

7.6 Myers-Briggs Type Indicator

7.6.1 Summary

Preference framework. The Myers-Briggs Type Indicator (MBTI) describes preferences in how people energize, gather information, decide, and adapt to the outer world; it is a sorting tool, not a measure of skill or ability [157].

Self-awareness. The goal is non-judgmental feedback that improves self-awareness and helps leaders accommodate different work-style preferences [157].

Team impact. Similar types communicate quickly but can miss perspectives; diverse types reduce error risk when leaders build understanding and trust [157].

7.6.2 History and Purpose

Origins. Carl Jung described predictable patterns in how people prefer to engage the world; Isabel Myers and Katherine Briggs translated those preferences into the MBTI to help people match skills, desires, and career opportunities [157].

Use case. The MBTI sorts preferences to support self-awareness and leadership effectiveness, not to evaluate performance or select personnel [157].

7.6.3 The MBTI Dichotomies

Energizing. Extraversion (outer world) vs. introversion (inner world) preferences define where people direct and receive energy [157].

Collecting data. Sensing focuses on facts and experience; intuition focuses on patterns and possibilities [157].

Evaluating. Thinking emphasizes logic and objectivity; feeling emphasizes values and impact on people [157].

Adapting. Judging favors structure and closure; perceiving favors flexibility and openness to options [157].

7.6.4 Self-Select and Best-Fit Types

Self-select vs. reported. Compare self-estimated type with reported MBTI results; if different, review both descriptions and decide which fits best [157].

Best-fit hierarchy. Best-fit type is the goal; reported type and self-estimated type inform the decision, but you decide what fits [157].

Why differences occur. Mindset, changes over time, and familiarity with questions can shift responses; differences are not a cause for concern [157].

7.6.5 MBTI Step II Insights

Preference confidence. The preference confidence index and category indicate how consistently a person selected one side of a dichotomy; they suggest, but do not determine, accuracy [157].

7.6.6 Team Dynamics and Problem Solving

Team composition. High similarity can accelerate understanding but risks incomplete viewpoints; diverse types can reduce conflict once members learn to value differences [157].

Problem-solving flow. Effective teams balance sensing and intuition for data collection, and thinking and feeling for evaluation to avoid blind spots [157].

Self-discovery. The Johari Window reminds teams that feedback and transparency reduce blind spots and build trust [157].

7.6.7 Quick Review

- MBTI describes preferences, not competence or ability [157].
- Leaders should adapt communication to different preferences to improve performance [157].
- Diverse types improve decision quality when teams value differences and share information [157].

7.7 California Psychological Inventory 260

7.7.1 Summary

Purpose. CPI 260 provides objective, comparative feedback on leadership tendencies and comfort zones to highlight strengths and potential blind spots [158].

Reports. The Client Feedback Report shows scores across scales and the Coaching Report for Leaders translates those scores into leadership development insights [158].

Action focus. Results are hypotheses, not truths; leaders are expected to interpret, corroborate, and build an action plan [158].

7.7.2 History and Methodology

Origins. The CPI family dates to 1956 (CPI 480); CPI 260 launched in 2002 and has been used in the senior course since 2003 [158].

Purpose. CPI 260 describes you from the perspective of knowledgeable, objective others, capturing comfort zones that may reveal blind spots [158].

Data foundation. Scale definitions rely on observed trends and comparative analysis (not theory alone), with demographic screening to retain the most discriminating scales [158].

7.7.3 Client Feedback Report

Metrics baseline. The Client Feedback Report displays CPI 260 scores across 29 scales and compares them to the U.S. adult population [158].

Screening results. Response screening flags overly favorable, overly negative, or random patterns that require further investigation [158].

7.7.4 Lifestyle Scales and Management Styles

Orientation. Lifestyle scales capture how you engage others and relate to norms, ranging from interactive and expressive to reflective and reserved [158].

Quadrants. Management styles map to implementer, supporter, innovator, and visualizer quadrants with distinct strengths and risks in change and execution [158].

Satisfaction scale. Satisfaction strongly influences overall results; higher scores tend to raise other scales while lower scores suppress them [158].

7.7.5 Scale Families

Dealing with others. Dominance, sociability, social presence, self-acceptance, independence, and empathy compare you to a national norm [158].

Self-management. Responsibility, social conformity, self-control, good impression, communality, well-being, and tolerance track self-regulation tendencies [158].

Motivations and thinking. Achievement via conformance, achievement via independence, and conceptual fluency provide insight into how you pursue goals [158].

Personal characteristics. Insightfulness, flexibility, and sensitivity signal how you process interpersonal and organizational dynamics [158].

Work-related measures. Managerial potential, work orientation, creative temperament, leadership, amicability, and law enforcement orientation capture applied leadership behaviors [158].

7.7.6 Interpreting Results

Letś talk approach. Treat results as approximations, compare to comfort zones, corroborate with other feedback, and define follow-through actions [158].

No ideal profile. There is no perfect set of results; each situation is unique and change is possible and worthwhile [158].

Johari window. Feedback reduces blind spots and improves self-awareness in leadership behavior [158].

7.7.7 Coaching Report for Leaders

Purpose. The Coaching Report for Leaders converts CPI scale data into leadership strengths, developmental opportunities, and areas to evaluate further [158].

Executive norm group. Leadership characteristics compare CPI scale pairs to an executive norm group to flag strengths, developmental opportunities, or undetermined areas [158].

Next steps. Focus on two or three key priorities, leverage strengths, and address blind spots with targeted action [158].

7.7.8 Quick Review

- CPI 260 is a self-awareness tool that highlights leadership tendencies and blind spots [158].
- The Client Feedback Report provides scale scores; the Coaching Report for Leaders turns them into development guidance [158].
- Results are hypotheses that require corroboration and action planning [158].

7.8 High Consequence Event Management

7.8.1 Summary

Risk-aware culture. A deliberate culture that promotes risk-aware behaviors and confronts human element weaknesses reduces the likelihood of high consequence events [159].

Human element focus. Technical strength alone is insufficient; human element weaknesses often enable failures and must be addressed explicitly [159].

HCEPF alignment. The High Consequence Events Prevention Framework links risk-aware behaviors to Get Real, Get Better and learning teams to sustain safe, effective performance [159].

7.8.2 Why High Consequence Event Prevention

Culture first. Technically strong organizations still fail when cultural flaws go unaddressed; a strong culture plus technical foundation preempts catastrophic mistakes [159].

Fukushima lesson. The Fukushima disaster illustrates how human and cultural factors compound technical risk and drive catastrophic outcomes [159].

7.8.3 Risk-Aware Culture

Definition. A risk-aware culture uses daily risk-aware behaviors, expects accountability for human element weaknesses, and empowers teams to identify and mitigate mission risks [159].

Deliberate design. Organizational habits must be intentionally designed rather than left to chance [159].

7.8.4 Heinrich's Triangle

Human error link. Major accidents often share the same causal human element weaknesses seen in minor accidents and near-misses [159].

Prevention strategy. Track and correct small failures to reduce the likelihood of apex events [159].

7.8.5 Performance Focus

Three domains. Effective risk management requires attention to technology, systems/processes, and people behaviors; overreliance on any single domain creates gaps [159].

Barriers. Programmatic system barriers can fail when risks align and decision errors defeat safeguards [159].

7.8.6 High Consequence Events Prevention Framework

Framework scope. HCEPF is a structured cultural initiative with 24 risk-aware behaviors, each paired to a human element weakness to track learning and performance [159].

Get Real, Get Better linkage. HCEPF behaviors are the mechanism for getting real, getting better, and building learning teams within Strategic Systems Program (SSP) [159].

Tools. The HCEPF tools reference guide provides a menu of practices to strengthen behaviors at the individual, team, and organizational levels [159].

7.8.7 Balancing Risk Aversion

Competing pressures. Production, cost, and schedule pressures compete with effectiveness, security, and safety; leaders must manage the tension and avoid drifting toward unsafe extremes [159].

7.8.8 Key Finding

Technical + human strengths. Organizations with cross-functional technical skill and strong human element behaviors prevent high consequence events and respond with resilience when surprises occur [159].

Root cause discipline. Technical errors must be traced to underlying human element causes to avoid recurrence [159].

7.8.9 Case Study Prompts

- Evaluate human element weaknesses in real-world incidents (e.g., OceanGate Titan, Kobe Bryant helicopter crash, Apollo 13 oxygen tank specification error, Grenfell Tower) to identify behaviors to strengthen [159].

7.8.10 Quick Review

- A risk-aware culture is deliberately designed, reinforced daily, and essential to high consequence event prevention [159].
- Human element weaknesses enable technical failures to cascade into disasters [159].
- HCEPF behaviors link to Get Real, Get Better and provide tools for continuous improvement [159].

7.9 Defense Acquisition Workforce Improvement Act (DAWIA)

7.9.1 Summary

Purpose. Defense Acquisition Workforce Improvement Act (DAWIA) professionalizes the DoW acquisition workforce through education, training, experience, and certification requirements [160].

Positions and functional areas. Acquisition billets are positions where at least 50 percent of duties are acquisition-related; the functional area and level in your orders or Position Description (PD) drive certification requirements, and most EDO billets are coded for program management [160].

Critical and key leadership roles. Critical Acquisition Positions (CAPs) and Key Leadership Positions (KLPs) carry tenure obligations and require Acquisition Professional Membership (APM) (or a waiver) to ensure experienced leadership continuity [160].

Certification path. DAWIA offers foundational, practitioner, and advanced levels; certification is requested in Electronic Defense Acquisition Career Manager (eDACM) after meeting education, training, and experience standards [160].

Continuous learning. Workforce members complete at least 80 hours of continuous learning every two years, with points requested through eDACM and no carryover between cycles [160].

7.9.2 DAWIA Background and Back-to-Basics

Origins. DAWIA was enacted in 1990 (Public Law 101-510) to raise the professionalism of the acquisition workforce through standardized education, training, and experience requirements [160].

Back-to-Basics. In 2020, USW(A&S) launched the Back-to-Basics initiative to modernize DoW implementation, streamline certification, and emphasize lifelong learning [160].

7.9.3 Acquisition Positions and Functional Areas

Acquisition positions. Acquisition positions are the largest part of the Acquisition Workforce (AWF) and require that at least half of assigned responsibilities are acquisition-related [160].

Coding and level. For officers, acquisition coding appears in orders; for civilians, it is in the PD. The functional area and level (often embedded in the Additional Qualification Designation (AQD)) drive the required education, training, and experience standards [160].

Functional areas. DAWIA functional areas include program management, engineering and technical management, contracting, life-cycle logistics, business financial management and cost estimating, and test and evaluation [160].

7.9.4 Critical Acquisition Positions and Key Leadership Positions

Critical Acquisition Positions (CAPs). CAPs are senior supervisory or managerial positions that require APM, carry a minimum three-year tenure obligation, and for military billets typically require O-5 or above [160].

Key Leadership Positions (KLPs). KLPs are the most senior DAWIA roles (less than one percent of the AWF), designated by ASN(RD&A) with USW(A&S) attention, and usually filled by O-6 or GS-15 equivalents or higher with three-to-four year tenures [160].

7.9.5 Certification Levels and Requirements

Levels. DAWIA certifications are offered at foundational, practitioner, and advanced levels across all functional areas [160].

Orders-driven timelines. EDO acquisition orders typically require practitioner certification in the program management functional area within 60 months of reporting; advanced certification is expected for senior program management billets [160].

Certification request. After completing education, training, and experience standards, submit a certification request in eDACM. Commands may certify O-6 and below when criteria are met, while Director of Acquisition Talent Management (DATM) certifies Flag and SES personnel [160].

7.9.6 Training and Experience Credit

Getting started. Use eDACM to set up a profile, search DAU training, and enroll in courses; new offerings typically open in August and early sign-up improves scheduling options [160].

Funding. Selecting "parent command will fund travel and per diem" can improve course acceptance if the command agrees and DATM funding is available for priority students [160].

Experience credit. Graduate education may be credited toward experience requirements when pursuing certification or APM, with credit caps of 12 months (up to 18 months for subspecialty 5100P and 24 months for 5100N) and no reuse for multiple certifications [160].

7.9.7 Acquisition Professional Membership

Purpose. APM provides a pool of qualified personnel for CAPs and KLPs; membership (or a waiver) is required for assignment to those positions [160].

Eligibility. Requirements include a baccalaureate degree with at least 12 business-related credits, DAWIA certification in any functional area, LCDR or above, and 48 months of documented acquisition experience (with limited graduate-study credit allowed) [160].

Package. APM requests are submitted in eDACM with a DAC application (NAVPERS-1301-86), eDACM transcript, college transcripts, and FITREPs when documenting non-acquisition coded time [160].

7.9.8 Continuous Learning

Requirement. Complete at least 80 hours of continuous learning every two years; points do not carry over to the next cycle, and members must request credit in eDACM [160].

7.9.9 Quick Review

- DAWIA sets education, training, experience, and certification standards for the DoW acquisition workforce [160].
- Acquisition billets are coded in orders or PD and require certification tied to functional area and level [160].
- CAPs and KLPs require APM and carry multi-year tenure obligations [160].
- Certification requests, training enrollment, and continuous learning credits flow through eDACM [160].
- APM eligibility requires education, certification, rank, and documented acquisition experience [160].

7.10 Engineering Duty Officer Qualification Program (EDQP)

7.10.1 Summary

Purpose. The Engineering Duty Officer Qualification Program (EDQP) ensures each EDO completes technical education, masters core competencies, and demonstrates readiness for the 144X designator through a structured qualification process [161].

Governance. The EDQP hierarchy includes the designator advisor (Commander, NAVSEA), the Engineering Duty Qualification Board (EDQB), the EDO School, certifying and counseling officers, and each candidate [161].

Milestones. Five milestones drive completion: acceptance, postgraduate education, basic course graduation, qualification tour, and certification via oral examination [161].

Candidate duties. Candidates complete command-specific certification programs, process improvement requirements, required DAU courses, and the oral board within prescribed timelines [161].

Special qualifications. The ED Dolphin and Information Warfare Officer (IWO) qualifications are optional but structured programs with defined prerequisites and timelines for eligible EDOs [161].

7.10.2 EDQP Objectives

Professional education. Ensure each EDO completes a technical master's degree and earns an approved subspecialty code [161].

Core competencies. Build mastery in leadership, technical acumen, financial management, contracting, DoW acquisition, civilian personnel management, and BDAR fundamentals [161].

Mentorship and professionalism. Provide senior EDO mentorship, consistent community experience, and encourage acquisition workforce certification and professional membership [161].

7.10.3 EDQP Hierarchy and Roles

Designator advisor. The Commander, NAVSEA establishes EDQP requirements and acts as the technical review authority for training strategies [161].

Engineering Duty Qualification Board. The EDQB provides senior community oversight, recommends program changes, and supports curriculum review for the EDO School [161].

Certifying officer and counseling officer. The certifying officer (typically the candidate's CO) sets written guidance, while the counseling officer monitors progress and assists the certifying officer [161].

Candidate. Each candidate owns completion of the EDQP and is accountable for meeting milestone timelines [161].

7.10.4 EDQP Milestones and Certification

Milestone sequence. Acceptance, postgraduate education, basic course graduation, qualification tour (minimum one year), and certification via oral board are the required phases [161].

Qualification tour requirements. Candidates complete command-specific training, process-improvement requirements, an introduction to civilian supervision, DAU courses ACQ 0030 and PMT 0141, and demonstrate EDO core competency fluency [161].

Oral board. The certification board is chaired by the certifying officer with at least three senior EDOs; candidates must demonstrate technical, organizational, and community knowledge and complete certification within two years of the basic course [161].

7.10.5 Candidate Responsibilities

Core completion tasks. Candidates complete entrance requirements, technical master's degree, EDO basic course, command certification program, process improvement requirements, and required DAU courses [161].

Performance and timelines. Candidates perform satisfactorily during the qualification tour, complete the oral board within two years, and request extensions when circumstances warrant [161].

Feedback. Candidates provide lessons learned to improve the EDO School curriculum and EDQP process [161].

7.10.6 Lessons Learned

Engaged leadership. Qualification success requires active certifying and counseling officer participation, frequent training touchpoints, and civilian subject matter expertise when available [161].

Common challenges. Candidates should plan early for docking observer requirements, select a process-improvement topic early enough to gather data, and schedule required courses in advance [161].

Board preparation. Oral boards reward clear communication, thorough study of the basic course, and the ability to reason through problems under pressure [161].

Extensions and waivers. Extension requests require timely coordination through the chain of command with adequate justification; do not wait until deadlines are imminent [161].

7.10.7 ED Dolphin Program

Program intent. The ED Dolphin program prepares EDOs without prior submarine qualification for submarine maintenance and acquisition roles and is voluntary [161].

Duration and scope. Completion typically takes 18 to 24 months and culminates in authorization to wear the ED Dolphin warfare insignia [161].

Key requirements. Requirements include submarine physicals, formal training, at least sixteen weeks of SSBN/SSN experience, an ED Dolphin journal, a submarine walkthrough examination, and an oral board of submarine-qualified officers [161].

7.10.8 Information Warfare Officer Qualification

Purpose. IWO qualification recognizes expertise in Information Warfare (IW) fields and is primarily intended for officers in information warfare communities [161].

Eligibility. EDOs may qualify when assigned to IW billets, pre-register intent with NAVIFOR, and document qualifying IW experience and duties [161].

Completion. Candidates complete IWO Personnel Qualification Standard (PQS) and an oral qualification board within 36 months, with AQD prerequisites met when required [161].

7.10.9 Quick Review

- EDQP aligns EDO education, core competencies, and mentorship with qualification standards [161].
- Five milestones guide the path from acceptance to certification, including a minimum one-year qualification tour [161].
- Candidates must complete process improvement requirements, required DAU courses, and the oral board on time [161].
- Lessons learned emphasize proactive scheduling, engaged leadership, and strong board preparation [161].
- ED Dolphin and IWO qualifications are optional programs with defined prerequisites and timelines [161].

7.11 Career Topics

7.11.1 Summary

FITREP fundamentals. Strong Fitness Report (FITREP) counseling, honest feedback, and clear ranking narratives are essential because boards rely on FITREPs as the primary performance signal [162].

OSR/PSR focus. Selection boards review the Officer Summary Record (OSR) and Performance Summary Report (PSR) for record continuity, competitive ranking, and trends in trait averages and breakout language [162].

Board procedures. Administrative and statutory boards follow different authorities, but all boards use precepts, convening orders, and structured screening processes to select the best qualified [162].

Promotion context. Defense Officer Personnel Management Act (DOPMA) governs promotion zones and selection cohorts, while recent changes emphasize merit and technical performance over seniority [162].

Record ownership. Officers are responsible for reviewing records, correcting gaps, and preparing for boards that are largely staffed by Unrestricted Line (URL) members [162].

7.11.2 FITREP Counseling and Performance Signals

Mid-term counseling. Mid-term counseling around the six-month point improves performance, clarifies expectations, and provides the feedback needed to improve FITREP outcomes [162].

Reporting senior expectations. Officers should ask their Reporting Senior (RS) how rankings and trait grades are determined, where they stand among peers, and what is needed to improve performance narratives [162].

Competitive signals. Boards look for high rankings, trait averages above Reporting Senior Cumulative Average (RSCA), and progression toward the right over time, with strong breakout language in FITREP narratives [162].

7.11.3 OSR/PSR and Record Continuity

Board view. The OSR and PSR are the only documents every board member sees, so officers should regularly review these records for accuracy and continuity [162].

Letters to the board. Letters can clarify missing or late information, but should be concise and submitted before the board convenes so record briefers can address them [162].

Photos and files. Officer photographs remain part of the Official Military Personnel File (OMPF), even though selection boards no longer display them during review [162].

7.11.4 Selection Board Types and Authorities

Administrative vs. statutory. Administrative boards are convened by policy while statutory boards are convened by SECNAV and governed by law, with membership and procedures set by statute and instruction [162].

DOPMA framework. DOPMA standardizes promotion rules, establishes grade ceilings, and drives selection board cohorts and separation timelines [162].

Zones and lineal numbers. Promotion zones are determined by projected requirements and lineal seniority, with annual updates published for in-zone and below-zone eligibility [162].

7.11.5 Board Processes

Recorders and briefers. Recorders check record continuity and briefers annotate OSR/PSR entries so board members can quickly assess records during deliberations [162].

Voting flow. Boards brief records, vote confidence levels, and apply "crunch" reviews to refine selections until authorized numbers are met [162].

7.11.6 Promotion Board Changes and Merit Reorder

Merit emphasis. Recent changes allow higher performers to move up promotion lists, reduce seniority bias, and reward strong technical performance for acquisition officers [162].

EDO merit signals. Sustained top performance, demanding EDO assignments, and proven leadership roles are valued indicators for merit reorder decisions [162].

7.11.7 Lessons Learned

Own the record. Review your record regularly, verify FITREP continuity, and ensure awards, education, and qualifications are current before board eligibility [162].

Know the audience. Most selection board members are URL officers; write FITREP narratives and accomplishments so they resonate with that audience [162].

Seek mentorship. Engage detailers and senior EDOs early to identify and address gaps before selection board reviews [162].

7.11.8 Quick Review

- FITREP counseling and breakout language drive board perceptions of performance and potential [162].
- The OSR/PSR are the universal board references, so record continuity matters [162].
- DOPMA sets promotion zones, cohorts, and baseline promotion rules [162].
- Selection boards use structured brief, vote, and crunch processes to reach authorized selects [162].
- Most board members are URL officers, so FITREP narratives must speak to that audience [162].

7.12 Mentoring Tips

7.12.1 Summary

Definition. Mentoring provides personal and professional development advice and career counseling; it is not detailing or formal performance counseling [163].

Cadence. Formal mentoring is required annually, but informal mentoring is encouraged and can occur with any trusted advisor regardless of rank or mentor group [163].

EDO policy. The Leader Development Framework (LDF) sets mentoring expectations and guiding principles focused on knowledge transfer, alignment, and long-term community investment [163].

Preparation. Effective sessions require read-ahead materials (career planner, OSR/PSR) and clear questions sent in advance to maximize mentor feedback [163].

Follow-through. The most important step is follow-up: capture guidance, update plans, and maintain contact to build long-term professional networks [163].

7.12.2 Mentoring Definition and Steps

What mentoring is. Mentoring focuses on coaching, development, and career guidance that helps EDOs make informed decisions across professional and personal milestones [163].

What mentoring is not. Mentoring does not replace detailing, and it is not solely a review of performance counseling or fitness reports [163].

Three-step cycle. Prepare, conduct, and follow up; the follow-up step is emphasized as the most important for growth and accountability [163].

7.12.3 EDO Mentoring Policy and Mentor Groups

Leader Development Framework. The LDF anchors formal mentoring and reinforces character, competence, and culture goals for the EDO community [163].

Guiding principles. Mentoring should transfer community knowledge, align professional and personal goals, cultivate potential, and invest in future leaders [163].

Mentor groups. Mentor groups are sponsored by senior EDOs, led by captains in key billets, and administered by a mentor group secretary to track participation and assignments [163].

7.12.4 Career Planner Guidance

Purpose. The career planner helps visualize career paths, communicate preferences, and support mentoring discussions with flag officers, mentor group leads, and detailers [163].

Content. Maintain several viable paths toward crown jewel jobs or command, and update the planner as assignments evolve and mentoring feedback is received [163].

Personal considerations. Include geographic preferences and family considerations to ensure mentors can provide realistic guidance and options [163].

7.12.5 Prepare and Conduct Mentoring

Prepare early. Provide read-ahead materials at least 48 hours in advance (career planner, OSR/PSR, and questions) and review your mentor's background to focus the discussion [163].

Core topics. Typical discussions include qualification status, assignment planning, promotion readiness (DAWIA level, APM status), and personal considerations such as family or Exceptional Family Member (EFM) impacts [163].

Mindset. Mentoring advice is autobiographical; be open, take notes, and treat sessions as a professional interview and learning opportunity [163].

7.12.6 Follow-Up

Sustain the relationship. Send a thank-you note, share updated planners, and provide periodic updates outside formal windows to maintain momentum [163].

Pay it forward. Share mentor resources with peers and mentor others as you would like to be mentored [163].

7.12.7 Quick Review

- Formal mentoring is required annually, but informal mentoring is encouraged throughout the year [163].
- The LDF and mentor groups guide community standards for mentoring and development [163].
- Effective mentoring requires preparation, focused discussion, and disciplined follow-up [163].
- Career planners and OSR/PSR reviews anchor the session and sharpen promotion readiness [163].
- Mentoring works best when advice is internalized and shared forward with others [163].

7.13 Warrior Toughness

7.13.1 Summary

Purpose. Warrior Toughness (WT) integrates stress management and human performance optimization so Sailors can perform under pressure in combat and daily operations [164].

Leadership link. WT reinforces the Navy core attributes of integrity, accountability, initiative, and toughness as emphasized in leadership development guidance [164].

Warrior mindset. The warrior mindset blends mind, body, and soul with a cycle of commitment, preparation, execution, and reflection [164].

Training tools. Mindfulness, goal setting, mental rehearsal, energy management, and intentional recovery are core techniques for sustaining peak performance [164].

Resilience focus. WT emphasizes recovering quickly from setbacks and maintaining optimism and persistence under adversity [164].

7.13.2 Warrior Toughness and Resilience

Definition. Toughness is the ability to take a hit and keep going by tapping all sources of strength and resilience while sustaining warfighting excellence [164].

Core attributes. WT aligns with the Navy core attributes and values, reinforcing integrity, accountability, initiative, and toughness across the force [164].

EDO alignment. The LDF describes resilience as handling pressure effectively, remaining optimistic, and recovering quickly from setbacks [164].

7.13.3 Warrior Mindset

Mind, body, soul. The warrior mindset balances cognitive focus, physical readiness, and personal conviction to sustain performance under stress [164].

Execution cycle. Commitment, preparation, execution, and reflection provide a repeatable loop for learning and improvement across operational scenarios [164].

7.13.4 Training and Performance Techniques

Performance psychology tools. Mindfulness, goal setting, recalibration, mental rehearsal, peer support, and energy management help regulate stress and maintain readiness [164].

Stress optimization. Success requires stress at the right level; training builds the capacity to operate effectively across varying stress intensities [164].

Recovery. Intentional recovery practices support sustainable performance and resilience during long operational cycles [164].

7.13.5 Mindfulness Benefits

Regulation. Mindfulness improves attention control, emotional regulation, and stress recovery, supporting the WT mindset [164].

7.13.6 Quick Review

- WT is the Navy's program for building resilience and performance under stress [164].
- The warrior mindset integrates mind, body, and soul with a commitment-preparation-execution-reflection cycle [164].
- Mindfulness and performance psychology tools help regulate stress and sustain readiness [164].
- WT supports the core attributes of integrity, accountability, initiative, and toughness [164].
- Resilience emphasizes fast recovery, optimism, and persistence during adversity [164].

7.14 Critical Conversations

7.14.1 Summary

Purpose. Critical conversations build a culture of awareness and candor that improves organizational performance and decision quality [165].

Bias and assumptions. Unchecked assumptions and cognitive biases distort judgment and undermine the Get Real, Get Better (GRGB) mindset [165].

Emotional intelligence. Emotional Intelligence (EQ) enables leaders to manage emotions, reduce conflict, and sustain trust during difficult discussions [165].

Preparation. Effective conversations require a safe space, clear intent, and understanding of human motivation and reactions [165].

Techniques. Use neutral tone, shared values, active listening, and solution-focused framing to keep dialogue constructive [165].

7.14.2 Principles and Rules of Engagement

Consideration and candor. Leaders should engage with transparency, invite participation, and reinforce non-attribution discussion norms when appropriate [165].

Assumptions. Recognize that different perceptions can be valid; test assumptions early to prevent flawed conclusions [165].

7.14.3 Emotional Intelligence and Bias Awareness

Emotional intelligence. EQ supports empathy, self-management, and constructive conflict resolution across teams [165].

Common biases. Halo and horn effects, confirmation bias, anchoring, and authority bias can distort reasoning and must be actively challenged [165].

7.14.4 Preparing for Difficult Conversations

Create the conditions. Schedule a convenient time, establish a safe environment, and approach the discussion with openness and vulnerability [165].

Know the dynamics. Anticipate human motivation (heart, head, hand) and reactions (fight, flight, freeze, fawn, flop) to better guide the conversation [165].

7.14.5 Techniques for Constructive Dialogue

Build trust. Use shared values, active listening, and respect to establish common ground and reduce defensiveness [165].

Use "I" statements. Frame concerns around observations and impact rather than blame to preserve dignity and encourage problem solving [165].

Focus on solutions. Discuss future actions, avoid assuming full resolution in one session, and reframe if the conversation becomes confrontational [165].

7.14.6 Quick Review

- Critical conversations require candor, inclusivity, and explicit engagement rules [165].

- Assumptions and biases must be surfaced to prevent flawed decisions [165].
- EQ enables leaders to manage emotion and maintain team cohesion during conflict [165].
- Preparation and safe-space conditions increase the odds of productive dialogue [165].
- Trust, empathy, and solution framing help move contentious issues toward resolution [165].

7.15 Leader Development Framework

7.15.1 Summary

Navy framework. The Navy Leader Development Framework (NLDF) supports NDS objectives and the Navy design to develop leaders who improve themselves and their teams faster than adversaries [166].

EDO framework. The LDF aligns EDO development with character, competence, and culture outcomes and embeds mentoring as a core community practice [166].

Development lanes. Education, on-the-job training, and self-guided learning build balanced leaders across technical, character, and team dimensions [166].

Mentoring and coaching. Career counseling focuses on milestones, mentoring builds long-term relationships, and coaching targets discrete development needs [166].

Command readiness. Command screening emphasizes mission focus, EQ, character, communication, and competence as predictors of command success [166].

7.15.2 Navy Leader Development Framework

Strategic alignment. The NLDF supports the NDS and Navy design by maximizing the performance of Sailors through deliberate leader development [166].

Leader expectations. Effective leaders commit to continuous growth, uphold Navy core values, and build competence, character, and culture within their teams [166].

7.15.3 Leader Development Lanes

Schools. Education and certification programs provide foundational knowledge and professional credibility [166].

On-the-job training. Qualifications, EDQP, and command-sponsored development translate learning into operational competence [166].

Self-guided learning. Reading, mentoring, and professional development activities sustain growth and reinforce the GRGB mindset [166].

7.15.4 EDO Leader Development Framework

Community strategy. The LDF provides a structured path to develop EDO leaders for major command and equivalent positions while aligning to CNO guidance [166].

Mentoring policy. Formal mentoring is embedded in the LDF and supports character, competence, and culture development over the notional career path [166].

7.15.5 Mentoring, Coaching, and Career Counseling

Career counseling. Focuses on milestones and timing for assignments and promotions [166].

Mentoring. Builds ongoing relationships that integrate professional and personal development beyond milestones [166].

Coaching. Uses time-bound engagements to target specific development needs or skills [166].

7.15.6 Command Screening Attributes

Command readiness. Command screening evaluates mission focus, EQ, character, communication, and competence as predictors of success in command [166].

7.15.7 Quick Review

- The NLDF and LDF align leader growth with Navy strategic priorities and community needs [166].
- Development lanes include schools, on-the-job training, and self-guided learning [166].
- Mentoring, coaching, and career counseling serve different development purposes [166].
- EDQP and mentoring are central to EDO leader development planning [166].
- Command screening emphasizes mission, EQ, character, communication, and competence [166].

8 Civilian Personnel

8.1 Civilian Personnel Overview

8.1.1 Summary

Why it matters. Civilian Personnel (CIVPERS) management is a major EDO responsibility and directly impacts readiness across NAVSEA and warfare center commands [167].

Myths vs. reality. Common myths about civil service (e.g., poor performance or inflexibility) are often inaccurate; effective leaders treat civilians as equal teammates while following the governing rules [167].

Guidance sources. Civilian personnel policy flows from Congress, executive orders, OPM, the DoW, SECNAV, NAVSEA guidance, and negotiated labor agreements [167].

HRO focus. Human Resources Office (HRO) primary functions are classification, staffing, employee relations, and labor relations; HRO and Equal Employment Opportunity (EEO) are distinct lanes, but local offices may coordinate process management, so use the servicing HRO/EEO structure [167].

EDQP link. EDQP requires completion of a local "Supervision of Civilian Personnel" course, reinforcing the need to understand CIVPERS management basics [167].

8.1.2 Why CIVPERS matters

Workforce scale. Civilian personnel dominate the manning at key technical commands and shipyards, making civilian leadership a core readiness driver for EDOs [167].

Table LVII: Illustrative civilian manning levels at selected activities

Activity	Military	Civilian
NIWC LANT	120	4,970
NIWC PAC	200	4,800
Norfolk NSY	810	10,600
NSWC Carderock	10	3,600

Continued on next page

Table LVII: Illustrative civilian manning levels at selected activities **(Continued)**

Activity	Military	Civilian
NSWC Port Hueneme	70	2,790
NSWC Panama City	45	1,410
Pearl Harbor NSY and IMF	490	5,270
Puget Sound NSY	35	12,000
Southwest Regional Maintenance Center	750	1,150

8.1.3 Myths and guidance sources

Common myths. Misconceptions include believing civil servants are unaccountable, unions block mission success, and private industry rules always apply; leaders should verify facts and engage partners early [167].

Guidance stack. Key sources include Congress, the President, OPM, OSW, the Department of the Navy, NAVSEA, Human Resources Operation Center and Service Center channels, local HRO guidance, and negotiated labor agreements [167].

8.1.4 HRO primary functions

Classification. Determine position titles and pay bands/grades using OPM classification standards [167, 168].

Staffing. Fill vacancies and manage hiring actions under applicable hiring authorities [167, 169].

Employee relations. Guide supervisors and employees on performance, discipline, and workplace issues [167].

Labor relations. Manage relationships and agreements with labor organizations [167].

8.1.5 Workforce shaping and classification

Classification basics. Classification assigns a pay plan, occupational series, grade/band, and title; supervisors define duties while HRO applies OPM standards to determine the final classification [168, 170].

Pay systems. The most common systems are Wage Grade (WG)/Wage Leader (WL)/Wage Supervisor (WS), General Schedule (GS), demonstration pay plans (e.g., Personnel Demonstration Project (DEMO)/Acquisition Workforce Personnel Demonstration Project (ACQDEMO)), Senior Scientist and Technical Manager (SSTM), and SES; understand grade/band structures before recruiting [170, 171].

Other pay and FLSA. Locality pay, premium pay, and Fair Labor Standards Act (FLSA) exempt/non-exempt status depend on position duties; ensure the PD matches actual responsibilities [170, 172].

Filling vacancies. Options include management reassignment, merit competition, competitive hiring, outsourcing, and mandatory placement programs; use Direct Hire Authority (DHA) only when a valid authority applies and plan timelines early [169, 170].

Downsizing levers. Workforce reductions rely on Reduction in Force (RIF), separation incentives, Voluntary Early Retirement Authority (VERA), and furloughs; sequence actions carefully to preserve critical skills [170].

Table LVIII: Federal Wage System appropriated-fund pay-plan cues

Pay plan	Grade structure	Board cue
WG	15 nonsupervisory grades	Trades, craft, and labor positions performing the work.
WL	15 leader grades	Lead positions that guide others while remaining closer to the work than a formal supervisor.
WS	19 supervisory grades	Wage-grade supervisors responsible for planning, assigning, and supervising trade work.

Source: *5 CFR 532.203: Structure of Regular Wage Schedules*, 2026 [171]

8.1.6 Equal Employment Opportunity anchors

Bases. Protected EEO categories include race, color, religion, sex (including pregnancy, sexual orientation, gender identity, and transgender status), national origin, age (40+), disability, genetic information, and retaliation for protected EEO activity; equal-pay claims are normally handled as sex-based wage discrimination rather than a separate board category [173, 174, 175].

Theories. Primary theories are disparate treatment, adverse impact, and denial of reasonable accommodation (disability or religion) unless undue hardship applies [173].

Sexual harassment. Quid pro quo and hostile work environment are the two primary types; hostile environments require severe or pervasive conduct that interferes with work [173, 174].

Complaint process. The federal-sector process includes informal counseling, a formal complaint, investigation, hearing or agency decision, and appeal; normally the employee must contact an EEO counselor within 45 days, file a formal complaint within 15 days after notice, and the agency has 180 days to investigate if the complaint is accepted [173, 175].

Management responsibilities. Leaders must contact the EEO office, preserve records, cooperate with the process, take appropriate action, and prevent reprisal; Navy harassment response timelines should be checked against the current command instruction [173, 175, 176].

8.1.7 Employee relations essentials

Supervisor and employee roles. Supervisors set expectations, reward good performance, and address poor conduct; employees are responsible for acceptable performance, attendance, and workplace behavior [177].

Work schedules. Common schedules include basic, Compressed Work Schedule (CWS) (e.g., 5-4/9 or 4/10), and flexible schedules; overtime and telework are governed by local policy and FLSA status [172, 177].

Leave categories. Annual, sick, Leave Without Pay (LWOP), Absent Without Leave (AWOL), Family and Medical Leave Act (FMLA), paid parental leave, military and court leave, the Voluntary Leave Transfer Program, and administrative excusal each carry specific approval and documentation rules [177].

Performance management. Defense Performance Management and Appraisal Program (DPMAP) or ACQDEMO processes require clear performance elements and standards, employee input opportunity, supervisor approval, performance discussions, at least one documented progress review, and Performance Improvement Plans (PIPs) or equivalent action for unacceptable performance; probation and trial periods vary by appointment authority and should be verified with HRO [177, 178, 179].

Probationary/trial period rule. Current OPM guidance implementing E.O. 14284 treats probationary and trial periods as extensions of the hiring process; agencies must certify continued employment in the public interest before an employee continues beyond the period. For board answers, this is the expedited lane for new/probationary employees, not a waiver of due process for tenured employees [179].

Awards and discipline. Incentive awards include monetary, time-off, and honorary recognition; discipline is progressive and must be timely, fair, and focused on correcting behavior [177].

8.1.8 Within-grade increases and GS step timing

Acceptable performance gates. Within-Grade Increases (WGIs) require an “acceptable level of competence” determination (typically a “Fully Successful” rating); maintain clear standards and timely appraisals so eligibility decisions are defensible [180].

Table LIX: GS within-grade increase waiting periods

Advancement to steps	Required waiting period at previous step
Steps 2, 3, and 4	52 weeks (1 year)
Steps 5, 6, and 7	104 weeks (2 years)
Steps 8, 9, and 10	156 weeks (3 years)

Assuming sustained acceptable performance, advancing from Step 1 to Step 10 takes 18 years (three years for Steps 1–4, six for Steps 4–7, and nine for Steps 7–10) [180, 181]. Exceptional GS performers may receive a Quality Step Increase (QSI), but OPM limits it to employees with a Level 5 or highest-summary rating and sustained high-quality performance significantly above Fully Successful [182].

8.1.9 Performance vs. conduct pathways

Supervisor baseline. Set and communicate standards, evaluate against them, and partner with HRO early; use coaching to reinforce desired behaviors [177].

Table LX: Differentiating performance and conduct issues

Aspect	Unacceptable performance	Misconduct
Definition	Failure to perform assigned duties at an acceptable level	Violation of a rule, law, or regulation (e.g., policy violations, absence without leave (AWOL), insubordination, security breaches)
Focus of action	Improve performance; not intended to punish	Correct behavior; penalties may be more severe depending on the offense
Standard of proof	Chapter 43: substantial evidence; Chapter 75: preponderance of the evidence	Preponderance of the evidence
Guiding principles	Chapter 43 requires an opportunity period or PIP; Chapter 75 does not require a PIP but requires efficiency-of-service nexus, due process, and Douglas Factor penalty analysis	Chapter 75 penalty must be reasonable; apply the Douglas Factors when selecting a penalty

Handling unacceptable performance. Start with clear standards, specific examples, assistance, and closer feedback. If performance remains unacceptable, coordinate with HRO and counsel on the proper Chapter 43 or Chapter 75 path. Under DPMAP, a Chapter 43 PIP identifies the deficient elements, required Fully Successful performance, assistance, consequences, and a reasonable demonstration period that is generally no more than 30 calendar days unless a longer period is necessary; a Chapter 43 reduction or removal also requires advance written notice, an opportunity to answer, and a written decision. Current Department-level guidance directs Components, subject to collective-bargaining obligations, to use a Chapter 75 removal process for unacceptable-performance separations under 5 U.S.C. 7513; that process does not require a PIP, requires at least 30 days' advance notice and a reasonable answer period of at least 7 days, uses the preponderance standard, and applies Douglas Factors to the removal decision. For probationary or trial employees, current OPM guidance gives agencies a separate certification/termination lane tied to whether continued employment advances the public interest; coordinate closely with HRO because appointment authority, bargaining-unit status, veteran preference, and local procedures can change appeal rights. A WGI is not a progressive-discipline tool; it is delayed or denied when the employee lacks an acceptable level of competence or receives an Unacceptable rating, and persistent failure may lead to reassignment, change to lower grade, or removal [177, 178, 179, 180, 183, 184].

Addressing misconduct. Apply progressive discipline calibrated to the offense: verbal or written counseling, reprimand, suspension, change to lower grade, or removal. The chosen penalty must be reasonable in light of the Douglas Factors (offense severity, job level, record, consistency, rehabilitation potential, and mission impact) [177, 185].

Employee rights and documentation. Honor Weingarten rights when a bargaining-unit employee is examined by agency representatives as part of an investigation, reasonably believes discipline may result, and requests union representation; if representation is requested, either provide it, postpone, or end the interview [186, 187]. Keep actions non-discriminatory, coordinate on any EEO complaints, and document counseling, training, and decisions to support subsequent actions [173, 175, 177].

8.1.10 Labor relations touchpoints

Definitions and coverage. Collective bargaining units are represented by an exclusive representative under a negotiated CBA; management officials, supervisors, confidential employees, certain HR and security roles, contractors, and uniformed members are excluded [186].

Rights. Employees may join or refrain from unions, unions have a duty of fair representation, and management retains rights to mission, budget, organization, and assignments [186].

Bargaining obligations. Agencies must bargain procedures and appropriate arrangements for management actions; impact and implementation bargaining requires notice, and unions must request bargaining or waive the right [186].

Past practice. Established practices that are clear, consistent, long-standing, and mutually accepted can become enforceable; changes require notice-and-bargain steps [186, 188].

Formal discussions and Weingarten. Unions may attend formal discussions on conditions of employment, and bargaining-unit employees may request union representation during investigatory examinations they reasonably believe may lead to discipline [186, 187].

Official time and ULPs. Official time covers representational activities but not internal union business; Unfair Labor Practice (ULP) charges generally must be filed with Federal Labor Relations Authority (FLRA) within six months, subject to statutory exceptions [186, 189].

8.1.11 CIVPERS wrap-up

Core takeaways. Civilians are teammates with different experiences and expectations; do not assume familiarity with uniformed service norms, but do assume professionalism and commitment to the mission [190].

Leader engagement. Invest in development, involve civilians in awards and performance processes, and communicate expectations consistently and transparently [190].

Use your partners. Leverage HRO and senior civilians for guidance, and document decisions related to hiring, performance, and conduct [190].

Practical application. Case studies reinforce how to apply rules on overtime, past practice, and workplace conduct while preserving fairness and mission focus [190].

8.1.12 Coursebook cue: CIVPERS applied rules

Coursebooks 6.1.2 through 6.1.6 test whether students can apply civilian personnel rules to realistic supervisor scenarios, not just define the terms [170, 173, 177, 186, 190].

Table LXI: Vacancy-fill method selection cues.

Method	When it fits
Competitive hiring	Best when the command needs a broad applicant pool or permanent placement.
Merit promotion	Best when filling from qualified federal employees under merit-system rules.
Reassignment or detail	Best for near-term mission need when the employee already has the required grade and qualification fit.
Temporary promotion	Best when higher-graded duties are temporary and properly classified.
Direct-hire or special authority	Best only when a valid authority applies and documentation supports its use.

For FLSA, do not assume every salaried federal employee is exempt. The practical supervisor question is whether the position is exempt or non-exempt and whether overtime was required or permitted beyond the basic work week [170, 172].

Table LXII: EEO participant rights board cue.

Participant	Supervisor cue
Complainant	Has protected access to the EEO process and must not face reprisal for using it.
Responsible Management Official	Must cooperate, preserve records, avoid retaliation, and keep the response factual and professional.
Witness or advisor	Must be protected from interference or reprisal tied to participation.

Table LXIII: Leave-use scenario cues.

Leave type	Supervisor cue
Annual leave	Schedule subject to mission needs and applicable policy; avoid arbitrary denial.
Sick leave	Use for qualifying medical, family-care, bereavement, adoption, or exposure situations under governing rules.
Leave without pay	Discretionary unless an entitlement applies; document the basis and duration.
Family and medical leave	Check eligibility, covered reason, notice, certification, and job-protection rules.
Administrative leave	Use only when authorized; do not treat it as a general substitute for discipline or convenience.

Table LXIV: Labor-relations scenario cues.

Issue	Supervisor cue
Official time	Some representational activity is statutory; other use depends on contract, bargaining, or negotiated procedure.
Formal discussion	Union notice and opportunity to attend may be required when conditions meet the statutory test.
Weingarten right	A bargaining-unit employee may request union representation during an agency investigatory examination when the employee reasonably fears discipline.
ULP risk	Bypassing the union, refusing to bargain, retaliation, coercive statements, unilateral changes, or late filings can create unfair-labor-practice exposure.

Board cue. For supervisor scenarios, identify the employee category, authority, required process, documentation, employee or union right, and the behavior that would create reprisal, EEO, FLSA, or ULP risk.

Appendices

A Platinum Card

The *Platinum Card* is a primary study tool that will have most of the information required for your boards. Sources: EDO Coursebook Modules 3.1.1, 3.1.3, 3.1.4, and 3.1.6. Be able to draw and talk through Figures A.1–A.9.

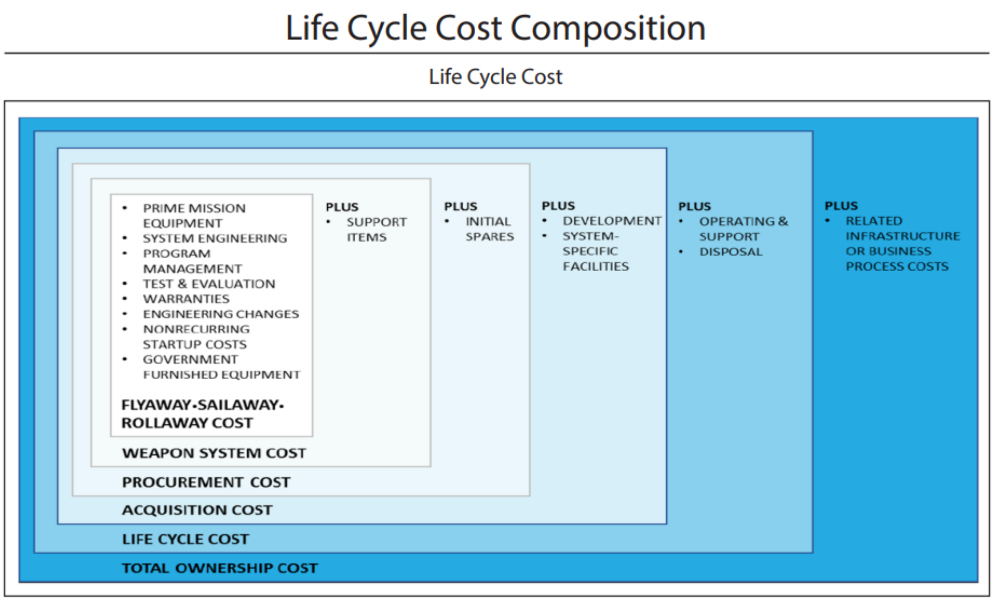

Figure A.1. *Platinum Card: Life Cycle Cost Composition*

Cost Estimating Requirements

	ACAT IB, IC & ID (MDAP)	ACAT IA
POE	Program initiation & all subsequent milestones, including FRP DR	Program initiation & all subsequent milestones - Includes Economic Analysis
CARD	MS A & all subsequent milestones including FRP DR - Draft 180 days prior to OIPT/milestone - Final 45 days prior to OIPT/milestone	MS A & whenever Economic Analysis is required - Draft 180 days prior to OIPT/milestone - Final 45 days prior to OIPT/milestone
CCE	MS A and all subsequent milestones including FRP DR	MS A and whenever an Economic Analysis is required
ICE	Required by law for all MDAP programs * - Prepared by OSD CAPE for ACAT ID, and ACAT IC at discretion of USD(A&S) - Prepared by component cost agency (AFCAA, DASA-CE, NCCA) for ACAT IC and ACAT IB (if no CAPE estimate) - In advance of any certification under Title 10, USC, Section 2366a (MS A) and Section 2366b (MS B) - In advance of any decision to enter low-rate initial production (LRIP) (MS C) or full-rate production (FRP DR) - In advance of any certification of MDAPs that experience critical cost growth (Title 10, USC, Sec 2433a)	

*ICE statutory requirement (Title 10, US Code, Section 2434) and P.L. 111-23, May 22, 2009
Source: DoDI 5000.02 January 7, 2015, Incorporating Change 3, August 10, 2017 and Weapon Systems Acquisition Reform Act of 2009

ACAT II & ACAT III: POE (and, at MDA discretion, an independent cost estimate prepared by the component cost agency) at program initiation and all subsequent milestones

AFCAA = Air Force Cost Analysis Agency
ACAT = Acquisition Category
CAPE = Cost Assessment & Program Evaluation
CARD = Cost Analysis Requirements Description
CCE = Component Cost Estimate
DASA-CE = Deputy Assistant Secretary of Army (Cost & Economics)
FRP DR = Full Rate Production Decision Review
ICE = Independent Cost Estimate
MDA = Milestone Decision Authority
MDAP = Major Defense Acquisition Program
NCCA = Naval Center for Cost Analysis
OIPT = Overarching Integrated Product Team
POE = Program Office Estimate
USD(A&S) = Under Secretary of Defense (Acquisition and Sustainment)

Figure A.2. *Platinum Card: Cost Estimating Requirements*

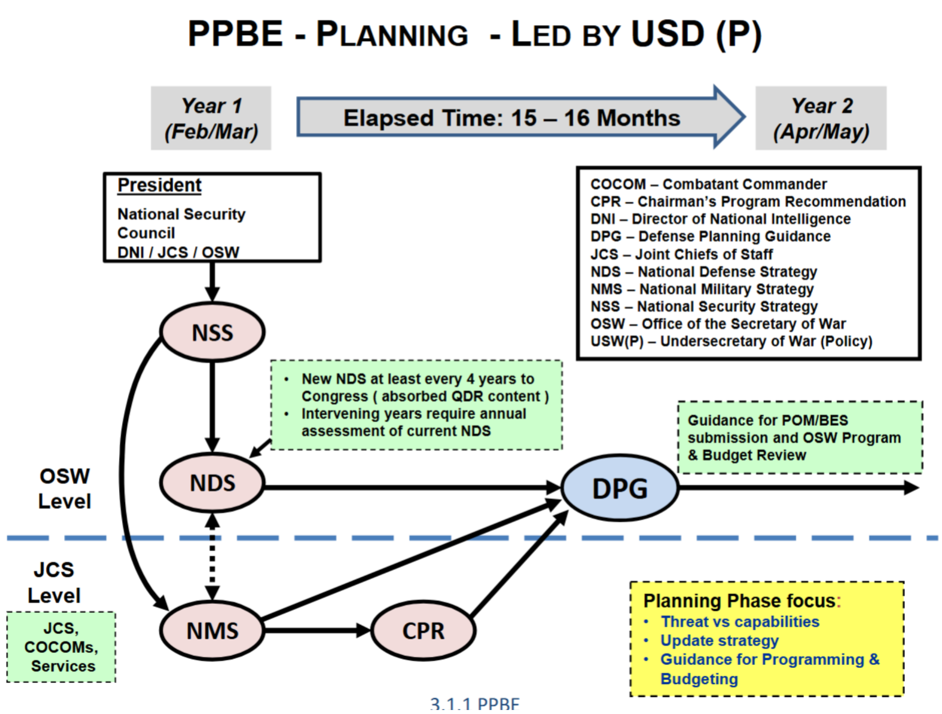

Figure A.3. *Platinum Card: PPBE Planning Phase (led by USW(P))*

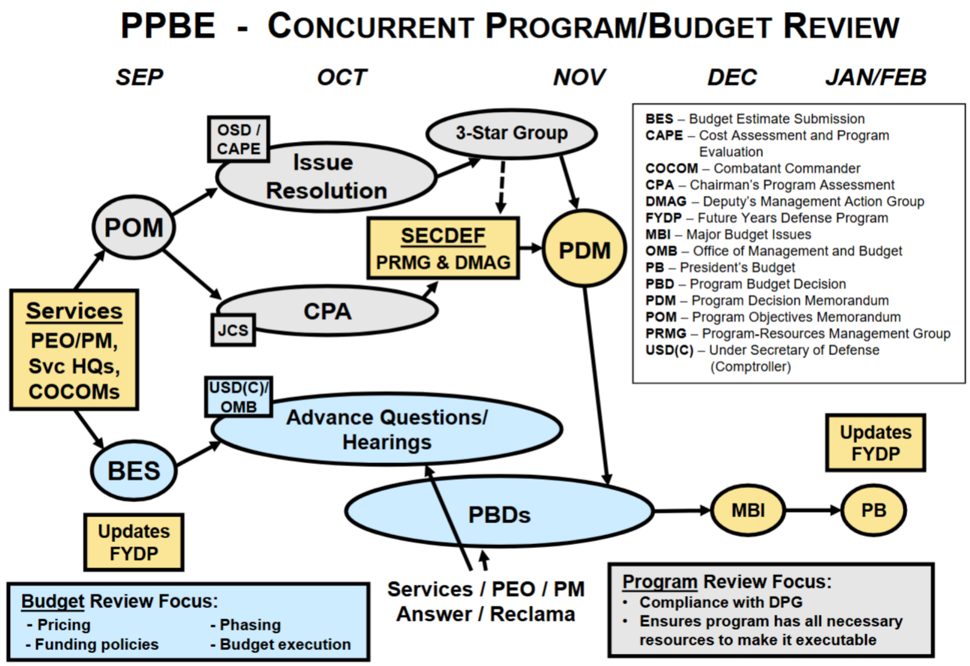

Figure A.4. *Platinum Card: Concurrent Program/Budget Review*

Below Threshold Reprogramming

Amounts are cumulative over Entire Period of Obligation Availability

APPRN	MAX INTO	MAX OUT	LEVEL OF CONTROL	OBL AVAIL
RDT&E	Lesser of +$15M or + 20%	Lesser of - $15M or - 20%	Program Element	2 Years
PROC	Lesser of + 15 M or + 20%	Lesser of - $15M or - 20%	Line Item	3 years SCN: 5 Years
O&M	+ $15M	- $15M	Budget Activity (or Defense Agency) Some Sub-Activity Limitations on Decreases (see reference below)	1 Year
MILPERS	+ $15M	- $15M	Budget Activity	1 Year
MILCON	Lesser of +$2M or + 25%	No Specific Congressional Restriction	Project	5 Years

Reference Sources: DoDFMR 7000.14-R, Volume 3, Chapter 6 (Sept 2015) and Chapter 7 (Mar 2011) Joint Explanatory Statement of Congress Making Appropriations for the Department of Defense for Fiscal Year 2024

Figure A.5. *Platinum Card: Below Threshold Reprogramming*

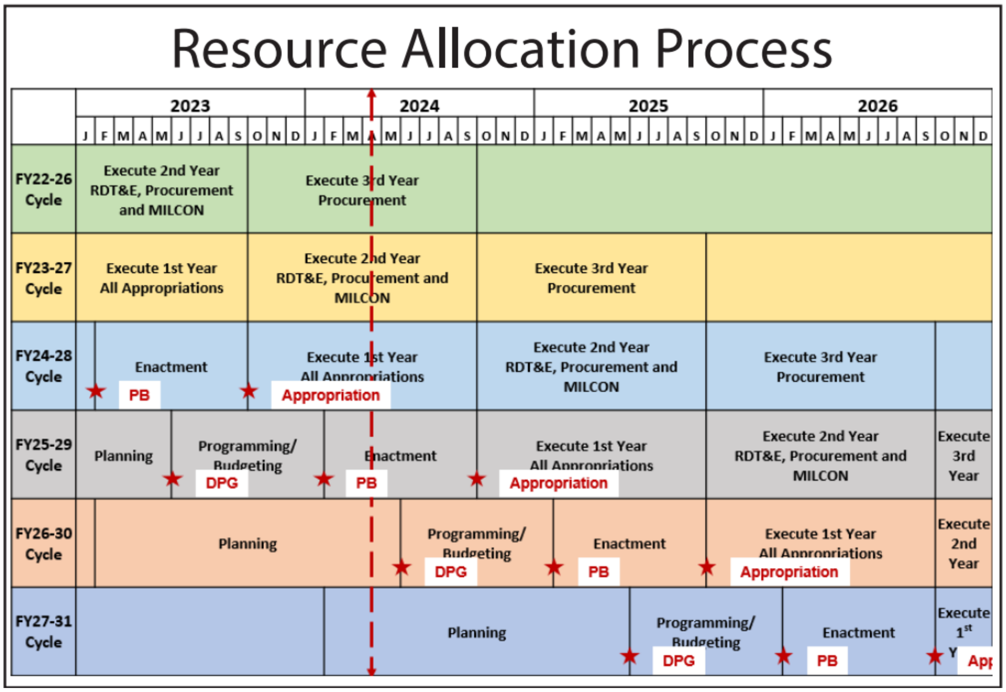

Figure A.6. *Platinum Card: Resource Allocation Process*

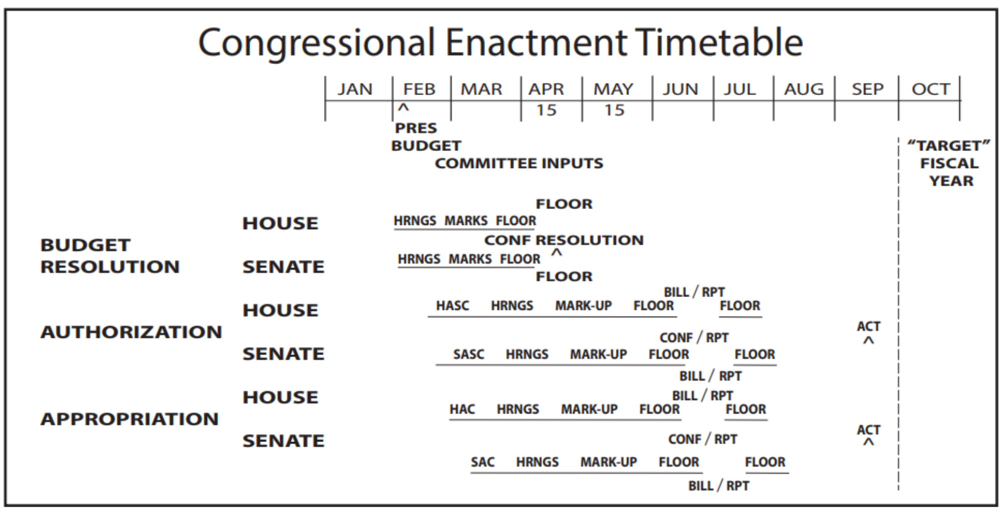

Figure A.7. *Platinum Card: Congressional Enactment Timeline*

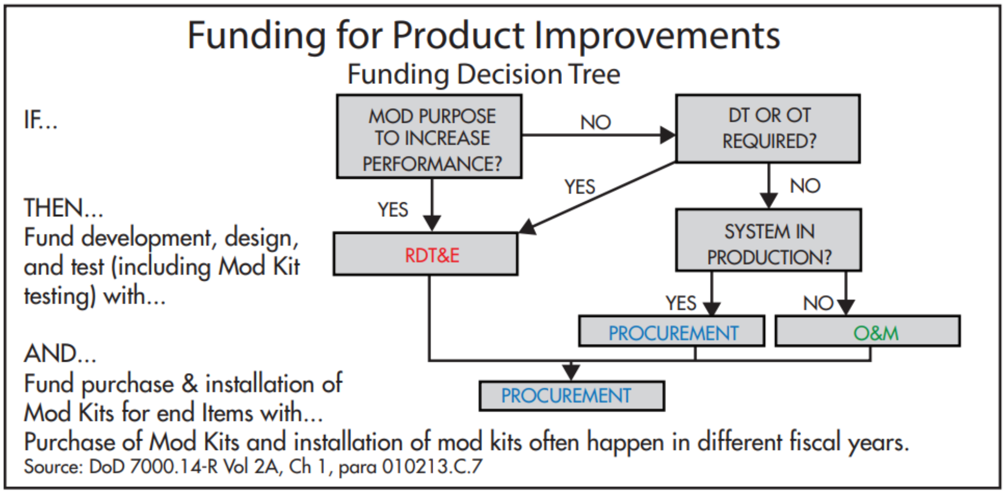

Figure A.8. *Platinum Card: Funding for Product Improvements*

RDT&E. Engineering changes requiring development/test.

Procurement. Production/installation of approved mods/end-items.

O&M. Minor mods/installation labor when authorized and not creating a new end-item.

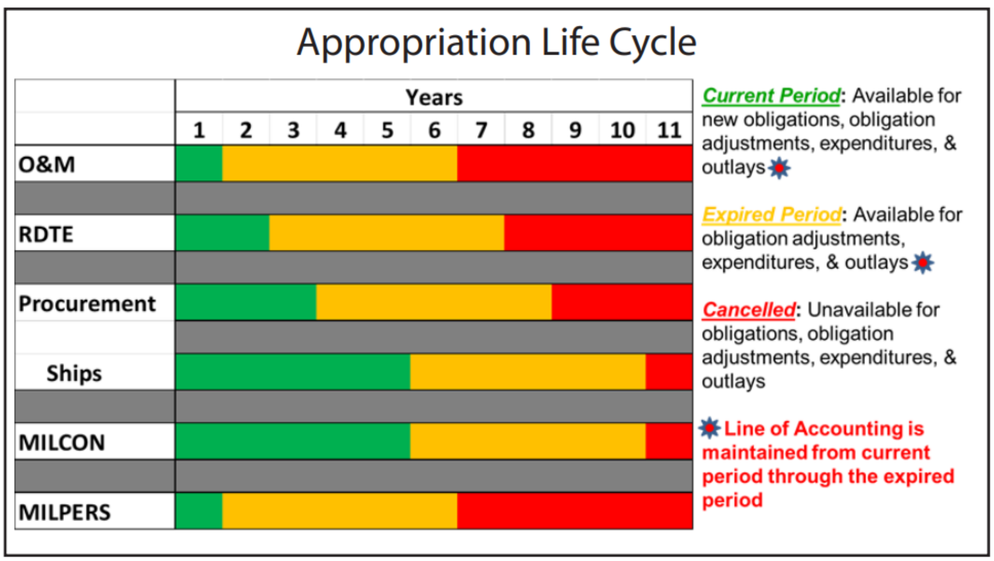

Figure A.9. *Platinum Card: Appropriation Life Cycle*

B EDO Roles in Acquisition and Modernization

B.1 Key Roles in Acquisition and Modernization

Department acquisition governance.

Sources: SECNAVINST 5400.15D; SECNAVINST 5000.2G; DoWI 5000.85.

- ASN(RD&A), the SAE and CAE for the DON, sets acquisition policy, charters PEOs / DRPMs, and adjudicates milestone decisions when delegated by the Milestone Decision Authority.
- PEOs and DRPMs hold delegated program authority, establish baselines, and are accountable for translating capability needs into executable acquisition strategies.
- Current as of 2026-06-04, DON has publicly established nine PAEs. PAE Mission Systems spans PEO C4I, PEO Digital, PEO IWS, PEO MLB, DRPM Overmatch, DRPM LRNFO, DRPM MILE, Minotaur from PMA-290, and mission-system elements across NAVWAR, NAVSEA, NAVAIR, and Marine Corps Systems Command [11, 21].
- SYSCOM Commanders provide the workforce, infrastructure, and TA warrants that underpin PEO execution inside the Navy matrix construct.

Joint requirements governance.

Sources [90, 191].

JROC: Validates joint capability requirements, prioritizes gaps, and issues Joint Requirements Oversight Council Memorandum (JROCM) direction that becomes binding input to Gate 2 and milestone documentation.

Joint Staff Gatekeeper/JCB: Controls the Knowledge Management and Decision Support (KM/DS) queue, assigns Joint Staffing Designators, and routes most Navy submissions for JCB-level validation to keep JROC time focused on the highest-risk issues.

FCBs: Chair certification working groups (Net-Ready, Force Protection, System Survivability), monitor DOTMLPF-P implementation, and elevate Configuration Steering Board actions when requirement trades alter program baselines.

PAEs and Capability Trade Councils (CTCs): The 2025 Acquisition Transformation Strategy empowers PAEs to convene Capability Trade Councils that record requirement/cost/schedule trades in real time so PEO teams keep the DAS synchronized with evolving demand signals.

Combatant command sponsors.

Sources [1, 191].

- **CCDRs** own warfighting problems, sign JUONs/JEONs, and direct lead Services or agencies to prosecute capability gaps that demand rapid solutions.
- **CCMD staffs** conduct mission-thread analysis and architecture sprints with Fleets, labs, and industry to translate operational risk into measurable capability requirements before the Gatekeeper accepts a package.
- Commanders remain voting participants during validation reviews and oversee fielding to ensure delivered capability solves the theater problem that generated the requirement.

Joint Chiefs leadership.

Sources [1, 191, 192].

CJCS: Principal military adviser to the President/SECWAR, co-chairs the JROC, and leads joint concept and readiness development that frames every Capability-Based Assessment.

VCJCS: Chairs the JCB, co-leads the Deputy's Management Action Group (DMAG), and arbitrates cross-Service requirement or resourcing disputes before they escalate to the Deputies or Secretary level.

OPNAV corporate leadership.

Source [1].

CNO: Serves as Service Chief, chairs the Navy corporate board, and approves Navy positions for Gate 0/1 decisions before submissions move to the Joint Staff.

OPNAV resource integrators and sponsors (N8, N82, N9, Integrated Warfare Directorate / resource sponsor code 9I (N9I)/9I): Translate Fleet priorities into the POM/FYDP, enforce two-pass/seven-gate rigor, and synchronize requirements, budgets, and PEO execution.

Acquisition decision authorities.

Sources [9, 84, 193].

DAE / USW(A&S): Issues acquisition policy, chairs the Defense Acquisition Board, and retains MDA for ACAT ID/Major Automated Information System (legacy ACAT I equivalent) (IAM) programs unless delegated.

SAE/ASN(RD&A): Assigns Navy MDA for ACAT II and below, approves Service-tailored governance (e.g., two-pass/seven-gate), and charters PEOs/DRPMs.

Delegated MDA (PEO/DRPM): Signs Acquisition Decision Memoranda, manages Configuration Steering Boards, and keeps KPP/APB trades synchronized with requirements direction and statutory certifications.

Program Authority chain.

Sources: SECNAVINST 5000.2G; DoWI 5000.85.

PEO/PA: Owns program outcomes, charters PMs, approves acquisition strategies and baselines.

DRPM: Direct-report program leads for special access or priority portfolios with the same milestone authority as PEOs.

SPM/PM: Accountable for cost/schedule/performance, leads risk management, orchestrates IPTs.

SHAPM: Delivers new hulls; coordinates design/build/activation sequencing.

SLM: Drives sustainment and modernization packages for in-service assets.

PARM/SPD: Sponsors platform-level upgrades and alteration packages.

Technical Authority chain.

Source: SECNAVINST 5400.15D.

Ultimate Technical Authority (UTA) / CHENG: Issues enterprise technical policy, owns warrants.

TWHs: Certifies design compliance, adjudicates departures/concurrence packages.

Engineering Agents (ISEA, DA, AEA, SIA, TDA): Execute lifecycle engineering; provide in-service engineering decisions under delegated authority.

Operational test oversight.

Sources [84, 194].

DOT&E: Independent OSW authority that concurs on TEMPs, monitors operational test realism, and reports suitability/effectiveness outcomes to Congress.

Service Operational Test Agencies. COMOPTEVFOR (COTF), Air Force Operational Test and Evaluation Center (AFOTEC), Army Test and Evaluation Command (ATEC), and Marine Corps Operational Test and Evaluation Activity (MCOTEA) plan and execute IOT&E/FOT&E, render the OTRR recommendation, and provide Service suitability findings to the MDA.

Operational test readiness and fleet release.

Sources [126, 128].

COTF/COMOPTEVFOR: Chairs OTRR events, certifies whether OT&E adequacy and logistics/training enablers support Fleet introduction, and can recommend restricted Fleet release when Cat I deficiencies remain.

MDA/PEO: Disposition Cat I/II findings, resource fixes, and may defer FRPDR until OT&E adequacy and deficiency burn-downs meet Title 10 standards.

TYCOM: Accepts Fleet risk, sets operating restrictions, and updates tactics and training based on OT&E and LFT&E findings.

Developmental test governance.

Sources [126].

DASD(T&E): Designates Chief Developmental Testers and Lead Developmental Test and Evaluation Organizations for ACAT I programs, adjudicates test resource shortfalls, and co-chairs the Defense T&E WIPT.

CDT: Synchronizes the integrated test schedule, deficiency management cadence, and readiness evidence required for each TRR/milestone.

LDTO: Provides instrumentation, modeling & simulation accreditation, range execution, and authoritative data packages for developmental tests.

Live-fire statutory leads.

Sources [106, 126].

DOT&E Live Fire Test Director: Approves plans, monitors execution realism, and delivers the statutory lethality/vulnerability report to Congress.

Program Manager/CHENG: Funds threat-representative articles, integrates survivability design fixes, and aligns LFT&E data with the Acquisition Program Baseline and sustainment plans.

Service Safety Centers and Warfare Centers: Provide casualty data, threat surrogates, and post-test forensic analysis that feed both live-fire reports and Fleet tactics updates.

NAVSEA headquarters roles.

Source: EDO Coursebook Module 2.1.2 (NAVSEA Organization, 2025 edition).

NAVSEA headquarters staff code (SEA) 01 Comptroller: BSO lead for NAVSEA financial governance and funds control.

SEA 02 Contracts: Enterprise contracting authority, policy, and warrant management.

SEA 03 Cyber Engineering and Digital: Drives cyber resiliency, digital engineering, and data transformation initiatives.

SEA 04 Industrial Operations: Oversees public shipyards, maintenance execution, and quality assurance.

SEA 05 CHENG: Chief TA; issues warrants, certifies designs, and maintains technical standards.

SEA 06 Sustainment: Manages product support strategies and lifecycle logistics integration.

SEA 07 Undersea Warfare: Dual-hatted as Program Executive Office, Undersea Warfare Systems (PEO UWS); oversees undersea combat systems sustainment.

SEA 08 Nuclear Propulsion: Three-hatted nuclear propulsion authority across DON and Department of Energy (DOE) roles.

SEA 09 Safety and Regulatory Compliance: Aligns enterprise safety governance and reporting.

SEA 10 Total Force and Corporate Ops: Manages workforce planning, corporate services, and governance.

SEA 21 In-Service Ships/CNRMC: Leads surface-ship sustainment and modernization (PMS 321/326/339/421/443/451, SEA 21I, SURFMEPP).

NAVWAR enterprise directorates.
Source: EDO Coursebook Module 2.1.3 (NAVWAR Enterprise, 2024 edition).

Code 1.0 Comptroller : Budget formulation/execution, BSO duties, and funds certification.

2.0 Contracts : Contracting policy, strategy reviews, and award/administration oversight.

3.0 Counsel : Acquisition law, ethics, protests, and claim resolution.

4.0 Logistics and Fleet Support : Lifecycle logistics, technical data, and fleet distance support integration.

5.0 CHENG/TA: Enterprise architectures, interoperability, and cyber certification authority.

6.0 Program Management : Portfolio governance, milestone preparation, and PEO integration.

7.0 Science and Technology : S&T portfolio management, prototyping, and technology transition.

8.0 Corporate Operations : Workforce, CIO services, facilities, security, and public affairs.

Fleet Readiness Directorate (FRD)-100 Fleet Support: Deployed engineering assistance and sustainment response.

FRD-200 Installations: Shore/afloat C4I installation planning and cutover execution.

Program Executive Offices (Navy portfolios).
Sources: SECNAVINST 5400.15D; EDO Coursebook Modules 2.1.4 and 3.1.5.

PEO CVN: Designs, builds, and sustains nuclear-powered aircraft carriers (e.g., PMS 312/378/379).

PEO IWS: Develops and sustains ship, submarine, aircraft, and allied combat-system portfolios; mission-aligned Integrated Warfare Systems (IWS) directorates cover sensors, weapons, C2, and allied integration.

PEO Ships: Oversees surface combatant and amphibious ship construction and modernization.

PEO USC: Leads LCS, Guided-Missile Frigate (FFG) 62, expeditionary, and unmanned surface/undersea portfolios.

Team Submarines (PEO SSN, Program Executive Office, SSBN (PEO SSBN), Program Executive Office, Columbia-Class Submarines (PEO Columbia), PEO UWS): Manages attack/strategic submarine acquisition, in-service support, and undersea combat systems.

PEO C4I: Delivers fleet C4I; Program Management Warfare (NAVWAR PEO C4I/NAVAIR PMS series) (PMW) 1XX focus on capability development, PMW 7XX on platform integration.

PEO Digital: Provides enterprise digital services (e.g., Flank Speed) across the DON.

PEO MLB: Modernizes manpower, logistics, and business IT systems.

PAE Mission Systems cue: Remember the established PAE as mission-system integration across NAVWAR/NAVSEA/NAVAIR/Marine Corps Systems Command portfolios; do not use it to delete still-current PEO recall unless official pages have caught up.

Contracting authority (KO family).
Sources: FAR; DFARS; NMCARS; EDO Coursebook Module 3.2.1.

PCO: Plans the acquisition, synchs with PMs before solicitation, awards and signs contracts/mods.

ACO: Oversees post-award performance, surveillance, and payment/contract administration (often DCMA / NAVSEA field activities).

TCO: Leads partial/full terminations, settlement negotiations, and equitable adjustments.

KO warrant: Defines the dollar/authority limits; only the warranted KO can bind or obligate the Government.

COR / assistant PM / engineering support: Provide technical surveillance and acceptance recommendations; cannot direct work or obligate funds (FAR Parts 1 and 42).

HCA: Approves actions such as letter contracts and high-value single-award IDIQs; may delegate no lower than flag/Senior Executive levels within the DON per FAR 1.601 and DFARS/NMCARS supplements.

SAE, ASN(RD&A): Signs D&Fs for multiyear contracting, extraordinary relief, and other actions reserved to the Service Acquisition Executive under FAR Subparts 1.7 and 17.1.

Program Manager / KO partnership.
Sources: DoWI 5000.85; EDO Coursebook Module 3.2.1.

- PM integrates warfighter need, technical baseline, and budget; KO Competition in Contracting Act (CICA), incentives, and change management before RFP release or modification execution.

Source selection governance.
Sources: FAR Parts 1, 5, and 15; DoW Source Selection Procedures; NAVSEA Source Selection Guide (2022); EDO Coursebook Module 3.2.2.

SSA: Senior official who approves the Source Selection Plan, receives SSAC/SSEB recommendations, and signs the best-value decision memorandum.

SSAC: Advisory council, when established or required by DoW/agency source-selection procedure, that synthesizes SSEB findings, compares proposals across factors, and briefs the SSA on trade-offs.

SSEB: Multi-disciplinary evaluators (technical, management, past performance, cost/price) who rate proposals against the solicitation evaluation factors, including Section M when the uniform contract format is used.

Small Business Professional / Competition Advocate : Confirms set-aside decisions, reviews subcontracting plans, and endorses synopsis waivers.

Cost/Price Analyst & Legal Counsel : Validate reasonableness determinations, alignment of Sections L/M, and clause sufficiency before release.

Award Fee Determining Official & Award Fee Board : Plans award-fee periods, chairs performance reviews, and signs the determination memo authorizing or withholding fee payments (FAR Part 16; NAVSEA Source Selection Guide).

Fiscal control & certification.

Source: Department of War Financial Management Regulation (DoW FMR) 7000.14-R Volume 3.

Funds Certifying Official/Comptroller : Verifies purpose/time/amount before obligation; Anti-Deficiency Act (ADA) safeguard.

Program/Budget Analyst : Tracks execution, monitors reprogramming thresholds, prepares obligation/expenditure burn-down.

Resource allocation chain : OUSW(C) apportions to the Navy; OPNAV (N82/FMB) allocates to BSOs; SYSCOM comptrollers (e.g., SEA 01) issue suballocations to executing activities.

PPBE resource sponsors.

Sources: EDO Coursebook Module 3.1.1 (PPBE, 2025 edition), NAVMAC Activity Manpower Management Guide Section 24, and current public OPNAVINST readiness/assessment instructions.

CAPE: Independent analytic staff supporting Program Review decisions and ICE/Program Office Estimate (POE) for major programs [49, 55].

N8: Navy PPBE resource integrator ("checkbook"): builds a balanced Navy POM across portfolios; do not confuse this with the NAVMAC resource-sponsor code list.

OPNAV Programming Division (N80): Runs the POM build and integration; adjudicates sponsor submissions.

OPNAV Assessment Division (N81): Provides force-structure and analytic assessments supporting POM trades.

N82: Budget formulation/execution chain (FMB); manages allocation, execution controls, and reprogramming posture.

OPNAV Fleet Readiness (N83): Fleet Readiness; OPNAV point of contact for ship maintenance and fleet-readiness matters, and lead on CNO N8 ship-maintenance responsibilities.

N9: Warfare-systems sponsor family ("demand signal"): validates and advocates warfare requirements.

N9I/9I: Cross-domain warfare integration; current public manpower coding lists this function as 9I Warfare Integration. Treat N91 as an older or alternate public-reference label, not the current resource-sponsor code.

N94 : Test and Evaluation sponsor in the public NAVMAC resource-sponsor code list.

Expeditionary Warfare Directorate (N95)/Surface Warfare Directorate (N96)/Undersea Warfare Directorate (N97)/Air Warfare Directorate (N98): Expeditionary/surface/undersea/air warfare sponsors under N9.

Unmanned Warfare Systems (N99): Unmanned warfare systems sponsor; integrates unmanned air/surface/undersea capability portfolios and defends their investments in PPBE.

Resources and Requirements Review Board (R3B)/Naval Capabilities Board (NCB): Navy governance boards that align requirements and resourcing decisions feeding the POM.

Congressional resource chain.
Source: EDO Coursebook Module 3.1.2 (Congressional Enactment, 2025 edition).

HASC / SASC: Authorize defense programs and policy in the annual NDAA.

HAC / SAC: Produce appropriations bills that provide BA to execute Navy programs.

HBC / SBC: Set topline guidance and 302 allocations through the budget resolution process.

CBO, Congressional Research Service (CRS), GAO: Supply independent scoring, research, and oversight that shape committee deliberations.

Cyber/Authorizations (when IT/C2 in scope).
Source: SECNAVINST 5000.2G.

AO: Grants Authority to Operate; enforces RMF controls that drive design and integration requirements.

B.2 PAE Construct Mapping

Current as of 2026-06-04.

B.2.1 Executive Summary

- The PAE construct is now the public DON acquisition-reform baseline under ASN(RD&A). It is not a 1:1 relabeling of SYSCOMs or PEOs; it is a portfolio-accountability model intended to align authority, responsibility, technical support, contracting support, sustainment, and industrial-base management around warfighting outcomes [10, 11].
- Public releases now identify nine PAEs: Robotics and Autonomous Systems; Industrial Operations; Marine Corps; Maritime; Strategic Systems Programs; Undersea / DRPM Submarines; Aviation; Mission Systems; and Munitions [10, 11].
- The old “source-limited” caveat for PAE Mission Systems is no longer current. DON publicly established PAE Mission Systems on 11 May 2026 and named Mr. Jim Day as interim PAE [11, 21].
- The user-provided PAE chart is a useful transition snapshot, but it predates the May 2026 addition of PAE Aviation, PAE Mission Systems, and PAE Munitions. Use the nine-PAE answer for board recall unless a newer official chart supersedes it.

- A board answer should not say SYSCOMs disappeared. The public releases say PAEs will have direct authority over program offices and associated technical, contracting, and sustainment functions, and that about 70 percent of these associated functions and personnel will move from SYSCOMs into PAEs. Technical authority, safety, certification, and engineering standards still need to be handled deliberately through the current technical authority chain until formal directives say otherwise [11].

B.2.2 How to Draw the Current Board Chart

Top.

SECNAV at the top, with CNO and CMC as Service leadership and requirements/readiness context.

Acquisition executive.

ASN(RD&A) under SECNAV; current PTDO ASN(RD&A) is William F. Mahan, with Jason L. Potter as Principal Civilian Deputy and VADM Seiko Okano as PMILDEP [19].

Portfolio layer.

The nine PAEs sit under the acquisition executive as portfolio-accountable officials.

Support and authorities.

SYSCOMs, warfare centers, shipyards, RMCs, NIWCs, contracting organizations, and sustainment organizations provide the technical, industrial, and execution depth being realigned into or in support of PAEs. Do not confuse portfolio accountability with operational command of forces.

B.2.3 Current Public PAE Roster

Table LXV: Current public PAE roster

PAE	Public leader / cue	Core portfolio	Board memory hook
Aviation	VADM John Dougherty	Naval aviation; public deputy lanes are Carrier Strike, Marine Corps Aviation, and Maritime ISR/NC3.	Aviation = air portfolio plus mission engineering, Rapid Capability Cell, and disruptive technology development.
Industrial Operations	VADM James P. Downey	Industrial capacity, naval shipyards, regional maintenance, maintenance contracting, and fleet maintenance throughput.	Industrial Operations = Foundry, shipyards, RMCs, maintenance backlog, industrial-base execution.
Marine Corps	LtGen Eric Austin	Marine Corps ground systems and acquisition support functions.	PAE Marine Corps replaced PEO Land Systems and incorporates MCSC programs/support functions.

Continued on next page

Table LXV: Current public PAE roster **(Continued)**

PAE	Public leader / cue	Core portfolio	Board memory hook
Maritime	Christopher Miller	Conventional surface ships, including aircraft carriers, combatants, expeditionary and mine warfare, auxiliaries, and modernization/sustainment.	Maritime = conventional surface-ship delivery and shipbuilding accountability.
Mission Systems	Jim Day	C4I, enterprise IT, business systems, integrated warfare mission systems, Overmatch, LRNFO, MILE, Minotaur, and cross-SYSCOM mission-system elements.	Mission Systems = central mission integrator and kill-chain closure.
Munitions	Paul Mann	Munitions industrial base, air weapons, surface weapons, advanced capabilities, and innovation.	Munitions = weapons production capacity and weapons portfolio integration.
Robotics and Autonomous Systems	Public leader not confirmed in this refresh	Unmanned and autonomous hedge capabilities across air, surface, and undersea domains.	RAS = unmanned/autonomy marketplace, rapid fielding, and commercial solutions.
Strategic Systems Programs	VADM Johnny Wolfe	Strategic weapons and sea-based deterrence.	SSP = Trident II D5LE, D5LE2, SLCM-N, CPS, strategic deterrence.
Undersea / DRPM Submarines	VADM Robert Gaucher	Nuclear-powered attack submarines, ballistic-missile submarines, guided-missile submarines, and undersea industrial-base execution.	Undersea = DRPM SUBS and the one Columbia/two Virginia production-rate problem.

Source: *Navy Reshapes Warfighting Acquisition System*, *Navy Advances Acquisition Reform Strategy: Appoints Three New Portfolio Acquisition Executives*, *Portfolio Acquisition Executive Maritime*, *Department of the Navy Launches PAE Mission Systems to Accelerate Warfighting Capability at Speed*, *Navy Establishes Portfolio Acquisition Executive for Munitions*, *Navy Establishes Portfolio Acquisition Executive for Aviation*, *About Us*, *U.S. Navy Announces Seven Companies Selected for MUSV Marketplace At-Sea Demonstrations*, *Our Organization*, *U.S. Navy Shipbuilding Plan*, *PAE SSP Employees Bring Their Children to Work*, 2026, 2026, 2026, 2026, 2026, 2026, 2026, 2026, 2026, 2026, 2026 [10, 11, 20, 21, 22, 23, 24, 25, 195, 196, 197]

B.2.4 Portfolio Notes

PAE Aviation

PAE Aviation was publicly established on 11 May 2026 and is led by VADM John Dougherty. Its public deputy lanes are Carrier Strike, Marine Corps Aviation, and Maritime ISR/NC3. PAE Aviation also oversees Mission Engineering, a Rapid Capability Cell linked to the DON Rapid Capabilities Office, and Disruptive Technology Development. The board-level reform cue is that PAEs are directed to prioritize commercial solutions, Modular Open Systems Approach (MOSA), government-purpose-rights interfaces, modular competition, supply-chain resiliency, and faster contracting mechanisms such as Other Transaction Authoritys (OTAs) [23].

PAE Industrial Operations

PAE Industrial Operations was announced on 16 Mar 2026 with VADM James P. Downey as interim PAE [10]. The May 2026 shipbuilding plan describes PAE Industrial Operations as combining control of the Navy Regional Maintenance Centers, the NAVSEA Industrial Operations Directorate, the four naval shipyards, and all Navy Regional Maintenance Centers to connect maintenance contracting with fleet responsibility [196]. The board answer is Foundry execution: shipyards, RMCs, maintenance backlog, ship repair throughput, and industrial-base management.

PAE Marine Corps

PAE Marine Corps was formally established on 10 Apr 2026. It replaced PEO Land Systems and incorporated programs from Marine Corps Systems Command, while MCSC assumed an expanded role as the workforce provider for the Force Development Enterprise [198]. Public Marine Corps PAE pages describe the PAE model as a single leader with cradle-to-grave responsibility for delivering defined mission capability, aligning technical, contracting, and sustainment personnel into the acquisition chain of command [24]. Public capability portfolios are C5ISRT, Protect, Sustain/Enable, Maneuver, and Fires [195].

PAE Maritime

PAE Maritime is the single accountable organization for delivering conventional surface ships for the U.S. Navy. Public portfolio lanes include Aircraft Carriers, Combatants, Expeditionary and Mine Warfare, Auxiliaries, and Modernization and Sustainment [20]. Public leadership listed by PAE Maritime includes Christopher Miller as PAE, RADM Casey Moton as Deputy PAE for Aircraft Carriers, RDML Brian Metcalf as Deputy PAE for Combatants, Melissa Kirkendall as Acting Deputy PAE for Expeditionary and Mine Warfare, CAPT Cedric McNeal as Acting Deputy PAE for Auxiliaries, and RDML Andrew Biehn as Deputy PAE for Modernization and Sustainment [20]. The May 2026 shipbuilding plan says PAE Maritime is intended to empower surface shipbuilding program leaders with subject matter experts and authority to drive program outcomes, safety, quality, resource use, producibility, and timely delivery [196].

PAE Mission Systems

PAE Mission Systems was publicly established on 11 May 2026 and Mr. Jim Day was named interim PAE [11, 21]. The public launch release says PAE Mission Systems pulls together mission-system elements from PEO C4I, PEO Digital, PEO IWS, PEO MLB, DRPM Overmatch, DRPM Long Range Naval Fires Office, DRPM Manpower IT Logistics ERP, the Minotaur program from PMA 290, NAVWAR, NAVSEA, NAVAIR, and MCSC [21]. The board answer is: PAE Mission Systems is the central mission integrator for kill-chain closure, value-chain optimization, interoperability, and warfighting efficiency. Do not reduce it to NAVWAR; it spans information warfare, combat systems, digital/business systems, and cross-platform mission integration.

PAE Munitions

PAE Munitions was publicly established on 11 May 2026 and Mr. Paul Mann was named interim PAE [11, 22]. Public pillars are Weapons Industrial Base, Air Weapons, Surface Weapons, and Advanced Capabilities and Innovation. The organization combines former PEO Unmanned Aviation and Strike Weapons offices PMA-201, PMA-208, PMA-242, PMA-259, PMA-280, and PMA-281 with former PEO IWS weapons offices IWS 3.0, IWS 11.0, and IWS 12.0. The public release also notes SM-3 is expected to transition from MDA in FY 2028 for production and sustainment [22]. The board answer is munitions industrial-base scale, weapons production capacity, and cross-platform weapons portfolio execution.

PAE Robotics and Autonomous Systems

PAE Robotics and Autonomous Systems predates the Mar 2026 PAE expansion; the 16 Mar 2026 DON release identifies it as established in Dec 2025 alongside the DON Rapid Capabilities Office [10]. Current public examples include the MUSV marketplace, where at-sea testing was scheduled to begin in June 2026 and complete by Oct 2026; the public mission statement for PAE RAS is to deliver hedge capabilities that expand naval power, increase operational persistence, and impose operational dilemmas that degrade adversary tempo and freedom of action [25]. The board answer is unmanned/autonomous hedge capability using commercial marketplace and rapid fielding logic.

PAE Strategic Systems Programs

PAE Strategic Systems Programs was announced on 16 Mar 2026 with VADM Johnny Wolfe as interim PAE [10]. SSP public material now uses the PAE SSP label and describes the command as the Navy's trusted expert for maintaining and sustaining the Trident II D5LE strategic weapon system. It also identifies SSP responsibilities for Ohio-class SSBN strategic weapon sustainment, D5LE integration on Columbia-class SSBNs, D5LE2 development, SLCM-N development, and Conventional Prompt Strike [197]. The board answer is sea-based strategic deterrence and strategic weapons lifecycle execution.

PAE Undersea / DRPM Submarines

PAE Undersea / DRPM Submarines was announced on 16 Mar 2026 with VADM Robert Gaucher as interim PAE [10]. The May 2026 shipbuilding plan states that PAE Undersea is dual-hatted as the Direct Reporting Portfolio Manager for Submarines and will direct all activities to deliver nuclear-powered attack submarines, ballistic-missile submarines, and guided-missile submarines. The plan links the construct to the Deputy Secretary of War goal of reaching at least one Columbia SSBN and two Virginia SSNs per year by FY 2031 [196]. The board answer is submarine production, AUKUS/undersea enterprise alignment, and industrial-base execution.

B.2.5 What Changed from Older Study Notes

- Count changed from six baseline PAEs to nine public PAEs after 11 May 2026.
- PAE Mission Systems changed from source-limited/public-indicator status to a formal public DON announcement.
- PAE Aviation and PAE Munitions are no longer "to be announced" lanes.
- PAE Marine Corps formally sunsetted PEO Land Systems and moved responsibilities to PAE-MC.
- Current PTDO ASN(RD&A) is William F. Mahan, not Jason Potter; Potter is Principal Civilian Deputy as of the 26 May 2026 DON release [19].

- The board distinction is no longer simply “PEOs report to ASN(RD&A) while SYSCOMs support.” The current answer is “PAEs are portfolio-accountable officials under ASN(RD&A); legacy PEO/SYSCOM lanes still matter for program recall, technical authority, workforce, contracting, and sustainment execution during transition.”

B.2.6 Board Bottom Line

The PAE restructuring moves Navy acquisition from a program-by-program, SYSCOM/PEO matrix toward accountable capability portfolios. PAEs own portfolio outcomes and industrial-base risk; SYSCOMs and warfare centers remain the technical depth and authority infrastructure; PMs still need contracting authority and technical authority alignment to execute.

B.3 Combatant Commands (Reference)

AOR: The geographic (or astrographic) area assigned to a CCDR for missions and forces. (Title 10)

Geographic (7):

- United States Africa Command (USAFRICOM)
- United States Central Command (USCENTCOM)
- United States European Command (USEUCOM)
- United States Indo-Pacific Command (USINDOPACOM)
- United States Northern Command (USNORTHCOM)
- United States Southern Command (USSOUTHCOM)
- United States Space Command (USSPACECOM)[1]

Functional (4):

- USCYBERCOM
- United States Special Operations Command (USSOCOM)
- United States Strategic Command (USSTRATCOM)
- United States Transportation Command (USTRANSCOM)

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[1] USSPACECOM is treated by DoW as a **geographic (astrographic)** command with an AOR beginning at the Kármán line (∼100 km)

C EDO Billet Distribution

C.1 November 2025 Main Slate Distribution

Current as of 2025-11-01. This appendix is a historical billet-density snapshot from the November 2025 main slate; use Appendix C.2 for the current flag roster.

Purpose.
Quick reference to where EDOs had billets on the November 2025 main slate, grouped by community and sorted by location with billet counts by rank.

Method.
Parsed the "SLATE_EDO_MAIN" worksheet in *EDO Main Slate November 2025.xlsx* by mentor group (community), location (AHPORT), and billet rank (BRANK); rank order follows flag to junior officer seniority (VADM, RADM, RDML, CAPT, CDR, LCDR, LT, LTJG).

Location expansions.
Location codes were expanded best-effort based on billet context (e.g., ANNAP → Annapolis, MD; MONTEY → Monterey, CA). Confirmations are appreciated for potentially ambiguous codes listed after the table.

Table LXVI: EDO billet distribution by community, location, and billet rank (November 2025 main slate)

Community	Location	Billet ranks (count)
ANY	Annapolis, MD	CAPT 1 / CDR 1 / LCDR 2 / LT 4
ANY	Cambridge, MA	CAPT 1 / LCDR 1
ANY	Crane, IN	CAPT 1
ANY	MacDill AFB (Tampa, FL)	CAPT 1 / CDR 1
ANY	Millington, TN	CAPT 1 / LCDR 3
ANY	Monterey, CA	CDR 1
ANY	Port Hueneme, CA	CAPT 1 / CDR 1 / LCDR 2
CC	Adelaide, Australia	CDR 1
CC	Arlington, VA	CDR 1

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Table LXVI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
CC	Bath, ME	LCDR 1
CC	China Lake, CA	LCDR 1
CC	Dahlgren, VA	CAPT 4 / CDR 6 / LCDR 7
CC	Dam Neck, VA	CDR 1
CC	Fort McNair, DC	CDR 1
CC	Moorestown, NJ	CAPT 2
CC	Norfolk, VA	CDR 3 / LCDR 2 / LT 3
CC	Pascagoula, MS	LCDR 1
CC	Pearl Harbor, HI	CDR 1
CC	Point Mugu, CA	CAPT 1 / CDR 2
CC	Port Hueneme, CA	CAPT 1 / CDR 4 / LCDR 3 / LT 2
CC	San Diego, CA	LCDR 1
CC	Tucson, AZ	CDR 1 / LCDR 1 / LT 1
CC	Wallops Island, VA	CAPT 2
CC	Washington Navy Yard, DC	LCDR 2
CC	Washington, DC	CAPT 12 / CDR 8 / LCDR 9 / LT 1
CC	White Sands, NM	CDR 1
CVN-C	NAS North Island (San Diego, CA)	CAPT 1 / CDR 3 / LCDR 1
CVN-C	Newport News, VA	CAPT 1 / CDR 4 / LCDR 5 / LT 5
CVN-C	Norfolk, VA	CAPT 1 / CDR 3 / LCDR 3
CVN-C	Point Mugu, CA	CAPT 1 / CDR 1
CVN-C	Portsmouth, VA	CAPT 1
CVN-C	San Diego, CA	CAPT 2 / CDR 5 / LCDR 3
CVN-C	Washington, DC	CAPT 4 / CDR 4 / LCDR 3
CVN-C	Yokosuka, Japan	LCDR 1
DIVER	Little Creek, VA	LCDR 1
DIVER	Manama, Bahrain	CDR 1 / LT 1
DIVER	NAS North Island (San Diego, CA)	LCDR 1

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Table LXVI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
DIVER	Panama City, FL	CDR 1 / LT 1
DIVER	Pearl Harbor, HI	CDR 1 / LCDR 1
DIVER	Rota, Spain	LT 1
DIVER	Washington, DC	CAPT 1 / LCDR 4 / LT 1
FLAG	Pearl Harbor, HI	RDML 1
FLAG	San Diego, CA	RDML 1
FLAG	Washington Navy Yard, DC	VADM 1
FLAG	Washington, DC	VADM 1 / RADM 2 / RDML 5 / LCDR 1
IMG	Agana Heights, Guam	CDR 3
IMG	Apra Harbor, Guam	CDR 1
IMG	Bangor, WA	CDR 1
IMG	Bremerton, WA	CAPT 4 / CDR 5 / LCDR 6 / LT 13
IMG	Fort Monroe, VA	CAPT 1
IMG	Groton, CT	CDR 1
IMG	Kaneohe Bay, HI	CDR 2
IMG	Keyport, WA	CAPT 1
IMG	Norfolk, VA	RADM 1 / CAPT 2 / CDR 3 / LCDR 2 / LT 1
IMG	Pearl Harbor, HI	CAPT 6 / CDR 5 / LCDR 6 / LT 10
IMG	Perth, Australia	LCDR 1
IMG	Point Loma, CA	CDR 1
IMG	Point Mugu, CA	CAPT 1 / CDR 1
IMG	Portsmouth, NH	CAPT 4 / CDR 3 / LCDR 8 / LT 8
IMG	Portsmouth, VA	CAPT 4 / CDR 5 / LCDR 5 / LT 11
IMG	Washington, DC	CAPT 3 / CDR 2 / LCDR 1
IWE	Agana Heights, Guam	LCDR 1
IWE	Bremerton, WA	LCDR 1
IWE	Chantilly, VA	CDR 1 / LCDR 6 / LT 7
IWE	Charleston, SC	CAPT 1 / CDR 1 / LCDR 1 / LT 3

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Table LXVI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
IWE	Coronado, CA	LCDR 1
IWE	Falls Church, VA	CAPT 1
IWE	Fort Meade, MD	LT 1
IWE	Manama, Bahrain	LCDR 1
IWE	Naples, Italy	CDR 1
IWE	New Orleans, LA	CDR 1
IWE	Norfolk, VA	CAPT 1 / CDR 1 / LCDR 2
IWE	Pearl Harbor, HI	CAPT 1
IWE	Point Mugu, CA	LCDR 1
IWE	Portsmouth, VA	LCDR 4 / LT 2
IWE	San Diego, CA	RADM 1 / CAPT 7 / CDR 14 / LCDR 21 / LT 5
IWE	Suffolk, VA	LT 1
IWE	Washington Navy Yard, DC	CDR 1
IWE	Washington, DC	CDR 2 / LCDR 3
IWE	Yokosuka, Japan	CDR 1 / LCDR 2
NR	Arlington, VA	CAPT 4 / CDR 4 / LCDR 9 / LT 9 / LTJG 1
NR	Bremerton, WA	LT 1
NR	Groton, CT	LCDR 1
NR	Pearl Harbor, HI	LCDR 2
NR	Washington, DC	CDR 1
NR	Yokosuka, Japan	LCDR 1
RESERVE	Bremerton, WA	LCDR 1
RESERVE	Pearl Harbor, HI	LCDR 1
RESERVE	Portsmouth, VA	LT 1
RESERVE	Washington, DC	CAPT 1 / LT 1
SMMR	Al Jubayl, Saudi Arabia	LCDR 1
SMMR	Arlington, VA	CAPT 1
SMMR	Coronado, CA	CAPT 1 / CDR 2 / LCDR 6 / LT 2

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Table LXVI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
SMMR	Everett, WA	CAPT 1 / CDR 1 / LT 3
SMMR	Little Creek, VA	LCDR 1
SMMR	Manama, Bahrain	CDR 2 / LCDR 1 / LT 1
SMMR	Mayport, FL	CAPT 2 / CDR 2 / LCDR 6 / LT 2
SMMR	Naples, Italy	CAPT 2
SMMR	Norfolk, VA	CAPT 6 / CDR 9 / LCDR 10 / LT 9
SMMR	Pearl Harbor, HI	CAPT 1 / LCDR 2
SMMR	Point Mugu, CA	CDR 1
SMMR	Riyadh, Saudi Arabia	CDR 1
SMMR	Rota, Spain	CDR 1 / LT 1
SMMR	San Diego, CA	CAPT 4 / CDR 5 / LCDR 5 / LT 6
SMMR	Sasebo, Japan	CDR 2 / LCDR 1 / LT 3
SMMR	Singapore	CDR 1
SMMR	Washington, DC	LCDR 3
SMMR	Yokosuka, Japan	CAPT 1 / CDR 3 / LCDR 4 / LT 5 / LTJG 1
SSP	Cape Canaveral, FL	CDR 1
SSP	Kaneohe Bay, HI	CAPT 1 / CDR 1 / LCDR 1 / LT 1
SSP	Little Rock, AR	CAPT 1 / CDR 1 / LCDR 2 / LT 1
SSP	Magna, UT	CDR 1
SSP	Pittsfield, MA	CAPT 1 / LCDR 1
SSP	Silverdale, WA	CAPT 1 / CDR 1 / LCDR 1 / LT 1
SSP	Sunnyvale, CA	LCDR 1
SSP	Titusville, FL	LCDR 1
SSP	Uniondale, NY	LT 1
SSP	Washington Navy Yard, DC	CAPT 5 / CDR 8 / LCDR 7 / LT 2
SURFPK	Bath, ME	CAPT 1 / CDR 1 / LCDR 1
SURFPK	Bethesda, MD	CAPT 1
SURFPK	Marine Corps Base Camp Lejeune, NC	CDR 2 / LCDR 3 / LT 2

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Table LXVI: EDO billet distribution by community, location, and billet rank (November 2025 main slate) **(Continued)**

Community	Location	Billet ranks (count)
SURFPK	Mobile, AL	CDR 1 / LCDR 2
SURFPK	New Orleans, LA	CDR 2 / LCDR 1 / LT 1
SURFPK	Norfolk, VA	LCDR 2
SURFPK	Pascagoula, MS	CAPT 2 / CDR 6 / LCDR 4 / LT 4
SURFPK	Philadelphia, PA	CAPT 1 / LCDR 1
SURFPK	Point Mugu, CA	CDR 1 / LCDR 1
SURFPK	San Diego, CA	CDR 1 / LCDR 1 / LT 1
SURFPK	Washington, DC	CAPT 10 / CDR 22 / LCDR 7
UEDO	Arlington, VA	LCDR 1
UEDO	Groton, CT	CAPT 1 / CDR 4 / LCDR 10 / LT 4
UEDO	Newport News, VA	CDR 2 / LCDR 3 / LT 5
UEDO	Norfolk, VA	LCDR 1
UEDO	Point Mugu, CA	CDR 1
UEDO	Portsmouth, NH	CAPT 1
UEDO	San Diego, CA	LCDR 1
UEDO	Washington, DC	CAPT 7 / CDR 9 / LCDR 18

C.2 Current EDO Flag Officer Roster

Current as of 2026-06-04. This appendix uses the June 2026 public Navy flag roster as the primary source for current grades and billets. Navy public biography headings that state "Rear Admiral" are not sufficient to distinguish RADM, RADM(sel), and RDML; command biographies are used only to explain live title conflicts. [85, 199, 200]

Scope.

Includes active and reserve EDO flag officers who currently hold Navy or joint billets in service. Retired officers and aerospace engineering duty officers are excluded.

Use.

Treat this as the current board-prep roster. For broader billet density by location and community, use Appendix C.1.

Table LXVII: Current engineering duty officer flag officers and billets

Rank	Name	Current billet	One-line job summary
VADM	Johnny R. Wolfe Jr.	Director for Strategic Systems Programs	Leads the Navy's sea-based strategic deterrence and strategic weapons enterprise.
VADM	James P. Downey	Commander, Naval Sea Systems Command	Leads the Navy systems command responsible for ships, submarines, and associated technical authority.
VADM	Elizabeth S. "Seiko" Okano	Principal Military Deputy Assistant Secretary of the Navy (Research, Development and Acquisition)	Serves as the senior uniformed acquisition deputy supporting Navy and Marine Corps research, development, and acquisition governance.
RADM	Casey J. Moton	Program Executive Officer for Aircraft Carriers	Oversees aircraft carrier acquisition, modernization, and major lifecycle availability portfolios.
RADM	J. Kurt Rothenhaus	Commander, Naval Information Warfare Systems Command	Leads naval information warfare, cyber, communications, and digital systems delivery.
RADM	Jonathan E. Rucker	Program Executive Officer, Attack Submarines	Oversees attack submarine acquisition and lifecycle support for Team Submarines.
RADM	Douglas L. Williams	Director for Test, Missile Defense Agency	Leads missile defense test planning, execution, and assessment across the joint enterprise.
RADM(sel)	Scott M. Brown	Director, Industrial Operations (NAVSEA 04)	Oversees naval shipyard and industrial operations that underpin fleet maintenance and industrial-base readiness.
RDML	Dianna Wolfson	Fleet Maintenance Officer, U.S. Fleet Forces Command	Integrates Atlantic fleet maintenance execution and readiness across type commanders and sustainment organizations.
RDML	Daniel W. Ettlich	Director, Fleet Maintenance, U.S. Pacific Fleet	Leads Pacific fleet maintenance readiness and execution for forward and west-coast forces.
RDML	Peter D. Small	Chief Engineer and Deputy Commander for Ship Design, Integration and Naval Engineering, SEA-05, Naval Sea Systems Command	Serves as NAVSEA's chief engineer for ship design, integration, and naval engineering technical authority.

Continued on next page

Table LXVII: Current engineering duty officer flag officers and billets **(Continued)**

Rank	Name	Current billet	One-line job summary
RDML	Daniel L. Lannamann	Commander, Navy Regional Maintenance Center	Oversees the regional maintenance center enterprise and major depot availability execution.
RDML	Brian A. Metcalf	Program Executive Officer for Ships	Oversees acquisition portfolios for surface combatants, amphibious ships, and auxiliaries.
RDML	Michael S. Richman (USNR)	Deputy Director for Hypersonics, Strategic Systems Programs	Supports Navy strategic strike and hypersonic capability development within Strategic Systems Programs.
RDML	Jonathan J. Jett-Parmer, PE (USNR)	Reserve Deputy, Naval Sea Systems Command	Provides reserve flag leadership and reserve integration across Naval Sea Systems Command headquarters.

Roster interpretation notes.

Scott Brown appears in the June 2026 roster under “Rear Admiral (selectee)” as “RADM(S)”; this appendix lists him as RADM(sel), not RDML, to preserve the board-relevant grade count. Richman’s official SSP leadership page uses “Deputy Director, Regional Strike Systems,” while the June 2026 roster uses “Deputy Director for Hypersonics.” Jett-Parmer’s official Navy bio still lists “Deputy Commander, Supervisor of Shipbuilding, Conversion and Repair,” while the June 2026 roster lists “Reserve Deputy, Naval Sea Systems Command.” This appendix follows the June 2026 roster for the displayed billet titles [85, 199, 200].

D CNO Strategic Messaging

D.1 CNO 34 Day One Message (Foundry–Fleet–Fight)

NAVADMIN 180/25; issued upon assuming duties as the 34th CNO [201].

Mission.

The Navy shall be organized, trained, and equipped to deliver peace through strength, securing our national interests and prosperity. We exist for prompt and sustained combat at sea, preserving freedom of navigation, safeguarding global commerce, and deterring those who would challenge America's values and sovereignty.

Vision.

We stand at an inflection point—an era marked by great power competition, proliferating threats, rapid technological convergence, and an increasingly contested maritime domain. To prevail, we must build and sustain a Navy that is ready to fight and win—today, tomorrow, and well into the future. At the center of this vision is the Sailor. Our men and women are the Navy's main weapon system—their skill, resilience, and courage remain our greatest asymmetric advantage. We will put them first by relentlessly pursuing full-spectrum readiness, deepening cross-domain integration, harnessing innovation at speed, and strengthening our warrior ethos. Our Navy must be resilient, agile, globally present, and combat credible. We will align tightly with national defense priorities, expand our ability to integrate with Joint Force and Allied partners, and preserve our enduring advantage at sea. In doing so, we will guarantee freedom of navigation, deter aggression, and project power where and when it matters most.

Priorities.

We will view everything we do through an operational lens focused on the Foundry, the Fleet, and the way we Fight.

Foundry.

The Foundry represents the enduring sum of our total force, shore infrastructure, maintenance depots, schoolhouses, industrial base, and intellectual capital required to generate, sustain, and modernize naval power. Every ounce of sea power we wield originates in the Foundry; it is the bedrock of our Navy. We will not neglect the global network of infrastructure and capacity that enables our Sailors to excel. From new ship construction to depot-level repair, from weapons production to the training pipelines that develop warfighters, the Foundry underwrites combat credibility. To remain the world's most formidable fighting force, we will operationalize readiness, accelerate shipbuilding and repair at scale, and modernize our force-generation model. This means strengthening public-private partnerships, investing in resilient supply chains, expanding workforce development, and ensuring our training systems deliver Sailors fully prepared for the high-end fight. Simply put—without a strong Foundry, there is no Fleet.

Fleet.

The Fleet is forged in the Foundry and is our Nation's decisive instrument of maritime power. Comprised of people, platforms, and payloads, it is designed to deter aggression, project power, and secure our national interests on and from the sea. No image symbolizes American resolve more than a U.S. Navy ship on the horizon—lethal, resilient, and ready to fight. The Fleet's unique mobility, persistence, and global reach enable us to impose costs on adversaries, reassure allies, and guarantee the free flow of commerce. Tempered by our resolve, our Fleet is world-class, battle-ready, and second to none. With our Sailors at the helm, supported by strong Navy families, there is no challenge we cannot overcome. We will continue to build an integrated, all-domain Fleet—one that fuses maritime (surface, air, and undersea), cyber, space, and strategic capabilities while leveraging Joint Force and Allied partners to dominate across the competition continuum. Interoperability and interchangeability with allies and partners will be our force multiplier. By enhancing collective maritime security, we will magnify our forward presence and strengthen deterrence worldwide.

Fight.

Fighting is our ultimate honor and sacred responsibility—delivering violence on behalf of the Nation when called. For nearly 250 years, the U.S. Navy has honed its toughness, tenacity, and ingenuity to secure victory. Those who challenge us in combat will be met with overwhelming force and sent to the bottom. But we must fight differently in the future. In today's rapidly evolving landscape, we will innovate, adapt, and integrate cutting-edge technologies at the speed of relevance. We will deliver and refine the Navy Warfighting Concept and develop a new Navy Deterrence Concept, both anchored in peace through strength. We will accelerate the integration of artificial intelligence, robotic and autonomous systems, resilient Command, Control, and Communications (C3) architectures, and synchronized effects—ensuring decision advantage, survivability, and combat overmatch. We will ruthlessly pursue a Future Fleet Design—comprised of unmatched conventionally manned ships, deeply integrated with asymmetric hedge forces that expand our reach and impose costs more effectively across every domain. Our charge is clear: To ensure the Navy remains the most lethal, survivable, and globally dominant maritime force.

Theory of victory.

At the end of my tenure as CNO, success will be measured by results:

- Platforms delivered and repaired on time.
- Fully manned and combat-ready ships.
- Ordnance production meeting contracted demand.
- Backlogs in repair parts eliminated.
- Sailors trained to the highest levels of mastery.

Through disciplined execution, we will increase our lethality and preserve our sustained advantage. To achieve this, we are calibrating our sights, refining our course, and transforming our approach of solving our Navy's toughest problems. Guided by the Navy Warfighting and Deterrence Concepts, we will develop and execute a holistic Campaign Plan that delivers the warfighting capability and capacity necessary to win in the 21st century. To maintain transparency and unity of effort, I will release a series of "CNO Notes"—or "C-Notes"—providing guidance and direction to the Fleet. These will be grounded in my vision, informed by your feedback, and driven by operational necessity. They will require your trust, ownership, and commitment to be successful. I am proud to serve alongside you—America's Sailors—as we embark on this next chapter in our Navy's history. Built in the Foundry—Tempered in the Fleet—Forged to Fight.

Daryl L. Caudle
Admiral, United States Navy
34th Chief of Naval Operations

D.2 CNO 34 C-Notes and Charge of Command

NAVADMINs 186/25, 203/25, 241/25, and 225/25 [202, 203, 204, 205].

D.2.1 Executive summary of priorities

Overview.

ADM Caudle's initial set of C-Notes and his Charge of Command reinforce the Foundry–Fleet–Fight construct: care for Sailors, strengthen the Navy–industrial team, and deliver a combat-credible fleet that can deter and win. Leaders are directed to build trust, enforce standards, and maintain a visible warfighting mindset across every command.

C-Note #1: Sailors First.

The Fleet's advantage is a trained, supported Total Force. Priorities include modern housing and galley upgrades, streamlined uniform requirements, resilient digital connectivity ashore, and pay/benefits accuracy so every Sailor can focus on the fight [202].

C-Note #2: Foundry Always.

The Foundry—people, infrastructure, and materiel—forges naval power. Caudle calls for relentless training (Career Training Continuum), expanded live-virtual-constructive integration, stronger shipyard shifts, revitalized intermediate maintenance activities, and modernized shore platforms and supply chains to deliver ready forces at scale [203].

C-Note #3: World Class Fleet.

The Fleet is the decisive instrument of national power, integrating people, platforms, and payloads across all domains. The note introduces a Hedge Strategy (tailored offsets and tailored forces) to outpace threats, modular payload investments, resilient C2/AI/unmanned integration, clarified administrative/operational control, and undersea C2 realignments (Commander, Task Force (CTF)-54/74) [204].

Charge of Command.

Command is an action and a privilege. Commanding Officers are accountable for warfighting readiness, disciplined risk management, mission command, continuous learning, and visible ownership of their Sailors' welfare and performance. Immediate superiors will review the Charge with subordinate COs to ensure shared understanding and mentorship [205].

D.2.2 Message texts

C-Note #1: Sailors First (NAVADMIN 186/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 031453Z SEP 25 MID180002011151U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 186/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/SEP//
SUBJ/C-Note #1: Sailors First//
RMKS/1. The lifeblood of our Navy is the Sailor: well-trained, connected,
supported and fit to fight. In every ocean around the world, our national
security interests are challenged by increasingly complex threats.
Underpinned by our warrior ethos, we will rise to that challenge. For 250
years, we have been a world-class fighting Navy, and our strength lies in our
Sailors. Our teamwork and esprit de corps are strengthened by our core
values and devotion to our mission.
2. We must resource the Navy our Nation Needs. I am committed to the well-
being of our Sailors and their families - providing them with state-of-the-
art platforms, world-class facilities and dependable support through our
Total Sailor and Family Action Plan.
We will leverage the insights and lessons learned from the past to improve
and rebrand our Culture of Excellence initiative to its new focus - Total
Sailor: Fit to Fight.
3. Today, I have directed your Navy leadership to take the steps necessary
to ensure that No Sailor Will Live Afloat. We will invest in unaccompanied
and family housing in addition to optimizing our new BAH authority to ensure
every Sailor resides in housing that is clean, comfortable and safe.
4. During my term, we will transform our galley facilities to ensure they
are designed to meet the nutritional needs and choices of our Sailor's
modern-day lifestyle. It is paramount that we modernize food service to
deliver healthy and flexible options.
5. Effective immediately, the Secretary of the Navy has directed we convene
a focused assessment of your uniforms and seabag. We will streamline uniform
requirements to ensure you have affordable, available and durable service
clothing that you are proud to wear.
The goal will be to reduce the number of uniforms while respecting our hard-
earned traditions that makes our Navy so special.
6. You have grown up in a digitally connected world where disconnection
means isolation. Therefore, I have mandated efforts to improve cell service
on our installations and bring free Wi-Fi to all of our barracks.
7. I have also ordered changes to the NAVADMIN process ensuring our Sailors
are able to find clear and correct information in one singular location. We
will not use NAVADMINs to work around the effort required to properly change
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the base instructions any longer.
8. Additionally, I've directed Navy leadership to revisit prior efforts such
as Sailor pay to ensure our previous corrective actions are still working so
that each Sailor is receiving the correct amount of pay and entitlements in a
timely manner.
9. You are my priority and I need you focused on the fight. This is the
first of many efforts to improve your Quality of Life - so, stand by for
what's coming next.
10. Built in the Foundry - Tempered in the Fleet - Forged to Fight.
11. ADM Daryl Caudle, 34th Chief of Naval Operations sends.
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CLASSIFICATION: UNCLASSIFIED/
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C-Note #2: Foundry Always (NAVADMIN 203/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 222218Z SEP 25 MID120012080004U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 203/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/SEP//
SUBJ/C-NOte #2: Foundry Always//
RMKS/1. The Foundry is the engine that drives our warfighting advantage. It
transforms raw inputs and forges them into lethal
outputs: ready warfighters, platforms and sustained readiness, at the speed
and scale necessary to win.
2. The Foundry builds, generates, sustains and modernizes naval power
through three critical components: PEOPLE, INFRASTRUCTURE and MATERIEL.
3. People: Our Total Force, comprised of Sailors, civilians and the
industrial workforce, is our most decisive warfighting advantage.
To prevail in conflict, we must recruit, train, educate and retain a
technically proficient and resilient team. Investing in our people's
development and well-being is not separate from combat power, it is the very
foundation of our Warrior Ethos, which is not exclusive to those in uniform.
It's a mindset marked by honor, integrity, resilience and a relentless
commitment to excellence and service.
That ethos must live across the spectrum: from our schoolhouses and combat
centers to the workforce that lays our keels and develops our intellectual
property. Every role our Total Force fills is vital by design.
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4. To keep pace with the proliferation of technology and the global evolution of threats, we are enhancing the quality of our curricula. Relentless improvement, technical competence and tactical mastery are all tenets demanded of a confident warfighter. Empowered by timely action and responses from our Navy's top leaders, we will inculcate a culture of continuous learning that delivers world-class, modernized and battle-ready Sailors. Therefore, we will rename Ready Relevant Learning to "Career Training Continuum" (CTC), the Navy's enterprise program of record for individual training and career-long learning.

5. Through a series of sprints, we will also expand the Military Learning Continuum and Civilian Workforce Development. With built-in feedback loops and mechanisms in place, we will rapidly identify, assess, respond to and correct gaps in our Sailor's training journey.

I have also directed that our Live-Virtual-Constructive (LVC) efforts be expanded past the current use cases (e.g., integrated C2X training, Fallon airwing events, etc.) to include a more generalized view of LVC that spans from unit level training to schoolhouse utilization to tailored wargaming.

6. Battle effectiveness is not achieved by only afloat platforms or commands. Simply put, without a strong Foundry, there is no effective Fleet. Therefore, I am working to rewrite the Battle "E" instruction to recognize shore installations with exceptional readiness and performance.

7. Infrastructure: Shore platforms generate naval power. Our global network of infrastructure, shipyards, airfields, barracks, electrical grids and piers are foundational to projecting that power. Modernization ensures our readiness can scale across the spectrum of conflict. We will prioritize and strengthen our shore platforms to support Fleet-wide operations and ensure our People take pride and ownership in the places that they work in and live.

8. Expanding the Fleet's deployable platforms is a whole-of-Navy evolution, and while we are not short on ideas to get there, our execution has not met expectations. The levers: innovation, oversight, optimization and recapitalization are known, but the fastest gains in production will come from our People and their elbow-grease. To meet operational timelines, we must prioritize workforce development, labor capacity and world-class project management to increase our efficiency.

9. Accordingly, we will strengthen second and third shifts in our public shipyards to increase throughput and on-time delivery. Only by improving shipyard training, workplace conditions and organizational support can we reduce attrition, accelerate experience and retain the workforce, both private and public, responsible for delivering our Fleet.

10. I will release a future C-NOte on Esprit de Corps Ashore, emphasizing the critical role our infrastructure plays in forging warfighting readiness.

11. Materiel: Reliable supply chains, accurate Coordinated Shipboard Allowance Lists (COSALs), comprehensive maintenance capabilities and

coordinated logistics are instrumental to keeping our Fleet in the fight.
Winning in conflict demands a strong Navy-Industrial Base partnership,
grounded in transparency, common understanding and intellectual honesty.
12. Our current maintenance structure lacks the right balance of Sailors,
civilians, artisans and engineers. Years of depot and activity level
consolidation have reduced opportunities for our Sailors to master the skill
sets needed to repair and sustain at sea.
13. For these reasons, I am pursuing the stand-up of Shore Intermediate
Maintenance Activities (SIMA) in Norfolk and San Diego.
These facilities will provide our Sailors with hands-on training in advanced
ship repair and expose them to modern capabilities such as AI/ML, advanced
manufacturing (e.g., 3-D printing, CAD/CAM, etc.), workflow monitoring
technologies and robotic systems.
14. We are also reviewing Class Maintenance Plans. To maintain a combat
surge-ready posture, our maintenance models must evolve.
Introducing shorter, more frequent and efficient availabilities will
increase our readiness, improve on-time completion and reduce risk at a
comparable cost.
15. Foundry is more than a metaphor; it is the process, standard and
discipline that underpins our Navy's ability to deliver ready, lethal and
resilient forces, on time, on cost and at scale.
16. Built in the Foundry - Tempered in the Fleet - Forged to Fight.
17. ADM Daryl Caudle, 34th Chief of Naval Operations sends.//
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CLASSIFICATION: UNCLASSIFIED/

C-Note #3: World Class Fleet (NAVADMIN 241/25).

CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 012011Z DEC 25 MID120012255382U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 241/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/DEC//
SUBJ/C-Note #3: World-Class Fleet//
RMKS/1. History proves that great nations are built on the strength of great
navies. And today, our Fleet stands as the world's premier warfighting force
- ready, capable, and unmatched. But we are more than ships, aircraft, and
weapons. Our Sailors - battle ready, trained, resilient, and forged in the

Foundry are the engine that empowers our Fleet. Through their strength, we command the seas, deter aggression, defend our homeland, and when called, deliver decisive combat power. This is what makes us a world-class Navy - second to none.

2. The Fleet is our most decisive instrument of national power and serves as the differentiated value our Navy brings to the Joint Force: expeditionary reach, unmatched mobility, and persistent presence that promotes our vital interests, protects the global commons, and wins our Nation's wars. Comprised of People, Platforms, and Payloads, the Fleet spans across all domains. To our Allies, it serves to reassure. To our Adversaries, it deters and shapes behavior. To all, it is a signal that freedom of the seas is non-negotiable and will never be surrendered.

3. As threats evolve and the world grows more complex, we will continue to build an integrated, all-domain Fleet - capable of conducting sea denial and, when needed, sea control within the battlespace and at decisive global chokepoints. This means fusing maritime, cyber, space, and undersea capabilities, and integrating Joint and combined forces to dominate across the spectrum of conflict. In short, we will not only design our forces to fight and win, but to outpace emerging threats through smart and cost-effective "hedges" to lower the risks faced by our general purposes forces while increasing survivability and lethality.

4. To that end, building a Fleet to cover every possible threat is too expensive, unrealistic, and sub-optimized. Instead, I have directed the development of a new Navy Strategy. Built on the fundamentals from our Navy Warfighting and Deterrence Concepts, this new effort introduces an enduring "Hedge Strategy" - designed as an approach that reduces operational risk by investing in Tailored Offsets and Tailored Forces as a way to amplify and synergize our general purposes forces to more effectively address low probability-high consequence contingencies.

5. At its core, our Navy is already a collection of "hedge" forces and concepts: risk-informed decisions about where to invest and how to fight while weighing the probability of occurrence in terms of severity and loss. For example, "Hellscape" is a use case of the generalized Hedge Strategy. Naval Special Warfare is our hedge against low intensity and irregular warfare threats. SSBNs and E-6Bs are our hedge that deters strategic attack with less than eight percent of the Navy. And our Navy-Marine Corps team itself, has and will continue to serve as the Joint Force's global hedge - forward deployed, postured, and ready to respond to a variety of threats at a time and place of our choosing.

6. Tailored Forces are logical groupings of tailored offsets that strengthen our general purpose forces. The Global Maritime Response Plan (GMRP) and associated Readiness Conditions (RESCON) provide tailored force packages that are Combat Surge Ready (CSR) in support of our Combatant Commanders through tailored certifications to execute specific missions. This is in run today as we continue to work to achieve our 80 percent CSR goals for all platforms.

7. Further, we will also invest in platforms, sensors, and weapon systems built with modularity that are scalable and ready for rapid upgrade. We will accelerate the delivery of integrated, networked capabilities - including unmanned systems, Artificial Intelligence, and resilient Command and Control (C2) architectures to ensure decision advantage and guarantee our overmatch in contested environments around the world.

8. Many of our Navy's toughest challenges cannot be solved in one year or even across one five-year Future Years Defense Program (FYDP) budgeting cycle. These types of longer-range problems (e.g., spare parts, critical munitions, foundry improvements, shipbuilding, and depot repair enhancements) require time horizons and funding profiles with trajectories that balance fiscal realities to operational risk.

Therefore, I have directed the development and use of multi-year/multi-FYDP "funding curves" to ensure sustained investment in the capabilities at the required capacity our warfighters need.

With clear "commitment to the curve" - I will prioritize the required sustained investments necessary to solve many of the problems that have hampered our Navy for decades.

9. The Administrative Chain of Command (ADCON) is comprised of subject matter experts who form the best force generators in the Navy. Over time we have drifted into commingling force generation and force employment with overlapping C2 structures that lack clear accountability. Therefore, we will review existing C2 structures from an operational task perspective and align C2 to the risk owner.

Additionally, we will review and align force generation responsibilities to ensure we are following the "Standard Model" of excellence. This will ensure that we clearly delineate the role of the appropriate Type Commander (TYCOM) to the Fleet Commander with clean and accountable hand-offs between force generation and force employment of their forces.

10. Additionally, to most effectively counter the pacing threat that the People's Republic of China presents, CTF-54 will be separated from CTF-74 allowing each to hone their focus on specific undersea mission sets, particularly in our most challenging threat environments. CTF-54 will be dual-hatted as Commander, Submarine Squadron 21 in Bahrain. To support the new CTF-54, primary broadcast control authority and submarine operating authority duties (e.g., waterspace management) will be shifted to Commander, Submarine Group 8 and CTF-69 in Naples, Italy. To support CTF-74 more fully across the Western Pacific Area of Responsibility, we will be assessing the establishment of Task Groups to enable clear and unambiguous C2 lines of operational accountability.

11. Fleet Improvement Offices (FIOs) are driving a culture of continuous improvement, intellectually honest assessments, disciplined execution, as well as improved operational risk management and accountability. We are codifying this through a new instruction, under the Office of Warfighting Advantage (OWA), to scale FIO practices across our Foundry, Fleet, and Fight

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priorities.
12. Built in the Foundry - Tempered in the Fleet - Forged to Fight.
13. ADM Daryl Caudle, 34th Chief of Naval Operations sends.//
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Charge of Command (NAVADMIN 225/25).

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CLASSIFICATION: UNCLASSIFIED/
ROUTINE
R 302016Z OCT 25 MID120012171056U
FM CNO WASHINGTON DC
TO NAVADMIN
INFO CNO WASHINGTON DC
BT
UNCLAS
NAVADMIN 225/25
MSGID/NAVADMIN/CNO WASHINGTON DC/CNO/OCT//
SUBJ/CHARGE OF COMMAND - ANNOUNCEMENT TO THE FLEET//
RMKS/1. As the 34th Chief of Naval Operations (CNO), I've released my Charge
of Command for officers selected for command, emphasizing the unique
responsibilities and privileges inherent in leading warfighters at a
strategic inflection point. Commanding Officers (COs) hold the highest
operational and moral accountability for their units, and their leadership
directly shapes combat readiness, culture, and mission success.
2. In the Charge of Command, I characterize command as "an action, not a
position," highlighting the essential attributes of integrity, courage,
humility, and ownership. The Charge emphasizes that the Navy's decisive
advantage is its Sailors, and COs must invest in their people with the same
intensity as operational readiness, fostering trust, accountability, and a
warfighting culture that thrives under pressure.
3. Command inherently involves risk. The Charge directs COs to manage
operational risk deliberately, provide clear Commander's intent, and empower
teams under Mission Command principles.
Disciplined risk-taking, sound judgment, and adaptability are required to
ensure mission success while cultivating initiative and learning throughout
the command.
4. The Charge further stresses the importance of preparation, assessment,
and continuous learning. COs are expected to plan, rehearse, test
assumptions, and innovate faster than potential adversaries. Excellence is
achieved through initiative, critical thinking, disciplined failure as a
learning pathway, and fostering a culture of resilience and innovation.
5. COs are the single accountable officer for their units and are expected
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to lead decisively, mentor future leaders, and uphold the Navy's enduring
values and warfighting mindset. COs and their commands must always be ready
to Fight Tonight. As Fleet Admiral Chester Nimitz once said, "When you're in
command, command."
6. Immediate Superior in Command commanders are expected to review the
charge with their subordinate COs, ensuring they understand, internalize, and
apply its principles. This review supports mentorship, reinforces
accountability, and strengthens unit readiness across the fleet.
7. The full Charge of Command letter, including detailed guidance and
reference materials are available at: https://www.navy.mil/
All personnel are encouraged to read, understand, and support the principles
outlined to strengthen their commands and the Navy as a whole.
8. Built in the Foundry - Tempered in the Fleet - Forged to Fight!
9. ADM Daryl Caudle, 34th Chief of Naval Operations sends.//
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CLASSIFICATION: UNCLASSIFIED/
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E Current Events

E.1 Current Events Relevant to EDOs and Acquisition

E.1.1 Project Overmatch and Digital Modernization

Five Eyes Project Arrangement.

In Oct 2024 Project Overmatch signed a Project Arrangement with Australia, Canada, New Zealand, and the United Kingdom, embedding Cooperative Project Personnel in San Diego to co-develop resilient coalition kill-chain networks ahead of Rim of the Pacific Exercise (RIMPAC) 26; Capt. Remil Capili (PM), VADM Seiko Okano (steering chair/PMILDEP), Shayna Bond (international deputy), and David Flowers (international lead) direct the effort while a coalition lab stands up to support continuous testing [206].

Open DAGIR pilot.

By 18 Jun 2025 Project Overmatch, NAVWAR, the Chief Digital and Artificial Intelligence Office, and were onboarding commercial Artificial Intelligence and Machine Learning (AI/ML) applications through the Open Data and Applications Government-owned Interoperable Repositories (DAGIR) pipeline, so Fleet commanders could visualize readiness data ("Maven Smart Systems") across the hybrid fleet in weeks instead of years [207].

Leadership signal.

VADM Seiko Okano continues to serve as the direct reporting program manager and steering committee chair while serving as PMILDEP to ASN(RD&A), keeping Overmatch digital architecture tied to Program Decision Meetings and coalition governance [9, 206].

E.1.2 EDO Community Updates

NAVWAR change of command.

Rear Adm. (now VADM) Seiko Okano relieved Rear Adm. Doug Small on 9 Aug 2024, assuming command of NAVWAR and direct oversight of Project Overmatch; she emphasized information warfare dominance and continued alliance engagement with industry and academia [208].

Role of the PMILDEP.

As PMILDEP to ASN(RD&A), Okano now chairs Program Decision Meetings when delegated and synchronizes PAEs and SYSCOM technical authority, a critical linkage as Capability Trade Councils and portfolio scorecards begin replacing JCIDS artifacts [9, 90].

Tycom maintenance focus.

USFF and U.S. Pacific Fleet (PACFLT) maintenance flag officers continue to drive execution of regional maintenance center reforms and the Naval Sustainment System–Shipyards initiatives highlighted in the 2025 coursebook; these billets remain key board discussion topics because they link EDO leadership to depot performance [88].

E.1.3 Acquisition Policy and Industrial Base Highlights

Authority stack for the RFO.

FAR-overhaul policy direction begins with the presidential action, "Restoring Common Sense to Federal Procurement" [91]. OMB Memo M-25-26 then provides implementation direction to agencies and the FAR Council [92]. FAR Council deviation guidance supports interim execution while final rulemaking progresses [93].

Live implementation tracker.

Acquisition.gov maintains the live part-by-part deviation tracker for the Revolutionary FAR Overhaul (RFO); use it as the operational status source for released text and agency adoption [98, 99].

Context vs. controlling references.

The White House fact sheet and DAU RFO announcements provide useful explanatory and timeline context, but controlling implementation references to remain the presidential action, OMB memo, and FAR Council guidance [100, 101]. Where needed, the Federal Register publication can be used as a metadata anchor for the presidential action record [209].

Current DoD execution.

DoD has published RFO class deviations for FAR Parts 6 (competition), 10 (market research), 12 (commercial products and services), and 18 (emergency acquisitions), all effective Feb 1, 2026; these deviations update DFARS crosswalks and solicitation prescriptions and should be reflected in acquisition strategies and templates now [94, 95, 96, 97].

RFO practical effects.

The RFO is shifting working practice before final rulemaking: rewritten parts arrive first as model deviation text, agencies adopt them through class deviations, and useful but non-mandatory material moves into Strategic Acquisition Guidance, buying guides, and practitioner resources [98, 210]. Board answer: acquisition teams must check both the codified FAR/DFARS and the active deviation/adoption status before reusing old templates.

Threshold changes.

Effective Oct 1, 2025, Acquisition.gov lists inflation adjustments that raise the major-system threshold from $2.5M to $3.0M, the micro-purchase threshold from $10K to $15K, and the Simplified Acquisition Threshold (SAT) from $250K to $350K [211]. The FY 2026 NDAA separately raises the certified cost-or-pricing-data threshold for defense contracts entered into after Jun 30, 2026, from $2M to $10M and directs CAS threshold increases, including full CAS coverage from $50M to $100M [212].

ACAT/MDAP limits.

Section 1804 of the FY 2026 NDAA resets Title 10 major-system and MDAP dollar thresholds into FY 2024 constant dollars: major-system RDT&E/procurement thresholds move to $275M/$1.3B, and MDAP RDT&E/procurement thresholds move to $1.0B/$4.5B [212]. Do not confuse these statutory ACAT-screening thresholds with the FAR major-system threshold used for contracting clauses and procedures.

WAS.

The Acquisition Transformation Strategy replaces JCIDS with Capability Portfolio Management and Capability Trade Councils, empowering PAEs to adjudicate cost/schedule/performance trades in real time and document decisions in portfolio scorecards [90].

New Navy PAEs.

Public DON releases now identify nine PAE organizations: Robotics and Autonomous Systems; Industrial Operations; Marine Corps; Maritime; Strategic Systems Programs; Undersea / DRPM Submarines; Aviation; Mission Systems; and Munitions [10, 11]. The board-level shift is single-accountable portfolio ownership with authority over program offices plus associated technical, contracting, sustainment, and industrial-base functions; DON states about 70 percent of those functions and associated personnel are moving from SYSCOMs into PAEs [11].

Industrial base actions.

SECWAR directed the Department to expand multi-year procurements, establish an Industrial Base Consortium, and reform bid protest incentives so the Munitions Acceleration Council and similar bodies can stabilize demand and attract private capital [90].

PPBE integration.

Under Secretary of War (Comptroller) (USW(C)) and USW(A&S) are piloting portfolio budgets and flexible execution authorities so PAEs can move funds inside approved controls while maintaining transparency with OMB and Congress—a major shift boards will expect candidates to explain [90].

Security cooperation realignment.

Defense Security Cooperation Agency (DSCA) and Defense Technology Security Administration (DTSA) now fall under USW(A&S), aligning exportability, Foreign Military Sales, and industrial-base planning so allied orders and U.S. production trades can be brokered inside Capability Trade Councils [213].

National Security Capital Forum.

Section 1092 of the FY 2025 NDAA created the National Security Capital Forum (NSCF), chaired by the Office of Strategic Capital, to coordinate public/private financing decisions that affect defense materiel transactions—EDOs should highlight how their programs leverage or are scrutinized by the forum [214].

E.1.4 Current Navy Acquisition Issues

2026 NDS release.

The latest NDS, released 23 Jan 2026, replaces the 2022 strategy as the current defense-strategy baseline and aligns to the Nov 2025 NSS [215, 216]. Its four headline lines of effort are homeland defense, deterring China in the Indo-Pacific, increasing allied burden-sharing, and supercharging the U.S. defense industrial base; for EDOs, the acquisition translation is faster fielding, stronger organic and private industrial capacity, and explicit Indo-Pacific sustainment risk management [215].

Golden Fleet and FY27 budget signal.

The DON FY 2027 budget request frames the Golden Fleet initiative as a strategy-driven budget, emphasizing shipbuilding, industrial-base revitalization, PAEs, the Rapid Capabilities Office, and disciplined requirements [217]. The public brief lists

$377.5B for DON, $150B for procurement, 34 total ships, 123 aircraft, $17B for ship maintenance, and $1.8B for public shipyards [217].

Make cybersecurity part of our DNA.

The Department of the Navy Cyber Strategy frames cybersecurity as a warfighting requirement and calls for a cyber-ready culture across the enterprise, reinforcing that programs must treat cyber risk as mission risk [218]. DoDI 5000.90 directs program managers to include and decompose cybersecurity requirements from early system development through the full acquisition and sustainment life cycle, including supply chain risk considerations [134].

Legacy NAVPLAN cues.

NAVPLAN 2022 and Force Design 2045 remain useful for board context on the hybrid-fleet ambition and Get Real, Get Better culture, but use the 2026 NDS, DON FY27 budget request, WAS, and PAE stand-up as the current-event baseline [10, 215, 217, 219].

Current as of 2026-04-27.

E.2 Shipbuilding and Sustainment Updates

E.2.1 Columbia-Class (SSBN) Delays

The Columbia-class ballistic missile submarine program, the Navy's highest acquisition priority, experienced significant schedule pressure in 2025. Navy officials reported an estimated 12- to 17-month delay in the delivery of the lead boat, USS *District of Columbia* (SSBN-826), pushing its first planned patrol into 2030 [220]. The delays were attributed to a combination of factors, including shipyard workforce shortages, supply chain disruptions for critical components like the bow dome and main turbine generators, and the inherent complexities of a first-of-class build [220, 221]. By late 2025, General Dynamics Electric Boat reported the lead vessel was approximately 60% complete, but the tight schedule margins remain a significant concern for maintaining the nation's continuous strategic deterrent capability [222].

E.2.2 Amphibious Warship Block Buy (LPD-17)

In a significant move to stabilize the amphibious ship industrial base, the Navy executed a multi-ship "block buy" procurement for four amphibious warships from Huntington Ingalls Industries (HII) Ingalls in late 2024 and early 2025 [223]. This procurement, valued at nearly $1 billion in savings, includes three San Antonio-class amphibious transport docks (LPD-33, 34, and 35) and one America-class amphibious assault ship (LHA-10) to be built between FY25 and FY29 [223]. The block buy provides workload stability for the shipyard and its suppliers, enabling more efficient production and ensuring the Navy meets the legally mandated 31-ship amphibious force structure [224]. This decision followed congressional pressure and signals a commitment to the amphibious fleet after a period of debate over future requirements.

E.2.3 Shipyard Infrastructure Optimization Program (SIOP) Status

The Shipyard Infrastructure Optimization Program (Shipyard Infrastructure Optimization Program (SIOP)), a 20-year, $21 billion effort to modernize the Navy's four public shipyards, reached a critical phase in 2025. A key milestone was the award of a $377.7 million contract for seismic upgrades to Dry Dock #4 at Puget Sound Naval Shipyard, which is essential for servicing nuclear submarines, including the future Columbia-class SSBNs [225]. However, the program faces significant cost growth due to inflation and unforeseen construction complexities, with annual construction cost escalation exceeding 6% [226]. In July 2025, program officials announced a comprehensive review to revise cost and schedule estimates for the entire SIOP portfolio [226]. Despite these challenges, the Navy continues to invest approximately $2.7 billion annually, with the goal of reducing submarine maintenance periods by up to 15% and returning ships to the fleet faster [226].

E.2.4 West Coast and Pacific Shipyard Development

West Coast and Pacific infrastructure are now explicit current-event examples of the Foundry/industrial-base story. At Puget Sound Naval Shipyard & Intermediate Maintenance Facility, a new 25-ton portal crane arrived in July 2025 as part of a $67M contract for four cranes supporting maintenance, repair, modernization, and retirement of U.S. naval vessels [227]. In Hawaii, the future Pearl Harbor Dry Dock 5 remains a major SIOP effort to modernize nuclear-powered submarine maintenance capacity for the Pacific Fleet [228, 229]. In Guam, NAVFAC Pacific awarded a combined $15B design-build multiple-award construction contract in Sept 2025 for Guam and other Pacific projects; the official release ties the award to resilient, modern, strategically important facilities supporting operational readiness and regional stability [230]. Board answer: shipyard development is no longer only a facilities topic; it is a naval-operations enabler tied to Indo-Pacific posture, PAE Industrial Operations, and NDS industrial-base priorities [10, 215].

References

[1] Engineering Duty Officer School, “2.1.1. Command Structures,” Engineering Duty Officer (EDO) School, Coursebook Module 2.1.1, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[2] Goldwater-Nichols Department of War Reorganization Act of 1986, Pub. L. 99-433, Oct. 1, 1986, Hosted by Naval Postgraduate School Digital Archive. Accessed: 2026-01-11. [Online]. Available: https://dair.nps.edu/bitstream/123456789/11182/1/GoldwaterNicholsAct1986.pdf

[3] 10 U.S.C. § 113, Secretary of War, 2023. Accessed: 2026-01-11. [Online]. Available: https://www.govinfo.gov/content/pkg/USCODE-2023-title10/html/USCODE-2023-title10-subtitleA-partI-chap2-sec113.htm

[4] 10 U.S.C. § 162, Combatant command: assigned forces; chain of command, 2023. Accessed: 2026-01-11. [Online]. Available: https://www.govinfo.gov/content/pkg/USCODE-2023-title10/html/USCODE-2023-title10-subtitleA-partI-chap6-sec162.htm

[5] 10 U.S.C. § 8032, Office of the Chief of Naval Operations, 2023. Accessed: 2026-01-11. [Online]. Available: https://www.govinfo.gov/content/pkg/USCODE-2023-title10/html/USCODE-2023-title10-subtitleC-partI-chap805-sec8032.htm

[6] “National Security Strategy,” The White House, Oct. 12, 2022, Hosted copy of the White House NSS. Accessed: 2026-01-11. [Online]. Available: https://georgetownsecuritystudiesreview.org/wp-content/uploads/2022/10/Biden-Harris-Administrations-National-Security-Strategy-10.2022.pdf

[7] “2022 National Defense Strategy of the United States of America,” U.S. Department of War, Oct. 27, 2022, Combined NDS/NPR/MDR publication. Accessed: 2026-01-11. [Online]. Available: https://media.defense.gov/2022/Oct/27/2003103845/-1/-1/1/2022-NATIONAL-DEFENSE-STRATEGY-NPR-MDR.PDF

[8] *SECNAVINST 5400.15D, Department of the Navy Research, Development and Acquisition—Organizational Relationships and Responsibilities*, SECNAVINST 5400.15D, Instruction, Department of the Navy, Jan. 19, 2021. Accessed: 2025-09-29. [Online]. Available: https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5400.15D.pdf

[9] *SECNAVINST 5000.2G, Defense Acquisition System and JCIDS Implementation in the Department of the Navy*, SECNAVINST 5000.2G, Instruction, Department of the Navy, Apr. 8, 2022. Accessed: 2025-09-29. [Online]. Available: https://www.secnav.navy.mil/doni/SECNAVInstructions/05000-05999/5000.2G.pdf

[10] Department of the Navy. “Navy Reshapes Warfighting Acquisition System,” Accessed: 2026-04-27. [Online]. Available: https://www.navy.mil/Press-Office/Press-Releases/display-pressreleases/Article/4435370/navy-reshapes-warfighting-acquisition-system/

[11] Department of the Navy. “Navy Advances Acquisition Reform Strategy: Appoints Three New Portfolio Acquisition Executives,” Accessed: 2026-06-04. [Online]. Available: https://www.navy.mil/Press-Office/Press-Releases/display-pressreleases/Article/4483312/navy-advances-acquisition-reform-strategy-appoints-three-new-portfolio-acquisit/

[12] MyNavy HR. “May 2026 Public Roster.” United States Navy Flag Officers, public roster, Accessed: 2026-05-09. [Online]. Available: https://www.mynavyhr.navy.mil/Portals/55/Career/Detailing/Flag/May%202026%20Public%20Roster.pdf?ver=7l9mYE94L1dRstF-xR7gug%3D%3D

[13] U.S. Navy. “Vice Admiral Seiko Okano.” Principal Military Deputy, ASN(RD&A), Accessed: 2025-12-05. [Online]. Available: https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/2283053/vice-admiral-seiko-okano/

[14] U.S. Navy. “Rear Admiral J. Kurt Rothenhaus.” Commander, Naval Information Warfare Systems Command, Accessed: 2025-12-05. [Online]. Available: https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/2236520/rear-admiral-j-kurt-rothenhaus/

[15] U.S. Navy. “Rear Admiral Brian Metcalf.” Program Executive Officer, Ships, Accessed: 2025-12-05. [Online]. Available: https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/4234702/rear-admiral-brian-metcalf/

[16] U.S. Navy. “Rear Admiral Casey Moton.” Program Executive Officer, Aircraft Carriers, Accessed: 2025-12-05. [Online]. Available: https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/2236471/rear-admiral-casey-moton/

[17] U.S. Navy. “Rear Admiral Jonathan Rucker.” Program Executive Officer, Attack Submarines, Accessed: 2025-12-05. [Online]. Available: https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/3052230/rear-admiral-jonathan-rucker/

[18] NAVSEA. “NAVSEA Organization Chart (December 2025).” NAVSEA Organization Chart (December 2025); verify current fills, Accessed: 2025-12-13. [Online]. Available: https://www.navsea.navy.mil/Portals/103/Documents/Command%5C%20Org%5C%20Charts/NAVSEA_Org_Chart_Dec2025.pdf

[19] Department of the Navy. “Department of the Navy Names New Service Acquisition Executive,” Accessed: 2026-06-04. [Online]. Available: https://www.navy.mil/Press-Office/News-Stories/display-news/Article/4501092/department-of-the-navy-names-new-service-acquisition-executive/

[20] Portfolio Acquisition Executive Maritime. “Portfolio Acquisition Executive Maritime,” Accessed: 2026-06-04. [Online]. Available: https://www.paemaritime.navy.mil/

[21] Department of the Navy. “Department of the Navy Launches PAE Mission Systems to Accelerate Warfighting Capability at Speed,” Accessed: 2026-05-27. [Online]. Available: https://www.navwar.navy.mil/Media/Article-Display/Article/4483415/department-of-the-navy-launches-pae-mission-systems-to-accelerate-warfighting-c/

[22] Department of the Navy. “Navy Establishes Portfolio Acquisition Executive for Munitions,” Accessed: 2026-06-04. [Online]. Available: https://www.navy.mil/Press-Office/News-Stories/display-news/Article/4483399/navy-establishes-portfolio-acquisition-executive-for-munitions-postured-to-harn/

[23] Naval Air Systems Command. “Navy Establishes Portfolio Acquisition Executive for Aviation,” Accessed: 2026-06-04. [Online]. Available: https://www.navair.navy.mil/news/Navy-Establishes-Portfolio-Acquisition-Executive-Aviation/Mon-05112026-0943

[24] Portfolio Acquisition Executive Marine Corps. “About Us,” Accessed: 2026-06-04. [Online]. Available: https://www.pae.marines.mil/About-Us/

[25] Department of the Navy. “U.S. Navy Announces Seven Companies Selected for MUSV Marketplace At-Sea Demonstrations,” Accessed: 2026-06-04. [Online]. Available: https://www.navy.mil/Press-Office/Press-Releases/display-pressreleases/Article/4503917/us-navy-announces-seven-companies-selected-for-musv-marketplace-at-sea-demonstr/

[26] "Mission and Functions," U.S. Fleet Forces Command, Accessed: 2026-01-11. [Online]. Available: https://www.usff.navy.mil/About-Us/Mission/

[27] Engineering Duty Officer School, "2.1.5. Technical Authority and Engineering Agents," Engineering Duty Officer (EDO) School, Coursebook Module 2.1.5, version 6.0, Dec. 10, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[28] MyNavy HR. "April 2026 Roster Public." United States Navy Flag Officers, public roster, Accessed: 2026-04-09. [Online]. Available: https://www.mynavyhr.navy.mil/Portals/55/Career/Detailing/Flag/April%202026%20Roster%20Public.pdf?ver=3DQoHU45rRXfYDTrp7jgcg%3D%3D

[29] Engineering Duty Officer School, "2.1.2. NAVSEA Organization," Engineering Duty Officer (EDO) School, Coursebook Module 2.1.2, version 6.0, Dec. 9, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[30] Naval Sea Systems Command. "PEO Integrated Warfare Systems," Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Home/PEO-IWS/

[31] Naval Sea Systems Command. "About PEO IWS," Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Home/PEO-IWS/About-PEO-IWS/

[32] Naval Sea Systems Command. "PEO IWS Industry Engagement," Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Home/PEO-IWS/PEO-IWS-Industry-Engagement/

[33] Naval Sea Systems Command. "PEO IWS X Program Overview and ICS Development," Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Portals/103/Documents/Exhibits/SAS2022/1100_ICS%20Engagement%20brief%2003302022_SAS_Distro%20A.pdf

[34] Engineering Duty Officer School, "2.1.4. Naval Warfare Centers," Engineering Duty Officer (EDO) School, Coursebook Module 2.1.4, version 6.0, Dec. 10, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[35] Engineering Duty Officer School, "2.1.3. NAVWAR Enterprise," Engineering Duty Officer (EDO) School, Coursebook Module 2.1.3, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[36] Program Executive Office Manpower, Logistics and Business Solutions. "PEO MLB Portfolio Book," Accessed: 2026-05-27. [Online]. Available: https://www.peomlb.navy.mil/Portals/97/Documents/PEO%20MLB%20Portfolio%20Book%20%28January%202025%29.pdf?ver=YMas-_iMjHcOUK5OHn-lsQ%3D%3D

[37] Naval Information Warfare Systems Command. "Who We Are," Accessed: 2026-05-27. [Online]. Available: https://www.navwar.navy.mil/About/Who-We-Are/

[38] Program Executive Office Command, Control, Communications, Computers, and Intelligence. "Program Offices," Accessed: 2026-05-27. [Online]. Available: https://www.peoc4i.navy.mil/Program-Offices/

[39] Program Executive Office Command, Control, Communications, Computers, and Intelligence. "PMW 120 Overview Tearsheet," Accessed: 2026-05-27. [Online]. Available: https://www.peoc4i.navy.mil/Portals/98/PMW120_Overview_Tearsheet.pdf

[40] Program Executive Office Command, Control, Communications, Computers, and Intelligence. "PMW 150 Tear Sheet," Accessed: 2026-05-27. [Online]. Available: https://www.peoc4i.navy.mil/Portals/98/2026_PMW%20150_Tear%20Sheet_FINAL_JAN2026.pdf

[41] Program Executive Office Command, Control, Communications, Computers, and Intelligence. "PMW 160 Tear Sheet," Accessed: 2026-05-27. [Online]. Available: https://www.peoc4i.navy.mil/Portals/98/Documents/Tear-Sheets/2025_PMW%20160_Tear%20Sheet_v01312025.pdf

[42] Program Executive Office Command, Control, Communications, Computers, and Intelligence. "PMW 170 Tear Sheet," Accessed: 2026-05-27. [Online]. Available: https://www.peoc4i.navy.mil/Portals/98/Documents/Tear-Sheets/2025_PMW%20170%20Tear%20Sheet_v01312025.pdf

[43] Program Executive Office Command, Control, Communications, Computers, and Intelligence. “PMW 790 Tear Sheet,” Accessed: 2026-05-27. [Online]. Available: https://www.peoc4i.navy.mil/Portals/98/Documents/Tear-Sheets/2025_PMW%20790_Tear%20Sheet_FINAL.pdf?timestamp=1744284354501&ver=-t1PGfkpkm1-N99IGFvsiQ%3D%3D

[44] Program Executive Office Digital. “PEO Digital Portfolio Overview,” Accessed: 2026-05-27. [Online]. Available: https://www.peodigital.navy.mil/Portals/96/Documents/PEO_Digital_Portfolio_Overview_June_2023.pdf?ver=0AiN3W5JCb1OMv4FhVfNKQ%3D%3D

[45] Defense Acquisition University. “MDAP Program Manager Changes,” Accessed: 2026-05-27. [Online]. Available: https://www.dau.edu/library/damag/may-june2024/mdapchanges

[46] Program Executive Office Command, Control, Communications, Computers, and Intelligence. “PMA/PMW 101 MIDS Tear Sheet,” Accessed: 2026-05-30. [Online]. Available: https://www.peoc4i.navy.mil/Portals/98/Documents/Tear-Sheets/2025_PMA%20PMW%20101_Tear%20Sheet_v01312025.pdf

[47] Naval Air Systems Command. “PMA/PMW-101,” Accessed: 2026-06-01. [Online]. Available: https://www.navair.navy.mil/orgterm/PMAPMW-101

[48] Naval Air Systems Command. “PMW/A-170,” Accessed: 2026-06-01. [Online]. Available: https://www.navair.navy.mil/organization/PMWA-170

[49] Engineering Duty Officer School, “3.1.1. PPBE (Planning, Programming, Budgeting, and Execution),” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.1, version 6.1, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[50] *Activity Manpower Management Guide*, AMMG, Guide, Activity Manpower Management Guide February 2025; Section 24, Resource Sponsor code table, Navy Manpower Analysis Center, Mar. 7, 2025. Accessed: 2026-05-27. [Online]. Available: https://www.mynavyhr.navy.mil/Portals/55/NAVMAC/Documents/AMMG%20FEB_2025.pdf?ver=gaTk8q6UbpYJVJYxFWmzig%3D%3D

[51] *OPNAVINST 4700.7N, Maintenance Policy for United States Navy Ships*, OPNAVINST 4700.7N, Instruction, Maintenance Policy for United States Navy Ships, Office of the Chief of Naval Operations, Dec. 5, 2024. Accessed: 2026-04-24. [Online]. Available: https://www.secnav.navy.mil/doni/Directives/04000%20Logistical%20Support%20and%20Services/04-700%20General%20Maintenance%20and%20Construction%20Support/4700.7N.pdf

[52] *OPNAVINST 3050.27A*, OPNAVINST 3050.27A, Instruction, Battle Force Structure Assessments, Office of the Chief of Naval Operations, Feb. 12, 2024. Accessed: 2026-04-24. [Online]. Available: https://www.secnav.navy.mil/doni/Directives/03000%20Naval%20Operations%20and%20Readiness/03-00%20General%20Operations%20and%20Readiness%20Support/3050.27A.pdf

[53] U.S. Department of War. “Flag Officer Assignments.” Release identifying the same officer as Deputy Assistant Secretary of the Navy for Budget (FMB) and Director, Fiscal Management Division, N82, Office of the Chief of Naval Operations, Accessed: 2026-05-02. [Online]. Available: https://www.war.gov/News/Releases/Release/Article/2178509/flag-officer-assignments/

[54] Engineering Duty Officer School, “3.1.6. Cost Estimating,” Engineering Duty Officer (EDO) School, Coursebook Module 3.1.6, version 6.1, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[55] 10 U.S.C. § 139c, Director of Cost Assessment and Program Evaluation, 2023. Accessed: 2025-12-16. [Online]. Available: https://uscode.house.gov/

[56] “OMB Circular A-11: Preparation, Submission, and Execution of the Budget,” Circular, Aug. 29, 2025. Accessed: 2025-12-16. [Online]. Available: https://www.whitehouse.gov/wp-content/uploads/2025/08/a11.pdf

[57] *OPNAVINST 3111.17C*, OPNAVINST 3111.17C, Instruction, Navy Mission Essential Task List Program, Office of the Chief of Naval Operations, May 14, 2024. Accessed: 2026-04-24. [Online]. Available: https://www.secnav.navy.mil/doni/Directives/03000%20Naval%20Operations%20and%20Readiness/03-100%20Naval%20Operations%20Support/3111.17C.pdf

[58] *OPNAVINST 5420.117A*, OPNAVINST 5420.117A, Instruction, Resources and Requirements Review Board and Naval Capabilities Board, Office of the Chief of Naval Operations, Apr. 16, 2021. Accessed: 2026-05-27. [Online]. Available: https://www.secnav.navy.mil/doni/Directives/05000%20General%20Management%20Security%20and%20Safety%20Services/05-400%20Organization%20and%20Functional%20Support%20Services/5420.117A.pdf

[59] *SECNAVINST 5460.4*, SECNAVINST 5460.4, Instruction, Mission, Functions, and Tasks of the Department of the Navy Special Access Program Central Office, Department of the Navy, Oct. 30, 2018. Accessed: 2026-05-27. [Online]. Available: https://www.secnav.navy.mil/doni/Directives/05000%20General%20Management%20Security%20and%20Safety%20Services/05-400%20Organization%20and%20Functional%20Support%20Services/5460.4.pdf

[60] Engineering Duty Officer School, "3.1.5. Field Activity Financial Management," Engineering Duty Officer (EDO) School, Coursebook Module 3.1.5, version 6.1, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[61] Engineering Duty Officer School, "3.1.2. Congressional Enactment," Engineering Duty Officer (EDO) School, Coursebook Module 3.1.2, version 6.1, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[62] Engineering Duty Officer School, "3.1.3. Program Funding," Engineering Duty Officer (EDO) School, Coursebook Module 3.1.3, version 6.1, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[63] Engineering Duty Officer School, "3.1.4. Budget Execution," Engineering Duty Officer (EDO) School, Coursebook Module 3.1.4, version 6.1, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[64] 31 U.S.C. § 1301(a), Application — Purpose Statute, 2018. [Online]. Available: https://uscode.house.gov/

[65] Defense Acquisition University. "Misappropriation Act," Accessed: 2026-06-01. [Online]. Available: https://www.dau.edu/acquipedia-article/misappropriation-act

[66] 31 U.S.C. § 1341; § 1517, Anti-Deficiency Act, 2018. [Online]. Available: https://uscode.house.gov/

[67] 31 U.S.C. § 1502(a), Bona Fide Needs Rule, 2018. [Online]. Available: https://uscode.house.gov/

[68] Defense Acquisition University. "Bona Fide Need," Accessed: 2026-06-01. [Online]. Available: https://www.dau.edu/index.php/acquipedia-article/bona-fide-need

[69] Engineering Duty Officer School, "3.2.1. Intro to Contracting," Engineering Duty Officer (EDO) School, Coursebook Module 3.2.1, version 6.1, Mar. 13, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[70] "Federal Acquisition Regulation (FAR)," Regulation, 2025. Accessed: 2025-09-29. [Online]. Available: https://www.acquisition.gov/far

[71] Engineering Duty Officer School, "3.2.2. Solicitation Preparation," Engineering Duty Officer (EDO) School, Coursebook Module 3.2.2, version 6.3, Mar. 18, 2026, CIN A-4N-0021/-0034; Version 6.3; coursebook PDF provided by author.

[72] 10 U.S.C. § 4021, Prototype projects: other transaction authority, 2023. Accessed: 2025-10-12. [Online]. Available: https://uscode.house.gov/

[73] 10 U.S.C. § 4022, Follow-on production contracts or transactions, 2023. Accessed: 2025-10-12. [Online]. Available: https://uscode.house.gov/

[74] "Work With Us," Defense Innovation Unit, Accessed: 2026-02-01. [Online]. Available: https://www.diu.mil/work-with-us

[75] U.S. Government Accountability Office, "Defense Acquisitions: DOD's Use of Other Transactions for Prototype Projects Has Increased," Report GAO-20-84, Nov. 2019. Accessed: 2026-02-01. [Online]. Available: https://www.gao.gov/assets/gao-20-84.pdf

[76] Engineering Duty Officer School, “3.2.3. Source Selection Process,” Engineering Duty Officer (EDO) School, Coursebook Module 3.2.3, version 6.2, Mar. 11, 2026, CIN A-4N-0021/-0034; Version 6.2; coursebook PDF provided by author.

[77] Engineering Duty Officer School, “3.2.4. Cost and Price Evaluation,” Engineering Duty Officer (EDO) School, Coursebook Module 3.2.4, version 6.1, Mar. 11, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[78] Engineering Duty Officer School, “3.2.5. Source Selection Practical,” Engineering Duty Officer (EDO) School, Coursebook Module 3.2.5, version 6.1, Dec. 16, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[79] *Department of Defense Source Selection Procedures*, Procedures, Aug. 20, 2022. Accessed: 2026-05-27. [Online]. Available: https://www.acq.osd.mil/dpap/policy/policyvault/USA000740-22-DPC.pdf

[80] Engineering Duty Officer School, “3.2.6. Contract Administration,” Engineering Duty Officer (EDO) School, Coursebook Module 3.2.6, version 6.0, Dec. 10, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[81] Engineering Duty Officer School, “3.6.1. Introduction to EVM,” Engineering Duty Officer (EDO) School, Coursebook Module 3.6.1, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[82] Engineering Duty Officer School, “3.6.2. EVM Data Analysis,” Engineering Duty Officer (EDO) School, Coursebook Module 3.6.2, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[83] “Class Deviation from the Federal Acquisition Regulation and Defense Federal Acquisition Regulation Supplement Regarding Earned Value Management System,” Office of the Assistant Secretary of War, Class deviation 2026-O0011, Feb. 1, 2026. Accessed: 2026-05-27. [Online]. Available: https://www.acq.osd.mil/dpap/dars/classdev/DFARS_RFO/Part-234/2026-O0011_TAB_A_Deviation_Memo_DFARS_234.pdf

[84] *DoWI 5000.85, Major Capability Acquisition*, DoWI 5000.85, Instruction, Change(s) as applicable, Department of War, Aug. 6, 2020. Accessed: 2025-09-29. [Online]. Available: https://www.esd.whs.mil/Directives/issuances/dodissuances/

[85] U.S. Navy. “Rear Admiral Jonathan J. Jett-Parmer.” Deputy Commander, Supervisor of Shipbuilding (NAVSEA), Accessed: 2025-12-05. [Online]. Available: https://www.navy.mil/Leadership/Flag-Officer-Biographies/BioDisplay/Article/4312456/rear-admiral-jonathan-j-jett-parmer-pe/

[86] “Integrated Program Management Data and Analysis Report (IPMDAR),” Defense Logistics Agency ASSIST QuickSearch, Data Item Description DI-MGMT-81861D, Mar. 12, 2026. Accessed: 2026-05-27. [Online]. Available: https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=278901

[87] Engineering Duty Officer School, “3.6.3. EVM Exercise,” Engineering Duty Officer (EDO) School, Coursebook Module 3.6.3, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[88] Engineering Duty Officer School, “3.3.1. Acquisition Policy and Players,” Engineering Duty Officer (EDO) School, Coursebook Module 3.3.1, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[89] Secretary of War, “Transforming the Defense Acquisition System into the Warfighting Acquisition System to Accelerate Fielding of Urgently Needed Capabilities to Our Warriors,” Memorandum, Nov. 7, 2025. Accessed: 2025-11-14. [Online]. Available: WAS/Transforming%5C%20the%5C%20Defense%5C%20Acquisitions%5C%20System%5C%20into%5C%20the%5C%20Warfighting%5C%20Acquisition%5C%20System%5C%20to%5C%20Accelerate%5C%20Fielding%5C%20of%5C%20Urgently%5C%20Needed%5C%20C.pdf

[90] Office of the Under Secretary of War for Acquisition & Sustainment, “Acquisition Transformation Strategy,” Strategy, Nov. 10, 2025, Implements Executive Order 14265 and Warfighting Acquisition System reforms. Accessed: 2025-11-12. [Online]. Available: https://media.defense.gov/2025/Nov/10/2003819441/-1/-1/1/ACQUISITION-TRANSFORMATION-STRATEGY.PDF

[91] “Restoring Common Sense to Federal Procurement.” Presidential action establishing FAR-overhaul policy direction (referenced as Executive Order 14275 in OMB M-25-26)., The White House, Accessed: 2026-02-06. [Online]. Available: https://www.whitehouse.gov/presidential-actions/2025/04/restoring-common-sense-to-federal-procurement/

[92] "Overhauling the Federal Acquisition Regulation," Memorandum M-25-26, May 2, 2025. Accessed: 2026-02-06. [Online]. Available: https://www.acquisition.gov/sites/default/files/page_file_uploads/M-25-26-Overhauling-the-Federal-Acquisition-Regulation-002.pdf

[93] "Deviation Guidance to Support the Overhaul of the Federal Acquisition Regulation," Guidance memorandum, May 2, 2025. Accessed: 2026-02-06. [Online]. Available: https://www.acquisition.gov/sites/default/files/page_file_uploads/FAR-Council-Deviation-Guidance-on-FAR-Overhaul.pdf

[94] "DoD RFO Class Deviation: FAR Part 6 (Competition Requirements)," Class deviation, Feb. 1, 2026. Accessed: 2026-01-22. [Online]. Available: https://www.acquisition.gov/sites/default/files/page_file_uploads/DoD_RFO_Deviation_Part-6.pdf

[95] "DoD RFO Class Deviation: FAR Part 10 (Market Research)," Class deviation, Feb. 1, 2026. Accessed: 2026-01-22. [Online]. Available: https://www.acquisition.gov/sites/default/files/page_file_uploads/DoD_RFO_Deviation_Part-10.pdf

[96] "DoD RFO Class Deviation: FAR Part 12 (Acquisition of Commercial Products and Commercial Services)," Class deviation, Feb. 1, 2026. Accessed: 2026-01-22. [Online]. Available: https://www.acquisition.gov/sites/default/files/page_file_uploads/DoD_RFO_Deviation_Part-12.pdf

[97] "DoD RFO Class Deviation: FAR Part 18 (Emergency Acquisitions)," Class deviation, Feb. 1, 2026. Accessed: 2026-01-22. [Online]. Available: https://www.acquisition.gov/sites/default/files/page_file_uploads/DoD_RFO_Deviation_Part-18.pdf

[98] "FAR Overhaul - FAR Part Deviation Guidance," Acquisition.gov, Accessed: 2026-01-22. [Online]. Available: https://www.acquisition.gov/far-overhaul/far-part-deviation-guide

[99] "Revolutionary FAR Overhaul (RFO)," Acquisition.gov, Accessed: 2026-01-22. [Online]. Available: https://www.acquisition.gov/far-overhaul

[100] "Fact Sheet: President Donald J. Trump Restores Common Sense to Federal Procurement." Informative context, not a controlling implementation instrument., The White House, Accessed: 2026-02-06. [Online]. Available: https://www.whitehouse.gov/fact-sheets/2025/04/fact-sheet-president-donald-j-trump-restores-common-sense-to-federal-procurement/

[101] "Revolutionary FAR Overhaul: Announcements." Announcement timeline metadata and links; use as context while relying on EO, OMB, FAR Council, and Acquisition.gov for controlling implementation text., Defense Acquisition University, Accessed: 2026-02-06. [Online]. Available: https://www.dau.edu/revolutionary-far-overhaul/announcements

[102] Engineering Duty Officer School, "3.3.2. Acquisition Milestones and Phases," Engineering Duty Officer (EDO) School, Coursebook Module 3.3.2, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[103] *CJCSM 5123.01*, CJCSM 5123.01, Manual, Manual for the Joint Requirements Oversight Council and Joint Force Requirements Process, Joint Chiefs of Staff, Jan. 15, 2026. Accessed: 2026-04-27. [Online]. Available: https://www.jcs.mil/Portals/36/Documents/Library/Manuals/CJCSM%205123.01.pdf

[104] O. of the Under Secretary of War for Research and Engineering, "Technology Readiness Assessment Guide," Guide, Apr. 2020. Accessed: 2025-11-02. [Online]. Available: https://www.cto.mil/tra/

[105] 10 U.S.C. § 4171, Operational Test and Evaluation of defense acquisition programs, 2026. Accessed: 2026-04-27. [Online]. Available: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title10-section4171&num=0&edition=prelim

[106] 10 U.S.C. § 4172, Major systems and munitions programs: survivability testing and lethality testing required before full-scale production, 2026. Accessed: 2026-04-27. [Online]. Available: https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title10-section4172&num=0&edition=prelim

[107] Engineering Duty Officer School, “3.3.3. Joint Capabilities Integration Development System,” Engineering Duty Officer (EDO) School, Coursebook Module 3.3.3, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[108] Engineering Duty Officer School, “3.3.4. Research and Development,” Engineering Duty Officer (EDO) School, Coursebook Module 3.3.4, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[109] Engineering Duty Officer School, “3.3.6. Risk Management and Problem-Solving,” Engineering Duty Officer (EDO) School, Coursebook Module 3.3.6, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034; Version 5.2; coursebook PDF provided by author.

[110] Engineering Duty Officer School, “3.3.7. Environmental Requirements and Responsibilities,” Engineering Duty Officer (EDO) School, Coursebook Module 3.3.7, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[111] Engineering Duty Officer School, “3.4.1. Software Acquisition Environment,” Engineering Duty Officer (EDO) School, Coursebook Module 3.4.1, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[112] Engineering Duty Officer School, “3.4.2. Software Acquisition Management,” Engineering Duty Officer (EDO) School, Coursebook Module 3.4.2, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[113] Engineering Duty Officer School, “3.4.3. Cybersecurity Awareness,” Engineering Duty Officer (EDO) School, Coursebook Module 3.4.3, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[114] Engineering Duty Officer School, “3.5.1. Systems Engineering Overview,” Engineering Duty Officer (EDO) School, Coursebook Module 3.5.1, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[115] Engineering Duty Officer School, “3.5.2. Systems Engineering Analysis and Control: Work Breakdown Structure,” Engineering Duty Officer (EDO) School, Coursebook Module 3.5.2, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[116] Engineering Duty Officer School, “3.5.3. Systems Engineering Analysis and Control: Configuration Management and Technical Reviews,” Engineering Duty Officer (EDO) School, Coursebook Module 3.5.3, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[117] *DoWI 5000.88, Engineering of Defense Systems*, DoWI 5000.88, Instruction, Incorporating Change 2, 2024-04-04; verify latest revision prior to use, Department of War, Nov. 4, 2020. Accessed: 2025-11-19. [Online]. Available: https://www.esd.whs.mil/Directives/issuances/dodi/500088p/

[118] *Engineering of Defense Systems Guidebook*, Guidebook, February 2022 guidebook incorporating Change 2, October 2024, Office of the Under Secretary of Defense for Research and Engineering, Oct. 7, 2024. Accessed: 2026-04-24. [Online]. Available: https://www.cto.mil/wp-content/uploads/2024/10/Eng-Def-Sys-Change2-7October2024-v3.pdf

[119] *Test and Evaluation Enterprise Guidebook*, Guidebook, Director, Operational Test and Evaluation, Aug. 2, 2022. Accessed: 2026-04-24. [Online]. Available: https://www.dote.osd.mil/Portals/97/pub/policies/2022/TE%5C%20Enterprise%5C%20Guidebook%5C%208.02.pdf

[120] Engineering Duty Officer School, “3.5.4. Acquisition Logistics,” Engineering Duty Officer (EDO) School, Coursebook Module 3.5.4, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[121] Engineering Duty Officer School, “3.7.1. Manufacturing and Producibility,” Engineering Duty Officer (EDO) School, Coursebook Module 3.7.1, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[122] Engineering Duty Officer School, “3.7.2. Production, Quality and Manufacturing,” Engineering Duty Officer (EDO) School, Coursebook Module 3.7.2, version 6.1, Mar. 12, 2026, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[123] Engineering Duty Officer School, "3.7.3. Introduction to Continuous Process Improvement (CPI)," Engineering Duty Officer (EDO) School, Coursebook Module 3.7.3, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[124] Engineering Duty Officer School, "3.8.1-4. Integrated Case Study Introduction and Modules," Engineering Duty Officer (EDO) School, Coursebook Module 3.8.1-4, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[125] Engineering Duty Officer School, "3.5.5. Test and Evaluation Part I," Engineering Duty Officer (EDO) School, Coursebook Module 3.5.5, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034; coursebook PDF provided by author.

[126] *DoWI 5000.89, Test and Evaluation*, DoWI 5000.89, Instruction, Incorporating Change 2, 2023-11-21; verify latest revision prior to use, Department of War, Nov. 4, 2020. Accessed: 2025-11-12. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500089p.pdf

[127] *DoWI 5000.02, Operation of the Adaptive Acquisition Framework*, DoWI 5000.02, Instruction, Incorporating Change 3, 2023-10-18, Department of War, Aug. 31, 2022. Accessed: 2025-10-12. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500002p.pdf

[128] Engineering Duty Officer School, "3.5.6. Test and Evaluation Part II," Engineering Duty Officer (EDO) School, Coursebook Module 3.5.6, version 5.2, Mar. 26, 2025, CIN A-4N-0021/-0034; coursebook PDF provided by author.

[129] 10 U.S.C. § 139, Director of Operational Test and Evaluation, 2025. Accessed: 2025-11-12. [Online]. Available: https://uscode.house.gov/

[130] *DoWM 5000.100, Test and Evaluation Master Plans and Test and Evaluation Strategies*, DoWM 5000.100, Manual, Part of the December 9, 2024 DOT&E manual set, Department of War, Dec. 9, 2024. Accessed: 2026-04-26. [Online]. Available: https://www.dote.osd.mil/Portals/97/pub/policies/2024/DoDM%205000.100%20-%20TEMP%20and%20T%26E%20Strategy.pdf

[131] *DoWI 5000.98, Operational Test & Evaluation and Live Fire Test & Evaluation*, DoWI 5000.98, Instruction, Effective December 9, 2024; supersedes OT&E and LFT&E material previously in DoDI 5000.89, Department of War, Dec. 9, 2024. Accessed: 2026-04-26. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500098p.pdf

[132] *DoWM 5000.103, Cyber Developmental Test and Evaluation*, DoWM 5000.103, Manual, Effective February 25, 2026; Cyber Developmental Test and Evaluation, Department of War, Feb. 25, 2026. Accessed: 2026-04-26. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodm/5000103m.pdf

[133] *DoWM 5000.96, Operational and Live Fire Test & Evaluation of Software*, DoWM 5000.96, Manual, Part of the December 9, 2024 DOT&E manual set, Department of War, Dec. 9, 2024. Accessed: 2026-04-26. [Online]. Available: https://www.dote.osd.mil/Portals/97/pub/policies/2024/DoDM%205000.96%20-%20Software%20OT%26E%20and%20LFT%26E.pdf

[134] *DoWI 5000.90, Cybersecurity for Acquisition Decision Authorities and Program Managers*, DoWI 5000.90, Instruction, Verify current revision prior to use, Department of War, Jul. 13, 2023. Accessed: 2026-01-16. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500090p.PDF

[135] Commander, Operational Test and Evaluation Force. "COMOPTEVFOR Official Website," U.S. Navy, Accessed: 2026-04-26. [Online]. Available: https://www.cnf.navy.mil/COMOPTEVFOR/

[136] Joint Interoperability Test Command. "Joint Interoperability Test Command," Defense Information Systems Agency, Accessed: 2026-04-26. [Online]. Available: https://jitc.disa.mil/

[137] *DoWM 5000.102, Modeling and Simulation Verification, Validation and Accreditation for OT&E and LFT&E*, DoWM 5000.102, Manual, Part of the December 9, 2024 DOT&E manual set, Department of War, Dec. 9, 2024. Accessed: 2026-04-26. [Online]. Available: https://www.dote.osd.mil/Portals/97/pub/policies/2024/DoDM%205000.102%20-%20M%26S%20VV%26A%20for%20OT%26E%20and%20LFT%26E.pdf

[138] "DoW Financial Management Regulation 7000.14—R, Volume 3," Office of the Under Secretary of War (Comptroller), Regulation Vol. 3, Jul. 1, 2025. Accessed: 2025-09-29. [Online]. Available: https://comptroller.defense.gov/fmr/

[139] Engineering Duty Officer School, "3.5.7. Ship Trials (T&E Part III)," Engineering Duty Officer (EDO) School, Coursebook Module 3.5.7, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[140] Engineering Duty Officer School, "4.1.1. Fleet Maintenance Assets," Engineering Duty Officer (EDO) School, Coursebook Module 4.1.1, version 6.0, Dec. 8, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[141] Engineering Duty Officer School, "4.1.2. Navy Modernization Program and Alteration Installation Team," Engineering Duty Officer (EDO) School, Coursebook Module 4.1.2, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[142] Engineering Duty Officer School, "4.1.3. Fleet Maintenance Policy and Implementation," Engineering Duty Officer (EDO) School, Coursebook Module 4.1.3, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[143] Engineering Duty Officer School, "4.1.4. AIT Fundamentals and Lessons Learned," Engineering Duty Officer (EDO) School, Coursebook Module 4.1.4, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[144] Engineering Duty Officer School, "4.1.5. Waterfront Organizations, Work Controls, and Safety," Engineering Duty Officer (EDO) School, Coursebook Module 4.1.5, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[145] Engineering Duty Officer School, "4.1.6. Battle Damage Assessment and Repair," Engineering Duty Officer (EDO) School, Coursebook Module 4.1.6, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[146] Engineering Duty Officer School, "4.2.1. CNO Availability Planning," Engineering Duty Officer (EDO) School, Coursebook Module 4.2.1, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[147] *Joint Fleet Maintenance Manual*, COMUSFLTFORCOM/COMPACFLTINST 4790.3, Instruction, Joint Fleet Maintenance Manual (JFMM), Rev E; public detailed text may require ServiceNow access, Commander, U.S. Fleet Forces; Commander, U.S. Pacific Fleet, Apr. 27, 2026. Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Home/RMC/CNRMC/Our-Programs/JFMM/

[148] Engineering Duty Officer School, "4.2.2. CNO Availability Execution," Engineering Duty Officer (EDO) School, Coursebook Module 4.2.2, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[149] *NAVSEA Standard Item 009-001, FY26*, 009-001, Standard Item, General Criteria, Naval Sea Systems Command, Oct. 1, 2024. Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Portals/103/Documents/SSRAC/NSI/FY%2026/009-001_FY26.pdf

[150] *NAVSEA Standard Item 009-060, FY26*, 009-060, Standard Item, Schedule of Events for Contractor-Responsible Items in a Chief of Naval Operations Availability, Naval Sea Systems Command, Oct. 1, 2024. Accessed: 2026-05-27. [Online]. Available: https://www.navsea.navy.mil/Portals/103/Documents/SSRAC/NSI/FY%2026/009-060%20FY26_001_pkg.pdf

[151] *TS9090-310, Alterations Installed by Alteration Installation Teams*, TS9090-310, Technical Specification, Alterations Installed by Alteration Installation Teams; CAC-enabled, Naval Sea Systems Command, Jun. 1, 2023. Accessed: 2025-11-14. [Online]. Available: https://nde.navy.mil

[152] Engineering Duty Officer School, "5.1.1. Naval Justice & Standards," Engineering Duty Officer (EDO) School, Coursebook Module 5.1.1, version 2025.10, Dec. 19, 2025, Intermediate Leader Legal Course master slides (Oct 2025); coursebook PDF provided by author.

[153] Engineering Duty Officer School, "5.1.3. Ethics and Acquisition," Engineering Duty Officer (EDO) School, Coursebook Module 5.1.3, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[154] Engineering Duty Officer School, “5.1.4. Presentation Techniques,” Engineering Duty Officer (EDO) School, Coursebook Module 5.1.4, version 6.0, Dec. 9, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[155] Engineering Duty Officer School, “5.1.5. Team Building,” Engineering Duty Officer (EDO) School, Coursebook Module 5.1.5, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[156] Engineering Duty Officer School, “5.1.6. Leading People,” Engineering Duty Officer (EDO) School, Coursebook Module 5.1.6, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[157] Engineering Duty Officer School, “5.1.8. Myers-Briggs Type Indicator,” Engineering Duty Officer (EDO) School, Coursebook Module 5.1.8, version 6.0, Dec. 10, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[158] Engineering Duty Officer School, “5.1.9. California Psychological Inventory 260,” Engineering Duty Officer (EDO) School, Coursebook Module 5.1.9, version 6.0, Dec. 11, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[159] Engineering Duty Officer School, “5.1.10. High Consequence Event Management,” Engineering Duty Officer (EDO) School, Coursebook Module 5.1.10, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[160] Engineering Duty Officer School, “5.2.1. Defense Acquisition Workforce Improvement Act,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.1, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[161] Engineering Duty Officer School, “5.2.2. Engineering Duty Officer Qualification Program,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.2, version 6.0, Dec. 17, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[162] Engineering Duty Officer School, “5.2.3. Career Topics,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.3, version 6.2, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.2; coursebook PDF provided by author.

[163] Engineering Duty Officer School, “5.2.6. Mentoring Tips,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.6, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[164] Engineering Duty Officer School, “5.2.8. Warrior Toughness,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.8, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[165] Engineering Duty Officer School, “5.2.9. Critical Conversations,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.9, version 6.0, Dec. 12, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[166] Engineering Duty Officer School, “5.2.10. Leader Development Framework,” Engineering Duty Officer (EDO) School, Coursebook Module 5.2.10, version 6.0, Dec. 18, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[167] Engineering Duty Officer School, “6.1.1. Civilian Personnel Overview,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.1, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[168] U.S. Office of Personnel Management. “Classifying general schedule positions,” Accessed: 2026-05-12. [Online]. Available: https://www.opm.gov/policy-data-oversight/classification-qualifications/classifying-general-schedule-positions/

[169] U.S. Office of Personnel Management. “Direct hire authority,” Accessed: 2026-05-12. [Online]. Available: https://www.opm.gov/policy-data-oversight/hiring-information/direct-hire-authority/

[170] Engineering Duty Officer School, “6.1.2. Workforce Shaping and Classification,” Engineering Duty Officer (EDO) School, Coursebook Module 6.1.2, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[171] Office of Personnel Management. “5 CFR 532.203: Structure of Regular Wage Schedules,” Accessed: 2026-05-30. [Online]. Available: https://www.ecfr.gov/current/title-5/part-532/section-532.203

[172] U.S. Office of Personnel Management. “How to compute flsa overtime pay,” Accessed: 2026-05-12. [Online]. Available: https://www.opm.gov/policy-data-oversight/pay-leave/pay-administration/fact-sheets/how-to-compute-flsa-overtime-pay/

[173] Engineering Duty Officer School, "6.1.3. Equal Employment Opportunity," Engineering Duty Officer (EDO) School, Coursebook Module 6.1.3, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[174] U.S. Equal Employment Opportunity Commission. "Harassment," Accessed: 2026-05-12. [Online]. Available: https://www.eeoc.gov/harassment

[175] U.S. Equal Employment Opportunity Commission. "Overview of federal sector eeo complaint process," Accessed: 2026-05-12. [Online]. Available: https://www.eeoc.gov/federal-sector/overview-federal-sector-eeo-complaint-process

[176] *OPNAVINST 5354.1H: Navy Equal Opportunity Program Manual*, Instruction, Apr. 25, 2023. Accessed: 2025-12-18. [Online]. Available: https://www.mynavyhr.navy.mil/Support-Services/Equal-Opportunity/Instruction-Manuals/

[177] Engineering Duty Officer School, "6.1.4. Employee Relations," Engineering Duty Officer (EDO) School, Coursebook Module 6.1.4, version 6.0, Dec. 15, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[178] *DoD Civilian Personnel Management System: Performance Management and Appraisal Program*, Instruction, DoDI 1400.25, Volume 431, Change 4, Aug. 18, 2025. Accessed: 2026-05-12. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/140025/140025_vol431.PDF

[179] U.S. Office of Personnel Management. "Updated Guidance on President Trump's Executive Order Strengthening Probationary Periods in the Federal Service," Accessed: 2026-05-30. [Online]. Available: https://www.opm.gov/chcoc/latest-memos/updated-guidance-on-president-trump-s-executive-order-strengthening-probationary-periods-in-the-federal-service.pdf

[180] U.S. Office of Personnel Management. "Fact sheet: Within-grade increases," Accessed: 2025-12-17. [Online]. Available: https://www.opm.gov/policy-data-oversight/pay-leave/pay-administration/fact-sheets/within-grade-increases/

[181] *National guard technician personnel handbook*, Handbook, 2023. Accessed: 2025-12-17. [Online]. Available: https://tmo.ngb.army.mil/public/

[182] U.S. Office of Personnel Management. "Who can get a quality step increase?" Accessed: 2026-05-12. [Online]. Available: https://www.opm.gov/frequently-asked-questions/new-performance-management-faq/performance-management-step-increases/who-can-get-a-quality-step-increase/

[183] *Separation of Employees with Unacceptable Performance*, Memorandum, Two-year exception to certain DoDI 1400.25, Volume 431 requirements for unacceptable-performance removals under 5 U.S.C. 7513, Sep. 30, 2025. Accessed: 2026-05-14. [Online]. Available: https://media.defense.gov/2025/Sep/30/2003812417/-1/-1/1/SEPARATION-OF-EMPLOYEES-WITH-UNACCEPTABLE-PERFORMANCE.PDF

[184] U.S. Merit Systems Protection Board. "Performance-based actions under chapters 43 and 75 of title 5: Similarities and differences," Accessed: 2026-05-12. [Online]. Available: https://www.mspb.gov/studies/adverse_action_report/4_Performance-Based%20Actions.htm

[185] U.S. Merit Systems Protection Board. "Adverse actions: Determining the penalty," Accessed: 2026-05-12. [Online]. Available: https://www.mspb.gov/studies/adverse_action_report/10_DeterminingthePenalty.htm

[186] Engineering Duty Officer School, "6.1.5. Labor Relations," Engineering Duty Officer (EDO) School, Coursebook Module 6.1.5, version 6.1, Dec. 19, 2025, CIN A-4N-0021/-0034; Version 6.1; coursebook PDF provided by author.

[187] Federal Labor Relations Authority. "The Statute: Section 7114. Representation Rights and Duties," Accessed: 2026-05-12. [Online]. Available: https://www.flra.gov/resources-training/resources/statute-and-regulations/statute/statute-subchapter-ii-rights-and-3

[188] Federal Labor Relations Authority, *Department of the Treasury, Internal Revenue Service, Kansas City Service Center, 3 FLRA 513*, Decision, Establishes past-practice criteria: clear, consistent, long-standing, and mutually accepted, 1980. Accessed: 2025-12-18. [Online]. Available: https://www.flra.gov/decisions/v03/03-082.html

[189] Federal Labor Relations Authority. "The Statute: Section 7118. Prevention of Unfair Labor Practices," Accessed: 2026-05-12. [Online]. Available: https://www.flra.gov/resources-training/resources/statute-and-regulations/statute/statute-subchapter-ii-rights-and-7

[190] Engineering Duty Officer School, "6.1.6. CIVPERS Wrap-up," Engineering Duty Officer (EDO) School, Coursebook Module 6.1.6, version 6.0, Dec. 18, 2025, CIN A-4N-0021/-0034; Version 6.0; coursebook PDF provided by author.

[191] Joint Staff J-8, *Manual for the Operation of the Joint Capabilities Integration and Development System*, Manual, Distribution A; includes updates published by DAU on 2024-01-12, Oct. 30, 2021. Accessed: 2025-11-12. [Online]. Available: https://www.dau.edu/sites/default/files/2024-01/Manual%5C%20-%5C%20JCIDS%5C%20Oct%5C%202021.pdf

[192] 10 U.S.C. § 181, Joint Requirements Oversight Council authorities and procedures, 2023. Accessed: 2025-11-12. [Online]. Available: https://uscode.house.gov/

[193] *DoWD 5000.01, The Defense Acquisition System*, DoWD 5000.01, Directive, Incorporating Change 1, 2023-11-09, Department of War, Sep. 9, 2020. Accessed: 2025-10-12. [Online]. Available: https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodd/500001p.pdf

[194] *Defense Acquisition Guidebook*, Guidebook, Living document; recency check pending because the DAU site requires an interactive browser challenge as of 2025-11-02, Department of War. Accessed: 2025-11-02. [Online]. Available: https://www.dau.edu/guidebooks/Shared%5C%20Documents/DAG.pdf

[195] Portfolio Acquisition Executive Marine Corps. "Our Organization," Accessed: 2026-06-04. [Online]. Available: https://www.pae.marines.mil/Organization/

[196] Department of the Navy, "U.S. Navy Shipbuilding Plan," Department of the Navy, Report, May 2026. Accessed: 2026-06-04. [Online]. Available: https://media.defense.gov/2026/May/11/2003928909/-1/-1/1/NAVY%20SHIPBUILDING%20PLAN%20MAY%202026.PDF

[197] Strategic Systems Programs. "PAE SSP Employees Bring Their Children to Work," Accessed: 2026-06-04. [Online]. Available: https://www.ssp.navy.mil/STEM/STEM-News/Article/4477482/pae-ssp-employees-bring-their-children-to-work/

[198] Marine Corps Combat Development Command. "Department of the Navy Creates Portfolio Acquisition Executive-Marine Corps to Accelerate Modernization," Accessed: 2026-06-04. [Online]. Available: https://www.29palms.marines.mil/Articles/Article/4460140/department-of-the-navy-creates-portfolio-aqcuisition-executive-marine-corps-to/

[199] MyNavy HR. "June 2026 Public Roster." United States Navy Flag Officers, public roster, Accessed: 2026-06-04. [Online]. Available: https://www.mynavyhr.navy.mil/Portals/55/Career/Detailing/Flag/June%202026%20Public%20Roster.pdf?ver=r_dOGV6Pq72YA2B_QzvbeA%3D%3D

[200] Strategic Systems Programs. "Rear Admiral Michael S. Richman." Deputy Director, Regional Strike Systems, Accessed: 2026-04-09. [Online]. Available: https://www.ssp.navy.mil/About-Us/Leadership/Biographies/Article/3963294/rear-admiral-michael-s-richman/

[201] D. L. Caudle, *CNO Day One Message: Foundry–Fleet–Fight*, NAVADMIN message to the fleet, Day One message issued upon assuming duties as the 34th Chief of Naval Operations; PDF mirrored at tex/assets/reference-docs/CNO34-DAY-ONE-MESSAGE-FOUNDRY-FLEET-FLIGHT.pdf, Sep. 3, 2025. Accessed: 2025-12-12. [Online]. Available: https://media.defense.gov/2025/Sep/03/2003791814/-1/-1/1/CNO34-DAY-ONE-MESSAGE-FOUNDRY-FLEET-FLIGHT.PDF

[202] D. L. Caudle, *C-Note #1: Sailors First*, NAVADMIN 186/25 message to the Fleet, C-Note #1: Sailors First, Sep. 3, 2025. Accessed: 2025-12-18.

[203] D. L. Caudle, *C-Note #2: Foundry Always*, NAVADMIN 203/25 message to the Fleet, C-Note #2: Foundry Always, Sep. 22, 2025. Accessed: 2025-12-18.

[204] D. L. Caudle, *C-Note #3: World-Class Fleet*, NAVADMIN 241/25 message to the Fleet, C-Note #3: World-Class Fleet, Dec. 1, 2025. Accessed: 2025-12-18.

[205] D. L. Caudle, *Charge of Command*, NAVADMIN 225/25 message to the Fleet, Charge of Command announcement for Commanding Officers, Oct. 30, 2025. Accessed: 2025-12-18.

[206] Naval Information Warfare Systems Command. "Project Overmatch and Five Eyes Coalition Partners Strengthen Ties." Story ID 551112, Accessed: 2025-12-05. [Online]. Available: https://www.dvidshub.net/news/551112/project-overmatch-and-five-eyes-coalition-partners-strengthen-ties

[207] L. Chen. "Commercial Tech Partnerships Drive Unprecedented Progress for Project Overmatch and Navy Capability," Naval Information Warfare Systems Command, Accessed: 2025-11-13. [Online]. Available: https://www.dvidshub.net/news/501030/commercial-tech-partnerships-drive-unprecedented-progress-project-overmatch-and-navy-capability

[208] L. Chen. "NAVWAR's New Commander: Rear Adm. Okano Takes Charge as Rear Adm. Small Retires After Distinguished Service," Naval Information Warfare Systems Command, Accessed: 2025-11-13. [Online]. Available: https://www.dvidshub.net/news/478261/navwars-new-commander-rear-adm-okano-takes-charge-rear-adm-small-retires-after-distinguished-service

[209] "Restoring Common Sense to Federal Procurement." Publication anchor; direct access may require an anti-bot browser challenge in some environments., Federal Register, Accessed: 2026-02-06. [Online]. Available: https://www.federalregister.gov/documents/2025/04/17/2025-06763/restoring-common-sense-to-federal-procurement

[210] Acquisition.gov. "FAR Overhaul - FAQs," Accessed: 2026-04-27. [Online]. Available: https://www.acquisition.gov/far-overhaul/faqs

[211] Acquisition.gov. "Threshold Changes - October 1st, 2025," Accessed: 2026-04-27. [Online]. Available: https://www.acquisition.gov/threshold-changes

[212] National Defense Authorization Act for Fiscal Year 2026, Public Law 119-60, Dec. 18, 2025. Accessed: 2026-04-27. [Online]. Available: https://www.govinfo.gov/content/pkg/PLAW-119publ60/pdf/PLAW-119publ60.pdf

[213] Secretary of War, "Unifying the Department's Arms Transfer and Security Cooperation Enterprise to Improve Efficiency and Enable Burden-Sharing," Memorandum OSD012502-25, Nov. 7, 2025. Accessed: 2025-11-14. [Online]. Available: WAS/Unifying%5C%20the%5C%20Department's%5C%20Arms%5C%20Transfer%5C%20and%5C%20Security%5C%20Cooperation%5C%20Enterprise%5C%20to%5C%20Improve%5C%20Efficiency%5C%20and%5C%20Enable%5C%20Burden-Sharing%5C%20OSD0.pdf

[214] Secretary of War, "Establishment of the National Security Capital Forum," Memorandum OSD010312-25, Nov. 10, 2025. Accessed: 2025-11-14. [Online]. Available: WAS/Establishment%5C%20of%5C%20the%5C%20National%5C%20Security%5C%20Capital%5C%20Forum%5C%20OSD010312-25%5C%20RES%5C%20FINAL.pdf

[215] "2026 National Defense Strategy," Department of War, Strategy, Jan. 23, 2026. Accessed: 2026-04-27. [Online]. Available: https://media.defense.gov/2026/Jan/23/2003864773/-1/-1/0/2026-NATIONAL-DEFENSE-STRATEGY.PDF

[216] "National Security Strategy of the United States of America," The White House, Strategy, Nov. 2025. Accessed: 2026-04-27. [Online]. Available: https://www.whitehouse.gov/wp-content/uploads/2025/12/2025-National-Security-Strategy.pdf

[217] Department of the Navy. "Navy FY27 Budget Request," Accessed: 2026-04-27. [Online]. Available: https://www.navy.mil/Press-Office/Speeches/display-speech/Article/4466519/navy-fy27-budget-request/

[218] Department of the Navy, "Department of the Navy Cyber Strategy," Strategy, Nov. 17, 2023, DVIDS release, Doc ID 8152628. Accessed: 2026-01-16. [Online]. Available: https://d1ldvf68ux039x.cloudfront.net/thumbs/photos/2311/8152628/1200w/4.pdf

[219] Chief of Naval Operations, "CNO Navigation Plan 2022," Strategy, Jul. 26, 2022, Official PDF; verify current version. Accessed: 2026-01-16. [Online]. Available: https://media.defense.gov/2022/Jul/26/2003042389/-1/-1/1/navigation%20plan%202022_signed.pdf

[220] Congressional Research Service, "Navy columbia-class (ssbn-826) program: Background and issues," Mar. 2025. Accessed: 2025-12-24. [Online]. Available: https://www.congress.gov/crs-product/R41129

[221] Z. B. Fletcher, "Delays in navy's next-gen submarine threaten us seapower, report says," *Defense News*, Jul. 2025. Accessed: 2025-12-24. [Online]. Available: https://www.defensenews.com/naval/2025/07/11/delays-in-navys-next-gen-submarine-threaten-us-seapower-report-says/

[222] S. LaGrone, "First columbia-class sub 60 percent complete, next year 'pivotal,' says gd electric boat," *USNI News*, Oct. 2025. Accessed: 2025-12-24. [Online]. Available: https://news.usni.org/2025/10/24/first-columbia-class-sub-60-complete-next-year-pivotal-says-gd-electric-boat

[223] Staff, "Navy secures multi-ship amphib buy, saving estimated $1b: Lawmaker," *Breaking Defense*, Aug. 2024. Accessed: 2025-12-24. [Online]. Available: https://breakingdefense.com/2024/08/navy-secures-multi-ship-amphib-buy-saving-estimated-1b-lawmaker/

[224] U.S. GAO, "Amphibious warfare fleet: Navy needs to ensure ships are available for marines," Dec. 2024. Accessed: 2025-12-24. [Online]. Available: https://www.gao.gov/products/gao-25-106728

[225] Staff, "Navy awards $377 million for next phase of psns dry dock 4 seismic upgrades," *Kitsap Sun*, Sep. 2025. Accessed: 2025-12-24. [Online]. Available: https://www.kitsapeda.org/news/p/item/64485/navy-awards-377-million-for-next-phase-of-psns-dry-dock-4-seismic-upgrades

[226] J. Katz, "Navy's shipyard revamp strategy evolves amid obstacles, years into $20b effort," *Breaking Defense*, Jul. 2025. Accessed: 2025-12-24. [Online]. Available: https://breakingdefense.com/2025/07/navys-shipyard-revamp-strategy-evolves-amid-obstacles-years-into-20b-effort/

[227] Naval Sea Systems Command. "First of 2 New Portal Cranes Arrives to Support PSNS & IMF Operations," Accessed: 2026-04-27. [Online]. Available: https://www.navsea.navy.mil/Media/News/Article-View/Article/4260306/first-of-2-new-portal-cranes-arrives-to-support-psns-imf-operations/

[228] Naval Facilities Engineering Systems Command. "Chief of Naval Operations Visits the Future Dry Dock 5," Accessed: 2026-04-27. [Online]. Available: https://www.navfac.navy.mil/Home/News-Detail/Article/4346140/chief-of-naval-operations-visits-the-future-dry-dock-5/

[229] Naval Sea Systems Command. "NAVFAC Pacific Awards $2.8-Billion Contract Task Order for Pearl Harbor Dry Dock Replacement," Accessed: 2026-04-27. [Online]. Available: https://www.navsea.navy.mil/Media/News/Article/3327775/navfac-pacific-awards-28-billion-contract-task-order-for-pearl-harbor-dry-dock/

[230] Naval Facilities Engineering Systems Command. "NAVFAC Awards $15 Billion Contract for Critical Enhancements in the Pacific," Accessed: 2026-04-27. [Online]. Available: https://www.navfac.navy.mil/Home/News-Detail/Article/4305539/navfac-awards-15-billion-contract-for-critical-enhancements-in-the-pacific/